# ISC Class XII Mathematics

## Chapter 5
# Differentiation

## 350 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

### Topics Covered



Compiled Answer Key – Step-by-step ISC board-style solutions

# Contents



# Index of Problems

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| **Exercise 5.1 – Continuity: Checking Continuity at a Point** | | | | |
| 1 | *(see parts below)* | 1 | Numerical | 5.1 |
| (i) | Examine the following function for continuity at the indicated point: … | 1 | Numerical | 5.1 |
| (ii) | Examine the following function for continuity at the indicated point: … | 1 | Numerical | 5.1 |
| 2 | If $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$ and $f(x) = 2$ when $x = 1$, show that … | 1 | Proof | 5.1 |
| 3 | A function $f$ is defined as $f(x) =$ (piecewise definition). Show that $f$ is … | 1 | Proof | 5.1 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 4 | Is the function $f$ defined by $f(x) =$ (piecewise definition) continuous at (i) ... | 2 | Numerical | 5.1 |
| 5 | Is the function $f$ defined by $f(x) =$ (piecewise definition) continuous at ... | 1 | Numerical | 5.1 |
| 6 | Is the function $f$ defined by $f(x) =$ (piecewise definition) continuous at ... | 1 | Numerical | 5.1 |
| 7 | If $f(x) =$ (piecewise definition), find $k$ so that $f$ may be continuous at ... | 3 | Numerical | 5.1 |
| 8 | If $f(x) =$ (piecewise definition), find $k$ so that $f$ may be continuous at ... | 3 | Numerical | 5.1 |
| 9 | If $f(x) =$ (piecewise definition), find $\lambda$ so that $f$ may be continuous at ... | 3 | Numerical | 5.1 |
| 10 | *(see parts below)* | 4 | Numerical | 5.1 |
| (i) | $f(x) = \frac{x^2-9}{x-3}$ is not defined at $x = 3$. What type of removable ... | 4 | Numerical | 5.1 |
| (ii) | If $f(x) =$ (piecewise definition), find $k$ so that the function $f$ may be ... | 3 | Numerical | 5.1 |
| (iii) | Determine the value of '$k$' for which the following function is continuous at $x = 3$: ... | 3 | Numerical | 5.1 |
| (iv) | If $f(x) =$ (piecewise definition), find $k$ so that the function $f$ may be ... | 3 | Numerical | 5.1 |
| 11 | If $f(x) =$ (piecewise definition), find $k$ so that the function $f$ may be ... | 3 | Numerical | 5.1 |
| 12 | Is the function $f$ defined by $f(x) = \tan x$ continuous at $x = \frac{\pi}{2}$? | 1 | Numerical | 5.1 |
| 13 | Is the function $f$ defined by $f(x) = \lvert x\rvert$ continuous at $x = 0$? | 5 | Numerical | 5.1 |
| 14 | Is the function $f$ defined by $f(x) = \lvert x-1\rvert$ continuous at $x = 1$? | 5 | Numerical | 5.1 |
| 15 | Is the function $f$ defined by $f(x) = x - \lvert x\rvert$ continuous at $x = 0$? | 5 | Numerical | 5.1 |
| 16 | *(see parts below)* | 2 | Numerical | 5.1 |
| (i) | Examine the following function for continuity at the indicated point: ... | 2 | Numerical | 5.1 |
| (ii) | Examine the following function for continuity at the indicated point: ... | 2 | Numerical | 5.1 |
| 17 | Is the function $f$ defined by $f(x) =$ (piecewise definition) continuous at ... | 2 | Numerical | 5.1 |
| 18 | *(see parts below)* | 5 | Numerical | 5.1 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| (i) | Examine the following function for continuity at $x = 0$: ... | 5 | Numerical | 5.1 |
| (ii) | Examine the following function for continuity at $x = 4$: ... | 5 | Numerical | 5.1 |
| 19 | *(see parts below)* | 6 | Numerical | 5.1 |
| (i) | Examine the following function for continuity at $x = 0$: ... | 6 | Numerical | 5.1 |
| (ii) | Examine the following function for continuity at $x = 0$: ... | 6 | Numerical | 5.1 |
| (iii) | Examine the following function for continuity at $x = 0$: ... | 6 | Numerical | 5.1 |
| 20 | *(see parts below)* | 3 | Numerical | 5.1 |
| (i) | If $f(x) =$ (piecewise definition) is continuous at $x = 5$, find the value of $k$. | 3 | Numerical | 5.1 |
| (ii) | Find the value of $k$ so that the function $f(x) =$ (piecewise definition) is ... | 3 | Numerical | 5.1 |
| (iii) | For what value of $k$ is the following function continuous at $x = 2$? ... | 3 | Numerical | 5.1 |
| (iv) | Find the value of $k$ so that the function $f$ defined by ... | 3 | Numerical | 5.1 |
| (v) | For what value of $k$ is the function $f(x) =$ (piecewise definition) continuous at ... | 3 | Numerical | 5.1 |
| (vi) | Find the value of $k$ for which the function $f$ given as ... | 3 | Numerical | 5.1 |
| (vii) | Find the value(s) of '$\lambda$' for which the function $f$ given as ... | 3 | Numerical | 5.1 |
| 21 | Find the relationship between $a$ and $b$ so that the function $f$ defined by ... | 3 | Numerical | 5.1 |
| 22 | If the function $f$ defined by $f(x) =$ (piecewise definition) is continuous at ... | 3 | Numerical | 5.1 |
| 23 | If $f(x) =$ (piecewise definition) is continuous at $x = 0$, then find the values ... | 3 | Numerical | 5.1 |
| 24 | *(see parts below)* | 4 | Numerical | 5.1 |
| (i) | Consider the function defined as follows, for $x \neq 0$: ... | 4 | Numerical | 5.1 |
| (ii) | Consider the function defined as follows, for $x \neq 0$: ... | 4 | Numerical | 5.1 |
| 25 | Examine the function $f(x) =$ (piecewise definition) for continuity at $x = 0$. | 7 | Numerical | 5.1 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 26 | Show that the function $f(x) = \|x\| + \|x-1\|, x \in \mathbb{R}$, is continuous both at … | 5 | Proof | 5.1 |
| 27 | Examine the function $f(x) =$ (piecewise definition) for continuity at $x = 0$. | 8 | Numerical | 5.1 |

**Exercise 5.2 – Continuity: Piecewise Functions and Finding Constants**

| | | | | |
|---|---|---|---|---|
| 1 | Is the function $f$ defined by $f(x) = [x]$ continuous at: | 1 | Numerical | 5.2 |
| (i) | $x = 0$ | 1 | Numerical | 5.2 |
| (ii) | $x = 1$ | 1 | Numerical | 5.2 |
| (iii) | $x = \frac{1}{2}$? | 1 | Numerical | 5.2 |
| 2 | *(see parts below)* | 2 | Numerical | 5.2 |
| (i) | Examine the following function for continuity: $f(x) =$ (piecewise definition) | 2 | Numerical | 5.2 |
| (ii) | Examine the following function for continuity: $f(x) =$ (piecewise definition) | 2 | Numerical | 5.2 |
| 3 | *(see parts below)* | 2 | Numerical | 5.2 |
| (i) | Examine the following function for continuity: $f(x) =$ (piecewise definition) | 2 | Numerical | 5.2 |
| (ii) | Examine the following function for continuity: $f(x) =$ (piecewise definition) | 2 | Numerical | 5.2 |
| 4 | *(see parts below)* | 5 | Numerical | 5.2 |
| (i) | Examine the continuity of $f(x) = \|x\|$. | 5 | Numerical | 5.2 |
| (ii) | Examine the continuity of $f(x) = \|x-5\|$. | 5 | Numerical | 5.2 |
| 5 | *(see parts below)* | 1 | Numerical | 5.2 |
| (i) | Examine the continuity of $f(x) = \frac{x^2-25}{x+5}$. | 1 | Numerical | 5.2 |
| (ii) | Examine the continuity of $f(x) = \frac{1}{x-5}$. | 1 | Numerical | 5.2 |
| 6 | *(see parts below)* | 6 | Numerical | 5.2 |
| (i) | Examine the following function for continuity: $f(x) =$ (piecewise definition) | 6 | Numerical | 5.2 |
| (ii) | Examine the following function for continuity: $f(x) =$ (piecewise definition) | 5 | Numerical | 5.2 |
| 7 | *(see parts below)* | 2 | Numerical | 5.2 |
| (i) | Examine the following function for continuity: $f(x) =$ (piecewise definition) | 2 | Numerical | 5.2 |
| (ii) | Examine the continuity of $f(x) = \tan x$ for $0 \le x \le \frac{\pi}{4}$. | 1 | Numerical | 5.2 |
| 8 | *(see parts below)* | 2 | Numerical | 5.2 |
| (i) | Locate the points of discontinuity (if any) of the function: … | 2 | Numerical | 5.2 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| (ii) | Locate the points of discontinuity (if any) of the function: ... | 2 | Numerical | 5.2 |
| 9 | *(see parts below)* | 1 | Numerical | 5.2 |
| (i) | Locate the points of discontinuity (if any) of the function: ... | 1 | Numerical | 5.2 |
| (ii) | Locate the points of discontinuity (if any) of the function: ... | 6 | Numerical | 5.2 |
| 10 | *(see parts below)* | 2 | Numerical | 5.2 |
| (i) | Locate the points of discontinuity (if any) of the function: ... | 2 | Numerical | 5.2 |
| (ii) | Locate the points of discontinuity (if any) of the function: ... | 2 | Numerical | 5.2 |
| 11 | Find the value of the constant $k$ so that the function $f$ defined by ... | 3 | Numerical | 5.2 |
| 12 | Determine the value of $k$ for which the following function is continuous at $x = 3$: ... | 3 | Numerical | 5.2 |
| 13 | *(see parts below)* | 3 | Numerical | 5.2 |
| (i) | For what choice of $a$ and $b$ is the following function continuous: ... | 3 | Numerical | 5.2 |
| (ii) | For what choice of $a$ and $b$ is the following function continuous: ... | 3 | Numerical | 5.2 |
| 14 | Consider $f(x) =$ (piecewise definition) | 4 | Numerical | 5.2 |
| (i) | Determine if there is a removable discontinuity at $x = 3$. | 4 | Numerical | 5.2 |
| (ii) | Classify it as missing point or isolated point. | 4 | Numerical | 5.2 |
| (iii) | Redefine the function to make it continuous. | 4 | Numerical | 5.2 |
| **Exercise 5.3 – Continuity: Algebra and Composition of Continuous Functions** | | | | |
| 1 | Is $\tan x$ a continuous function? | 9 | Numerical | 5.3 |
| 2 | State which of the following functions are continuous? | 9 | Numerical | 5.3 |
| (i) | $\sin x + \cos x$ | 9 | Numerical | 5.3 |
| (ii) | $\sin x - \cos x$ | 9 | Numerical | 5.3 |
| (iii) | $\sin x \cdot \cos x$ | 9 | Numerical | 5.3 |
| 3 | Is the function $f$ defined by $f(x) = 2 - 3x + \|x\|$ a continuous function? | 5 | Numerical | 5.3 |
| 4 | Is the function $f$ defined by $f(x) = \|\sin x\|$ a continuous function? | 10 | Numerical | 5.3 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 5 | Is the function $f$ defined by $f(x) = \cos\lvert x\rvert$ a continuous function? | 10 | Numerical | 5.3 |
| 6 | Prove that the following functions are continuous: | 9 | Proof | 5.3 |
| (i) | $\frac{2x^3-7x^2+3}{(x-1)(x+3)}$ | 9 | Proof | 5.3 |
| (ii) | $\lvert\sec x + \tan x\rvert$ | 9 | Proof | 5.3 |
| 7 | Examine the following functions for continuity: | 10 | Numerical | 5.3 |
| (i) | $\cos(x^2)$ | 10 | Numerical | 5.3 |
| (ii) | $\sin(3x^2 - 5)$ | 10 | Numerical | 5.3 |
| (iii) | $\tan^{-1}(2x^2 + 3)$ | 10 | Numerical | 5.3 |
| **Exercise 5.4 – Differentiability: Definition and Relation to Continuity** | | | | |
| 1 | If a function is derivable at a point, is it necessary that it must be continuous at that ... | 11 | Numerical | 5.4 |
| 2 | If a function is continuous at a point, is it necessary that it must be derivable at that ... | 11 | Numerical | 5.4 |
| 3 | Is the function $f(x) = \lvert x\rvert$ derivable at $x = 0$? | 12 | Numerical | 5.4 |
| 4 | Is the function $f(x) = \cot x$ derivable at $x = 0$? | 13 | Numerical | 5.4 |
| 5 | Is the function $f(x) = \sec x$ derivable at $x = \frac{\pi}{2}$? | 13 | Numerical | 5.4 |
| 6 | When a function is called derivable? | 14 | Numerical | 5.4 |
| 7 | If a function is derivable at $x = c$, then write the value of $\lim_{x\to c} f(x)$. | 11 | Numerical | 5.4 |
| 8 | Which of the following functions are derivable? | 13 | Numerical | 5.4 |
| (i) | $\sin x$ | 13 | Numerical | 5.4 |
| (ii) | $\cos x$ | 13 | Numerical | 5.4 |
| (iii) | $\tan x$ | 13 | Numerical | 5.4 |
| (iv) | $\cot x$ | 13 | Numerical | 5.4 |
| (v) | $\sec x$ | 13 | Numerical | 5.4 |
| (vi) | $\csc x$ | 13 | Numerical | 5.4 |
| 9 | Examine the function $f(x) =$ (piecewise definition) for differentiability at ... | 15 | Numerical | 5.4 |
| 10 | If $f(2) = 4$ and $f'(2) = 4$, then evaluate ... | 16 | Numerical | 5.4 |
| 11 | Examine the following function for continuity at $x = 1$ and differentiability at $x = 2$: ... | 15 | Numerical | 5.4 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 12 | Show that the function $f(x) = 2x - \|x\|$ is continuous at $x = 0$ but not differentiable ... | 12 | Proof | 5.4 |
| 13 | Show that $f(x) = \|x - 4\|$ is continuous but not differentiable at $x = 4$. | 12 | Proof | 5.4 |
| 14 | If the function $f(x) = \|x - 3\| + \|x - 4\|$, then show that $f$ is not differentiable at ... | 12 | Proof | 5.4 |
| 15 | Show that the function $f$ is continuous at $x = 1$ for all values of $a$ where ... | 17 | Numerical | 5.4 |
| 16 | Prove that the function $f(x) =$ (piecewise definition) is not differentiable at ... | 11 | Proof | 5.4 |
| 17 | If $f(x) =$ (piecewise definition), then show that $f$ is not differentiable at ... | 15 | Proof | 5.4 |
| 18 | If $f(x) =$ (piecewise definition) is a differentiable function in $(0, 2)$, then ... | 17 | Numerical | 5.4 |
| 19 | If the function $f(x) =$ (piecewise definition) is differentiable at $x = 3$, find ... | 17 | Numerical | 5.4 |
| 20 | *(see parts below)* | 18 | Numerical | 5.4 |
| (i) | Examine for continuity and differentiability: ... | 18 | Numerical | 5.4 |
| (ii) | Examine for continuity and differentiability: ... | 18 | Numerical | 5.4 |
| (iii) | Examine for continuity and differentiability: ... | 18 | Numerical | 5.4 |
| 21 | Find the values of $a$ and $b$ so that the following function is differentiable for all ... | 17 | Numerical | 5.4 |
| 22 | Check the differentiability of $f(x) = \|\cos x\|$ at $x = \frac{\pi}{2}$. | 12 | Numerical | 5.4 |
| **Exercise 5.5 – Derivatives of Standard and Absolute Value Functions** | | | | |
| 1 | *(see parts below)* | 19 | Numerical | 5.5 |
| (i) | If $f(x) = \sqrt{4 - x^2}, x \in (-2, 2)$, find $f'(x)$. | 19 | Numerical | 5.5 |
| (ii) | If $f(x) = \sqrt{1 - x^2}, x \in (0, 1)$, find $f'(x)$. | 19 | Numerical | 5.5 |
| 2 | *(see parts below)* | 19 | Numerical | 5.5 |
| (i) | Differentiate $\sqrt{2x - 1}$ w.r.t. $x$. | 19 | Numerical | 5.5 |
| (ii) | Differentiate $(3x^2 - 9x + 5)^9$ w.r.t. $x$. | 19 | Numerical | 5.5 |
| 3 | *(see parts below)* | 20 | Numerical | 5.5 |
| (i) | Differentiate $\sqrt{3x + 2} + \frac{1}{\sqrt{2x^2+4}}$ w.r.t. $x$. | 20 | Numerical | 5.5 |
| (ii) | Differentiate $\sin^3 x + \cos^6 x$ w.r.t. $x$. | 20 | Numerical | 5.5 |
| 4 | *(see parts below)* | 21 | Numerical | 5.5 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| (i) | Differentiate $\sin(x^2)$ w.r.t. $x$. | 21 | Numerical | 5.5 |
| (ii) | Differentiate $\cos\sqrt{x}$ w.r.t. $x$. | 21 | Numerical | 5.5 |
| 5 | *(see parts below)* | 19 | Numerical | 5.5 |
| (i) | Differentiate $\sin(ax+b)$ w.r.t. $x$. | 19 | Numerical | 5.5 |
| (ii) | Differentiate $\tan(2x+3)$ w.r.t. $x$. | 19 | Numerical | 5.5 |
| 6 | *(see parts below)* | 21 | Numerical | 5.5 |
| (i) | Differentiate $\cos(\sin x)$ w.r.t. $x$. | 21 | Numerical | 5.5 |
| (ii) | Differentiate $\sin(\cos(x^2))$ w.r.t. $x$. | 22 | Numerical | 5.5 |
| 7 | *(see parts below)* | 22 | Numerical | 5.5 |
| (i) | Differentiate $\cos(\tan\sqrt{x+1})$ w.r.t. $x$. | 22 | Numerical | 5.5 |
| (ii) | Differentiate $\sqrt{\tan\sqrt{x}}$ w.r.t. $x$. | 22 | Numerical | 5.5 |
| 8 | *(see parts below)* | 23 | Numerical | 5.5 |
| (i) | If $f(x) = x - 1$, find $\frac{d}{dx}((f \circ g)(x))$. | 23 | Numerical | 5.5 |
| (ii) | If $f(x) = x + 7$ and $g(x) = x - 7$, find $\frac{d}{dx}((f \circ f)(x))$. | 23 | Numerical | 5.5 |
| 9 | If ... | 24 | Proof | 5.5 |
| 10 | Find the derivative of $\lvert 2x^2 - 3\rvert$ with respect to $x$. | 25 | Numerical | 5.5 |
| 11 | If $f(x) = \lvert\cos x\rvert$, find $f'(\frac{3\pi}{4})$. | 26 | Numerical | 5.5 |
| 12 | The table below shows some of the values of differentiable functions $u$ and $v$ and their ... | 27 | Numerical | 5.5 |
| 13 | *(see parts below)* | 28 | Numerical | 5.5 |
| (i) | Differentiate $\sqrt{\frac{a^2-x^2}{a^2+x^2}}$ w.r.t. $x$. | 28 | Numerical | 5.5 |
| (ii) | Differentiate $\sin\sqrt{x} + \cos^2\sqrt{x}$ w.r.t. $x$. | 20 | Numerical | 5.5 |
| 14 | *(see parts below)* | 28 | Numerical | 5.5 |
| (i) | Differentiate $\frac{\sin(ax+b)}{\cos(cx+d)}$ w.r.t. $x$. | 28 | Numerical | 5.5 |
| (ii) | Differentiate $\frac{\sin x + x^2}{\cot 2x}$ w.r.t. $x$. | 28 | Numerical | 5.5 |
| 15 | *(see parts below)* | 20 | Numerical | 5.5 |
| (i) | Differentiate $\sin^m x \cos^n x$ w.r.t. $x$. | 20 | Numerical | 5.5 |
| (ii) | Differentiate $\sin^n(ax^2+bx+c)$ w.r.t. $x$. | 22 | Numerical | 5.5 |
| 16 | *(see parts below)* | 29 | Numerical | 5.5 |
| (i) | Differentiate $\frac{\sqrt{a+x}-\sqrt{a-x}}{\sqrt{a+x}+\sqrt{a-x}}$ w.r.t. $x$. | 29 | Numerical | 5.5 |
| (ii) | Differentiate $\frac{\sqrt{x^2+1}+\sqrt{x^2-1}}{\sqrt{x^2+1}-\sqrt{x^2-1}}$ w.r.t ... | 29 | Numerical | 5.5 |
| 17 | If $f(x) = \sqrt{\frac{\sec x - 1}{\sec x + 1}}$, find $f'(x)$. Also find ... | 30 | Numerical | 5.5 |
| 18 | *(see parts below)* | 31 | Numerical | 5.5 |
| (i) | Find the derivative of $\cos(2x + \frac{\pi}{3})$ at $x = \frac{\pi}{3}$. | 31 | Numerical | 5.5 |
| (ii) | Find the derivative of $\frac{1-\sin x}{1+\cos x}$ at $x = \frac{\pi}{2}$. | 28 | Numerical | 5.5 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 19 | If $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, then show that . . . | 32 | Proof | 5.5 |
| 20 | If $y = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$, prove that . . . | 30 | Proof | 5.5 |
| 21 | Differentiate $\lvert\cos x\rvert$ w.r.t. $x$. Is this function differentiable? What can you say . . . | 25 | Numerical | 5.5 |
| 22 | Prove that the derivative of an odd function is always an even function. | 33 | Proof | 5.5 |
| 23 | If $2f(x) + 3f(-x) = x^2 + x + 1$, find $f'(1)$. | 34 | Numerical | 5.5 |
| 24 | If $f(x) = \lvert\tan 2x\rvert$, then find the value of $f'(x)$ at $x = \frac{\pi}{3}$. | 26 | Numerical | 5.5 |
| 25 | If $y = \cos^3(\sec^2 2t)$, find $\frac{dy}{dt}$. | 22 | Numerical | 5.5 |
| 26 | The figure given below shows the graph of a function $y = f(x)$. What is the derivative of . . . | 25 | Numerical | 5.5 |
| **Exercise 5.6 – Derivatives of Inverse Trigonometric Functions** | | | | |
| 1 | Find the domain of differentiability of the following functions: | 35 | Numerical | 5.6 |
| (i) | $\sin^{-1} x$ | 35 | Numerical | 5.6 |
| (ii) | $\cos^{-1} x$ | 35 | Numerical | 5.6 |
| (iii) | $\sec^{-1} x$ | 35 | Numerical | 5.6 |
| (iv) | $\csc^{-1} x$ | 35 | Numerical | 5.6 |
| (v) | $\tan^{-1} x$ | 35 | Numerical | 5.6 |
| (vi) | $\cot^{-1} x$ | 35 | Numerical | 5.6 |
| 2 | Find the derivatives of the following functions: | 36 | Numerical | 5.6 |
| (i) | $\tan^{-1}(x\sqrt{x})$ | 36 | Numerical | 5.6 |
| (ii) | $\cos^{-1}(3x)$ | 36 | Numerical | 5.6 |
| (iii) | $\sqrt{\sin^{-1}(2x)}$ | 36 | Numerical | 5.6 |
| (iv) | $\sin(\tan^{-1} x)$ | 36 | Numerical | 5.6 |
| (v) | $\cot^{-1}\sqrt{x}$ | 36 | Numerical | 5.6 |
| (vi) | $(\tan^{-1} x)^2$ | 36 | Numerical | 5.6 |
| 3 | Find the derivatives of the following functions: | 37 | Numerical | 5.6 |
| (i) | $\sec^{-1}(\csc x)$ | 37 | Numerical | 5.6 |
| (ii) | $\tan^{-1}(\cot x)$ | 37 | Numerical | 5.6 |
| 4 | If $f(x) = \cot^{-1}(\tan x)$, show that $f'(x) = -1$. | 37 | Proof | 5.6 |
| 5 | If $y = \sin^{-1} x + \cos^{-1} x$, find $\frac{dy}{dx}$. | 37 | Numerical | 5.6 |
| 6 | If $f(x) = \sin^{-1} x + \sec^{-1}\frac{1}{x}$, find $f'(x)$. | 37 | Numerical | 5.6 |
| 7 | If $y = \tan^{-1} x + \cot^{-1}\frac{1}{x}$, $x > 0$, find $\frac{dy}{dx}$. | 37 | Numerical | 5.6 |
| 8 | If $y = \sin^{-1}(\cos x) + 3\csc^{-1}(\sec x)$, find $\frac{dy}{dx}$. | 37 | Numerical | 5.6 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 9 | If $y = 2\cos^{-1}(\sin x) + 3\cot^{-1}(\tan x)$, find $\frac{dy}{dx}$. | 37 | Numerical | 5.6 |
| 10 | Differentiate the following functions w.r.t. $x$: | 38 | Numerical | 5.6 |
| (i) | $x\tan^{-1}x$ | 38 | Numerical | 5.6 |
| (ii) | $x\cos^{-1}\sqrt{x}$ | 38 | Numerical | 5.6 |
| (iii) | $\frac{\sin^{-1}x}{x}$ | 38 | Numerical | 5.6 |
| 11 | Differentiate the following functions w.r.t. $x$: | 39 | Numerical | 5.6 |
| (i) | $\cos^{-1}(\frac{1-x}{1+x})$ | 39 | Numerical | 5.6 |
| (ii) | $\sec^{-1}(\frac{1}{x-1})$ | 39 | Numerical | 5.6 |
| (iii) | $\sin^{-1}(\frac{1}{\sqrt{x+1}})$ | 39 | Numerical | 5.6 |
| 12 | Differentiate the following functions w.r.t. $x$: | 37 | Numerical | 5.6 |
| (i) | $\sin^{-1}x + \sin^{-1}\sqrt{1-x^2}$ | 37 | Numerical | 5.6 |
| (ii) | $x(\sin^{-1}x)^2 + 2\sqrt{1-x^2}\sin^{-1}x - 2x$ | 37 | Numerical | 5.6 |
| 13 | Differentiate the following functions w.r.t. $x$: | 37 | Numerical | 5.6 |
| (i) | $\tan^{-1}(\sin x + \cos x)$ | 37 | Numerical | 5.6 |
| (ii) | $\tan^{-1}(\frac{\sin x}{1+\cos x})$ | 37 | Numerical | 5.6 |
| (iii) | $\tan^{-1}(\frac{1+\cos x}{\sin x})$ | 37 | Numerical | 5.6 |
| (iv) | $\tan^{-1}(\frac{\cos x}{1+\sin x})$ | 37 | Numerical | 5.6 |
| 14 | Differentiate the following functions w.r.t. $x$: | 37 | Numerical | 5.6 |
| (i) | $\cot^{-1}(\csc x + \cot x)$ | 37 | Numerical | 5.6 |
| (ii) | $\tan^{-1}(\sqrt{\frac{1-\cos x}{1+\cos x}})$ | 37 | Numerical | 5.6 |
| (iii) | $\cot^{-1}(\sqrt{\frac{1-\sin x}{1+\sin x}})$ | 37 | Numerical | 5.6 |
| (iv) | $\tan^{-1}(\frac{\cos x-\sin x}{\cos x+\sin x})$ | 37 | Numerical | 5.6 |
| 15 | See parts below: | 32 | Proof | 5.6 |
| (i) | If $y = \frac{\sin^{-1}x}{\sqrt{1-x^2}}$, prove that ... | 32 | Proof | 5.6 |
| (ii) | If $y = \sin^{-1}(\frac{2x}{1+x^2})$, prove that $(1+x^2)\frac{dy}{dx} = 2$. | 32 | Proof | 5.6 |
| 16 | Differentiate the following functions w.r.t. $x$: | 39 | Numerical | 5.6 |
| (i) | $\tan^{-1}(\frac{2x}{1-x^2})$ | 39 | Numerical | 5.6 |
| (ii) | $\cos^{-1}(\frac{1-x^2}{1+x^2})$ | 39 | Numerical | 5.6 |
| (iii) | $\cos^{-1}(1-2x^2)$ | 39 | Numerical | 5.6 |
| (iv) | $\sin^{-1}(2x\sqrt{1-x^2})$ | 39 | Numerical | 5.6 |
| 17 | Differentiate the following functions w.r.t. $x$: | 39 | Numerical | 5.6 |
| (i) | $\sin^{-1}(3x-4x^3)$ | 39 | Numerical | 5.6 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| (ii) | $\tan^{-1}(\frac{3x-x^3}{1-3x^2})$ | 39 | Numerical | 5.6 |
| (iii) | $\cos^{-1}(\frac{2x}{1+x^2})$ | 39 | Numerical | 5.6 |
| (iv) | $\sec^{-1}(\frac{1}{4x^3-3x})$ | 39 | Numerical | 5.6 |
| 18 | Differentiate the following functions w.r.t. $x$: | 39 | Numerical | 5.6 |
| (i) | $\sec^{-1}(\frac{x^2+1}{x^2-1})$ | 39 | Numerical | 5.6 |
| (ii) | $\sin^{-1}(\frac{1-x^2}{1+x^2})$ | 39 | Numerical | 5.6 |
| (iii) | $\tan^{-1}(\sqrt{1+x^2}-x)$ | 39 | Numerical | 5.6 |
| (iv) | $\cot^{-1}(\sqrt{1+x^2}+x)$ | 39 | Numerical | 5.6 |
| 19 | Differentiate the following functions w.r.t. $x$: | 39 | Numerical | 5.6 |
| (i) | $\tan^{-1}(\frac{x}{\sqrt{a^2-x^2}})$ | 39 | Numerical | 5.6 |
| (ii) | $\sin^{-1}(\frac{1}{\sqrt{1+x^2}})$ | 39 | Numerical | 5.6 |
| (iii) | $\cos^{-1}(\sqrt{\frac{1+x}{2}})$ | 39 | Numerical | 5.6 |
| (iv) | $\sin^{-1}(\frac{x}{\sqrt{x^2+a^2}})$ | 39 | Numerical | 5.6 |
| 20 | Differentiate the following functions w.r.t. $x$: | 39 | Numerical | 5.6 |
| (i) | $\sin^{-1}(\sqrt{\frac{1+x^2}{2}})$ | 39 | Numerical | 5.6 |
| (ii) | $\tan^{-1}(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}})$ | 39 | Numerical | 5.6 |
| (iii) | $\sin^2(\cot^{-1}\sqrt{\frac{1+x}{1-x}})$ | 39 | Numerical | 5.6 |
| (iv) | $\sin^{-1}(\frac{2\sec x}{1+\sec^2 x})$ | 39 | Numerical | 5.6 |
| 21 | Differentiate the following functions w.r.t. $x$: | 39 | Numerical | 5.6 |
| (i) | $\cos^{-1}(\frac{\sin x+\cos x}{\sqrt{2}})$, $-\frac{\pi}{4}<x<\frac{\pi}{4}$ ... | 39 | Numerical | 5.6 |
| (ii) | $\cos^{-1}(\frac{3\cos x-4\sin x}{5})$ | 39 | Numerical | 5.6 |
| (iii) | $\sin^{-1}(\frac{x+\sqrt{1-x^2}}{\sqrt{2}})$ | 39 | Numerical | 5.6 |
| (iv) | $\cos^{-1}(\frac{3x+4\sqrt{1-x^2}}{5})$ | 39 | Numerical | 5.6 |
| 22 | Differentiate the following functions w.r.t. $x$: | 39 | Numerical | 5.6 |
| (i) | $\tan^{-1}(\frac{4\sqrt{x}}{1-4x})$ | 39 | Numerical | 5.6 |
| (ii) | $\tan^{-1}(\frac{x^{1/3}+a^{1/3}}{1-x^{1/3}a^{1/3}})$ | 39 | Numerical | 5.6 |
| (iii) | $\sin(2\sin^{-1}x)$ | 39 | Numerical | 5.6 |
| (iv) | $\cot^{-1}(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}})$ ... | 39 | Numerical | 5.6 |
| 23 | If $y=\sin(\tan^{-1}e^x)$, then find $\frac{dy}{dx}$ at $x=0$. | 31 | Numerical | 5.6 |
| 24 | If $y=\csc(\cot^{-1}x)$, then prove that $\sqrt{1+x^2}\frac{dy}{dx}-x=0$. | 32 | Proof | 5.6 |

**Exercise 5.7 – Implicit Differentiation**

| | | | | |
|---|---|---|---|---|
| 1 | *(see parts below)* | 40 | Numerical | 5.7 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| (i) | Find $\frac{dy}{dx}$ in $x - y = \pi$. | 40 | Numerical | 5.7 |
| (ii) | Find $\frac{dy}{dx}$ in $x^2 + y^2 = 25$. | 40 | Numerical | 5.7 |
| 2 | *(see parts below)* | 40 | Numerical | 5.7 |
| (i) | Find $\frac{dy}{dx}$ in $xy = c^2$. | 40 | Numerical | 5.7 |
| (ii) | Find $\frac{dy}{dx}$ in $\frac{x^2}{a^2} + \frac{y^2}{b^2} = c^2$. | 40 | Numerical | 5.7 |
| 3 | *(see parts below)* | 40 | Numerical | 5.7 |
| (i) | Find $\frac{dy}{dx}$ in $\sqrt{x} + \sqrt{y} = \sqrt{c}$. | 40 | Numerical | 5.7 |
| (ii) | Find $\frac{dy}{dx}$ in $x^3 + y^3 = 3axy$. | 40 | Numerical | 5.7 |
| 4 | *(see parts below)* | 40 | Numerical | 5.7 |
| (i) | Find $\frac{dy}{dx}$ in $2x + 3y = \sin x$. | 40 | Numerical | 5.7 |
| (ii) | Find $\frac{dy}{dx}$ in $2x + 3y = \sin y$. | 40 | Numerical | 5.7 |
| 5 | *(see parts below)* | 40 | Numerical | 5.7 |
| (i) | Find $\frac{dy}{dx}$ when $(x^2 + y^2)^2 = xy$. | 40 | Numerical | 5.7 |
| (ii) | Find $\frac{dy}{dx}$ when $\sin(x + y) = \frac{2}{3}$. | 40 | Numerical | 5.7 |
| 6 | If $x^{2/3} + y^{2/3} = 2$, find $\frac{dy}{dx}$ at $(1, 1)$. | 41 | Numerical | 5.7 |
| 7 | *(see parts below)* | 42 | Proof | 5.7 |
| (i) | Use implicit differentiation to verify that $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1 \ldots$ | 42 | Proof | 5.7 |
| (ii) | Use implicit differentiation to verify that $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1 \ldots$ | 42 | Proof | 5.7 |
| 8 | *(see parts below)* | 40 | Numerical | 5.7 |
| (i) | Find $\frac{dy}{dx}$, when $y + \sin y = \cos x$. | 40 | Numerical | 5.7 |
| (ii) | Find $\frac{dy}{dx}$, when $ax + by^2 = \cos y$. | 40 | Numerical | 5.7 |
| (iii) | Find $\frac{dy}{dx}$, when $y \sec x + \tan x + x^2 y = 0$. | 40 | Numerical | 5.7 |
| (iv) | Find $\frac{dy}{dx}$, when $\sin^2 x + \cos^2 y = 1$. | 40 | Numerical | 5.7 |
| (v) | Find $\frac{dy}{dx}$, when $xy + y^2 = \tan x + y$. | 40 | Numerical | 5.7 |
| (vi) | Find $\frac{dy}{dx}$, when $y = \tan(x + y)$. | 40 | Numerical | 5.7 |
| (vii) | Find $\frac{dy}{dx}$, when $\sin(xy) + \frac{x}{y} = x^2 - y$. | 40 | Numerical | 5.7 |
| (viii) | Find $\frac{dy}{dx}$, when $\sec(x + y) = xy$. | 40 | Numerical | 5.7 |
| 9 | If $\sec(\frac{x+y}{x-y}) = a$, prove that $\frac{dy}{dx} = \frac{y}{x}$. | 43 | Proof | 5.7 |
| 10 | *(see parts below)* | 44 | Proof | 5.7 |
| (i) | If $y = x \sin(a + y)$, prove that ... | 44 | Proof | 5.7 |
| (ii) | If $\sin x = y \sin(x + a)$, prove that $\frac{dy}{dx} = \frac{\sin a}{\sin^2(x+a)}$. | 44 | Proof | 5.7 |
| (iii) | If $\sin y = x \sin(a + y)$, prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$. | 44 | Proof | 5.7 |
| 11 | If $\sqrt{1 - x^6} + \sqrt{1 - y^6} = a^3(x^3 - y^3)$, prove that ... | 39 | Proof | 5.7 |
| 12 | *(see parts below)* | 45 | Proof | 5.7 |
| (i) | If $y = \sqrt{x + \sqrt{x + \sqrt{x + \ldots \infty}}}$, prove that ... | 45 | Proof | 5.7 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| (ii) | If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \ldots \infty}}}$, prove that ... | 45 | Proof | 5.7 |
| (iii) | If $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \ldots \infty}}}$, prove that ... | 45 | Proof | 5.7 |
| 13 | If $y = x + \frac{1}{x + \frac{1}{x + \ldots \text{ to } \infty}}$, prove that ... | 45 | Proof | 5.7 |
| **Exercise 5.8 – Differentiation of Exponential and Logarithmic Functions** | | | | |
| 1 | Differentiate the following functions w.r.t. $x$: | 20 | Numerical | 5.8 |
| (i) | $\log x + \frac{1}{x}$ | 20 | Numerical | 5.8 |
| (ii) | $10^x + \frac{1}{3}e^x - 2\log x$ | 20 | Numerical | 5.8 |
| 2 | Differentiate the following functions w.r.t. $x$: | 19 | Numerical | 5.8 |
| (i) | $e^{-x}$ | 19 | Numerical | 5.8 |
| (ii) | $\sin(\log x), \quad x > 0$ | 19 | Numerical | 5.8 |
| 3 | Differentiate the following functions w.r.t. $x$: | 19 | Numerical | 5.8 |
| (i) | $e^{\cos x}$ | 19 | Numerical | 5.8 |
| (ii) | $e^{\sin^{-1} x}$ | 19 | Numerical | 5.8 |
| 4 | Differentiate the following functions w.r.t. $x$: | 22 | Numerical | 5.8 |
| (i) | $e^{x^3}$ | 22 | Numerical | 5.8 |
| (ii) | $\sqrt{e^{\sqrt{x}}}, \quad x > 0$ | 22 | Numerical | 5.8 |
| 5 | Differentiate the following functions w.r.t. $x$: | 19 | Numerical | 5.8 |
| (i) | $\log(\log x), \quad x > 1$ | 19 | Numerical | 5.8 |
| (ii) | $\log_7(\log x), \quad x > 1$ | 19 | Numerical | 5.8 |
| 6 | Differentiate the following functions w.r.t. $x$: | 22 | Numerical | 5.8 |
| (i) | $\log(\cos e^x)$ | 22 | Numerical | 5.8 |
| (ii) | $\tan^{-1}(e^{\sin x})$ | 22 | Numerical | 5.8 |
| 7 | Is the function $f(x) = \log(2x - 1)$ derivable at $x = 0$? | 35 | Numerical | 5.8 |
| 8 | If $f(x) = \log(\log x)$, find $f'(e)$. | 31 | Numerical | 5.8 |
| 9 | If $y = \log(\tan x)$, show that $\frac{dy}{dx} = 2\csc 2x$. | 33 | Proof | 5.8 |
| 10 | If $f(1) = 4$ and $f'(1) = 2$, find the value of the derivative of $\log f(e^x)$ w.r.t ... | 31 | Numerical | 5.8 |
| 11 | If $f(x) = e^{xg(x)}$, $g(0) = 2$ and $g'(0) = 1$, then find $f'(0)$. | 31 | Numerical | 5.8 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 12 | Differentiate the following functions w.r.t. $x$: | 28 | Numerical | 5.8 |
| (i) | $\frac{e^x}{\sin x}$ | 28 | Numerical | 5.8 |
| (ii) | $\frac{e^x}{1+\sin x}$ | 28 | Numerical | 5.8 |
| 13 | Differentiate the following functions w.r.t. $x$: | 38 | Numerical | 5.8 |
| (i) | $2x\tan^{-1}x - \log(1+x^2)$ | 38 | Numerical | 5.8 |
| (ii) | $\frac{\cos x}{\log x}, \quad x > 0$ | 38 | Numerical | 5.8 |
| 14 | Differentiate the following functions w.r.t. $x$: | 20 | Numerical | 5.8 |
| (i) | $e^{\sec^2 x} + 3\cos^{-1}x$ | 20 | Numerical | 5.8 |
| (ii) | $e^x + e^{x^2} + \cdots + e^{x^5}$ | 20 | Numerical | 5.8 |
| 15 | Differentiate the following functions w.r.t. $x$: | 28 | Numerical | 5.8 |
| (i) | $\frac{5^x \log x}{x^2+1}$ | 28 | Numerical | 5.8 |
| (ii) | $\cos(\log x + e^x)$ | 28 | Numerical | 5.8 |
| 16 | Differentiate the following functions w.r.t. $x$: | 30 | Numerical | 5.8 |
| (i) | $\log(\frac{x+\sqrt{x^2-a^2}}{x-\sqrt{x^2-a^2}})$ | 30 | Numerical | 5.8 |
| (ii) | $\log\sqrt{\frac{1-\cos x}{1+\cos x}}$ | 30 | Numerical | 5.8 |
| 17 | Differentiate the following functions w.r.t. $x$: | 36 | Numerical | 5.8 |
| (i) | $e^{\cot^{-1}x^2}$ | 36 | Numerical | 5.8 |
| (ii) | $x\sqrt{1+x^2} + \log(x+\sqrt{x^2+1})$ | 36 | Numerical | 5.8 |
| 18 | See parts below: | 32 | Proof | 5.8 |
| (i) | If $y = \frac{\log(x+\sqrt{x^2+1})}{\sqrt{x^2+1}}$, prove that ... | 32 | Proof | 5.8 |
| (ii) | If $y = e^{2\log x + 3x}$, prove that $\frac{dy}{dx} = x(2+3x)e^{3x}$. | 32 | Proof | 5.8 |
| 19 | Differentiate $\tan^{-1}(\frac{2^{x+1}}{1-4^x})$ w.r.t. $x$. | 39 | Numerical | 5.8 |
| 20 | Find $\frac{dy}{dx}$ when | 40 | Numerical | 5.8 |
| (i) | $xy + xe^{-y} + ye^x = x^2$ | 40 | Numerical | 5.8 |
| (ii) | $e^{x-y} = \log(\frac{x}{y})$ | 40 | Numerical | 5.8 |
| (iii) | $5^x + 5^y = 5^{x+y}$ | 40 | Numerical | 5.8 |
| 21 | If $\log(x^2+y^2) = 2\tan^{-1}(\frac{y}{x})$, show that ... | 40 | Proof | 5.8 |
| 22 | If $y\log x = x - y$, prove that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$. | 44 | Proof | 5.8 |
| 23 | Differentiate the following functions w.r.t. $x$: | 28 | Numerical | 5.8 |
| (i) | $\log_x(2x-3)$ | 28 | Numerical | 5.8 |
| (ii) | $\log_{\cos x}\sin x$ | 28 | Numerical | 5.8 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| **Exercise 5.9 – Logarithmic Differentiation** | | | | |
| 1 | *(see parts below)* | 46 | Numerical | 5.9 |
| (i) | Differentiate $(x+3)^2(x+4)^3(x+5)^4$ w.r.t. $x$. | 46 | Numerical | 5.9 |
| (ii) | Differentiate $\cos x \cos 2x \cos 3x$ w.r.t. $x$. | 46 | Numerical | 5.9 |
| 2 | *(see parts below)* | 46 | Numerical | 5.9 |
| (i) | Differentiate $e^x \cos^3 x \sin^2 x$ w.r.t. $x$. | 46 | Numerical | 5.9 |
| (ii) | Differentiate $\frac{e^{x^2}\tan^{-1} x}{\sqrt{1+x^2}}$ w.r.t. $x$. | 46 | Numerical | 5.9 |
| 3 | *(see parts below)* | 47 | Numerical | 5.9 |
| (i) | Differentiate $(\sin x)^{\cos x}, 0 < x < \pi$ w.r.t. $x$. | 47 | Numerical | 5.9 |
| (ii) | Differentiate $(\sin x)^{\sin x}, 0 < x < \pi$ w.r.t. $x$. | 47 | Numerical | 5.9 |
| (iii) | Differentiate $(\tan x)^{\tan x}, 0 < x < \frac{\pi}{2}$ w.r.t. $x$. | 47 | Numerical | 5.9 |
| 4 | *(see parts below)* | 47 | Numerical | 5.9 |
| (i) | Differentiate $x^{\sin x}, x > 0$ w.r.t. $x$. | 47 | Numerical | 5.9 |
| (ii) | Differentiate $(2x+3)^{x-5}, x > -\frac{3}{2}$ w.r.t. $x$. | 47 | Numerical | 5.9 |
| 5 | *(see parts below)* | 47 | Numerical | 5.9 |
| (i) | Differentiate $(\log x)^{\cos x}, x > 1$ w.r.t. $x$. | 47 | Numerical | 5.9 |
| (ii) | Differentiate $(\log x)^{\log x}, x > 1$ w.r.t. $x$. | 47 | Numerical | 5.9 |
| 6 | *(see parts below)* | 48 | Proof | 5.9 |
| (i) | If $y = x^y$, prove that $x\frac{dy}{dx} = \frac{y^2}{1-y\log x}$. | 48 | Proof | 5.9 |
| (ii) | If $x = e^{x/y}$, prove that $\frac{dy}{dx} = \frac{x-y}{x\log x}$. | 48 | Proof | 5.9 |
| (iii) | If $x^y = e^{x-y}$, prove that $\frac{dy}{dx} = \frac{\log x}{(\log xe)^2}$. | 48 | Proof | 5.9 |
| (iv) | If $x^{16}y^9 = (x^2+y)^{17}$, prove that $\frac{dy}{dx} = \frac{2y}{x}$. | 48 | Proof | 5.9 |
| 7 | Find the derivative of $x^x + a^x + x^a + a^a$ for some fixed $a > 0, x > 0$. | 49 | Numerical | 5.9 |
| 8 | *(see parts below)* | 49 | Numerical | 5.9 |
| (i) | Differentiate $x^{\log x} + (\log x)^x$ w.r.t. $x$. | 49 | Numerical | 5.9 |
| (ii) | Differentiate $(\sin x)^{\cos x} + x^{\sin x}$ w.r.t. $x$. | 49 | Numerical | 5.9 |
| (iii) | Differentiate $x^{\cos x} + (\cos x)^x$ w.r.t. $x$. | 49 | Numerical | 5.9 |
| (iv) | Differentiate $x^{\cos x} + (\sin x)^{\tan x}$ w.r.t. $x$. | 49 | Numerical | 5.9 |
| (v) | Differentiate $(\sin 2x)^x + \sin^{-1}\sqrt{3x}$ w.r.t. $x$. | 49 | Numerical | 5.9 |
| (vi) | Differentiate $e^{\sin x} + (\tan x)^x$ w.r.t. $x$. | 49 | Numerical | 5.9 |
| (vii) | Differentiate $x^x - 2^{\sin x}$ w.r.t. $x$. | 49 | Numerical | 5.9 |
| (viii) | Differentiate $x^{x\cos x} + \frac{x^2+1}{x^2-1}$ w.r.t. $x$. | 49 | Numerical | 5.9 |
| 9 | *(see parts below)* | 49 | Numerical | 5.9 |
| (i) | If $y = (\log x)^{\cos x} + \frac{x^2+1}{x^2-1}$, find $\frac{dy}{dx}$. | 49 | Numerical | 5.9 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| (ii) | If $y = e^{x^2 \cos x} + (\cos x)^x$, find $\frac{dy}{dx}$. | 49 | Numerical | 5.9 |
| 10 | *(see parts below)* | 48 | Numerical | 5.9 |
| (i) | Find $\frac{dy}{dx}$ when $x^y y^x = a^b$. | 48 | Numerical | 5.9 |
| (ii) | Find $\frac{dy}{dx}$ when $x^y + y^x = a^b$. | 49 | Numerical | 5.9 |
| (iii) | Find $\frac{dy}{dx}$ when $x^x + y^x = 1$. | 49 | Numerical | 5.9 |
| (iv) | Find $\frac{dy}{dx}$ when $(\sin x)^y = x + y$. | 48 | Numerical | 5.9 |
| (v) | Find $\frac{dy}{dx}$ when $xy = e^{x-y}$. | 48 | Numerical | 5.9 |
| (vi) | Find $\frac{dy}{dx}$ when $(\cos x)^y = (\cos y)^x$. | 48 | Numerical | 5.9 |
| 11 | If $y = x^{x^{x^{\cdot^{\cdot^{\infty}}}}}$, prove that ... | 45 | Proof | 5.9 |
| 12 | If $y = x^x$, then find $\frac{dy}{dx}$ at $x = 1$. | 50 | Numerical | 5.9 |
| (i) | Find $\frac{dy}{dx}$ if $y = (\log x)^{\cos x}$. | 61 | Numerical | 5.9 |
| (ii) | Find $\frac{dy}{dx}$ if $y = x^{e^{x^2} \cos x}$. | 61 | Numerical | 5.9 |
| (i) | Find $\frac{dy}{dx}$ if $x^y + y^x = a^b$. (Implicit differentiation / Logarithmic ... | 62 | Numerical | 5.9 |
| (ii) | Find $\frac{dy}{dx}$ if $x^{y-1} + y^{x-1} = 1$. | 62 | Numerical | 5.9 |
| (iii) | Find $\frac{dy}{dx}$ if $x^x + y^x = xy$. | 62 | Numerical | 5.9 |
| (iv) | Find $\frac{dy}{dx}$ if $(x + y)^y = \log(\sin x)$. | 61 | Numerical | 5.9 |
| (v) | Find $\frac{dy}{dx}$ if $y^x = x^y$. | 61 | Numerical | 5.9 |
| (vi) | Find $\frac{dy}{dx}$ if $y = \frac{y \tan x + \log \cos y}{x \tan y + \log \cos x}$ ... | 62 | Numerical | 5.9 |

**Exercise 5.10 – Parametric Differentiation**

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 1 | *(see parts below)* | 51 | Numerical | 5.10 |
| (i) | Find $\frac{dy}{dx}$ when $x = at^2$ and $y = 2at$. | 51 | Numerical | 5.10 |
| (ii) | Find $\frac{dy}{dx}$ when $x = 2at^2$ and $y = at^4$. | 51 | Numerical | 5.10 |
| 2 | *(see parts below)* | 51 | Numerical | 5.10 |
| (i) | Find $\frac{dy}{dx}$ when $x = a\cos\theta$ and $y = b\cos\theta$. | 51 | Numerical | 5.10 |
| (ii) | Find $\frac{dy}{dx}$ when $x = a\cos\theta$ and $y = a\sin\theta$. | 51 | Numerical | 5.10 |
| 3 | *(see parts below)* | 51 | Numerical | 5.10 |
| (i) | Find $\frac{dy}{dx}$ when $x = a\sec\theta$ and $y = b\tan\theta$. | 51 | Numerical | 5.10 |
| (ii) | Find $\frac{dy}{dx}$ when $x = \sin t$ and $y = \cos 2t$. | 51 | Numerical | 5.10 |
| 4 | *(see parts below)* | 51 | Numerical | 5.10 |
| (i) | Find $\frac{dy}{dx}$ when $x = 4t$ and $y = \frac{4}{t}$. | 51 | Numerical | 5.10 |
| (ii) | Find $\frac{dy}{dx}$ when $x = t + \frac{1}{t}$ and $y = t - \frac{1}{t}$. | 51 | Numerical | 5.10 |
| (iii) | Find $\frac{dy}{dx}$ when $x = a(t - \sin t)$ and $y = a(1 + \cos t)$. | 51 | Numerical | 5.10 |
| 5 | *(see parts below)* | 51 | Numerical | 5.10 |
| (i) | Find $\frac{dy}{dx}$ when $x = a\cos^3 t$ and $y = b\sin^3 t$. | 51 | Numerical | 5.10 |

| **Q. No.** | **Topic / Statement Summary** | **Type No.** | **Kind** | **Section** |
|---|---|---|---|---|
| (ii) | Find $\frac{dy}{dx}$ when $x = a(1 - \sin t)$ and $y = a(1 + \cos t)$. | 51 | Numerical | 5.10 |
| 6 | *(see parts below)* | 51 | Numerical | 5.10 |
| (i) | Find $\frac{dy}{dx}$ when $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$. | 51 | Numerical | 5.10 |
| (ii) | Find $\frac{dy}{dx}$ when $x = a(1 - \cos\theta)$ and $y = a(\theta + \sin\theta)$. | 51 | Numerical | 5.10 |
| 7 | *(see parts below)* | 51 | Numerical | 5.10 |
| (i) | Find $\frac{dy}{dx}$ when $x = e^t(\sin t + \cos t)$ and $y = e^t(\sin t - \cos t)$. | 51 | Numerical | 5.10 |
| (ii) | Find $\frac{dy}{dx}$ when $x = \cos\theta - \cos 2\theta$ and ... | 51 | Numerical | 5.10 |
| 8 | *(see parts below)* | 51 | Numerical | 5.10 |
| (i) | Find $\frac{dy}{dx}$ when $x = a(\cos\theta + \cos 2\theta)$ and ... | 51 | Numerical | 5.10 |
| (ii) | Find $\frac{dy}{dx}$ when $x = \frac{1+\log t}{t^2}$ and $y = \frac{3+2\log t}{t}$. | 51 | Numerical | 5.10 |
| 9 | Find $\frac{dy}{dx}$ when $x = 3\cos\theta - 2\cos^3\theta$ and ... | 51 | Numerical | 5.10 |
| 10 | *(see parts below)* | 52 | Numerical | 5.10 |
| (i) | If $x = 2\cos t - \cos 2t$ and $y = 2\sin t - \sin 2t$, find $\frac{dy}{dx}$ at ... | 52 | Numerical | 5.10 |
| (ii) | If $x = 3\sin t - \sin 3t$ and $y = 3\cos t - \cos 3t$, find $\frac{dy}{dx}$ at ... | 52 | Numerical | 5.10 |
| (iii) | If $x = a\sin 2t$ and $y = a(\cos 2t + \log\tan t)$, find $\frac{dy}{dx}$. | 51 | Numerical | 5.10 |
| 11 | *(see parts below)* | 52 | Numerical | 5.10 |
| (i) | If $x = \frac{2bt}{1+t^2}$ and $y = \frac{a(1-t^2)}{1+t^2}$, find $\frac{dy}{dx}$ at ... | 52 | Numerical | 5.10 |
| (ii) | If $x = ae^\theta(\sin\theta - \cos\theta)$ and ... | 52 | Numerical | 5.10 |
| 12 | *(see parts below)* | 53 | Numerical | 5.10 |
| (i) | Differentiate $\frac{x^2}{1-x^2}$ with respect to $x^2$. | 53 | Numerical | 5.10 |
| (ii) | Differentiate $\sin x^2$ with respect to $x^3$. | 53 | Numerical | 5.10 |
| (iii) | Differentiate $\cot^3(2x + 1)$ with respect to $x^2 + 1$. | 53 | Numerical | 5.10 |
| (iv) | Differentiate $\sin^2 x$ with respect to $e^{\cos x}$. | 53 | Numerical | 5.10 |
| 13 | *(see parts below)* | 54 | Numerical | 5.10 |
| (i) | Differentiate $\sin^{-1}(\frac{2x}{1+x^2})$ with respect to $\tan^{-1} x$. | 54 | Numerical | 5.10 |
| (ii) | Differentiate $\tan^{-1}(\frac{2x}{1-x^2})$ with respect to $\tan^{-1} x$. | 54 | Numerical | 5.10 |
| 14 | Differentiate $\cos^{-1}(\frac{1}{\sqrt{1+t^2}})$ with respect to ... | 53 | Numerical | 5.10 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 15 | *(see parts below)* | 54 | Numerical | 5.10 |
| (i) | Differentiate $\sin^{-1}(\frac{2x}{1+x^2})$ with respect to ... | 54 | Numerical | 5.10 |
| (ii) | Differentiate $\tan^{-1}(\frac{3x-x^3}{1-3x^2})$ with respect to ... | 54 | Numerical | 5.10 |
| 16 | *(see parts below)* | 54 | Numerical | 5.10 |
| (i) | Differentiate $\tan^{-1}(\frac{\sqrt{1+x^2}-1}{x})$ with respect to $\tan^{-1} x$. | 54 | Numerical | 5.10 |
| (ii) | Differentiate $\tan^{-1}(\frac{\sqrt{1-x^2}}{x})$ with respect to ... | 54 | Numerical | 5.10 |
| (iii) | Differentiate $\sec^{-1}(\frac{1}{\sqrt{1-x^2}})$ with respect to ... | 54 | Numerical | 5.10 |
| 17 | Prove that the derivative of $\tan^{-1}(\frac{x}{1+\sqrt{1-x^2}})$ with respect to ... | 54 | Proof | 5.10 |
| 18 | Differentiate $x^x$ with respect to $x \log x$. | 53 | Numerical | 5.10 |
| 19 | If $x = \sqrt{a^{\tan^{-1} t}}$ and $y = \sqrt{a^{\cot^{-1} t}}$, show that ... | 43 | Proof | 5.10 |

**Exercise 5.11 – Second-Order Derivatives (Miscellaneous)**

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 1 | *(see parts below)* | 63 | Numerical | 5.11 |
| (i) | Find the second order derivative of $x^3 + \tan x$. | 63 | Numerical | 5.11 |
| (ii) | Find the second order derivative of $\log x$. | 63 | Numerical | 5.11 |
| (iii) | Find the second order derivative of $\tan^{-1} x$. | 63 | Numerical | 5.11 |
| 2 | *(see parts below)* | 63 | Numerical | 5.11 |
| (i) | If $y = \log(x-2)$, $x > 2$, find $\frac{d^2y}{dx^2}$. | 63 | Numerical | 5.11 |
| (ii) | If $y = \cot x$, find $\frac{d^2y}{dx^2}$ at $x = \frac{\pi}{4}$. | 63 | Numerical | 5.11 |
| (iii) | If $x = at^2$ and $y = 2at$, then find $\frac{d^2y}{dx^2}$. | 64a | Numerical | 5.11 |
| 3 | *(see parts below)* | 63 | Numerical | 5.11 |
| (i) | Find the second order derivatives of $\sin^{-1} x$. | 63 | Numerical | 5.11 |
| (ii) | Find the second order derivatives of $x \cos x$. | 65 | Numerical | 5.11 |
| (iii) | Find the second order derivatives of $x \sin 2x$. | 65 | Numerical | 5.11 |
| (iv) | Find the second order derivatives of $e^x \sin 5x$. | 65 | Numerical | 5.11 |
| (v) | Find the second order derivatives of $e^{2x} \sin 3x$. | 65 | Numerical | 5.11 |
| (vi) | Find the second order derivatives of $\log(\log x)$. | 63 | Numerical | 5.11 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| (vii) | Find the second order derivatives of $\frac{\log x}{x}$. | 66 | Numerical | 5.11 |
| (viii) | Find the second order derivatives of $x^2 \log \lvert \cos x \rvert$. | 65 | Numerical | 5.11 |
| (ix) | Find the second order derivatives of $\frac{2x+1}{2x+3}$. | 66 | Numerical | 5.11 |
| 4 | *(see parts below)* | 63 | Numerical | 5.11 |
| (i) | Find the second order derivative of $\sec ax$. | 63 | Numerical | 5.11 |
| (ii) | Find the second order derivative of $\cot(1-2x)$. | 63 | Numerical | 5.11 |
| (iii) | Find the second order derivative of $\sin 3x \cos 5x$. | 65 | Numerical | 5.11 |
| (iv) | Find the second order derivative of $\sin^3 x$. | 63 | Numerical | 5.11 |
| (v) | Find the second order derivative of $\cos(2x^2-1)$. | 63 | Numerical | 5.11 |
| (vi) | Find the second order derivative of $\sqrt{1-x^2}$. | 63 | Numerical | 5.11 |
| 5 | If $y = \cos^{-1} x$, find $\frac{d^2y}{dx^2}$ in terms of $y$ alone. | 67 | Numerical | 5.11 |
| 6 | *(see parts below)* | 68a | Proof | 5.11 |
| (i) | If $y = \cot x$, prove that $\frac{d^2y}{dx^2} + 2y\frac{dy}{dx} = 0$. | 68a | Proof | 5.11 |
| (ii) | If $y = 5\cos x - 3\sin x$, prove that $\frac{d^2y}{dx^2} + y = 0$. | 68a | Proof | 5.11 |
| 7 | *(see parts below)* | 68a | Proof | 5.11 |
| (i) | If $y = x + \tan x$, prove that $\cos^2 x\frac{d^2y}{dx^2} - 2y + 2x = 0$. | 68a | Proof | 5.11 |
| (ii) | If $y = \tan x + \sec x$, prove that ... | 68a | Proof | 5.11 |
| 8 | *(see parts below)* | 68a | Proof | 5.11 |
| (i) | If $y = \sec x - \tan x$, prove that $\cos x\frac{d^2y}{dx^2} = y^2$. | 68a | Proof | 5.11 |
| (ii) | If $y = x + \cot x$, prove that $\sin^2 x\frac{d^2y}{dx^2} - 2y + 2x = 0$. | 68a | Proof | 5.11 |
| 9 | *(see parts below)* | 68a | Proof | 5.11 |
| (i) | If $y = ae^{mx} + be^{-mx}$, prove that $y_2 - m^2y = 0$. | 68a | Proof | 5.11 |
| (ii) | If $y = ae^{2x} + be^{-x}$, prove that $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0$. | 68a | Proof | 5.11 |
| 10 | *(see parts below)* | 68a | Proof | 5.11 |
| (i) | If $y = A\cos nx + B\sin nx$, show that $\frac{d^2y}{dx^2} + n^2y = 0$. | 68a | Proof | 5.11 |
| (ii) | If $y = A\sin 2x + B\cos 2x$ and $\frac{d^2y}{dx^2} - ky = 0$, find the value of $k$. | 69 | Numerical | 5.11 |

| Q. No. | Topic / Statement Summary | Type No. | Kind | Section |
|---|---|---|---|---|
| 11 | *(see parts below)* | 68a | Proof | 5.11 |
| (i) | If $y = \tan^{-1} x$, prove that $(1+x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 0$. | 68a | Proof | 5.11 |
| (ii) | If $y = \sin^{-1} x$, prove that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0$. | 68a | Proof | 5.11 |
| 12 | *(see parts below)* | 68a | Proof | 5.11 |
| (i) | If $y = \sin(\log x)$, prove that $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$. | 68a | Proof | 5.11 |
| (ii) | If $y = 3\cos(\log x) + 4\sin(\log x)$, prove that ... | 68a | Proof | 5.11 |
| 13 | If $y = x\cos x$, prove that ... | 68a | Proof | 5.11 |
| 14 | *(see parts below)* | 68a | Proof | 5.11 |
| (i) | If $y = (\cos^{-1} x)^2$, prove that $(1-x^2)y_2 - xy_1 = 2$. | 68a | Proof | 5.11 |
| (ii) | If $y = (\tan^{-1} x)^2$, prove that ... | 68a | Proof | 5.11 |
| (iii) | If $y = (\sec^{-1} x)^2, x > 1$, show that ... | 68a | Proof | 5.11 |
| 15 | *(see parts below)* | 68a | Proof | 5.11 |
| (i) | If $y = \log(x + \sqrt{x^2+1})$, prove that ... | 68a | Proof | 5.11 |
| (ii) | If $y = \log(x + \sqrt{x^2+a^2})$, prove that $(x^2+a^2)y_2 + xy_1 = 0$. | 68a | Proof | 5.11 |
| (iii) | If $y = (\log x)^2$, prove that $x^2y'' + xy' = 2$. | 68a | Proof | 5.11 |
| 16 | If $y = \cos(\sin x)$, prove that ... | 68b | Proof | 5.11 |
| 17 | *(see parts below)* | 68c | Proof | 5.11 |
| (i) | If $y = e^{\tan^{-1} x}$, prove that ... | 68c | Proof | 5.11 |
| (ii) | If $x = \tan(\frac{1}{a}\log y)$, then prove that ... | 68d | Proof | 5.11 |
| (iii) | If $y = e^{a\cos^{-1} x}, -1 < x < 1$, prove that ... | 68e | Proof | 5.11 |
| 18 | If $y = \log(\frac{x}{a+bx})^x$, prove that ... | 68f | Proof | 5.11 |
| 19 | If $\sqrt{x} + \sqrt{y} = \sqrt{a}$, find $\frac{d^2y}{dx^2}$ at $x = a$. | 75 | Numerical | 5.11 |
| 20 | If $x^my^n = (x+y)^{m+n}$, then prove that: (i) $\frac{dy}{dx} = \frac{y}{x}$ (ii) ... | 68g | Proof | 5.11 |
| 21 | If $x = a(\theta - \sin\theta)$, $y = a(1+\cos\theta)$, find $\frac{d^2y}{dx^2}$. | 64b | Numerical | 5.11 |
| 22 | If $x = \log t$ and $y = \frac{1}{t}$, prove that ... | 64c | Proof | 5.11 |
| 23 | If $x = a(\frac{1-t^2}{1+t^2})$ and $y = \frac{2at}{1+t^2}$, prove that ... | 64d | Proof | 5.11 |
| 24 | If $x = a\sin pt$ and $y = b\cos pt$, find the value of $\frac{d^2y}{dx^2}$ at $t = 0$. | 64e | Numerical | 5.11 |
| 25 | If $x = a\sec^3\theta$ and $y = a\tan^3\theta$, find $\frac{d^2y}{dx^2}$ at ... | 64f | Numerical | 5.11 |
| 26 | If $x = a(1+\cos t)$, $y = a(t+\sin t)$, find $\frac{d^2y}{dx^2}$ at ... | 64g | Numerical | 5.11 |

| **Q. No.** | **Topic / Statement Summary** | **Type No.** | **Kind** | **Section** |
|---|---|---|---|---|
| 27 | If $x = \cos t + \log(\tan \frac{t}{2})$, $y = \sin t$, then find $\frac{d^2y}{dx^2}$ at ... | 64h | Numerical | 5.11 |

## Formula & Standard Results Sheet

## I. Continuity (Ex 5.1 – 5.3)

### Continuity at a Point

$$f \text{ is continuous at } x = a \iff \lim_{x\to a} f(x) = f(a)$$

Requires the limit to exist and equal the function value at that point

### Continuity via Left- and Right-Hand Limits

$$\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a)$$

The standard three-part test used for piecewise functions at a junction point

### Continuity of Standard Functions

$$\text{Polynomial, } \sin x, \cos x, e^x \text{ are continuous everywhere; } \tan x, \sec x, \cot x, \text{c}$$

Discontinuities in trig functions occur only where the function itself is undefined

### Algebra of Continuous Functions

$$f, g \text{ continuous at } a \implies f \pm g,\ fg,\ \frac{f}{g}\,(g(a) \neq 0) \text{ are all continuous at } a$$

Continuity is preserved under the usual arithmetic operations

### Continuity of Composite Functions

$$f \text{ continuous at } a \text{ and } g \text{ continuous at } f(a) \implies g \circ f \text{ continuous at } a$$

Composing two continuous functions gives a continuous function

### Removable Discontinuity

$$\lim_{x\to a} f(x) \text{ exists but} \neq f(a) \text{ (or } f(a) \text{ undefined)}$$

Can be "removed" by redefining $f(a)$ to equal the limit

## II. Differentiability (Ex 5.4)

**Differentiability at a Point**

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}, \quad \text{provided the limit exists}$$

The derivative at a point is defined as this two-sided limit

**Left-Hand and Right-Hand Derivatives**

$$Lf'(a) = \lim_{h\to 0^-} \frac{f(a+h)-f(a)}{h}, \qquad Rf'(a) = \lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h}$$

$f$ is differentiable at $a$ only when $Lf'(a) = Rf'(a)$

**Differentiability Implies Continuity**

$$f \text{ differentiable at } a \implies f \text{ continuous at } a$$

The converse is false – continuity does not guarantee differentiability (e.g. $|x|$ at $x = 0$)

## III. Derivatives of Standard and Composite Functions (Ex 5.5 – 5.6)

**Chain Rule**

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

Differentiate the outer function first, then multiply by the derivative of the inner function

**Derivative of $|x|$**

$$\frac{d}{dx}|x| = \frac{x}{|x|} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}, \quad \text{not differentiable at } x = 0$$

Always split into cases based on the sign of the expression inside the modulus

**Standard Derivatives**

$$\frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x, \quad \frac{d}{dx}\tan x = \sec^2 x, \quad \frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\log x = \frac{1}{x}$$

The core building-block derivatives every chain-rule question relies on

**Derivatives of Inverse Trigonometric Functions**

$$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}\cos^{-1}x = \frac{-1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2}$$

Valid on the open interval $(-1, 1)$ for the first two; all reals for the third

**Trigonometric Substitution Before Differentiation**

$$\text{Substitute } x = \sin\theta, \cos\theta, \text{ or } \tan\theta \text{ to simplify an inverse-trig expression before differentia}$$

Converts a complicated nested inverse-trig expression into a simple linear one in $\theta$

## IV. Implicit Differentiation (Ex 5.7)

**Implicit Differentiation**

$$\text{Differentiate both sides of } F(x,y) = c \text{ w.r.t. } x, \text{ treating } y \text{ as a function of } x, \text{ then solve for } \frac{dy}{dx}$$

Every term containing $y$ picks up a factor of $\frac{dy}{dx}$ via the chain rule

**Product Rule (needed inside implicit differentiation)**

$$\frac{d}{dx}(uv) = u'v + uv'$$

Frequently needed when a term mixes $x$ and $y$ together, e.g. $xy$

## V. Logarithmic Differentiation (Ex 5.8 – 5.9)

**Logarithmic Differentiation**

$$\text{For } y = [f(x)]^{g(x)} : \text{ take } \log \text{ of both sides, then differentiate implicitly}$$

The standard method whenever both the base and the exponent are functions of $x$

**General Logarithmic Differentiation Formula**

$$y = [f(x)]^{g(x)} \implies \frac{dy}{dx} = y\left[g'(x)\log f(x) + \frac{g(x)f'(x)}{f(x)}\right]$$

Derived from $\log y = g(x)\log f(x)$, differentiated implicitly

**Logarithm Laws (needed before differentiating)**

$$\log(ab) = \log a + \log b, \quad \log\frac{a}{b} = \log a - \log b, \quad \log a^n = n\log a$$

Always simplify using these laws before differentiating – it avoids a messy quotient/product rule

## VI. Parametric Differentiation and Derivatives w.r.t. Another Function (Ex 5.10)

**Parametric Differentiation**

$$x = f(t),\ y = g(t) \implies \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Both derivatives are taken with respect to the parameter $t$, then divided

**Derivative of One Function w.r.t. Another**

$$\frac{du}{dv} = \frac{du/dx}{dv/dx}$$

Treat $x$ as the underlying parameter for both $u$ and $v$

## VII. Second-Order Derivatives (Ex 5.11)

**Second-Order Derivative**

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

Differentiate the first derivative again with respect to $x$

**Second-Order Derivative – Parametric Form**

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$$

Never differentiate $dy/dx$ directly with respect to $x$ in parametric form – always divide by $dx/dt$ again

**Second-Order Implicit Differentiation**

Differentiate $\dfrac{dy}{dx}$ (already in terms of $x, y$) implicitly a second time, substit

The first-derivative expression must be substituted back in before simplifying

**Verifying a Second-Order Differential Equation**

Compute $y'$, $y''$ from the given relation, then substitute into the given equ

The standard structure for every "prove that" second-derivative identity question

# 5.1 – Continuity: Checking Continuity at a Point

## Question 1 (i)

Examine the following function for continuity at the indicated point:

$$f(x) = \begin{cases} x^3 + 3, & x \neq 0 \\ 1, & x = 0 \end{cases} \quad \text{at } x = 0$$

**TYPE 1** **Checking Continuity at a Point**

**Given:** $f(x) = \begin{cases} x^3 + 3, & x \neq 0 \\ 1, & x = 0 \end{cases}$ at $x = 0$

**Concept / Formula Used:** $\lim_{x \to a} f(x) = f(a) \qquad x = a$

**Solution:**

First, evaluate the limit of $f(x)$ as $x \to 0$:

$$\begin{aligned} \lim_{x \to 0} f(x) &= \lim_{x \to 0} (x^3 + 3) \\ &= 0^3 + 3 \\ &= 3 \end{aligned}$$

Next, find the function value at $x = 0$:

$$f(0) = 1$$

Comparing the limit value with the point value:

$$\lim_{x \to 0} f(x) = 3 \neq f(0) = 1$$

Since the limit as $x \to 0$ does not equal $f(0)$, the function is discontinuous at $x = 0$.

**Final Answer**

$f(x)$ is discontinuous at $x = 0$ because $\lim_{x \to 0} f(x) = 3 \neq f(0) = 1$.

## Question 1 (ii)

Examine the following function for continuity at the indicated point:

$$f(x) = x^3 + 2x^2 - 1 \quad \text{at } x = 1$$

**TYPE 1** **Checking Continuity at a Point**

**Given:** $f(x) = x^3 + 2x^2 - 1$ at $x = 1$

**Concept / Formula Used:** $\lim_{x \to a} f(x) = f(a) \qquad x = a$

**Solution:**

Evaluate the limit of $f(x)$ as $x \to 1$:

$$\begin{aligned}\lim_{x \to 1} f(x) &= \lim_{x \to 1}(x^3 + 2x^2 - 1)\\ &= 1^3 + 2(1)^2 - 1\\ &= 1 + 2 - 1\\ &= 2\end{aligned}$$

Evaluate the function value at $x = 1$:

$$\begin{aligned}f(1) &= 1^3 + 2(1)^2 - 1\\ &= 1 + 2 - 1\\ &= 2\end{aligned}$$

Since $\lim_{x \to 1} f(x) = f(1) = 2$, the function $f(x)$ is continuous at $x = 1$.

**Final Answer**

$f(x)$ is continuous at $x = 1$.

## Question 2

If $f(x) = \dfrac{x^2 - 1}{x - 1}$ for $x \neq 1$ and $f(x) = 2$ when $x = 1$, show that the function is continuous at $x = 1$.

**TYPE 1** **Checking Continuity at a Point**

**To Prove:** $f(x)$ is continuous at $x = 1$

**Concept / Formula Used:** $\lim_{x \to a} f(x) = f(a) \qquad x = a$

**Solution:**

Calculate the limit of $f(x)$ as $x \to 1$:

$$\begin{aligned}\lim_{x \to 1} f(x) &= \lim_{x \to 1} \frac{x^2 - 1}{x - 1}\\ &= \lim_{x \to 1} \frac{(x - 1)(x + 1)}{x - 1}\\ &= \lim_{x \to 1}(x + 1)\\ &= 1 + 1\\ &= 2\end{aligned}$$

The given value of the function at $x = 1$ is:

$$f(1) = 2$$

Comparing both values:

$$\lim_{x\to 1} f(x) = 2 = f(1)$$

LHS equals RHS, hence $f(x)$ is continuous at $x = 1$.

**Hence Proved**

Since $\lim_{x\to 1} f(x) = f(1) = 2$, $f(x)$ is continuous at $x = 1$.

## Question 3

A function $f$ is defined as $f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x - 3}, & x \neq 3 \\ 5, & x = 3 \end{cases}$. Show that $f$ is continuous at $x = 3$.

**TYPE 1** **Checking Continuity at a Point**

**To Prove:** $f(x)$ is continuous at $x = 3$

**Concept / Formula Used:** $\lim_{x\to a} f(x) = f(a) \qquad x = a$

**Solution:**

Evaluate the limit of $f(x)$ as $x \to 3$:

$$\begin{aligned} \lim_{x\to 3} f(x) &= \lim_{x\to 3} \frac{x^2 - x - 6}{x - 3} \\ &= \lim_{x\to 3} \frac{(x-3)(x+2)}{x - 3} \\ &= \lim_{x\to 3} (x + 2) \\ &= 3 + 2 \\ &= 5 \end{aligned}$$

Given value at $x = 3$:

$$f(3) = 5$$

Thus:

$$\lim_{x\to 3} f(x) = f(3) = 5$$

Hence, $f$ is continuous at $x = 3$.

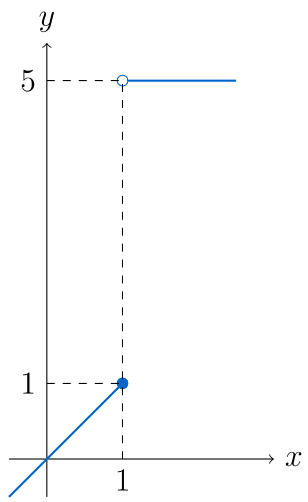
**Hence Proved**

Since $\lim_{x\to 3} f(x) = f(3) = 5$, the function $f$ is continuous at $x = 3$.

## Question 4

Is the function $f$ defined by $f(x) = \begin{cases} x, & \text{if } x \leq 1 \\ 5, & \text{if } x > 1 \end{cases}$ continuous at (i) $x = 0$ (ii) $x = 1$ (iii) $x = 2$?

**TYPE 2** **Continuity of Piecewise Functions**




*Graph of piecewise function $f(x)$ showing a step discontinuity at $x = 1$*

**Given:** $f(x) = \begin{cases} x, & x \leq 1 \\ 5, & x > 1 \end{cases}$

**Concept / Formula Used:** $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a) \qquad x = a$

**Solution:**

(i) At $x = 0$: In a small neighborhood of $x = 0$ ($x < 1$), $f(x) = x$.

$$\lim_{x\to 0} f(x) = \lim_{x\to 0}(x) = 0$$
$$f(0) = 0$$

Since $\lim_{x\to 0} f(x) = f(0) = 0$, $f$ is continuous at $x = 0$.

(ii) At $x = 1$:

$$\text{LHL} = \lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (x) = 1$$
$$\text{RHL} = \lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} (5) = 5$$

Since LHL $\neq$ RHL, $\lim_{x\to 1} f(x)$ does not exist. Hence, $f$ is discontinuous at $x = 1$.

(iii) At $x = 2$: In a small neighborhood of $x = 2$ $(x > 1)$, $f(x) = 5$.

$$\lim_{x \to 2} f(x) = \lim_{x \to 2} (5) = 5$$
$$f(2) = 5$$

Since $\lim_{x \to 2} f(x) = f(2) = 5$, $f$ is continuous at $x = 2$.

**Final Answer**

(i) Continuous at $x = 0$
(ii) Discontinuous at $x = 1$
(iii) Continuous at $x = 2$

## Question 5

Is the function $f$ defined by $f(x) = \begin{cases} 3x + 5, & \text{if } x \geq 2 \\ x^2, & \text{if } x < 2 \end{cases}$ continuous at $x = 2$?

**TYPE 1** **Checking Continuity at a Point**

**Given:** $f(x) = \begin{cases} 3x + 5, & x \geq 2 \\ x^2, & x < 2 \end{cases}$ at $x = 2$

**Concept / Formula Used:** $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a) \qquad x = a$

**Solution:**

Find the Left-Hand Limit (LHL) at $x = 2$:

$$\begin{aligned} \text{LHL} &= \lim_{x \to 2^-} f(x) \\ &= \lim_{x \to 2^-} (x^2) \\ &= 2^2 \\ &= 4 \end{aligned}$$

Find the Right-Hand Limit (RHL) at $x = 2$:

$$\begin{aligned} \text{RHL} &= \lim_{x \to 2^+} f(x) \\ &= \lim_{x \to 2^+} (3x + 5) \\ &= 3(2) + 5 \\ &= 6 + 5 \\ &= 11 \end{aligned}$$

Since LHL $= 4 \neq 11 =$ RHL, the limit does not exist. Hence, $f(x)$ is not continuous at $x = 2$.

**Final Answer**

No, $f(x)$ is not continuous at $x = 2$ because $\lim_{x\to 2^-} f(x) = 4 \neq 11 = \lim_{x\to 2^+} f(x)$.

## Question 6

Is the function $f$ defined by $f(x) = \begin{cases} \dfrac{2x^2 - 3x - 2}{x - 2}, & \text{if } x \neq 2 \\ 5, & \text{if } x = 2 \end{cases}$ continuous at $x = 2$?

**TYPE 1** **Checking Continuity at a Point**

**Given:** $f(x) = \begin{cases} \dfrac{2x^2 - 3x - 2}{x - 2}, & x \neq 2 \\ 5, & x = 2 \end{cases}$ at $x = 2$

**Concept / Formula Used:** $\lim_{x\to a} f(x) = f(a) \qquad x = a$

**Solution:**

Evaluate the limit of $f(x)$ as $x \to 2$:

$$\begin{aligned} \lim_{x\to 2} f(x) &= \lim_{x\to 2} \frac{2x^2 - 3x - 2}{x - 2} \\ &= \lim_{x\to 2} \frac{2x^2 - 4x + x - 2}{x - 2} \\ &= \lim_{x\to 2} \frac{2x(x-2) + 1(x-2)}{x - 2} \\ &= \lim_{x\to 2} \frac{(2x+1)(x-2)}{x - 2} \\ &= \lim_{x\to 2} (2x + 1) \\ &= 2(2) + 1 \\ &= 5 \end{aligned}$$

Given value of function at $x = 2$:

$$f(2) = 5$$

Since $\lim_{x\to 2} f(x) = f(2) = 5$, the function $f(x)$ is continuous at $x = 2$.

**Final Answer**

Yes, $f(x)$ is continuous at $x = 2$.

## Question 7

If $f(x) = \begin{cases} kx^2 + 5, & x \leq 1 \\ 2, & x > 1 \end{cases}$, find $k$ so that $f$ may be continuous at $x = 1$.

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} kx^2 + 5, & x \leq 1 \\ 2, & x > 1 \end{cases}$ continuous at $x = 1$

**Concept / Formula Used:** $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x) = f(1)$

**Solution:**

Calculate Left-Hand Limit (LHL) at $x = 1$:

$$\begin{aligned} \text{LHL} &= \lim_{x \to 1^-} (kx^2 + 5) \\ &= k(1)^2 + 5 \\ &= k + 5 \end{aligned}$$

Calculate Right-Hand Limit (RHL) at $x = 1$:

$$\begin{aligned} \text{RHL} &= \lim_{x \to 1^+} (2) \\ &= 2 \end{aligned}$$

Also, $f(1) = k(1)^2 + 5 = k + 5$. For $f$ to be continuous at $x = 1$, we must have:

$$\begin{aligned} \text{LHL} &= \text{RHL} \\ k + 5 &= 2 \\ k &= 2 - 5 \\ k &= -3 \end{aligned}$$

**Final Answer**

$k = -3$

## Question 8

If $f(x) = \begin{cases} 3x - 8, & x \leq 5 \\ 2k, & x > 5 \end{cases}$, find $k$ so that $f$ may be continuous at $x = 5$.

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} 3x - 8, & x \leq 5 \\ 2k, & x > 5 \end{cases}$ continuous at $x = 5$

**Concept / Formula Used:** $\lim_{x \to 5^-} f(x) = \lim_{x \to 5^+} f(x) = f(5)$

**Solution:**

Calculate Left-Hand Limit (LHL) at $x = 5$:

$$\begin{aligned}\text{LHL} &= \lim_{x\to 5^-}(3x-8)\\ &= 3(5)-8\\ &= 15-8\\ &= 7\end{aligned}$$

Calculate Right-Hand Limit (RHL) at $x = 5$:

$$\begin{aligned}\text{RHL} &= \lim_{x\to 5^+}(2k)\\ &= 2k\end{aligned}$$

For continuity at $x = 5$, equate LHL and RHL:

$$\begin{aligned}2k &= 7\\ k &= \frac{7}{2}\end{aligned}$$

**Final Answer**

$k = \frac{7}{2}$

## Question 9

If $f(x) = \begin{cases} 3x-4, & 0 \le x \le 2 \\ 2x+\lambda, & 2 < x \le 5 \end{cases}$, find $\lambda$ so that $f$ may be continuous at $x = 2$.

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} 3x-4, & 0 \le x \le 2 \\ 2x+\lambda, & 2 < x \le 5 \end{cases}$ continuous at $x = 2$

**Concept / Formula Used:** $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = f(2)$

**Solution:**

Calculate Left-Hand Limit (LHL) at $x = 2$:

$$\begin{aligned}\text{LHL} &= \lim_{x\to 2^-}(3x-4)\\ &= 3(2)-4\\ &= 6-4\\ &= 2\end{aligned}$$

Calculate Right-Hand Limit (RHL) at $x = 2$:

$$\begin{aligned}\text{RHL} &= \lim_{x\to 2^+}(2x+\lambda)\\ &= 2(2)+\lambda\\ &= 4+\lambda\end{aligned}$$

Also, $f(2) = 3(2) - 4 = 2$. For $f(x)$ to be continuous at $x = 2$:

$$\begin{aligned} \text{RHL} &= \text{LHL} \\ 4 + \lambda &= 2 \\ \lambda &= 2 - 4 \\ \lambda &= -2 \end{aligned}$$

**Final Answer**

$\lambda = -2$

## Question 10 (i)

$f(x) = \dfrac{x^2 - 9}{x - 3}$ is not defined at $x = 3$. What type of removable discontinuity is there? What value should be assigned to $f(3)$ for continuity of $f(x)$ at $x = 3$?

**TYPE 4 Removable Discontinuity Analysis**

**Given:** $f(x) = \dfrac{x^2 - 9}{x - 3}$ for $x \neq 3$

**Concept / Formula Used:** $\lim_{x \to a} f(x) = f(a)$

**Solution:**

Calculate the limit of $f(x)$ as $x \to 3$:

$$\begin{aligned} \lim_{x \to 3} f(x) &= \lim_{x \to 3} \frac{x^2 - 9}{x - 3} \\ &= \lim_{x \to 3} \frac{(x-3)(x+3)}{x - 3} \\ &= \lim_{x \to 3} (x + 3) \\ &= 3 + 3 \\ &= 6 \end{aligned}$$

Since $\lim_{x \to 3} f(x)$ exists finitely $(= 6)$ but $f(3)$ is undefined, the discontinuity is a **missing point removable discontinuity**. To redefine $f(x)$ so that it is continuous at $x = 3$, we must assign $f(3) = \lim_{x \to 3} f(x) = 6$.

**Final Answer**

Type of discontinuity: Removable (Missing Point) Discontinuity.
Value assigned to $f(3) = 6$.

## Question 10 (ii)

If $f(x) = \begin{cases} \dfrac{x^2 - 2x - 3}{x + 1}, & x \neq -1 \\ k, & x = -1 \end{cases}$, find $k$ so that the function $f$ may be continuous at $x = -1$.

**TYPE 3** **Finding Constants for Continuity**

★ **Likely Exam Question**

**Given:** $f(x) = \begin{cases} \dfrac{x^2 - 2x - 3}{x + 1}, & x \neq -1 \\ k, & x = -1 \end{cases}$ continuous at $x = -1$

**Concept / Formula Used:** $\lim_{x \to a} f(x) = f(a)$

**Solution:**

Find the limit of $f(x)$ as $x \to -1$:

$$\begin{aligned} \lim_{x \to -1} f(x) &= \lim_{x \to -1} \frac{x^2 - 2x - 3}{x + 1} \\ &= \lim_{x \to -1} \frac{(x + 1)(x - 3)}{x + 1} \\ &= \lim_{x \to -1} (x - 3) \\ &= -1 - 3 \\ &= -4 \end{aligned}$$

Given that $f(-1) = k$. For continuity at $x = -1$:

$$\begin{aligned} f(-1) &= \lim_{x \to -1} f(x) \\ k &= -4 \end{aligned}$$

**Final Answer**

$k = -4$

## Question 10 (iii)

Determine the value of '$k$' for which the following function is continuous at $x = 3$:

$$f(x) = \begin{cases} \dfrac{(x + 3)^2 - 36}{x - 3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} \dfrac{(x+3)^2 - 36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$ continuous at $x = 3$

**Concept / Formula Used:** $\lim_{x \to a} f(x) = f(a)$

**Solution:**

Evaluate the limit of $f(x)$ as $x \to 3$:

$$\begin{aligned} \lim_{x \to 3} f(x) &= \lim_{x \to 3} \frac{(x+3)^2 - 6^2}{x-3} \\ &= \lim_{x \to 3} \frac{[(x+3)-6][(x+3)+6]}{x-3} \\ &= \lim_{x \to 3} \frac{(x-3)(x+9)}{x-3} \\ &= \lim_{x \to 3}(x+9) \\ &= 3 + 9 \\ &= 12 \end{aligned}$$

Since $f(3) = k$, for $f(x)$ to be continuous at $x = 3$:

$$\begin{aligned} f(3) &= \lim_{x \to 3} f(x) \\ k &= 12 \end{aligned}$$

**Final Answer**

$k = 12$

## Question 10 (iv)

If $f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x^2 - 2x - 3}, & x \neq 3 \\ k, & x = 3 \end{cases}$, find $k$ so that the function $f$ may be continuous at $x = 3$.

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x^2 - 2x - 3}, & x \neq 3 \\ k, & x = 3 \end{cases}$ continuous at $x = 3$

**Concept / Formula Used:** $\lim_{x \to a} f(x) = f(a)$

**Solution:**

Factorize both numerator and denominator:

$$x^2 - x - 6 = (x-3)(x+2)$$
$$x^2 - 2x - 3 = (x-3)(x+1)$$

Evaluate the limit of $f(x)$ as $x \to 3$:

$$\begin{aligned}\lim_{x\to 3} f(x) &= \lim_{x\to 3} \frac{(x-3)(x+2)}{(x-3)(x+1)} \\ &= \lim_{x\to 3} \frac{x+2}{x+1} \\ &= \frac{3+2}{3+1} \\ &= \frac{5}{4}\end{aligned}$$

For continuity at $x = 3$:

$$f(3) = \lim_{x\to 3} f(x)$$
$$k = \frac{5}{4}$$

**Final Answer**

$k = \frac{5}{4}$

## Question 11

If $f(x) = \begin{cases} \dfrac{\tan 3x}{kx}, & x \neq 0 \\ 1, & x = 0 \end{cases}$, find $k$ so that the function $f$ may be continuous at $x = 0$.

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} \dfrac{\tan 3x}{kx}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ continuous at $x = 0$

**Concept / Formula Used:** $\lim_{\theta\to 0} \frac{\tan\theta}{\theta} = 1$

**Solution:**

Evaluate the limit as $x \to 0$:

$$\begin{aligned}\lim_{x\to 0} f(x) &= \lim_{x\to 0} \frac{\tan 3x}{kx} \\ &= \lim_{x\to 0} \left(\frac{\tan 3x}{3x} \cdot \frac{3}{k}\right) \\ &= \frac{3}{k} \cdot \lim_{3x\to 0} \frac{\tan 3x}{3x}\end{aligned}$$

Applying the standard result $\lim_{\theta\to 0}\dfrac{\tan\theta}{\theta} = 1$:

$$\begin{aligned}\lim_{x\to 0} f(x) &= \frac{3}{k}\cdot 1\\ &= \frac{3}{k}\end{aligned}$$

For $f(x)$ to be continuous at $x = 0$:

$$\begin{aligned}\lim_{x\to 0} f(x) &= f(0)\\ \frac{3}{k} &= 1\\ k &= 3\end{aligned}$$

**Final Answer**

$k = 3$

## Question 12

Is the function $f$ defined by $f(x) = \tan x$ continuous at $x = \dfrac{\pi}{2}$?

**TYPE 1** **Checking Continuity at a Point**

**Given:** $f(x) = \tan x$ at $x = \dfrac{\pi}{2}$

**Concept / Formula Used:** $\lim_{x\to a} f(x) = f(a) \qquad x = a$

**Solution:**

Evaluate the function at $x = \dfrac{\pi}{2}$:

$$f\left(\frac{\pi}{2}\right) = \tan\left(\frac{\pi}{2}\right) \quad \text{(undefined)}$$

Also, evaluating left-hand and right-hand limits:

$$\begin{aligned}\lim_{x\to(\pi/2)^-} \tan x &= +\infty\\ \lim_{x\to(\pi/2)^+} \tan x &= -\infty\end{aligned}$$

Since $f\left(\dfrac{\pi}{2}\right)$ is not defined and the limit does not exist, $f(x)$ is not continuous at $x = \dfrac{\pi}{2}$.

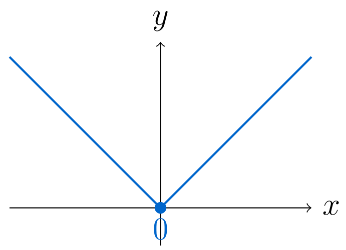
**Final Answer**

No, $f(x) = \tan x$ is not continuous at $x = \dfrac{\pi}{2}$ because $f\left(\dfrac{\pi}{2}\right)$ is undefined.

## Question 13

Is the function $f$ defined by $f(x) = |x|$ continuous at $x = 0$?

**TYPE 5** **Continuity of Absolute Value Functions**




*Graph of $f(x) = |x|$ continuous at the origin $x = 0$*

**Given:** $f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$ at $x = 0$

**Concept / Formula Used:** $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) = f(0)$

**Solution:**

Evaluate Left-Hand Limit (LHL) at $x = 0$:

$$\begin{aligned} \text{LHL} &= \lim_{x \to 0^-} |x| \\ &= \lim_{x \to 0^-} (-x) \\ &= -0 \\ &= 0 \end{aligned}$$

Evaluate Right-Hand Limit (RHL) at $x = 0$:

$$\begin{aligned} \text{RHL} &= \lim_{x \to 0^+} |x| \\ &= \lim_{x \to 0^+} (x) \\ &= 0 \end{aligned}$$

Evaluate function value at $x = 0$:

$$f(0) = |0| = 0$$

Since LHL $=$ RHL $= f(0) = 0$, $f(x) = |x|$ is continuous at $x = 0$.

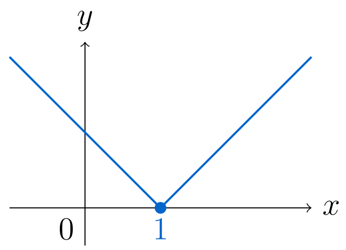
**Final Answer**

Yes, $f(x) = |x|$ is continuous at $x = 0$.

## Question 14

Is the function $f$ defined by $f(x) = |x - 1|$ continuous at $x = 1$?

**TYPE 5** **Continuity of Absolute Value Functions**




*Graph of $f(x) = |x - 1|$ continuous at $x = 1$*

**Given:** $f(x) = |x - 1| = \begin{cases} x - 1, & x \geq 1 \\ -(x - 1), & x < 1 \end{cases}$ at $x = 1$

**Concept / Formula Used:** $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x) = f(1)$

**Solution:**

Evaluate Left-Hand Limit (LHL) at $x = 1$:

$$\begin{aligned} \text{LHL} &= \lim_{x \to 1^-} |x - 1| \\ &= \lim_{x \to 1^-} [-(x - 1)] \\ &= \lim_{x \to 1^-} (1 - x) \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

Evaluate Right-Hand Limit (RHL) at $x = 1$:

$$\begin{aligned} \text{RHL} &= \lim_{x \to 1^+} |x - 1| \\ &= \lim_{x \to 1^+} (x - 1) \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

Evaluate function value at $x = 1$:

$$f(1) = |1 - 1| = 0$$

Since $\text{LHL} = \text{RHL} = f(1) = 0$, $f(x) = |x - 1|$ is continuous at $x = 1$.

**Final Answer**

Yes, $f(x) = |x - 1|$ is continuous at $x = 1$.

## Question 15

Is the function $f$ defined by $f(x) = x - |x|$ continuous at $x = 0$?

**TYPE 5** **Continuity of Absolute Value Functions**

**Given:** $f(x) = x - |x|$ at $x = 0$

**Concept / Formula Used:** $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = f(0)$

**Solution:**

Redefining $f(x)$ using the definition of $|x|$:

$$\begin{aligned} \text{For } x \geq 0, \quad |x| = x \implies f(x) = x - x = 0 \\ \text{For } x < 0, \quad |x| = -x \implies f(x) = x - (-x) = 2x \end{aligned}$$

Thus, $f(x) = \begin{cases} 0, & x \geq 0 \\ 2x, & x < 0 \end{cases}$.

Evaluate Left-Hand Limit (LHL) at $x = 0$:

$$\begin{aligned} \text{LHL} &= \lim_{x\to 0^-} f(x) \\ &= \lim_{x\to 0^-} (2x) \\ &= 2(0) \\ &= 0 \end{aligned}$$

Evaluate Right-Hand Limit (RHL) at $x = 0$:

$$\begin{aligned} \text{RHL} &= \lim_{x\to 0^+} f(x) \\ &= \lim_{x\to 0^+} (0) \\ &= 0 \end{aligned}$$

Evaluate function value at $x = 0$:

$$f(0) = 0 - |0| = 0$$

Since LHL $=$ RHL $= f(0) = 0$, $f(x)$ is continuous at $x = 0$.

**Final Answer**

Yes, $f(x) = x - |x|$ is continuous at $x = 0$.

## Question 16 (i)

Examine the following function for continuity at the indicated point:

$$f(x) = \begin{cases} x^2, & x \geq 0 \\ -x, & x < 0 \end{cases} \quad \text{at } x = 0$$

**TYPE 2** **Continuity of Piecewise Functions**

**Given:** $f(x) = \begin{cases} x^2, & x \geq 0 \\ -x, & x < 0 \end{cases}$ at $x = 0$

**Concept / Formula Used:** $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) = f(0)$

**Solution:**

Evaluate Left-Hand Limit (LHL) at $x = 0$:

$$\begin{aligned} \text{LHL} &= \lim_{x \to 0^-} (-x) \\ &= -0 \\ &= 0 \end{aligned}$$

Evaluate Right-Hand Limit (RHL) at $x = 0$:

$$\begin{aligned} \text{RHL} &= \lim_{x \to 0^+} (x^2) \\ &= 0^2 \\ &= 0 \end{aligned}$$

Evaluate function value at $x = 0$:

$$f(0) = 0^2 = 0$$

Since LHL $=$ RHL $= f(0) = 0$, $f(x)$ is continuous at $x = 0$.

**Final Answer**

$f(x)$ is continuous at $x = 0$.

## Question 16 (ii)

Examine the following function for continuity at the indicated point:

$$f(x) = \begin{cases} 5x - 4, & \text{if } x < 1 \\ 4x^2 - 3x, & \text{if } x \geq 1 \end{cases} \quad \text{at } x = 1$$

**TYPE 2** **Continuity of Piecewise Functions**

**Given:** $f(x) = \begin{cases} 5x - 4, & x < 1 \\ 4x^2 - 3x, & x \geq 1 \end{cases}$ at $x = 1$

**Concept / Formula Used:** $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x) = f(1)$

**Solution:**

Evaluate Left-Hand Limit (LHL) at $x = 1$:

$$\begin{aligned} \text{LHL} &= \lim_{x \to 1^-} (5x - 4) \\ &= 5(1) - 4 \\ &= 5 - 4 \\ &= 1 \end{aligned}$$

Evaluate Right-Hand Limit (RHL) at $x = 1$:

$$\begin{aligned} \text{RHL} &= \lim_{x \to 1^+} (4x^2 - 3x) \\ &= 4(1)^2 - 3(1) \\ &= 4 - 3 \\ &= 1 \end{aligned}$$

Evaluate function value at $x = 1$:

$$\begin{aligned} f(1) &= 4(1)^2 - 3(1) \\ &= 4 - 3 \\ &= 1 \end{aligned}$$

Since $\text{LHL} = \text{RHL} = f(1) = 1$, $f(x)$ is continuous at $x = 1$.

**Final Answer**

$f(x)$ is continuous at $x = 1$.

## Question 17

Is the function $f$ defined by

$$f(x) = \begin{cases} 2x^2 - 3, & \text{if } x \leq 1 \\ 5, & \text{if } x > 1 \end{cases}$$

continuous at $x = 0$? What about its continuity at $x = 1$ and at $x = 2$?

**TYPE 2** **Continuity of Piecewise Functions**

**Given:** $f(x) = \begin{cases} 2x^2 - 3, & \text{if } x \leq 1 \\ 5, & \text{if } x > 1 \end{cases}$

**Concept / Formula Used:** $f(x)$ $x = c$ $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = f(c)$

**Solution:**

**1. Continuity at $x = 0$:** Since $0 < 1$, in a neighbourhood of $x = 0$, the function is defined as $f(x) = 2x^2 - 3$.

$$\begin{aligned}\lim_{x \to 0} f(x) &= \lim_{x \to 0}(2x^2 - 3)\\ &= 2(0)^2 - 3\\ &= -3\end{aligned}$$

The value of the function at $x = 0$ is:

$$\begin{aligned}f(0) &= 2(0)^2 - 3\\ &= -3\end{aligned}$$

Since $\lim_{x \to 0} f(x) = f(0) = -3$, the function $f$ is continuous at $x = 0$.

**2. Continuity at $x = 1$:** Left-Hand Limit (LHL) at $x = 1$:

$$\begin{aligned}\text{LHL} &= \lim_{x \to 1^-} f(x)\\ &= \lim_{x \to 1^-} (2x^2 - 3)\\ &= 2(1)^2 - 3\\ &= 2 - 3\\ &= -1\end{aligned}$$

Right-Hand Limit (RHL) at $x = 1$:

$$\begin{aligned}\text{RHL} &= \lim_{x \to 1^+} f(x)\\ &= \lim_{x \to 1^+} (5)\\ &= 5\end{aligned}$$

Since LHL $\neq$ RHL (i.e., $-1 \neq 5$), $\lim_{x \to 1} f(x)$ does not exist. Hence, $f$ is discontinuous at $x = 1$.

**3. Continuity at $x = 2$:** Since $2 > 1$, in a neighbourhood of $x = 2$, the function is defined as $f(x) = 5$.

$$\begin{aligned}\lim_{x \to 2} f(x) &= \lim_{x \to 2}(5)\\ &= 5\end{aligned}$$

The value of the function at $x = 2$ is:

$$f(2) = 5$$

Since $\lim_{x \to 2} f(x) = f(2) = 5$, the function $f$ is continuous at $x = 2$.

**Final Answer**

The function $f$ is continuous at $x = 0$ and $x = 2$, but discontinuous at $x = 1$.

## Question 18 (i)

Examine the following function for continuity at $x = 0$:

$$f(x) = \begin{cases} \dfrac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

**TYPE 5** **Continuity of Absolute Value Functions**

**Given:** $f(x) = \begin{cases} \dfrac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ at $c = 0$

**Concept / Formula Used:** $|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$

**Solution:**

Left-Hand Limit (LHL) at $x = 0$: For $x < 0$, $|x| = -x$.

$$\begin{aligned} \text{LHL} &= \lim_{x \to 0^-} \frac{x}{|x|} \\ &= \lim_{x \to 0^-} \frac{x}{-x} \\ &= \lim_{x \to 0^-} (-1) \\ &= -1 \end{aligned}$$

Right-Hand Limit (RHL) at $x = 0$: For $x > 0$, $|x| = x$.

$$\begin{aligned} \text{RHL} &= \lim_{x \to 0^+} \frac{x}{|x|} \\ &= \lim_{x \to 0^+} \frac{x}{x} \\ &= \lim_{x \to 0^+} (1) \\ &= 1 \end{aligned}$$

Since LHL $\neq$ RHL ($-1 \neq 1$), the limit $\lim_{x \to 0} f(x)$ does not exist.

**Final Answer**

$f(x)$ is discontinuous at $x = 0$.

**Common Mistake:** Evaluating $\dfrac{0}{|0|}$ directly leads to an indeterminate form. Always evaluate left-hand and right-hand limits separately when absolute value functions are involved.

## Question 18 (ii)

Examine the following function for continuity at $x = 4$:

$$f(x) = \begin{cases} \dfrac{x-4}{2|x-4|}, & x \neq 4 \\ 0, & x = 4 \end{cases}$$

**TYPE 5** **Continuity of Absolute Value Functions**

**Given:** $f(x) = \begin{cases} \dfrac{x-4}{2|x-4|}, & x \neq 4 \\ 0, & x = 4 \end{cases}$ at $c = 4$

**Concept / Formula Used:** $|x-a| = \begin{cases} x-a, & x > a \\ -(x-a), & x < a \end{cases}$

**Solution:**

Left-Hand Limit (LHL) at $x = 4$: For $x < 4$, $(x-4) < 0$, so $|x-4| = -(x-4)$.

$$\begin{aligned} \text{LHL} &= \lim_{x \to 4^-} \frac{x-4}{2|x-4|} \\ &= \lim_{x \to 4^-} \frac{x-4}{2(-(x-4))} \\ &= \lim_{x \to 4^-} \left(-\frac{1}{2}\right) \\ &= -\frac{1}{2} \end{aligned}$$

Right-Hand Limit (RHL) at $x = 4$: For $x > 4$, $(x-4) > 0$, so $|x-4| = x-4$.

$$\begin{aligned} \text{RHL} &= \lim_{x \to 4^+} \frac{x-4}{2|x-4|} \\ &= \lim_{x \to 4^+} \frac{x-4}{2(x-4)} \\ &= \lim_{x \to 4^+} \left(\frac{1}{2}\right) \\ &= \frac{1}{2} \end{aligned}$$

Since LHL $\neq$ RHL (i.e., $-\frac{1}{2} \neq \frac{1}{2}$), $\lim_{x \to 4} f(x)$ does not exist.

**Final Answer**

$f(x)$ is discontinuous at $x = 4$.

## Question 19 (i)

Examine the following function for continuity at $x = 0$:

$$f(x) = \begin{cases} \dfrac{x}{\sin 3x}, & \text{when } x \neq 0 \\ 3, & \text{when } x = 0 \end{cases}$$

**TYPE 6 Continuity Using Trigonometric Limits**

**Given:** $f(x) = \begin{cases} \dfrac{x}{\sin 3x}, & x \neq 0 \\ 3, & x = 0 \end{cases}$ at $c = 0$

**Concept / Formula Used:** $\lim_{\theta \to 0} \dfrac{\sin \theta}{\theta} = 1$

**Solution:**

We evaluate the limit of $f(x)$ as $x \to 0$:

$$\begin{aligned} \lim_{x \to 0} f(x) &= \lim_{x \to 0} \frac{x}{\sin 3x} \\ &= \lim_{x \to 0} \frac{1}{3} \cdot \frac{3x}{\sin 3x} \\ &= \frac{1}{3} \cdot \frac{1}{\lim_{x \to 0} \dfrac{\sin 3x}{3x}} \end{aligned}$$

Using the standard result $\lim_{\theta \to 0} \dfrac{\sin \theta}{\theta} = 1$ with $\theta = 3x$:

$$\begin{aligned} \lim_{x \to 0} f(x) &= \frac{1}{3} \cdot \frac{1}{1} \\ &= \frac{1}{3} \end{aligned}$$

The given value of the function at $x = 0$ is:

$$f(0) = 3$$

Since $\lim_{x \to 0} f(x) = \dfrac{1}{3} \neq 3 = f(0)$, the function is not continuous at $x = 0$.

**Final Answer**

$f(x)$ is discontinuous at $x = 0$.

## Question 19 (ii)

Examine the following function for continuity at $x = 0$:

$$f(x) = \begin{cases} \dfrac{\sin 2x}{\sin 3x}, & \text{when } x \neq 0 \\ 2, & \text{when } x = 0 \end{cases}$$

**TYPE 6** **Continuity Using Trigonometric Limits**

**Given:** $f(x) = \begin{cases} \dfrac{\sin 2x}{\sin 3x}, & x \neq 0 \\ 2, & x = 0 \end{cases}$ at $c = 0$

**Concept / Formula Used:** $\lim_{\theta \to 0} \dfrac{\sin \theta}{\theta} = 1$

**Solution:**

Evaluating the limit of $f(x)$ as $x \to 0$:

$$\begin{aligned} \lim_{x \to 0} f(x) &= \lim_{x \to 0} \frac{\sin 2x}{\sin 3x} \\ &= \lim_{x \to 0} \frac{\frac{\sin 2x}{2x} \cdot 2x}{\frac{\sin 3x}{3x} \cdot 3x} \\ &= \lim_{x \to 0} \left( \frac{2}{3} \cdot \frac{\frac{\sin 2x}{2x}}{\frac{\sin 3x}{3x}} \right) \\ &= \frac{2}{3} \cdot \frac{\lim_{x \to 0} \frac{\sin 2x}{2x}}{\lim_{x \to 0} \frac{\sin 3x}{3x}} \end{aligned}$$

Applying the standard limit $\lim_{\theta \to 0} \dfrac{\sin \theta}{\theta} = 1$:

$$\begin{aligned} \lim_{x \to 0} f(x) &= \frac{2}{3} \cdot \frac{1}{1} \\ &= \frac{2}{3} \end{aligned}$$

Given value of $f(0)$:

$$f(0) = 2$$

Since $\lim_{x \to 0} f(x) = \dfrac{2}{3} \neq 2 = f(0)$, the function is not continuous at $x = 0$.

**Final Answer**

$f(x)$ is discontinuous at $x = 0$.

## Question 19 (iii)

Examine the following function for continuity at $x = 0$:

$$f(x) = \begin{cases} \dfrac{\cos 3x - \cos x}{x^2}, & \text{if } x \neq 0 \\ -4, & \text{if } x = 0 \end{cases}$$

**TYPE 6** **Continuity Using Trigonometric Limits**

**★ Likely Exam Question**

**Given:** $f(x) = \begin{cases} \dfrac{\cos 3x - \cos x}{x^2}, & x \neq 0 \\ -4, & x = 0 \end{cases}$ at $c = 0$

**Concept / Formula Used:** $\cos C - \cos D = -2\sin\left(\dfrac{C+D}{2}\right)\sin\left(\dfrac{C-D}{2}\right)$ $\lim_{\theta \to 0} \dfrac{\sin\theta}{\theta} = 1$

**Solution:**

Applying the identity $\cos 3x - \cos x = -2\sin\left(\dfrac{3x+x}{2}\right)\sin\left(\dfrac{3x-x}{2}\right)$:

$$\cos 3x - \cos x = -2\sin(2x)\sin(x)$$

Now evaluating the limit as $x \to 0$:

$$\begin{aligned}
\lim_{x\to 0} f(x) &= \lim_{x\to 0} \frac{-2\sin 2x \sin x}{x^2} \\
&= -2 \cdot \lim_{x\to 0}\left(\frac{\sin 2x}{x} \cdot \frac{\sin x}{x}\right) \\
&= -2 \cdot \lim_{x\to 0}\left(2 \cdot \frac{\sin 2x}{2x} \cdot \frac{\sin x}{x}\right) \\
&= -4 \cdot \left(\lim_{x\to 0}\frac{\sin 2x}{2x}\right) \cdot \left(\lim_{x\to 0}\frac{\sin x}{x}\right)
\end{aligned}$$

Using the standard result $\lim_{\theta\to 0}\dfrac{\sin\theta}{\theta} = 1$:

$$\begin{aligned}
\lim_{x\to 0} f(x) &= -4 \cdot (1) \cdot (1) \\
&= -4
\end{aligned}$$

The given functional value is $f(0) = -4$. Since $\lim_{x\to 0} f(x) = f(0) = -4$, the function is continuous at $x = 0$.

**Final Answer**

$f(x)$ is continuous at $x = 0$.

## Question 20 (i)

If $f(x) = \begin{cases} \dfrac{x^2 - 25}{x - 5}, & x \neq 5 \\ k, & x = 5 \end{cases}$ is continuous at $x = 5$, find the value of $k$.

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} \dfrac{x^2 - 25}{x - 5}, & x \neq 5 \\ k, & x = 5 \end{cases}$ is continuous at $x = 5$

**Concept / Formula Used:** $f(x)$ $x = c$ $\lim_{x \to c} f(x) = f(c)$

**Solution:**

Evaluating the limit of $f(x)$ as $x \to 5$:

$$\lim_{x \to 5} f(x) = \lim_{x \to 5} \frac{x^2 - 25}{x - 5}$$

Factoring the numerator: $x^2 - 25 = (x - 5)(x + 5)$.

$$\lim_{x \to 5} f(x) = \lim_{x \to 5} \frac{(x - 5)(x + 5)}{x - 5}$$

Since $x \neq 5$, we divide numerator and denominator by $(x - 5)$:

$$\begin{aligned} \lim_{x \to 5} f(x) &= \lim_{x \to 5} (x + 5) \\ &= 5 + 5 \\ &= 10 \end{aligned}$$

Given $f(5) = k$. For continuity at $x = 5$:

$$\begin{aligned} f(5) &= \lim_{x \to 5} f(x) \\ k &= 10 \end{aligned}$$

**Final Answer**

$k = 10$

## Question 20 (ii)

Find the value of $k$ so that the function $f(x) = \begin{cases} kx + 1, & \text{if } x \leq 5 \\ 3x - 5, & \text{if } x > 5 \end{cases}$ is continuous at $x = 5$.

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} kx+1, & x \le 5 \\ 3x-5, & x > 5 \end{cases}$ is continuous at $x = 5$

**Concept / Formula Used:** LHL = RHL = $f(5)$

**Solution:**

Left-Hand Limit (LHL) and functional value at $x = 5$:

$$\begin{aligned} \text{LHL} = f(5) &= \lim_{x \to 5^-} (kx+1) \\ &= k(5) + 1 \\ &= 5k + 1 \end{aligned}$$

Right-Hand Limit (RHL) at $x = 5$:

$$\begin{aligned} \text{RHL} &= \lim_{x \to 5^+} (3x-5) \\ &= 3(5) - 5 \\ &= 15 - 5 \\ &= 10 \end{aligned}$$

Since $f$ is continuous at $x = 5$, we equate LHL and RHL:

$$\begin{aligned} 5k + 1 &= 10 \\ 5k &= 10 - 1 \\ 5k &= 9 \\ k &= \frac{9}{5} \end{aligned}$$

**Final Answer**

$k = \frac{9}{5}$

## Question 20 (iii)

For what value of $k$ is the following function continuous at $x = 2$?

$$f(x) = \begin{cases} 2x+1, & x < 2 \\ k, & x = 2 \\ 3x-1, & x > 2 \end{cases}$$

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} 2x+1, & x < 2 \\ k, & x = 2 \\ 3x-1, & x > 2 \end{cases}$ is continuous at $x = 2$

**Concept / Formula Used:** LHL = RHL = $f(2)$

**Solution:**

Evaluating Left-Hand Limit (LHL) at $x = 2$:

$$\begin{aligned}\text{LHL} &= \lim_{x\to 2^-}(2x+1)\\ &= 2(2)+1\\ &= 4+1\\ &= 5\end{aligned}$$

Evaluating Right-Hand Limit (RHL) at $x = 2$:

$$\begin{aligned}\text{RHL} &= \lim_{x\to 2^+}(3x-1)\\ &= 3(2)-1\\ &= 6-1\\ &= 5\end{aligned}$$

Given $f(2) = k$. For continuity at $x = 2$:

$$\begin{aligned}f(2) &= \text{LHL} = \text{RHL}\\ k &= 5\end{aligned}$$

**Final Answer**

$k = 5$

## Question 20 (iv)

Find the value of $k$ so that the function $f$ defined by

$$f(x) = \begin{cases} kx+1, & \text{if } x \le \pi \\ \cos x, & \text{if } x > \pi \end{cases}$$

is continuous at $x = \pi$.

**TYPE 3** **Finding Constants for Continuity**

★ **Likely Exam Question**

**Given:** $f(x) = \begin{cases} kx+1, & x \le \pi \\ \cos x, & x > \pi \end{cases}$ is continuous at $x = \pi$

**Concept / Formula Used:** LHL = RHL = $f(\pi)$

**Solution:**

Left-Hand Limit (LHL) and value of function at $x = \pi$:

$$\begin{aligned} \text{LHL} = f(\pi) &= \lim_{x \to \pi^-} (kx + 1) \\ &= k\pi + 1 \end{aligned}$$

Right-Hand Limit (RHL) at $x = \pi$:

$$\begin{aligned} \text{RHL} &= \lim_{x \to \pi^+} \cos x \\ &= \cos \pi \end{aligned}$$

Using the trigonometric value $\cos \pi = -1$:

$$\text{RHL} = -1$$

Equating LHL and RHL for continuity:

$$\begin{aligned} k\pi + 1 &= -1 \\ k\pi &= -1 - 1 \\ k\pi &= -2 \\ k &= -\frac{2}{\pi} \end{aligned}$$

**Final Answer**

$k = -\dfrac{2}{\pi}$

## Question 20 (v)

For what value of $k$ is the function

$$f(x) = \begin{cases} \dfrac{\tan 5x}{\sin 2x}, & x \neq 0 \\ k, & x = 0 \end{cases}$$

continuous at $x = 0$?

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} \dfrac{\tan 5x}{\sin 2x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$

**Concept / Formula Used:** $\displaystyle\lim_{\theta \to 0} \frac{\tan \theta}{\theta} = 1 \qquad \lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1$

**Solution:**

Evaluating the limit of $f(x)$ as $x \to 0$:

$$\begin{aligned}
\lim_{x\to 0} f(x) &= \lim_{x\to 0} \frac{\tan 5x}{\sin 2x} \\
&= \lim_{x\to 0} \frac{\frac{\tan 5x}{5x}\cdot 5x}{\frac{\sin 2x}{2x}\cdot 2x} \\
&= \lim_{x\to 0} \left( \frac{5}{2} \cdot \frac{\frac{\tan 5x}{5x}}{\frac{\sin 2x}{2x}} \right) \\
&= \frac{5}{2} \cdot \frac{\lim_{x\to 0} \frac{\tan 5x}{5x}}{\lim_{x\to 0} \frac{\sin 2x}{2x}}
\end{aligned}$$

Using $\lim_{\theta\to 0} \frac{\tan\theta}{\theta} = 1$ and $\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$:

$$\begin{aligned}
\lim_{x\to 0} f(x) &= \frac{5}{2} \cdot \frac{1}{1} \\
&= \frac{5}{2}
\end{aligned}$$

For continuity at $x = 0$, $f(0) = \lim_{x\to 0} f(x)$:

$$k = \frac{5}{2}$$

**Final Answer**

$k = \frac{5}{2}$

## Question 20 (vi)

Find the value of $k$ for which the function $f$ given as

$$f(x) = \begin{cases} \dfrac{1-\cos x}{2x^2}, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$$

is continuous at $x = 0$.

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} \dfrac{1-\cos x}{2x^2}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$

**Concept / Formula Used:** $1 - \cos x = 2\sin^2\left(\frac{x}{2}\right)$ $\quad \lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$

**Solution:**

Substituting $1 - \cos x = 2\sin^2\left(\frac{x}{2}\right)$ into the limit expression:

$$\begin{aligned}
\lim_{x\to 0} f(x) &= \lim_{x\to 0} \frac{2\sin^2\left(\frac{x}{2}\right)}{2x^2} \\
&= \lim_{x\to 0} \frac{\sin^2\left(\frac{x}{2}\right)}{x^2} \\
&= \lim_{x\to 0} \left[\frac{\sin\left(\frac{x}{2}\right)}{x}\right]^2 \\
&= \lim_{x\to 0} \left[\frac{1}{2} \cdot \frac{\sin\left(\frac{x}{2}\right)}{\frac{x}{2}}\right]^2 \\
&= \frac{1}{4} \cdot \left[\lim_{x\to 0} \frac{\sin\left(\frac{x}{2}\right)}{\frac{x}{2}}\right]^2
\end{aligned}$$

Using the standard result $\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$:

$$\begin{aligned}
\lim_{x\to 0} f(x) &= \frac{1}{4} \cdot (1)^2 \\
&= \frac{1}{4}
\end{aligned}$$

For continuity, $f(0) = \lim_{x\to 0} f(x)$:

$$k = \frac{1}{4}$$

**Final Answer**

$k = \frac{1}{4}$

## Question 20 (vii)

Find the value(s) of 'λ' for which the function $f$ given as

$$f(x) = \begin{cases} \frac{\sin^2 \lambda x}{x^2}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$$

is continuous at $x = 0$.

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} \dfrac{\sin^2 \lambda x}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ is continuous at $x = 0$

**Concept / Formula Used:** $\lim\limits_{\theta \to 0} \dfrac{\sin \theta}{\theta} = 1$

**Solution:**

Evaluating the limit of $f(x)$ as $x \to 0$:

$$\begin{aligned} \lim_{x \to 0} f(x) &= \lim_{x \to 0} \frac{\sin^2 \lambda x}{x^2} \\ &= \lim_{x \to 0} \left( \frac{\sin \lambda x}{x} \right)^2 \\ &= \lim_{x \to 0} \left( \lambda \cdot \frac{\sin \lambda x}{\lambda x} \right)^2 \\ &= \lambda^2 \cdot \left( \lim_{x \to 0} \frac{\sin \lambda x}{\lambda x} \right)^2 \end{aligned}$$

Applying the standard limit $\lim\limits_{\theta \to 0} \dfrac{\sin \theta}{\theta} = 1$ with $\theta = \lambda x$:

$$\begin{aligned} \lim_{x \to 0} f(x) &= \lambda^2 \cdot (1)^2 \\ &= \lambda^2 \end{aligned}$$

For $f(x)$ to be continuous at $x = 0$:

$$\begin{aligned} \lim_{x \to 0} f(x) &= f(0) \\ \lambda^2 &= 1 \\ \lambda &= \pm 1 \end{aligned}$$

**Final Answer**

$\lambda = \pm 1$

## Question 21

Find the relationship between $a$ and $b$ so that the function $f$ defined by

$$f(x) = \begin{cases} ax + 1, & \text{if } x \leq 3 \\ bx + 3, & \text{if } x > 3 \end{cases}$$

is continuous at $x = 3$.

**TYPE 3** **Finding Constants for Continuity**

**Given:** $f(x) = \begin{cases} ax + 1, & x \leq 3 \\ bx + 3, & x > 3 \end{cases}$ is continuous at $x = 3$

**Concept / Formula Used:** $x = 3 \qquad \text{LHL} = \text{RHL} = f(3)$

**Solution:**

Evaluating Left-Hand Limit (LHL) and $f(3)$:

$$\begin{aligned} \text{LHL} = f(3) &= \lim_{x \to 3^-} (ax + 1) \\ &= a(3) + 1 \\ &= 3a + 1 \end{aligned}$$

Evaluating Right-Hand Limit (RHL) at $x = 3$:

$$\begin{aligned} \text{RHL} &= \lim_{x \to 3^+} (bx + 3) \\ &= b(3) + 3 \\ &= 3b + 3 \end{aligned}$$

Since the function $f$ is continuous at $x = 3$, we set LHL = RHL:

$$\begin{aligned} 3a + 1 &= 3b + 3 \\ 3a - 3b &= 3 - 1 \\ 3a - 3b &= 2 \end{aligned}$$

$$3a = 3b + 2 \quad \text{or} \quad a = b + \frac{2}{3}$$

**Final Answer**

$3a = 3b + 2 \quad \left(\text{or } a - b = \frac{2}{3}\right)$

## Question 22

If the function $f$ defined by

$$f(x) = \begin{cases} 3ax + b, & \text{if } x > 1 \\ 11, & \text{if } x = 1 \\ 5ax - 2b, & \text{if } x < 1 \end{cases}$$

is continuous at $x = 1$, find the values of $a$ and $b$.

**TYPE 3** **Finding Constants for Continuity**

★ **Likely Exam Question**

**Given:** $f(x) = \begin{cases} 3ax + b, & x > 1 \\ 11, & x = 1 \\ 5ax - 2b, & x < 1 \end{cases}$ is continuous at $x = 1$

**Concept / Formula Used:** $x = 1 \qquad \text{LHL} = \text{RHL} = f(1)$

**Solution:**

Left-Hand Limit (LHL) at $x = 1$:

$$\begin{aligned} \text{LHL} &= \lim_{x \to 1^-} (5ax - 2b) \\ &= 5a(1) - 2b \\ &= 5a - 2b \end{aligned}$$

Right-Hand Limit (RHL) at $x = 1$:

$$\begin{aligned} \text{RHL} &= \lim_{x \to 1^+} (3ax + b) \\ &= 3a(1) + b \\ &= 3a + b \end{aligned}$$

Given functional value $f(1) = 11$. Since $f$ is continuous at $x = 1$:

$$\begin{aligned} \text{RHL} = f(1) &\implies 3a + b = 11 \quad \text{— (Equation 1)} \\ \text{LHL} = f(1) &\implies 5a - 2b = 11 \quad \text{— (Equation 2)} \end{aligned}$$

Solving simultaneous equations step-by-step: From Equation 1, express $b$ in terms of $a$:

$$b = 11 - 3a$$

Substitute $b = 11 - 3a$ into Equation 2:

$$\begin{aligned} 5a - 2(11 - 3a) &= 11 \\ 5a - 22 + 6a &= 11 \\ 11a - 22 &= 11 \\ 11a &= 11 + 22 \\ 11a &= 33 \\ a &= 3 \end{aligned}$$

Substitute $a = 3$ back into $b = 11 - 3a$:

$$\begin{aligned} b &= 11 - 3(3) \\ b &= 11 - 9 \\ b &= 2 \end{aligned}$$

**Final Answer**

$a = 3, \quad b = 2$

## Question 23

If $f(x) = \begin{cases} \dfrac{\sin(a+1)x + 2\sin x}{x}, & x < 0 \\ 2, & x = 0 \\ \dfrac{\sqrt{1+bx}-1}{x}, & x > 0 \end{cases}$ is continuous at $x = 0$, then find the values of $a$ and $b$.

**TYPE 3** **Finding Constants for Continuity**

**★ Likely Exam Question**

**Given:** $f(x) = \begin{cases} \dfrac{\sin(a+1)x + 2\sin x}{x}, & x < 0 \\ 2, & x = 0 \\ \dfrac{\sqrt{1+bx}-1}{x}, & x > 0 \end{cases}$ is continuous at $x = 0$

**Concept / Formula Used:** $x = 0$ $\quad$ LHL $= f(0) =$ RHL

**Solution:**

**1. Evaluating Left-Hand Limit (LHL) at $x = 0$:**

$$\begin{aligned} \text{LHL} &= \lim_{x \to 0^-} \frac{\sin(a+1)x + 2\sin x}{x} \\ &= \lim_{x \to 0^-} \left( \frac{\sin(a+1)x}{x} + \frac{2\sin x}{x} \right) \\ &= \lim_{x \to 0^-} \left( (a+1) \cdot \frac{\sin(a+1)x}{(a+1)x} + 2 \cdot \frac{\sin x}{x} \right) \\ &= (a+1) \cdot \lim_{x \to 0^-} \frac{\sin(a+1)x}{(a+1)x} + 2 \cdot \lim_{x \to 0^-} \frac{\sin x}{x} \end{aligned}$$

Using the standard result $\lim\limits_{\theta \to 0} \dfrac{\sin\theta}{\theta} = 1$:

$$\begin{aligned} \text{LHL} &= (a+1)(1) + 2(1) \\ &= a + 1 + 2 \\ &= a + 3 \end{aligned}$$

Since $f$ is continuous at $x = 0$, LHL $= f(0)$:

$$\begin{aligned} a + 3 &= 2 \\ a &= 2 - 3 \\ a &= -1 \end{aligned}$$

**2. Evaluating Right-Hand Limit (RHL) at $x = 0$:**

$$\text{RHL} = \lim_{x \to 0^+} \frac{\sqrt{1+bx} - 1}{x}$$

Rationalizing the numerator by multiplying and dividing by $(\sqrt{1+bx} + 1)$:

$$\begin{aligned}
\text{RHL} &= \lim_{x \to 0^+} \frac{(\sqrt{1+bx} - 1)(\sqrt{1+bx} + 1)}{x(\sqrt{1+bx} + 1)} \\
&= \lim_{x \to 0^+} \frac{(1+bx) - 1}{x(\sqrt{1+bx} + 1)} \\
&= \lim_{x \to 0^+} \frac{bx}{x(\sqrt{1+bx} + 1)} \\
&= \lim_{x \to 0^+} \frac{b}{\sqrt{1+bx} + 1} \\
&= \frac{b}{\sqrt{1+b(0)} + 1} \\
&= \frac{b}{1+1} \\
&= \frac{b}{2}
\end{aligned}$$

Since $f$ is continuous at $x = 0$, RHL $= f(0)$:

$$\begin{aligned}
\frac{b}{2} &= 2 \\
b &= 4
\end{aligned}$$

**Final Answer**

$a = -1, \quad b = 4$

## Question 24 (i)

Consider the function defined as follows, for $x \neq 0$:

$$f(x) = \frac{\sqrt[3]{1+x} - 1}{x}$$

What choice (if any) of $f(0)$ will make the function continuous at $x = 0$?

**TYPE 4** **Removable Discontinuity Analysis**

**Given:** $f(x) = \dfrac{\sqrt[3]{1+x} - 1}{x}$ for $x \neq 0$

**Concept / Formula Used:** $\displaystyle\lim_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1} \qquad (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$

**Solution:**

For $f(x)$ to be continuous at $x = 0$, $f(0)$ must be equal to $\lim_{x\to 0} f(x)$.

Let $1 + x = y$. As $x \to 0$, $y \to 1$. Substituting $x = y - 1$:

$$\lim_{x\to 0} \frac{\sqrt[3]{1+x} - 1}{x} = \lim_{y\to 1} \frac{y^{1/3} - 1^{1/3}}{y - 1}$$

Using the standard result $\lim_{y\to a} \frac{y^n - a^n}{y - a} = na^{n-1}$ with $a = 1$ and $n = \frac{1}{3}$:

$$\begin{aligned}\lim_{y\to 1} \frac{y^{1/3} - 1^{1/3}}{y - 1} &= \frac{1}{3}(1)^{\frac{1}{3}-1} \\ &= \frac{1}{3}(1)^{-2/3} \\ &= \frac{1}{3}\end{aligned}$$

Thus, setting $f(0) = \frac{1}{3}$ will make $f(x)$ continuous at $x = 0$.

**Final Answer**

$f(0) = \dfrac{1}{3}$

## Question 24 (ii)

Consider the function defined as follows, for $x \neq 0$:

$$f(x) = \frac{5|x| - 3\tan x + \sin 2x}{x}$$

What choice (if any) of $f(0)$ will make the function continuous at $x = 0$?

**TYPE 4** **Removable Discontinuity Analysis**

**Given:** $f(x) = \dfrac{5|x| - 3\tan x + \sin 2x}{x}$ for $x \neq 0$

**Concept / Formula Used:** $|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$ $\quad \lim_{x\to 0} \dfrac{\tan x}{x} = 1, \lim_{x\to 0} \dfrac{\sin 2x}{2x} = 1$

**Solution:**

**1. Left-Hand Limit (LHL) at $x = 0$:** For $x < 0$, $|x| = -x$.

$$\begin{aligned}
\text{LHL} &= \lim_{x \to 0^-} \frac{5(-x) - 3\tan x + \sin 2x}{x} \\
&= \lim_{x \to 0^-} \left( \frac{-5x}{x} - \frac{3\tan x}{x} + \frac{\sin 2x}{x} \right) \\
&= \lim_{x \to 0^-} \left( -5 - 3 \cdot \frac{\tan x}{x} + 2 \cdot \frac{\sin 2x}{2x} \right) \\
&= -5 - 3(1) + 2(1) \\
&= -5 - 3 + 2 \\
&= -6
\end{aligned}$$

**2. Right-Hand Limit (RHL) at $x = 0$:** For $x > 0$, $|x| = x$.

$$\begin{aligned}
\text{RHL} &= \lim_{x \to 0^+} \frac{5x - 3\tan x + \sin 2x}{x} \\
&= \lim_{x \to 0^+} \left( \frac{5x}{x} - \frac{3\tan x}{x} + \frac{\sin 2x}{x} \right) \\
&= \lim_{x \to 0^+} \left( 5 - 3 \cdot \frac{\tan x}{x} + 2 \cdot \frac{\sin 2x}{2x} \right) \\
&= 5 - 3(1) + 2(1) \\
&= 5 - 3 + 2 \\
&= 4
\end{aligned}$$

Since LHL $\neq$ RHL (i.e., $-6 \neq 4$), the limit $\lim_{x \to 0} f(x)$ does not exist. Hence, no choice of $f(0)$ can make $f(x)$ continuous at $x = 0$.

**Final Answer**

No choice of $f(0)$ will make the function continuous at $x = 0$.

## Question 25

Examine the function

$$f(x) = \begin{cases} x \sin \dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

for continuity at $x = 0$.

**TYPE 7** **Squeeze Theorem for Continuity**

**★ Likely Exam Question**

**Given:** $f(x) = \begin{cases} x \sin \dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ at $c = 0$

**Concept / Formula Used:** $g(x) \leq f(x) \leq h(x)$ $\quad \lim_{x\to c} g(x) = \lim_{x\to c} h(x) = L$ $\quad \lim_{x\to c} f(x) = L$

**Solution:**

For any $x \neq 0$, the sine function is bounded:

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

Multiplying the inequality by $|x| > 0$:

$$-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|$$

Taking limits as $x \to 0$:

$$\lim_{x\to 0}(-|x|) = 0$$
$$\lim_{x\to 0}|x| = 0$$

By the Squeeze Theorem:

$$\lim_{x\to 0}\left(x \sin\frac{1}{x}\right) = 0$$

Thus:

$$\lim_{x\to 0} f(x) = 0$$

The defined value of the function at $x = 0$ is $f(0) = 0$. Since $\lim_{x\to 0} f(x) = f(0) = 0$, the function $f(x)$ is continuous at $x = 0$.

**Final Answer**

$f(x)$ is continuous at $x = 0$.

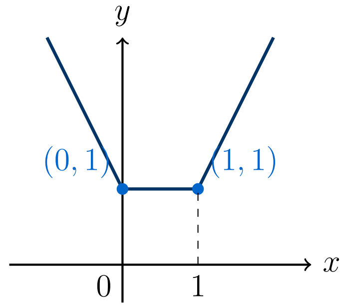
## Question 26

Show that the function $f(x) = |x| + |x - 1|, x \in \mathbb{R}$, is continuous both at $x = 0$ and $x = 1$.

**TYPE 5** Continuity of Absolute Value Functions

★ **Likely Exam Question**

**To Prove:** $f(x) = |x| + |x - 1|$ is continuous at $x = 0$ and $x = 1$.




*Graph of $f(x) = |x| + |x-1|$ showing continuity at corner points $x = 0$ and $x = 1$*

**Concept / Formula Used:** $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases} \qquad |x-1| = \begin{cases} x-1, & x \geq 1 \\ -(x-1), & x < 1 \end{cases}$

**Solution:**

Redefining $f(x) = |x| + |x-1|$ piecewise:

- Case 1: For $x < 0$, $|x| = -x$ and $|x-1| = -(x-1) = 1-x$.
$$f(x) = -x + 1 - x = 1 - 2x$$
- Case 2: For $0 \leq x < 1$, $|x| = x$ and $|x-1| = -(x-1) = 1-x$.
$$f(x) = x + 1 - x = 1$$
- Case 3: For $x \geq 1$, $|x| = x$ and $|x-1| = x-1$.
$$f(x) = x + x - 1 = 2x - 1$$

Thus:
$$f(x) = \begin{cases} 1-2x, & x < 0 \\ 1, & 0 \leq x < 1 \\ 2x-1, & x \geq 1 \end{cases}$$

**1. Continuity at $x = 0$:**

$$\text{LHL} = \lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (1-2x) = 1 - 2(0) = 1$$
$$\text{RHL} = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (1) = 1$$
$$f(0) = |0| + |0-1| = 0 + 1 = 1$$

Since LHL $=$ RHL $= f(0) = 1$, $f(x)$ is continuous at $x = 0$.

**2. Continuity at $x = 1$:**

$$\text{LHL} = \lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (1) = 1$$
$$\text{RHL} = \lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} (2x-1) = 2(1) - 1 = 1$$
$$f(1) = |1| + |1-1| = 1 + 0 = 1$$

Since LHL = RHL = $f(1) = 1$, $f(x)$ is continuous at $x = 1$.

**Hence Proved**

LHS = RHS in limit evaluations at $x = 0$ and $x = 1$. Thus, $f(x)$ is continuous at both $x = 0$ and $x = 1$.

## Question 27

Examine the function

$$f(x) = \begin{cases} e^{1/x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

for continuity at $x = 0$.

**TYPE 8** **Continuity of Exponential Functions**

**Given:** $f(x) = \begin{cases} e^{1/x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ at $c = 0$

**Concept / Formula Used:** $\lim_{y \to -\infty} e^y = 0 \qquad \lim_{y \to +\infty} e^y = +\infty$

**Solution:**

Evaluating Left-Hand Limit (LHL) at $x = 0$: Let $x = 0 - h$, where $h > 0$ and $h \to 0$.

$$\begin{aligned} \text{LHL} &= \lim_{h \to 0} e^{1/(0-h)} \\ &= \lim_{h \to 0} e^{-1/h} \\ &= \lim_{h \to 0} \frac{1}{e^{1/h}} \end{aligned}$$

As $h \to 0^+$, $\dfrac{1}{h} \to +\infty \implies e^{1/h} \to +\infty$.

$$\begin{aligned} \text{LHL} &= \frac{1}{\infty} \\ &= 0 \end{aligned}$$

Evaluating Right-Hand Limit (RHL) at $x = 0$: Let $x = 0 + h$, where $h > 0$ and $h \to 0$.

$$\begin{aligned} \text{RHL} &= \lim_{h \to 0} e^{1/(0+h)} \\ &= \lim_{h \to 0} e^{1/h} \end{aligned}$$

As $h \to 0^+$, $\dfrac{1}{h} \to +\infty \implies e^{1/h} \to +\infty$.

$$\text{RHL} = +\infty$$

Since LHL $= 0 \neq +\infty =$ RHL, $\lim_{x \to 0} f(x)$ does not exist. Hence, $f(x)$ is discontinuous at $x = 0$.

**Final Answer**

$f(x)$ is discontinuous at $x = 0$.

## 5.2 – Continuity: Piecewise Functions and Finding Constants

### Question 1

Is the function $f$ defined by $f(x) = [x]$ continuous at:

(i) $x = 0$

(ii) $x = 1$

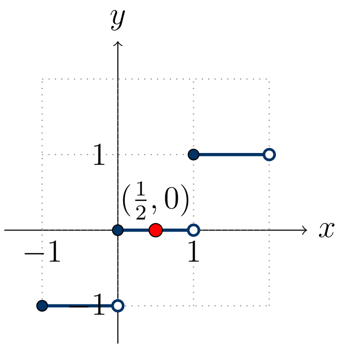
(iii) $x = \dfrac{1}{2}$?

**TYPE 1** **Checking Continuity at a Point**

**Given:** The function $f(x) = [x]$, where $[x]$ denotes the greatest integer less than or equal to $x$.

**To Find:** Check continuity at $x = 0$, $x = 1$, and $x = \dfrac{1}{2}$.

**Concept / Formula Used:** $f(x)$ $\quad x = c \quad$ $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = f(c)$




*Graph of Greatest Integer Function $f(x) = [x]$ showing step discontinuities at integers*

**Solution:**

**(i) Continuity at $x = 0$:**

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 0^-} [x] \\ &= -1 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 0^+} [x] \\ &= 0 \end{aligned}$$

Value of the function at $x = 0$:

$$f(0) = [0] = 0$$

Since LHL $\neq$ RHL, $\lim_{x \to 0} f(x)$ does not exist. Hence, $f(x)$ is not continuous at $x = 0$.

**(ii) Continuity at $x = 1$:**

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 1^-} [x] \\ &= 0 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 1^+} [x] \\ &= 1 \end{aligned}$$

Value of the function at $x = 1$:

$$f(1) = [1] = 1$$

Since LHL $\neq$ RHL, $\lim_{x \to 1} f(x)$ does not exist. Hence, $f(x)$ is not continuous at $x = 1$.

**(iii) Continuity at $x = \frac{1}{2}$:**

For any $x \in (0, 1)$, the value of $[x] = 0$.

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to (1/2)^-} [x] \\ &= 0 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to (1/2)^+} [x] \\ &= 0 \end{aligned}$$

Value of the function at $x = \frac{1}{2}$:

$$f\left(\frac{1}{2}\right) = \left[\frac{1}{2}\right] = 0$$

Since LHL = RHL = $f\left(\frac{1}{2}\right) = 0$, $f(x)$ is continuous at $x = \frac{1}{2}$.

**Final Answer**

(i) No, $f$ is not continuous at $x = 0$.
(ii) No, $f$ is not continuous at $x = 1$.
(iii) Yes, $f$ is continuous at $x = \frac{1}{2}$.

**Common Mistake:** Assuming that greatest integer function is discontinuous everywhere. It is discontinuous only at integer values, but continuous at all non-integer values.

## Question 2 (i)

Examine the following function for continuity:

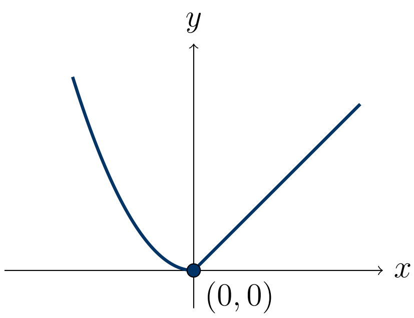
$$f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ x^2, & \text{if } x < 0 \end{cases}$$

**TYPE 2** **Continuity of Piecewise Functions**

**Given:** The function $f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ x^2, & \text{if } x < 0 \end{cases}$

**To Find:** Examine the continuity of $f(x)$ for all $x \in \mathbb{R}$.

**Concept / Formula Used:** $x = c$ $\quad \lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = f(c)$




*Graph of piecewise function $f(x) = x$ for $x \geq 0$ and $f(x) = x^2$ for $x < 0$*

**Solution:**

Case 1: For $x < 0$, $f(x) = x^2$, which is a polynomial function and hence continuous for all $x < 0$.

Case 2: For $x > 0$, $f(x) = x$, which is a polynomial function and hence continuous for all $x > 0$.

Case 3: At the boundary point $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x\to 0^-} f(x)\\ &= \lim_{x\to 0^-} (x^2)\\ &= 0^2\\ &= 0\end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x\to 0^+} f(x)\\ &= \lim_{x\to 0^+} (x)\\ &= 0\end{aligned}$$

Value of the function at $x = 0$:

$$f(0) = 0$$

Since LHL = RHL = $f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Combining all three cases, $f(x)$ is continuous at all points $x \in \mathbb{R}$.

**Final Answer**

The function $f(x)$ is continuous for all $x \in \mathbb{R}$.

## Question 2 (ii)

Examine the following function for continuity:

$$f(x) = \begin{cases} x+5, & \text{if } x \le 1 \\ x-5, & \text{if } x > 1 \end{cases}$$

**TYPE 2** **Continuity of Piecewise Functions**

**Given:** The function $f(x) = \begin{cases} x+5, & \text{if } x \le 1 \\ x-5, & \text{if } x > 1 \end{cases}$

**To Find:** Examine the continuity of $f(x)$ for all real numbers.

**Concept / Formula Used:** $x = 1$ $\lim_{x\to 1^-} f(x)$ $\lim_{x\to 1^+} f(x)$

**Solution:**

Case 1: For $x < 1$, $f(x) = x + 5$, which is a polynomial function and hence continuous.

Case 2: For $x > 1$, $f(x) = x - 5$, which is a polynomial function and hence continuous.

Case 3: At $x = 1$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x\to 1^-}(x+5) \\ &= 1+5 \\ &= 6 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x\to 1^+}(x-5) \\ &= 1-5 \\ &= -4 \end{aligned}$$

Value of the function at $x = 1$:

$$\begin{aligned} f(1) &= 1+5 \\ &= 6 \end{aligned}$$

Since LHL $\neq$ RHL, $\lim_{x\to 1} f(x)$ does not exist. Hence, $f(x)$ is discontinuous at $x = 1$.

**Final Answer**

The function $f(x)$ is continuous for all real numbers except at $x = 1$.

## Question 3 (i)

Examine the following function for continuity:

$$f(x) = \begin{cases} 2x-1, & \text{if } x < 2 \\ \frac{3}{2}x, & \text{if } x \geq 2 \end{cases}$$

**TYPE 2** **Continuity of Piecewise Functions**

**Given:** The function $f(x) = \begin{cases} 2x-1, & \text{if } x < 2 \\ \frac{3}{2}x, & \text{if } x \geq 2 \end{cases}$

**To Find:** Examine the continuity of $f(x)$.

**Concept / Formula Used:** $f(x)$ $x = 2$ $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = f(2)$

**Solution:**

For $x < 2$, $f(x) = 2x - 1$ is continuous as it is linear polynomial. For $x > 2$, $f(x) = \dfrac{3}{2}x$ is continuous as it is linear polynomial.

At the boundary point $x = 2$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 2^-} (2x - 1) \\ &= 2(2) - 1 \\ &= 4 - 1 \\ &= 3 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 2^+} \left(\frac{3}{2}x\right) \\ &= \frac{3}{2}(2) \\ &= 3 \end{aligned}$$

Value of the function at $x = 2$:

$$\begin{aligned} f(2) &= \frac{3}{2}(2) \\ &= 3 \end{aligned}$$

Since LHL = RHL = $f(2) = 3$, $f(x)$ is continuous at $x = 2$. Therefore, $f(x)$ is continuous for all $x \in \mathbb{R}$.

**Final Answer**

The function $f(x)$ is continuous for all $x \in \mathbb{R}$.

## Question 3 (ii)

Examine the following function for continuity:

$$f(x) = \begin{cases} \sin x - \cos x, & x \neq 0 \\ -1, & x = 0 \end{cases}$$

**TYPE 2** **Continuity of Piecewise Functions**

**Given:** The function $f(x) = \begin{cases} \sin x - \cos x, & x \neq 0 \\ -1, & x = 0 \end{cases}$

**To Find:** Examine the continuity of $f(x)$.

**Concept / Formula Used:** $f(x)$ $\quad x = 0 \quad$ $\lim_{x \to 0} f(x) = f(0)$

**Solution:**

For $x \neq 0$, $f(x) = \sin x - \cos x$, which is the difference of two continuous trigonometric functions, hence continuous for all $x \neq 0$.

At $x = 0$:

Limit as $x \to 0$:

$$\begin{aligned} \lim_{x \to 0} f(x) &= \lim_{x \to 0} (\sin x - \cos x) \\ &= \sin 0 - \cos 0 \\ &= 0 - 1 \\ &= -1 \end{aligned}$$

Value of the function at $x = 0$:

$$f(0) = -1$$

Since $\lim_{x \to 0} f(x) = f(0) = -1$, the function $f(x)$ is continuous at $x = 0$.

Hence, $f(x)$ is continuous for all $x \in \mathbb{R}$.

**Final Answer**

The function $f(x)$ is continuous for all $x \in \mathbb{R}$.

## Question 4 (i)

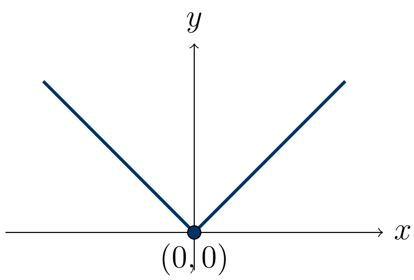
Examine the continuity of $f(x) = |x|$.

**TYPE 5 Continuity of Absolute Value Functions**

**Given:** The modulus function $f(x) = |x|$.

**To Find:** Examine the continuity of $f(x)$ for all $x \in \mathbb{R}$.

**Concept / Formula Used:** $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$




*Graph of modulus function $f(x) = |x|$*

**Solution:**

By definition, $f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$

Case 1: For $x < 0$, $f(x) = -x$, a polynomial function, continuous for all $x < 0$.

Case 2: For $x > 0$, $f(x) = x$, a polynomial function, continuous for all $x > 0$.

Case 3: At $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 0^-} (-x) \\ &= -0 \\ &= 0 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 0^+} (x) \\ &= 0 \end{aligned}$$

Value of the function at $x = 0$:

$$f(0) = |0| = 0$$

Since LHL $=$ RHL $= f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Therefore, $f(x) = |x|$ is continuous for all real numbers $x \in \mathbb{R}$.

**Final Answer**

The function $f(x) = |x|$ is continuous for all $x \in \mathbb{R}$.

## Question 4 (ii)

Examine the continuity of $f(x) = |x - 5|$.

**TYPE 5 Continuity of Absolute Value Functions**

**Given:** The modulus function $f(x) = |x - 5|$.

**To Find:** Examine the continuity of $f(x)$.

**Concept / Formula Used:** $|x - 5| = \begin{cases} x - 5, & x \geq 5 \\ -(x - 5), & x < 5 \end{cases}$

**Solution:**

Rewrite $f(x)$ as a piecewise function:

$$f(x) = \begin{cases} -(x - 5), & \text{if } x < 5 \\ x - 5, & \text{if } x \geq 5 \end{cases}$$

Case 1: For $x < 5$, $f(x) = -(x-5) = 5-x$, which is polynomial and hence continuous.

Case 2: For $x > 5$, $f(x) = x-5$, which is polynomial and hence continuous.

Case 3: At $x = 5$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x\to 5^-}(5-x) \\ &= 5-5 \\ &= 0 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x\to 5^+}(x-5) \\ &= 5-5 \\ &= 0 \end{aligned}$$

Value of the function at $x = 5$:

$$f(5) = |5-5| = 0$$

Since LHL = RHL = $f(5) = 0$, $f(x)$ is continuous at $x = 5$.

Hence, $f(x)$ is continuous for all real numbers $x \in \mathbb{R}$.

**Final Answer**

The function $f(x) = |x-5|$ is continuous for all $x \in \mathbb{R}$.

## Question 5 (i)

Examine the continuity of $f(x) = \dfrac{x^2-25}{x+5}$.

**TYPE 1** **Checking Continuity at a Point**

**Given:** The rational function $f(x) = \dfrac{x^2-25}{x+5}$.

**To Find:** Determine the domain and examine the continuity of $f(x)$.

**Concept / Formula Used:** $\dfrac{P(x)}{Q(x)} \qquad Q(x) \neq 0$

**Solution:**

The denominator $x+5 \neq 0 \implies x \neq -5$. Thus, the domain of $f(x)$ is $\mathbb{R} \setminus \{-5\}$.

Let $c$ be any arbitrary real number in the domain $(\mathbb{R} \setminus \{-5\})$, so $c \neq -5$.

Value of function at $x = c$:

$$f(c) = \frac{c^2 - 25}{c + 5}$$

Limit at $x = c$:

$$\begin{aligned}\lim_{x \to c} f(x) &= \lim_{x \to c} \frac{x^2 - 25}{x + 5} \\ &= \frac{c^2 - 25}{c + 5} \\ &= f(c)\end{aligned}$$

Since $\lim_{x \to c} f(x) = f(c)$ for every $c$ in the domain of $f$, $f(x)$ is continuous at all points in its domain.

**Final Answer**

The function $f(x)$ is continuous for all $x \in \mathbb{R} \setminus \{-5\}$.

## Question 5 (ii)

Examine the continuity of $f(x) = \dfrac{1}{x - 5}$.

**TYPE 1** **Checking Continuity at a Point**

**Given:** The function $f(x) = \dfrac{1}{x - 5}$.

**To Find:** Examine continuity in its domain.

**Solution:**

The denominator $x - 5 = 0 \implies x = 5$. So $f(x)$ is defined for all $x \in \mathbb{R} \setminus \{5\}$.

For any point $c \in \mathbb{R} \setminus \{5\}$:

$$\begin{aligned}\lim_{x \to c} f(x) &= \lim_{x \to c} \frac{1}{x - 5} \\ &= \frac{1}{c - 5} \\ &= f(c)\end{aligned}$$

Since $\lim_{x \to c} f(x) = f(c)$ for all $c \in \mathbb{R} \setminus \{5\}$, the function is continuous on its domain.

**Final Answer**

The function $f(x)$ is continuous for all $x \in \mathbb{R} \setminus \{5\}$.

## Question 6 (i)

Examine the following function for continuity:

$$f(x) = \begin{cases} \dfrac{\sin 2x}{x}, & \text{if } x > 0 \\ 2, & \text{if } x \leq 0 \end{cases}$$

**TYPE 6** **Continuity Using Trigonometric Limits**

**Given:** The piecewise function $f(x) = \begin{cases} \dfrac{\sin 2x}{x}, & \text{if } x > 0 \\ 2, & \text{if } x \leq 0 \end{cases}$

**To Find:** Examine the continuity of $f(x)$ at $x = 0$ and over $\mathbb{R}$.

**Concept / Formula Used:** $\lim\limits_{\theta \to 0} \dfrac{\sin \theta}{\theta} = 1$

**Solution:**

For $x < 0$, $f(x) = 2$ is a constant function, hence continuous. For $x > 0$, $f(x) = \dfrac{\sin 2x}{x}$ is a quotient of continuous trigonometric and polynomial functions with $x \neq 0$, hence continuous.

At $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 0^-} f(x) \\ &= \lim_{x \to 0^-} (2) \\ &= 2 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 0^+} f(x) \\ &= \lim_{x \to 0^+} \frac{\sin 2x}{x} \\ &= \lim_{x \to 0^+} \left(2 \cdot \frac{\sin 2x}{2x}\right) \end{aligned}$$

Using standard result $\lim\limits_{\theta \to 0} \dfrac{\sin \theta}{\theta} = 1$ with $\theta = 2x$:

$$\begin{aligned} &= 2 \times 1 \\ &= 2 \end{aligned}$$

Value of function at $x = 0$:

$$f(0) = 2$$

Since LHL = RHL = $f(0) = 2$, $f(x)$ is continuous at $x = 0$.

Hence, $f(x)$ is continuous for all $x \in \mathbb{R}$.

**Final Answer**

The function $f(x)$ is continuous for all $x \in \mathbb{R}$.

## Question 6 (ii)

Examine the following function for continuity:

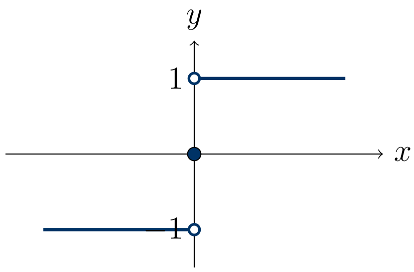
$$f(x) = \begin{cases} \dfrac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

**TYPE 5** **Continuity of Absolute Value Functions**

**Given:** The function $f(x) = \begin{cases} \dfrac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

**To Find:** Examine continuity for all $x \in \mathbb{R}$.

**Concept / Formula Used:** $|x| = x \quad x > 0 \quad |x| = -x \quad x < 0$




*Graph of signum-type function $f(x) = \frac{x}{|x|}$ showing jump discontinuity at $x = 0$*

**Solution:**

Rewrite $f(x)$ for $x < 0$ and $x > 0$:

For $x < 0$, $|x| = -x$, so:

$$f(x) = \frac{x}{-x} = -1$$

For $x > 0$, $|x| = x$, so:

$$f(x) = \frac{x}{x} = 1$$

At $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 0^-} (-1) \\ &= -1 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 0^+} (1) \\ &= 1 \end{aligned}$$

Value of function at $x = 0$:

$$f(0) = 0$$

Since LHL $\neq$ RHL, $\lim_{x \to 0} f(x)$ does not exist. Hence, $f(x)$ is discontinuous at $x = 0$.

**Final Answer**

The function $f(x)$ is continuous for all real numbers except at $x = 0$.

## Question 7 (i)

Examine the following function for continuity:

$$f(x) = \begin{cases} x^{10} - 1, & \text{if } x \leq 1 \\ x^2, & \text{if } x > 1 \end{cases}$$

**TYPE 2 Continuity of Piecewise Functions**

**Given:** The piecewise function $f(x) = \begin{cases} x^{10} - 1, & \text{if } x \leq 1 \\ x^2, & \text{if } x > 1 \end{cases}$

**To Find:** Examine the continuity of $f(x)$.

**Concept / Formula Used:** $f(x)$ $\quad x = 1 \quad$ $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x) = f(1)$

**Solution:**

For $x < 1$, $f(x) = x^{10} - 1$ is a polynomial, continuous everywhere on $(-\infty, 1)$. For $x > 1$, $f(x) = x^2$ is a polynomial, continuous everywhere on $(1, \infty)$.

At $x = 1$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 1^-} (x^{10} - 1) \\ &= 1^{10} - 1 \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 1^+} (x^2) \\ &= 1^2 \\ &= 1 \end{aligned}$$

Value of function at $x = 1$:

$$\begin{aligned} f(1) &= 1^{10} - 1 \\ &= 0 \end{aligned}$$

Since LHL $\neq$ RHL, $\lim_{x \to 1} f(x)$ does not exist. Hence, $f(x)$ is discontinuous at $x = 1$.

**Final Answer**

The function $f(x)$ is continuous for all $x \in \mathbb{R} \setminus \{1\}$.

## Question 7 (ii)

Examine the continuity of $f(x) = \tan x$ for $0 \leq x \leq \frac{\pi}{4}$.

**TYPE 1** **Checking Continuity at a Point**

**Given:** The function $f(x) = \tan x$ defined on the closed interval $\left[0, \frac{\pi}{4}\right]$.

**To Find:** Examine the continuity of $f(x)$ on $\left[0, \frac{\pi}{4}\right]$.

**Concept / Formula Used:** $\tan x = \dfrac{\sin x}{\cos x} \qquad \cos x \neq 0$

**Solution:**

We know $\tan x = \dfrac{\sin x}{\cos x}$. The function $\tan x$ is undefined only when $\cos x = 0$, which occurs at $x = \dfrac{\pi}{2} + n\pi, n \in \mathbb{Z}$.

On the given closed interval $\left[0, \frac{\pi}{4}\right]$: $\cos x \neq 0$ for all $x \in \left[0, \frac{\pi}{4}\right]$ since $\cos 0 = 1$ and $\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$.

For any point $c \in \left[0, \frac{\pi}{4}\right]$:

$$\lim_{x \to c} \tan x = \tan c = f(c)$$

Therefore, $f(x) = \tan x$ is continuous at every point in the interval $\left[0, \frac{\pi}{4}\right]$.

**Final Answer**

The function $f(x) = \tan x$ is continuous on the interval $\left[0, \frac{\pi}{4}\right]$.

## Question 8 (i)

Locate the points of discontinuity (if any) of the function:

$$f(x) = \begin{cases} 2x + 3, & \text{if } x \leq 2 \\ 2x - 3, & \text{if } x > 2 \end{cases}$$

**TYPE 2 Continuity of Piecewise Functions**

**Given:** The function $f(x) = \begin{cases} 2x + 3, & \text{if } x \leq 2 \\ 2x - 3, & \text{if } x > 2 \end{cases}$

**To Find:** Locate all points of discontinuity.

**Concept / Formula Used:** $x = c$ $\quad \lim_{x \to c^-} f(x) \neq \lim_{x \to c^+} f(x)$ $\quad \lim_{x \to c} f(x) \neq f(c)$

**Solution:**

For $x < 2$, $f(x) = 2x + 3$ is continuous as it is a polynomial function. For $x > 2$, $f(x) = 2x - 3$ is continuous as it is a polynomial function.

At $x = 2$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 2^-} (2x + 3) \\ &= 2(2) + 3 \\ &= 4 + 3 \\ &= 7 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 2^+} (2x - 3) \\ &= 2(2) - 3 \\ &= 4 - 3 \\ &= 1 \end{aligned}$$

Value of function at $x = 2$:

$$f(2) = 2(2) + 3 = 7$$

Since LHL $\neq$ RHL, $f(x)$ is not continuous at $x = 2$.

**Final Answer**

The point of discontinuity is $x = 2$.

## Question 8 (ii)

Locate the points of discontinuity (if any) of the function:

$$f(x) = \begin{cases} x^3 - 3, & \text{if } x \leq 2 \\ x^2 + 1, & \text{if } x > 2 \end{cases}$$

**TYPE 2** **Continuity of Piecewise Functions**

**Given:** The function $f(x) = \begin{cases} x^3 - 3, & \text{if } x \leq 2 \\ x^2 + 1, & \text{if } x > 2 \end{cases}$

**To Find:** Find all points of discontinuity.

**Concept / Formula Used:** $x = 2$ LHL RHL $f(2)$

**Solution:**

For $x < 2$, $f(x) = x^3 - 3$ is continuous (polynomial). For $x > 2$, $f(x) = x^2 + 1$ is continuous (polynomial).

At $x = 2$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 2^-} (x^3 - 3) \\ &= 2^3 - 3 \\ &= 8 - 3 \\ &= 5 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 2^+} (x^2 + 1) \\ &= 2^2 + 1 \\ &= 4 + 1 \\ &= 5 \end{aligned}$$

Value of function at $x = 2$:

$$\begin{aligned} f(2) &= 2^3 - 3 \\ &= 5 \end{aligned}$$

Since LHL = RHL = $f(2) = 5$, $f(x)$ is continuous at $x = 2$.

Thus, $f(x)$ is continuous for all real numbers.

**Final Answer**

There are no points of discontinuity.

## Question 9 (i)

Locate the points of discontinuity (if any) of the function:

$$f(x) = \begin{cases} \dfrac{x^4 - 16}{x - 2}, & x \neq 2 \\ 16, & x = 2 \end{cases}$$

**TYPE 1** **Checking Continuity at a Point**

**Given:** The function $f(x) = \begin{cases} \dfrac{x^4 - 16}{x - 2}, & x \neq 2 \\ 16, & x = 2 \end{cases}$

**To Find:** Find all points of discontinuity.

**Concept / Formula Used:** $x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$

**Solution:**

For $x \neq 2$, $f(x) = \dfrac{x^4 - 16}{x - 2}$ is continuous as a quotient of continuous polynomials.

At $x = 2$:

Evaluating the limit as $x \to 2$:

$$\begin{aligned} \lim_{x \to 2} f(x) &= \lim_{x \to 2} \frac{x^4 - 16}{x - 2} \\ &= \lim_{x \to 2} \frac{(x^2 - 4)(x^2 + 4)}{x - 2} \\ &= \lim_{x \to 2} \frac{(x - 2)(x + 2)(x^2 + 4)}{x - 2} \end{aligned}$$

Cancelling $(x-2)$ as $x \neq 2$:

$$\begin{aligned} &= \lim_{x\to 2}(x+2)(x^2+4) \\ &= (2+2)(2^2+4) \\ &= (4)(4+4) \\ &= 4\times 8 \\ &= 32 \end{aligned}$$

Value of function at $x = 2$:

$$f(2) = 16$$

Since $\lim_{x\to 2} f(x) = 32 \neq f(2) = 16$, the function is not continuous at $x = 2$.

**Final Answer**

The point of discontinuity is $x = 2$.

## Question 9 (ii)

Locate the points of discontinuity (if any) of the function:

$$f(x) = \begin{cases} \dfrac{\sin 3x}{x} + \cos x, & x > 0 \\ 4 - 3x, & x \leq 0 \end{cases}$$

**TYPE 6** Continuity Using Trigonometric Limits

**Given:** The function $f(x) = \begin{cases} \dfrac{\sin 3x}{x} + \cos x, & x > 0 \\ 4 - 3x, & x \leq 0 \end{cases}$

**To Find:** Locate points of discontinuity.

**Concept / Formula Used:** $\lim_{\theta\to 0} \dfrac{\sin\theta}{\theta} = 1$

**Solution:**

For $x < 0$, $f(x) = 4 - 3x$ is a polynomial function, continuous everywhere. For $x > 0$, $f(x) = \dfrac{\sin 3x}{x} + \cos x$ is a sum of continuous functions, continuous everywhere.

At $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x\to 0^-}(4-3x) \\ &= 4 - 3(0) \\ &= 4 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x\to 0^+}\left(\frac{\sin 3x}{x} + \cos x\right)\\ &= \lim_{x\to 0^+}\left(3\cdot\frac{\sin 3x}{3x} + \cos x\right)\end{aligned}$$

Using $\lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1$ with $\theta = 3x$:

$$\begin{aligned}&= 3(1) + \cos 0\\ &= 3 + 1\\ &= 4\end{aligned}$$

Value of function at $x = 0$:

$$\begin{aligned}f(0) &= 4 - 3(0)\\ &= 4\end{aligned}$$

Since LHL $=$ RHL $= f(0) = 4$, $f(x)$ is continuous at $x = 0$.

**Final Answer**

There are no points of discontinuity.

## Question 10 (i)

Locate the points of discontinuity (if any) of the function:

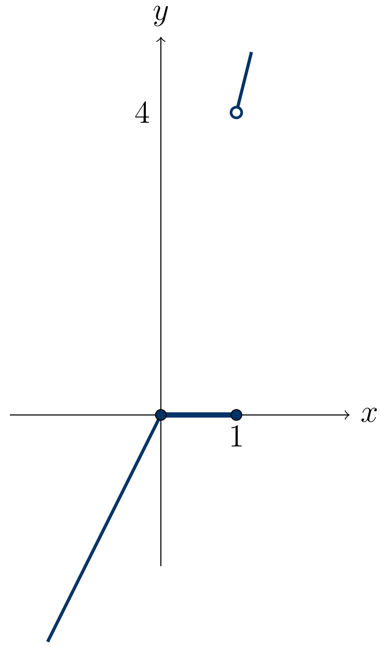
$$f(x) = \begin{cases} 2x, & \text{if } x < 0\\ 0, & \text{if } 0 \le x \le 1\\ 4x, & \text{if } x > 1\end{cases}$$

**TYPE 2** Continuity of Piecewise Functions

**Given:** The function $f(x) = \begin{cases} 2x, & \text{if } x < 0\\ 0, & \text{if } 0 \le x \le 1\\ 4x, & \text{if } x > 1\end{cases}$

**To Find:** Find all points of discontinuity.

**Concept / Formula Used:** $x = 0 \qquad x = 1$




*Graph of three-piece function showing continuity at $x = 0$ and jump discontinuity at $x = 1$*

**Solution:**

Check continuity at boundary point $x = 0$:

Left-Hand Limit (LHL at $x = 0$):

$$\begin{aligned} \text{LHL} &= \lim_{x\to 0^-}(2x) \\ &= 2(0) \\ &= 0 \end{aligned}$$

Right-Hand Limit (RHL at $x = 0$):

$$\begin{aligned} \text{RHL} &= \lim_{x\to 0^+}(0) \\ &= 0 \end{aligned}$$

Value of function $f(0) = 0$. Since LHL = RHL = $f(0) = 0$, $f(x)$ is continuous at $x = 0$.

Check continuity at boundary point $x = 1$:

Left-Hand Limit (LHL at $x = 1$):

$$\begin{aligned} \text{LHL} &= \lim_{x\to 1^-}(0) \\ &= 0 \end{aligned}$$

Right-Hand Limit (RHL at $x = 1$):

$$\begin{aligned} \text{RHL} &= \lim_{x\to 1^+}(4x) \\ &= 4(1) \\ &= 4 \end{aligned}$$

Value of function $f(1) = 0$. Since LHL $\neq$ RHL at $x = 1$, $f(x)$ is discontinuous at $x = 1$.

**Final Answer**

The point of discontinuity is $x = 1$.

## Question 10 (ii)

Locate the points of discontinuity (if any) of the function:

$$f(x) = \begin{cases} x+2, & x \leq 1 \\ x-2, & 1 < x < 2 \\ 0, & x \geq 2 \end{cases}$$

**TYPE 2** Continuity of Piecewise Functions

**Given:** The function $f(x) = \begin{cases} x+2, & x \leq 1 \\ x-2, & 1 < x < 2 \\ 0, & x \geq 2 \end{cases}$

**To Find:** Find all points of discontinuity.

**Concept / Formula Used:** $x = 1 \qquad x = 2$

**Solution:**

Check continuity at $x = 1$:

Left-Hand Limit (LHL at $x = 1$):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 1^-} (x+2) \\ &= 1 + 2 \\ &= 3 \end{aligned}$$

Right-Hand Limit (RHL at $x = 1$):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 1^+} (x-2) \\ &= 1 - 2 \\ &= -1 \end{aligned}$$

Since LHL $\neq$ RHL at $x = 1$, $f(x)$ is discontinuous at $x = 1$.

Check continuity at $x = 2$:

Left-Hand Limit (LHL at $x = 2$):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 2^-} (x-2) \\ &= 2 - 2 \\ &= 0 \end{aligned}$$

Right-Hand Limit (RHL at $x = 2$):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 2^+} (0) \\ &= 0 \end{aligned}$$

Value of function $f(2) = 0$. Since LHL = RHL = $f(2) = 0$, $f(x)$ is continuous at $x = 2$.

**Final Answer**

The point of discontinuity is $x = 1$.

## Question 11

Find the value of the constant $k$ so that the function $f$ defined by

$$f(x) = \begin{cases} kx^2, & x \leq 2 \\ x - 3, & x > 2 \end{cases}$$

may be continuous.

**TYPE 3** **Finding Constants for Continuity**

**Given:** The function $f(x) = \begin{cases} kx^2, & x \leq 2 \\ x - 3, & x > 2 \end{cases}$ is continuous everywhere.

**To Find:** The value of constant $k$.

**Concept / Formula Used:** $x = 2$ $\quad \lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x) = f(2)$

**Solution:**

Since $f(x)$ is given to be continuous, it must be continuous at $x = 2$.

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 2^-} f(x) \\ &= \lim_{x \to 2^-} (kx^2) \\ &= k(2)^2 \\ &= 4k \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 2^+} f(x) \\ &= \lim_{x \to 2^+} (x - 3) \\ &= 2 - 3 \\ &= -1 \end{aligned}$$

Value of function at $x = 2$:

$$f(2) = k(2)^2 = 4k$$

For $f(x)$ to be continuous at $x = 2$:

$$\begin{aligned} \text{LHL} &= \text{RHL} \\ 4k &= -1 \\ k &= -\frac{1}{4} \end{aligned}$$

**Final Answer**

$k = -\frac{1}{4}$

## Question 12

Determine the value of $k$ for which the following function is continuous at $x = 3$:

$$f(x) = \begin{cases} \frac{(x+3)^2 - 36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

★ **Likely Exam Question**

**TYPE 3 Finding Constants for Continuity**

**Given:** The function $f(x) = \begin{cases} \frac{(x+3)^2 - 36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$ is continuous at $x = 3$.

**To Find:** The value of $k$.

**Concept / Formula Used:** $f(x)$ $x = 3$ $\lim_{x \to 3} f(x) = f(3) = k$

**Solution:**

We evaluate $\lim_{x \to 3} f(x)$:

$$\lim_{x \to 3} f(x) = \lim_{x \to 3} \frac{(x+3)^2 - 36}{x-3}$$

Expanding the numerator $(x+3)^2 - 36$:

$$\begin{aligned} (x+3)^2 - 36 &= x^2 + 6x + 9 - 36 \\ &= x^2 + 6x - 27 \end{aligned}$$

Factorizing $x^2 + 6x - 27$:

$$\begin{aligned} x^2 + 6x - 27 &= x^2 + 9x - 3x - 27 \\ &= x(x+9) - 3(x+9) \\ &= (x-3)(x+9) \end{aligned}$$

Substituting back into the limit:

$$\lim_{x \to 3} f(x) = \lim_{x \to 3} \frac{(x-3)(x+9)}{x-3}$$

Cancelling $(x-3)$ as $x \neq 3$:

$$\begin{aligned} &= \lim_{x \to 3}(x+9) \\ &= 3 + 9 \\ &= 12 \end{aligned}$$

Since $f(x)$ is continuous at $x = 3$:

$$\begin{aligned} f(3) &= \lim_{x \to 3} f(x) \\ k &= 12 \end{aligned}$$

**Final Answer**

$k = 12$

## Question 13 (i)

For what choice of $a$ and $b$ is the following function continuous:

$$f(x) = \begin{cases} -5, & \text{if } x \leq -1 \\ ax - b, & \text{if } -1 < x < 3 \\ 7, & \text{if } x \geq 3 \end{cases}$$

**★ Likely Exam Question**

**TYPE 3 Finding Constants for Continuity**

**Given:** The function $f(x) = \begin{cases} -5, & \text{if } x \leq -1 \\ ax - b, & \text{if } -1 < x < 3 \\ 7, & \text{if } x \geq 3 \end{cases}$ is continuous everywhere.

**To Find:** The values of constants $a$ and $b$.

**Concept / Formula Used:** $x = -1 \qquad x = 3$

**Solution:**

Since $f(x)$ is continuous everywhere, it must be continuous at $x = -1$ and $x = 3$.

**Continuity at $x = -1$:**

Left-Hand Limit (LHL):

$$\text{LHL} = \lim_{x \to -1^-} (-5) = -5$$

Right-Hand Limit (RHL):

$$\begin{aligned}\text{RHL} &= \lim_{x \to -1^+} (ax - b) \\ &= a(-1) - b \\ &= -a - b\end{aligned}$$

Value of function $f(-1) = -5$.

Equating LHL and RHL for continuity at $x = -1$:

$$\begin{aligned}-a - b &= -5 \\ a + b &= 5 \quad \text{— (Equation 1)}\end{aligned}$$

**Continuity at $x = 3$:**

Left-Hand Limit (LHL):

$$\begin{aligned}\text{LHL} &= \lim_{x \to 3^-} (ax - b) \\ &= a(3) - b \\ &= 3a - b\end{aligned}$$

Right-Hand Limit (RHL):

$$\text{RHL} = \lim_{x \to 3^+} (7) = 7$$

Value of function $f(3) = 7$.

Equating LHL and RHL for continuity at $x = 3$:

$$3a - b = 7 \quad \text{— (Equation 2)}$$

**Solving Equation 1 and Equation 2 simultaneously:**

Add Equation 1 and Equation 2:

$$\begin{aligned}(a + b) + (3a - b) &= 5 + 7 \\ 4a &= 12 \\ a &= \frac{12}{4} \\ a &= 3\end{aligned}$$

Substitute $a = 3$ into Equation 1:

$$\begin{aligned} 3 + b &= 5 \\ b &= 5 - 3 \\ b &= 2 \end{aligned}$$

**Final Answer**

$a = 3, \quad b = 2$

## Question 13 (ii)

For what choice of $a$ and $b$ is the following function continuous:

$$f(x) = \begin{cases} x^2, & x \leq 0 \\ ax + b, & x > 0 \end{cases}$$

**TYPE 3** **Finding Constants for Continuity**

**Given:** The function $f(x) = \begin{cases} x^2, & x \leq 0 \\ ax + b, & x > 0 \end{cases}$ is continuous everywhere.

**To Find:** The values of $a$ and $b$.

**Concept / Formula Used:** $x = 0 \qquad \lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) = f(0)$

**Solution:**

Check continuity at $x = 0$:

Left-Hand Limit (LHL):

$$\begin{aligned} \text{LHL} &= \lim_{x \to 0^-} (x^2) \\ &= 0^2 \\ &= 0 \end{aligned}$$

Right-Hand Limit (RHL):

$$\begin{aligned} \text{RHL} &= \lim_{x \to 0^+} (ax + b) \\ &= a(0) + b \\ &= b \end{aligned}$$

Value of function at $x = 0$:

$$f(0) = 0^2 = 0$$

Equating LHL and RHL for continuity at $x = 0$:

$$b = 0$$

Since $ax + b$ becomes $ax$ when $b = 0$, $\lim_{x \to 0^+} ax = 0$ regardless of the value of $a$. Hence, $a$ can take any real value ($a \in \mathbb{R}$).

**Final Answer**

$b = 0$, and $a$ may be any real number ($a \in \mathbb{R}$).

## Question 14

Consider $f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3 \\ 5, & x = 3 \end{cases}$

(i) Determine if there is a removable discontinuity at $x = 3$.
(ii) Classify it as missing point or isolated point.
(iii) Redefine the function to make it continuous.

**TYPE 4 Removable Discontinuity Analysis**

**Given:** The function $f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3 \\ 5, & x = 3 \end{cases}$

**To Find:**
(i) Check for removable discontinuity at $x = 3$.
(ii) Classify the discontinuity.
(iii) Redefine $f(x)$ for continuity.

**Concept / Formula Used:** $x = c$ $\quad \lim_{x \to c} f(x) \quad f(c) \quad f(c)$

**Solution:**

**(i) Check for Removable Discontinuity at $x = 3$:**

Evaluate $\lim_{x \to 3} f(x)$:

$$\begin{aligned} \lim_{x \to 3} f(x) &= \lim_{x \to 3} \frac{x^2 - 9}{x - 3} \\ &= \lim_{x \to 3} \frac{(x - 3)(x + 3)}{x - 3} \end{aligned}$$

Cancelling $(x - 3)$ since $x \neq 3$:

$$\begin{aligned} &= \lim_{x \to 3} (x + 3) \\ &= 3 + 3 \\ &= 6 \end{aligned}$$

Value of function at $x = 3$:

$$f(3) = 5$$

Since $\lim_{x \to 3} f(x) = 6$ exists but $\lim_{x \to 3} f(x) \neq f(3)$, there is a removable discontinuity at $x = 3$.

**(ii) Classification:**

Since $f(3)$ is defined $(f(3) = 5)$ but does not match the limit $(\lim_{x \to 3} f(x) = 6)$, the discontinuity is an **isolated point discontinuity**.

**(iii) Redefining the function for continuity:**

To make $f(x)$ continuous at $x = 3$, we set $f(3) = \lim_{x \to 3} f(x) = 6$.

The redefined continuous function is:

$$f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3 \\ 6, & x = 3 \end{cases}$$

**Final Answer**

(i) Yes, there is a removable discontinuity at $x = 3$.
(ii) Isolated point discontinuity.
(iii) Redefined function: $f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3 \\ 6, & x = 3 \end{cases}$

# 5.3 – Continuity: Algebra and Composition of Continuous Functions

## Question 1

Is $\tan x$ a continuous function?

TYPE 9 **Algebra of Continuous Functions**

**Given:** $f(x) = \tan x$

**Concept / Formula Used:** $\frac{g(x)}{h(x)}$ $g(x)$ $h(x)$ $h(x) \neq 0$

**Solution:**

The given function is:

$$f(x) = \tan x = \frac{\sin x}{\cos x}$$

Let $g(x) = \sin x$ and $h(x) = \cos x$. Both $g(x)$ and $h(x)$ are standard trigonometric functions that are continuous for all real numbers $x \in \mathbb{R}$.

The domain of $f(x) = \tan x$ consists of all real numbers $x$ for which $\cos x \neq 0$:

$$\text{Domain}(f) = \mathbb{R} \setminus \left\{ (2k+1)\frac{\pi}{2} \;\middle|\; k \in \mathbb{Z} \right\}$$

By the quotient rule of continuous functions, if $g(x)$ and $h(x)$ are continuous at $x = c$, then $\frac{g(x)}{h(x)}$ is continuous at $x = c$ provided $h(c) \neq 0$.

Therefore, $f(x) = \tan x$ is continuous at every point in its domain.

**Final Answer**

Yes, $\tan x$ is a continuous function in its domain $\mathbb{R} \setminus \left\{ (2k+1)\frac{\pi}{2} \;\middle|\; k \in \mathbb{Z} \right\}$.

## Question 2

State which of the following functions are continuous?

(i) $\sin x + \cos x$
(ii) $\sin x - \cos x$
(iii) $\sin x \cdot \cos x$

**TYPE 9** **Algebra of Continuous Functions**

**Given:** The functions (i) $f_1(x) = \sin x + \cos x$, (ii) $f_2(x) = \sin x - \cos x$, and (iii) $f_3(x) = \sin x \cdot \cos x$.

**Concept / Formula Used:** $g(x)$ $h(x)$ $\mathbb{R}$ $(g+h)(x) = g(x) + h(x)$ $\mathbb{R}$ $(g-h)(x) = g(x) - h(x)$ $\mathbb{R}$ $(g \cdot h)(x) = g(x) \cdot h(x)$ $\mathbb{R}$

**Solution:**

Let $g(x) = \sin x$ and $h(x) = \cos x$. Both $g(x)$ and $h(x)$ are continuous functions for all $x \in \mathbb{R}$.

**Part (i):**

$$\begin{aligned} f_1(x) &= \sin x + \cos x \\ &= g(x) + h(x) \end{aligned}$$

Since the sum of two continuous functions is continuous, $f_1(x)$ is continuous for all $x \in \mathbb{R}$.

**Part (ii):**

$$\begin{aligned} f_2(x) &= \sin x - \cos x \\ &= g(x) - h(x) \end{aligned}$$

Since the difference of two continuous functions is continuous, $f_2(x)$ is continuous for all $x \in \mathbb{R}$.

**Part (iii):**

$$\begin{aligned} f_3(x) &= \sin x \cdot \cos x \\ &= g(x) \cdot h(x) \end{aligned}$$

Since the product of two continuous functions is continuous, $f_3(x)$ is continuous for all $x \in \mathbb{R}$.

**Final Answer**

All three functions (i), (ii), and (iii) are continuous for all $x \in \mathbb{R}$.

## Question 3

Is the function $f$ defined by $f(x) = 2 - 3x + |x|$ a continuous function?

**TYPE 5 Continuity of Absolute Value Functions**

**Given:** $f(x) = 2 - 3x + |x|$

**Concept / Formula Used:** $\mathbb{R}$ $g(x) = |x|$ $\mathbb{R}$ $\mathbb{R}$

**Solution:**

**Method 1: Using Algebra of Continuous Functions**

Let $p(x) = 2 - 3x$ and $q(x) = |x|$.

$$f(x) = p(x) + q(x)$$

- $p(x) = 2 - 3x$ is a polynomial function, so it is continuous for all $x \in \mathbb{R}$. - $q(x) = |x|$ is the absolute value function, which is continuous for all $x \in \mathbb{R}$.

Since $f(x)$ is the sum of two continuous functions, $f(x)$ is continuous for all $x \in \mathbb{R}$.

**Method 2: By Definition at $x = 0$**

Redefining $f(x)$:

$$f(x) = \begin{cases} 2 - 3x + x = 2 - 2x, & \text{if } x \geq 0 \\ 2 - 3x - x = 2 - 4x, & \text{if } x < 0 \end{cases}$$

At $x = 0$:

$$\begin{aligned} \text{LHL} &= \lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (2 - 4x) = 2 - 4(0) = 2 \\ \text{RHL} &= \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (2 - 2x) = 2 - 2(0) = 2 \\ f(0) &= 2 - 2(0) = 2 \end{aligned}$$

Since LHL = RHL = $f(0)$, $f(x)$ is continuous at $x = 0$. For $x > 0$ and $x < 0$, $f(x)$ is a linear polynomial and hence continuous.

**Final Answer**

Yes, $f(x) = 2 - 3x + |x|$ is a continuous function on $\mathbb{R}$.

## Question 4

Is the function $f$ defined by $f(x) = |\sin x|$ a continuous function?

**TYPE 10** **Continuity of Composite Functions**

★ **Likely Exam Question**

**Given:** $f(x) = |\sin x|$

**Concept / Formula Used:** $h(x)$ $c$ $g(x)$ $h(c)$ $(g \circ h)(x) = g(h(x))$ $c$

**Solution:**

Let $h(x) = \sin x$ and $g(x) = |x|$.

The composition $(g \circ h)(x)$ is given by:

$$\begin{aligned}(g \circ h)(x) &= g(h(x)) \\ &= g(\sin x) \\ &= |\sin x| \\ &= f(x)\end{aligned}$$

We know that: 1. $h(x) = \sin x$ is continuous for all $x \in \mathbb{R}$. 2. $g(x) = |x|$ is continuous for all $x \in \mathbb{R}$.

By the composite function theorem, since both $g$ and $h$ are continuous on $\mathbb{R}$, their composition $(g \circ h)(x) = |\sin x|$ is also continuous for all $x \in \mathbb{R}$.

**Final Answer**

Yes, $f(x) = |\sin x|$ is a continuous function on $\mathbb{R}$.

## Question 5

Is the function $f$ defined by $f(x) = \cos |x|$ a continuous function?

**TYPE 10** **Continuity of Composite Functions**

★ **Likely Exam Question**

**Given:** $f(x) = \cos |x|$

**Concept / Formula Used:** $h : \mathbb{R} \to \mathbb{R}$ $g : \mathbb{R} \to \mathbb{R}$ $(g \circ h)(x) = g(h(x))$ $\mathbb{R}$

**Solution:**

Let $h(x) = |x|$ and $g(x) = \cos x$.

The composite function $(g \circ h)(x)$ is defined as:

$$\begin{aligned}(g \circ h)(x) &= g(h(x)) \\ &= g(|x|) \\ &= \cos|x| \\ &= f(x)\end{aligned}$$

We check the continuity of individual functions: 1. $h(x) = |x|$ is continuous for all $x \in \mathbb{R}$. 2. $g(x) = \cos x$ is continuous for all $x \in \mathbb{R}$.

Since the composition of two continuous functions is continuous, $f(x) = (g \circ h)(x) = \cos|x|$ is continuous for all $x \in \mathbb{R}$.

**Final Answer**

Yes, $f(x) = \cos|x|$ is a continuous function on $\mathbb{R}$.

## Question 6

Prove that the following functions are continuous:

(i) $\dfrac{2x^3 - 7x^2 + 3}{(x-1)(x+3)}$

(ii) $|\sec x + \tan x|$

**TYPE 9 Algebra of Continuous Functions**

**To Prove:** That (i) $f(x) = \frac{2x^3-7x^2+3}{(x-1)(x+3)}$ and (ii) $g(x) = |\sec x + \tan x|$ are continuous in their respective domains.

**Concept / Formula Used:** $\frac{P(x)}{Q(x)}$ $\quad Q(x) \neq 0$

**Solution:**

**Part (i):** Let $f(x) = \frac{2x^3-7x^2+3}{(x-1)(x+3)}$.

Let $P(x) = 2x^3 - 7x^2 + 3$ and $Q(x) = (x-1)(x+3) = x^2 + 2x - 3$.

- $P(x)$ is a polynomial function, so it is continuous for all $x \in \mathbb{R}$. - $Q(x)$ is a polynomial function, so it is continuous for all $x \in \mathbb{R}$.

The domain of $f(x)$ is where the denominator is non-zero:

$$\begin{aligned}Q(x) &\neq 0 \\ (x-1)(x+3) &\neq 0 \\ x &\neq 1 \quad \text{and} \quad x \neq -3\end{aligned}$$

$$\text{Domain}(f) = \mathbb{R} \setminus \{1, -3\}$$

By the quotient property of continuous functions, $f(x) = \frac{P(x)}{Q(x)}$ is continuous at every point in its domain $\mathbb{R} \setminus \{1, -3\}$.

**Part (ii):** Let $g(x) = |\sec x + \tan x|$.

Let $u(x) = \sec x + \tan x$ and $v(y) = |y|$.

- $u(x) = \sec x + \tan x$ is continuous for all $x \in \mathbb{R} \setminus \left\{(2k+1)\frac{\pi}{2} \mid k \in \mathbb{Z}\right\}$. - $v(y) = |y|$ is continuous for all $y \in \mathbb{R}$.

The composite function is:

$$\begin{aligned}(v \circ u)(x) &= v(u(x)) \\ &= v(\sec x + \tan x) \\ &= |\sec x + \tan x| \\ &= g(x)\end{aligned}$$

By the composite function theorem, $g(x)$ is continuous at all points in its domain $\mathbb{R} \setminus \left\{(2k+1)\frac{\pi}{2} \mid k \in \mathbb{Z}\right\}$.

**Hence Proved**

Both functions are continuous in their respective domains, as proved above.

## Question 7

Examine the following functions for continuity:

(i) $\cos(x^2)$

(ii) $\sin(3x^2 - 5)$

(iii) $\tan^{-1}(2x^2 + 3)$

**TYPE 10** **Continuity of Composite Functions**

★ **Likely Exam Question**

**Given:** The functions (i) $f_1(x) = \cos(x^2)$, (ii) $f_2(x) = \sin(3x^2 - 5)$, and (iii) $f_3(x) = \tan^{-1}(2x^2 + 3)$.

**Concept / Formula Used:** $h : \mathbb{R} \to \mathbb{R}$ $\mathbb{R}$ $g : \mathbb{R} \to \mathbb{R}$ $\mathbb{R}$ $(g \circ h)(x) = g(h(x))$ $\mathbb{R}$

**Solution:**

**Part (i):** $f_1(x) = \cos(x^2)$ Let $h_1(x) = x^2$ and $g_1(x) = \cos x$.

$$(g_1 \circ h_1)(x) = g_1(h_1(x)) = \cos(x^2) = f_1(x)$$

- $h_1(x) = x^2$ is a polynomial function, hence continuous on $\mathbb{R}$. - $g_1(x) = \cos x$ is a trigonometric function, hence continuous on $\mathbb{R}$.

By the composite function theorem, $f_1(x) = \cos(x^2)$ is continuous for all $x \in \mathbb{R}$.

**Part (ii):** $f_2(x) = \sin(3x^2 - 5)$ Let $h_2(x) = 3x^2 - 5$ and $g_2(x) = \sin x$.

$$(g_2 \circ h_2)(x) = g_2(h_2(x)) = \sin(3x^2 - 5) = f_2(x)$$

- $h_2(x) = 3x^2 - 5$ is a polynomial function, hence continuous on $\mathbb{R}$. - $g_2(x) = \sin x$ is a trigonometric function, hence continuous on $\mathbb{R}$.

By the composite function theorem, $f_2(x) = \sin(3x^2 - 5)$ is continuous for all $x \in \mathbb{R}$.

**Part (iii):** $f_3(x) = \tan^{-1}(2x^2 + 3)$ Let $h_3(x) = 2x^2 + 3$ and $g_3(x) = \tan^{-1} x$.

$$(g_3 \circ h_3)(x) = g_3(h_3(x)) = \tan^{-1}(2x^2 + 3) = f_3(x)$$

- $h_3(x) = 2x^2 + 3$ is a polynomial function, hence continuous on $\mathbb{R}$. - $g_3(x) = \tan^{-1} x$ is an inverse trigonometric function, continuous for all $x \in \mathbb{R}$.

By the composite function theorem, $f_3(x) = \tan^{-1}(2x^2 + 3)$ is continuous for all $x \in \mathbb{R}$.

**Final Answer**

(i) $f(x) = \cos(x^2)$ is continuous for all $x \in \mathbb{R}$.
(ii) $f(x) = \sin(3x^2 - 5)$ is continuous for all $x \in \mathbb{R}$.
(iii) $f(x) = \tan^{-1}(2x^2 + 3)$ is continuous for all $x \in \mathbb{R}$.

## 5.4 – Differentiability: Definition and Relation to Continuity

### Question 1

If a function is derivable at a point, is it necessary that it must be continuous at that point?

**TYPE 11** Differentiability and Continuity Relationship

**Given:** A function $f(x)$ is derivable at a point $x = c$.

**To Find:** Whether $f(x)$ must be continuous at $x = c$.

**Concept / Formula Used:** $f'(c) = \lim_{x \to c} \dfrac{f(x) - f(c)}{x - c}$

**Solution:**

Let $f(x)$ be derivable at $x = c$. By definition, the limit

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists finitely. Now, consider the difference $f(x) - f(c)$ for $x \neq c$:

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} \cdot (x - c)$$

Taking the limit as $x \to c$ on both sides:

$$\begin{aligned}\lim_{x\to c}[f(x)-f(c)] &= \lim_{x\to c}\left[\frac{f(x)-f(c)}{x-c}\cdot(x-c)\right] \\ &= \left(\lim_{x\to c}\frac{f(x)-f(c)}{x-c}\right)\cdot\left(\lim_{x\to c}(x-c)\right) \\ &= f'(c)\cdot 0 \\ &= 0\end{aligned}$$

Therefore:

$$\lim_{x\to c} f(x)-f(c)=0 \implies \lim_{x\to c} f(x)=f(c)$$

This is the precise condition for $f(x)$ to be continuous at $x=c$.

**Final Answer**

Yes, if a function is derivable at a point, it is necessary that it must be continuous at that point.

## Question 2

If a function is continuous at a point, is it necessary that it must be derivable at that point?

**TYPE 11** **Differentiability and Continuity Relationship**

**Given:** A function $f(x)$ is continuous at a point $x=c$.

**To Find:** Whether $f(x)$ must be derivable at $x=c$.

**Solution:**

Consider the absolute value function $f(x)=|x|$ at the point $x=0$.

1. Checking continuity at $x=0$:

$$\begin{aligned}\lim_{x\to 0^-} f(x) &= \lim_{x\to 0^-}(-x)=0 \\ \lim_{x\to 0^+} f(x) &= \lim_{x\to 0^+} x=0 \\ f(0) &= |0|=0\end{aligned}$$

Since $\lim_{x\to 0^-} f(x)=\lim_{x\to 0^+} f(x)=f(0)=0$, $f(x)$ is continuous at $x=0$.

2. Checking derivability at $x=0$:

$$\begin{aligned}\text{LHD } &= Lf'(0)=\lim_{h\to 0^-}\frac{f(0+h)-f(0)}{h}=\lim_{h\to 0^-}\frac{|h|-0}{h}=\lim_{h\to 0^-}\frac{-h}{h}=-1 \\ \text{RHD } &= Rf'(0)=\lim_{h\to 0^+}\frac{f(0+h)-f(0)}{h}=\lim_{h\to 0^+}\frac{|h|-0}{h}=\lim_{h\to 0^+}\frac{h}{h}=1\end{aligned}$$

Since $Lf'(0)\neq Rf'(0)$, $f(x)$ is not derivable at $x=0$.

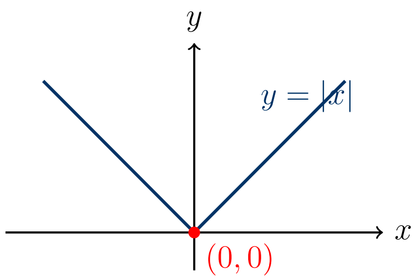
**Final Answer**

No, it is not necessary. A function may be continuous at a point without being derivable at that point (e.g., $f(x) = |x|$ at $x = 0$).

## Question 3

Is the function $f(x) = |x|$ derivable at $x = 0$?

**TYPE 12 Differentiability of Absolute Value Functions**




*Graph of $f(x) = |x|$ showing a sharp corner at $x = 0$*

**Given:** Function $f(x) = |x|$ at $x = 0$.

**To Find:** Whether $f(x)$ is derivable at $x = 0$.

**Concept / Formula Used:** $Lf'(c) = \lim_{h\to 0^-} \dfrac{f(c+h) - f(c)}{h}$ $Rf'(c) = \lim_{h\to 0^+} \dfrac{f(c+h) - f(c)}{h}$

**Solution:**

To test derivability at $x = 0$, we calculate the Left-Hand Derivative (LHD) and Right-Hand Derivative (RHD):

1. Left-Hand Derivative at $x = 0$:

$$\begin{aligned} Lf'(0) &= \lim_{h\to 0^-} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h\to 0^-} \frac{|0+h| - |0|}{h} \\ &= \lim_{h\to 0^-} \frac{-h}{h} \quad (\text{since } h < 0 \implies |h| = -h) \\ &= -1 \end{aligned}$$

2. Right-Hand Derivative at $x = 0$:

$$\begin{aligned}
Rf'(0) &= \lim_{h\to 0^+} \frac{f(0+h) - f(0)}{h} \\
&= \lim_{h\to 0^+} \frac{|0+h| - |0|}{h} \\
&= \lim_{h\to 0^+} \frac{h}{h} \quad (\text{since } h > 0 \implies |h| = h) \\
&= 1
\end{aligned}$$

Since $Lf'(0) = -1 \neq 1 = Rf'(0)$, the limit does not exist.

**Final Answer**

No, the function $f(x) = |x|$ is not derivable at $x = 0$.

## Question 4

Is the function $f(x) = \cot x$ derivable at $x = 0$?

**TYPE 13** **Differentiability of Trigonometric Functions**

**Given:** Function $f(x) = \cot x$ at $x = 0$.

**To Find:** Whether $f(x)$ is derivable at $x = 0$.

**Solution:**

We examine the definition and continuity of $f(x) = \cot x$ at $x = 0$:

$$\cot x = \frac{\cos x}{\sin x}$$

At $x = 0$:

$$\sin 0 = 0 \implies \cot 0 \text{ is undefined.}$$

Since $f(x) = \cot x$ is not defined at $x = 0$, it cannot be continuous at $x = 0$. Consequently, it cannot be derivable at $x = 0$.

**Final Answer**

No, $f(x) = \cot x$ is not derivable at $x = 0$ because it is undefined at $x = 0$.

## Question 5

Is the function $f(x) = \sec x$ derivable at $x = \frac{\pi}{2}$?

**TYPE 13** **Differentiability of Trigonometric Functions**

**Given:** Function $f(x) = \sec x$ at $x = \frac{\pi}{2}$.

**To Find:** Whether $f(x)$ is derivable at $x = \frac{\pi}{2}$.

**Solution:**

We evaluate $f(x) = \sec x$ at $x = \frac{\pi}{2}$:

$$\sec x = \frac{1}{\cos x}$$

At $x = \frac{\pi}{2}$:

$$\cos\left(\frac{\pi}{2}\right) = 0 \implies \sec\left(\frac{\pi}{2}\right) \text{ is undefined.}$$

Since $f(x) = \sec x$ is not defined at $x = \frac{\pi}{2}$, it is not continuous, and therefore not derivable at $x = \frac{\pi}{2}$.

**Final Answer**

No, $f(x) = \sec x$ is not derivable at $x = \frac{\pi}{2}$ because it is undefined at $x = \frac{\pi}{2}$.

## Question 6

When a function is called derivable?

**TYPE 14 Definition of Differentiability**

**Given:** A general real-valued function $f(x)$.

**To Find:** The definition of when a function is derivable at a point and on an interval.

**Concept / Formula Used:** $f'(c) = \lim_{h\to 0} \dfrac{f(c+h) - f(c)}{h}$

**Solution:**

1. **Derivability at a point:** A function $f(x)$ is called derivable (or differentiable) at a point $x = c$ in its domain if the limit

$$f'(c) = \lim_{h\to 0} \frac{f(c+h) - f(c)}{h} \quad \text{or equivalently} \quad \lim_{x\to c} \frac{f(x) - f(c)}{x - c}$$

exists finitely. This requires that both the Left-Hand Derivative ($Lf'(c)$) and Right-Hand Derivative ($Rf'(c)$) exist and are equal:

$$Lf'(c) = Rf'(c) < \infty$$

2. **Derivability on an interval:** A function is called derivable on an open interval $(a, b)$ if it is derivable at every point in $(a, b)$.

**Final Answer**

A function $f(x)$ is called derivable at $x = c$ if $\lim_{h\to 0} \dfrac{f(c+h) - f(c)}{h}$ exists finitely (i.e., LHD = RHD). It is derivable on an interval if it is derivable at every point of that interval.

## Question 7

If a function is derivable at $x = c$, then write the value of $\lim_{x\to c} f(x)$.

**TYPE 11** **Differentiability and Continuity Relationship**

**Given:** Function $f(x)$ is derivable at $x = c$.

**To Find:** The value of $\lim_{x\to c} f(x)$.

**Solution:**

Since $f(x)$ is derivable at $x = c$, it is necessarily continuous at $x = c$.

By definition of continuity at $x = c$:

$$\lim_{x\to c} f(x) = f(c)$$

**Final Answer**

$\lim_{x\to c} f(x) = f(c)$

## Question 8

Which of the following functions are derivable?

(i) $\sin x$
(ii) $\cos x$
(iii) $\tan x$
(iv) $\cot x$
(v) $\sec x$
(vi) $\csc x$

**TYPE 13** **Differentiability of Trigonometric Functions**

**Given:** Trigonometric functions: $\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$.

**To Find:** The derivability domain for each function.

**Solution:**

We evaluate each function over its domain:

1. $\sin x$ **and** $\cos x$: Defined and continuous for all $x \in \mathbb{R}$. Their derivatives $\cos x$ and $-\sin x$ exist everywhere on $\mathbb{R}$.

2. $\tan x$ **and** $\sec x$: Defined for all $x \in \mathbb{R} \setminus \left\{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\right\}$. They are derivable at all points in their domain.

3. $\cot x$ **and** $\csc x$: Defined for all $x \in \mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$. They are derivable at all points in their domain.

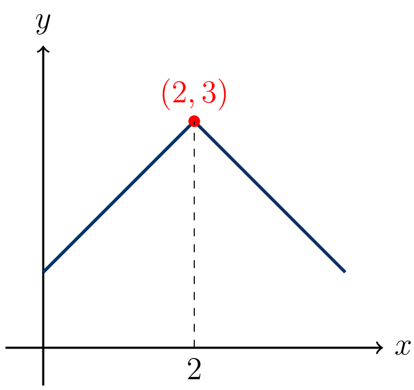
**Final Answer**

(i) $\sin x$: Derivable for all $x \in \mathbb{R}$.
(ii) $\cos x$: Derivable for all $x \in \mathbb{R}$.
(iii) $\tan x$: Derivable for all $x \in \mathbb{R} \setminus \left\{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\right\}$.
(iv) $\cot x$: Derivable for all $x \in \mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$.
(v) $\sec x$: Derivable for all $x \in \mathbb{R} \setminus \left\{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\right\}$.
(vi) $\csc x$: Derivable for all $x \in \mathbb{R} \setminus \{n\pi : n \in \mathbb{Z}\}$.

## Question 9

Examine the function $f(x) = \begin{cases} 1+x, & \text{if } x \le 2 \\ 5-x, & \text{if } x > 2 \end{cases}$ for differentiability at $x = 2$.

**TYPE 15 Differentiability of Piecewise Functions**




*Graph of $f(x)$ showing a non-differentiable corner at $x = 2$*

**Given:** Function $f(x) = \begin{cases} 1+x, & x \le 2 \\ 5-x, & x > 2 \end{cases}$ at $x = 2$.

**To Find:** Differentiability of $f(x)$ at $x = 2$.

**Concept / Formula Used:** $Lf'(c) = \lim_{h\to 0^-} \dfrac{f(c+h)-f(c)}{h}$ $Rf'(c) = \lim_{h\to 0^+} \dfrac{f(c+h)-f(c)}{h}$

**Solution:**

First, compute $f(2)$:

$$f(2) = 1 + 2 = 3$$

1. Left-Hand Derivative at $x = 2$:

$$Lf'(2) = \lim_{h \to 0^-} \frac{f(2+h) - f(2)}{h}$$

For $h < 0$, $2 + h < 2$, so $f(2 + h) = 1 + (2 + h) = 3 + h$:

$$\begin{aligned} Lf'(2) &= \lim_{h \to 0^-} \frac{(3+h) - 3}{h} \\ &= \lim_{h \to 0^-} \frac{h}{h} \\ &= 1 \end{aligned}$$

2. Right-Hand Derivative at $x = 2$:

$$Rf'(2) = \lim_{h \to 0^+} \frac{f(2+h) - f(2)}{h}$$

For $h > 0$, $2 + h > 2$, so $f(2 + h) = 5 - (2 + h) = 3 - h$:

$$\begin{aligned} Rf'(2) &= \lim_{h \to 0^+} \frac{(3-h) - 3}{h} \\ &= \lim_{h \to 0^+} \frac{-h}{h} \\ &= -1 \end{aligned}$$

Since $Lf'(2) = 1 \neq -1 = Rf'(2)$, $f(x)$ is not differentiable at $x = 2$.

**Final Answer**

The function $f(x)$ is not differentiable at $x = 2$.

## Question 10

If $f(2) = 4$ and $f'(2) = 4$, then evaluate $\lim_{x \to 2} \frac{xf(2) - 2f(x)}{x - 2}$.

**TYPE 16** **Differentiability Using Limits**

**★ Likely Exam Question**

**Given:** $f(2) = 4$ and $f'(2) = 4$.

**To Find:** $\lim_{x \to 2} \frac{xf(2) - 2f(x)}{x - 2}$

**Concept / Formula Used:** $f'(c) = \lim_{x\to c} \dfrac{f(x) - f(c)}{x - c}$

**Solution:**

Let $L = \lim\limits_{x\to 2} \dfrac{xf(2) - 2f(x)}{x - 2}$.

Add and subtract $2f(2)$ in the numerator:

$$\begin{aligned} L &= \lim_{x\to 2} \frac{xf(2) - 2f(2) + 2f(2) - 2f(x)}{x - 2} \\ &= \lim_{x\to 2} \frac{(x-2)f(2) - 2[f(x) - f(2)]}{x - 2} \end{aligned}$$

Split the fraction into two separate terms:

$$\begin{aligned} L &= \lim_{x\to 2} \left[ \frac{(x-2)f(2)}{x-2} - 2 \cdot \frac{f(x) - f(2)}{x - 2} \right] \\ &= \lim_{x\to 2} f(2) - 2 \cdot \lim_{x\to 2} \frac{f(x) - f(2)}{x - 2} \end{aligned}$$

Using the given definition $f'(2) = \lim_{x\to 2} \dfrac{f(x) - f(2)}{x - 2}$:

$$L = f(2) - 2f'(2)$$

Substitute the given values $f(2) = 4$ and $f'(2) = 4$:

$$\begin{aligned} L &= 4 - 2(4) \\ &= 4 - 8 \\ &= -4 \end{aligned}$$

**Final Answer**

$\lim\limits_{x\to 2} \dfrac{xf(2) - 2f(x)}{x - 2} = -4$

## Question 11

Examine the following function for continuity at $x = 1$ and differentiability at $x = 2$:

$$f(x) = \begin{cases} 5x - 4, & 0 < x < 1 \\ 4x^2 - 3x, & 1 \le x < 2 \\ 3x + 4, & x \ge 2 \end{cases}$$

**TYPE 15 Differentiability of Piecewise Functions**

**Given:** Function $f(x) = \begin{cases} 5x - 4, & 0 < x < 1 \\ 4x^2 - 3x, & 1 \le x < 2 \\ 3x + 4, & x \ge 2 \end{cases}$

**To Find:** Continuity at $x = 1$ and differentiability at $x = 2$.

**Concept / Formula Used:** $\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = f(c)$ $\quad Lf'(c) = Rf'(c)$

**Solution:**

**Part 1: Continuity at $x = 1$**

1. Left-Hand Limit (LHL) at $x = 1$:

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (5x - 4) = 5(1) - 4 = 1$$

2. Right-Hand Limit (RHL) at $x = 1$:

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} (4x^2 - 3x) = 4(1)^2 - 3(1) = 1$$

3. Value of function at $x = 1$:

$$f(1) = 4(1)^2 - 3(1) = 1$$

Since $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = f(1) = 1$, the function $f(x)$ is continuous at $x = 1$.

**Part 2: Differentiability at $x = 2$**

Value of function at $x = 2$:

$$f(2) = 3(2) + 4 = 10$$

1. Left-Hand Derivative at $x = 2$:

$$Lf'(2) = \lim_{h\to 0^-} \frac{f(2+h) - f(2)}{h}$$

For $h < 0$, $2 + h < 2$, so $f(2 + h) = 4(2 + h)^2 - 3(2 + h)$:

$$\begin{aligned}
Lf'(2) &= \lim_{h\to 0^-} \frac{4(4 + 4h + h^2) - 3(2 + h) - 10}{h} \\
&= \lim_{h\to 0^-} \frac{16 + 16h + 4h^2 - 6 - 3h - 10}{h} \\
&= \lim_{h\to 0^-} \frac{13h + 4h^2}{h} \\
&= \lim_{h\to 0^-} (13 + 4h) \\
&= 13
\end{aligned}$$

2. Right-Hand Derivative at $x = 2$:

$$Rf'(2) = \lim_{h\to 0^+} \frac{f(2+h) - f(2)}{h}$$

For $h > 0$, $2 + h > 2$, so $f(2 + h) = 3(2 + h) + 4$:

$$\begin{aligned} Rf'(2) &= \lim_{h\to0^+} \frac{[3(2+h)+4]-10}{h} \\ &= \lim_{h\to0^+} \frac{6+3h+4-10}{h} \\ &= \lim_{h\to0^+} \frac{3h}{h} \\ &= 3 \end{aligned}$$

Since $Lf'(2) = 13 \neq 3 = Rf'(2)$, $f(x)$ is not differentiable at $x = 2$.

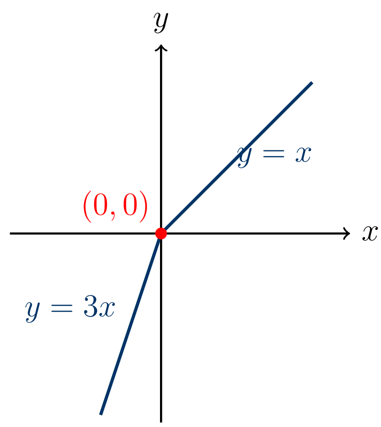
**Final Answer**

$f(x)$ is continuous at $x = 1$, but not differentiable at $x = 2$.

## Question 12

Show that the function $f(x) = 2x - |x|$ is continuous at $x = 0$ but not differentiable at $x = 0$.

**TYPE 12** **Differentiability of Absolute Value Functions**




*Graph of $f(x) = 2x - |x|$ with a sharp peak/corner at $x = 0$*

**To Prove:** $f(x) = 2x - |x|$ is continuous at $x = 0$ but not differentiable at $x = 0$.

**Concept / Formula Used:** $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

**Solution:**

Redefining $f(x)$ explicitly without absolute value:

$$f(x) = \begin{cases} 2x - x = x, & x \geq 0 \\ 2x - (-x) = 3x, & x < 0 \end{cases}$$

**Step 1: Proof of Continuity at $x = 0$**

$$\begin{aligned} \text{LHL} &= \lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (3x) = 3(0) = 0 \\ \text{RHL} &= \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} (x) = 0 \\ f(0) &= 2(0) - |0| = 0 \end{aligned}$$

Since LHL = RHL = $f(0) = 0$, $f(x)$ is continuous at $x = 0$.

**Step 2: Proof of Non-Differentiability at $x = 0$**

$$Lf'(0) = \lim_{h\to 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0^-} \frac{3h - 0}{h} = 3$$
$$Rf'(0) = \lim_{h\to 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0^+} \frac{h - 0}{h} = 1$$

Since $Lf'(0) = 3 \neq 1 = Rf'(0)$, $f(x)$ is not differentiable at $x = 0$.

**Hence Proved**

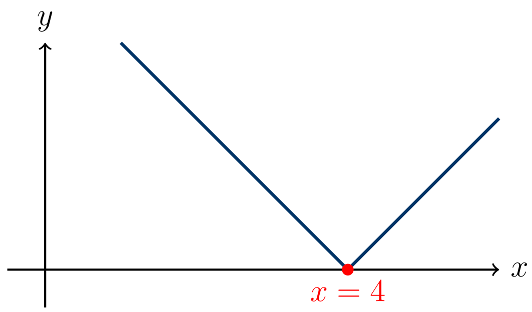
LHL = RHL = f(0) proves continuity at $x = 0$; LHD $\neq$ RHD proves non-differentiability at $x = 0$.

## Question 13

Show that $f(x) = |x - 4|$ is continuous but not differentiable at $x = 4$.

**TYPE 12** Differentiability of Absolute Value Functions

★ **Likely Exam Question**




*Graph of $f(x) = |x - 4|$ with a corner at $x = 4$*

**To Prove:** $f(x) = |x - 4|$ is continuous but not differentiable at $x = 4$.

**Concept / Formula Used:** $|x - 4| = \begin{cases} x - 4, & x \geq 4 \\ -(x-4) = 4 - x, & x < 4 \end{cases}$

**Solution:**

Redefining $f(x)$:

$$f(x) = \begin{cases} 4-x, & x<4 \\ x-4, & x \geq 4 \end{cases}$$

**Step 1: Proof of Continuity at $x = 4$**

$$\begin{aligned} \text{LHL} &= \lim_{x\to 4^-} f(x) = \lim_{x\to 4^-}(4-x) = 4-4 = 0 \\ \text{RHL} &= \lim_{x\to 4^+} f(x) = \lim_{x\to 4^+}(x-4) = 4-4 = 0 \\ f(4) &= |4-4| = 0 \end{aligned}$$

Since LHL $=$ RHL $= f(4) = 0$, $f(x)$ is continuous at $x = 4$.

**Step 2: Proof of Non-Differentiability at $x = 4$**

$$Lf'(4) = \lim_{h\to 0^-}\frac{f(4+h)-f(4)}{h} = \lim_{h\to 0^-}\frac{|4+h-4|-0}{h} = \lim_{h\to 0^-}\frac{|h|}{h} = \lim_{h\to 0^-}\frac{-h}{h} = -1$$
$$Rf'(4) = \lim_{h\to 0^+}\frac{f(4+h)-f(4)}{h} = \lim_{h\to 0^+}\frac{|4+h-4|-0}{h} = \lim_{h\to 0^+}\frac{|h|}{h} = \lim_{h\to 0^+}\frac{h}{h} = 1$$

Since $Lf'(4) = -1 \neq 1 = Rf'(4)$, $f(x)$ is not differentiable at $x = 4$.

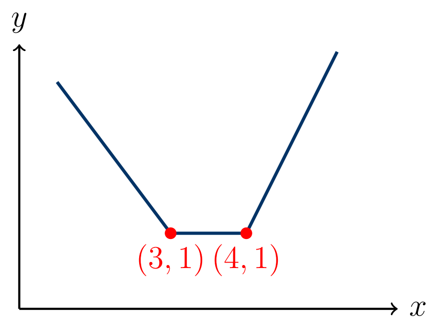
**Hence Proved**

$f(x) = |x-4|$ is continuous at $x = 4$ but not differentiable at $x = 4$.

## Question 14

If the function $f(x) = |x-3| + |x-4|$, then show that $f$ is not differentiable at $x = 3$ and $x = 4$.

**TYPE 12** **Differentiability of Absolute Value Functions**




*Graph of $f(x) = |x-3| + |x-4|$ showing corners at $x = 3$ and $x = 4$*

**To Prove:** $f(x) = |x-3| + |x-4|$ is not differentiable at $x = 3$ and $x = 4$.

**Concept / Formula Used:** $x = 3$ $x = 4$

**Solution:**

Expressing $f(x)$ explicitly in intervals:

$$f(x) = \begin{cases} -(x-3)-(x-4) = 7-2x, & x < 3 \\ (x-3)-(x-4) = 1, & 3 \leq x < 4 \\ (x-3)+(x-4) = 2x-7, & x \geq 4 \end{cases}$$

**Part 1: Non-Differentiability at $x = 3$**

Calculate $f(3) = |3-3| + |3-4| = 0 + 1 = 1$.

$$Lf'(3) = \lim_{h \to 0^-} \frac{f(3+h)-f(3)}{h} = \lim_{h \to 0^-} \frac{[7-2(3+h)]-1}{h} = \lim_{h \to 0^-} \frac{1-2h-1}{h} = -2$$
$$Rf'(3) = \lim_{h \to 0^+} \frac{f(3+h)-f(3)}{h} = \lim_{h \to 0^+} \frac{1-1}{h} = 0$$

Since $Lf'(3) = -2 \neq 0 = Rf'(3)$, $f(x)$ is not differentiable at $x = 3$.

**Part 2: Non-Differentiability at $x = 4$**

Calculate $f(4) = |4-3| + |4-4| = 1 + 0 = 1$.

$$Lf'(4) = \lim_{h \to 0^-} \frac{f(4+h)-f(4)}{h} = \lim_{h \to 0^-} \frac{1-1}{h} = 0$$
$$Rf'(4) = \lim_{h \to 0^+} \frac{f(4+h)-f(4)}{h} = \lim_{h \to 0^+} \frac{[2(4+h)-7]-1}{h} = \lim_{h \to 0^+} \frac{1+2h-1}{h} = 2$$

Since $Lf'(4) = 0 \neq 2 = Rf'(4)$, $f(x)$ is not differentiable at $x = 4$.

**Hence Proved**

$f(x)$ is not differentiable at $x = 3$ and $x = 4$.

## Question 15

Show that the function $f$ is continuous at $x = 1$ for all values of $a$ where

$$f(x) = \begin{cases} ax^2 + 1, & x \geq 1 \\ x + a, & x < 1 \end{cases}$$

Find its right and left hand derivatives at $x = 1$. Hence, find the condition for the existence of the derivative at $x = 1$.

**TYPE 17** **Finding Constants for Differentiability**

**Given:** Function $f(x) = \begin{cases} ax^2 + 1, & x \geq 1 \\ x + a, & x < 1 \end{cases}$

**To Find:** (i) Proof of continuity at $x = 1$ for all $a$, (ii) LHD and RHD at $x = 1$, (iii) Condition for derivative to exist at $x = 1$.

**Concept / Formula Used:** $Lf'(1) = Rf'(1)$

**Solution:**

**Step 1: Proof of Continuity at $x = 1$ for all $a$**

$$\begin{aligned}
\text{LHL} &= \lim_{x\to1^-} f(x) = \lim_{x\to1^-}(x+a) = 1+a \\
\text{RHL} &= \lim_{x\to1^+} f(x) = \lim_{x\to1^+}(ax^2+1) = a(1)^2+1 = a+1 \\
f(1) &= a(1)^2 + 1 = a+1
\end{aligned}$$

Since LHL = RHL = $f(1) = a+1$ for every real number $a$, $f(x)$ is continuous at $x = 1$ for all values of $a$.

**Step 2: Finding Left and Right Hand Derivatives at $x = 1$**

1. Left-Hand Derivative ($Lf'(1)$):

$$\begin{aligned}
Lf'(1) &= \lim_{h\to0^-} \frac{f(1+h)-f(1)}{h} \\
&= \lim_{h\to0^-} \frac{[(1+h)+a]-(a+1)}{h} \\
&= \lim_{h\to0^-} \frac{h}{h} \\
&= 1
\end{aligned}$$

2. Right-Hand Derivative ($Rf'(1)$):

$$\begin{aligned}
Rf'(1) &= \lim_{h\to0^+} \frac{f(1+h)-f(1)}{h} \\
&= \lim_{h\to0^+} \frac{[a(1+h)^2+1]-(a+1)}{h} \\
&= \lim_{h\to0^+} \frac{a(1+2h+h^2)+1-a-1}{h} \\
&= \lim_{h\to0^+} \frac{2ah+ah^2}{h} \\
&= \lim_{h\to0^+} (2a+ah) \\
&= 2a
\end{aligned}$$

**Step 3: Condition for existence of $f'(1)$**

The derivative $f'(1)$ exists if and only if $Lf'(1) = Rf'(1)$:

$$1 = 2a \implies a = \frac{1}{2}$$

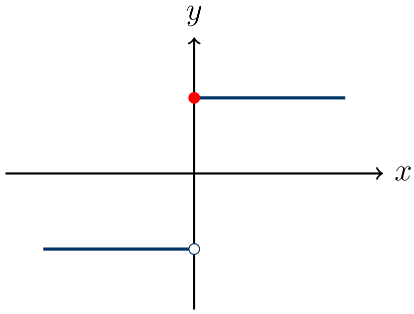
**Final Answer**

$Lf'(1) = 1$, $Rf'(1) = 2a$. The derivative $f'(1)$ exists if and only if $a = \dfrac{1}{2}$.

## Question 16

Prove that the function $f(x) = \begin{cases} \dfrac{x}{|x|}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ is not differentiable at $x = 0$.

**TYPE 11** **Differentiability and Continuity Relationship**




*Graph of the signum function $f(x)$ showing jump discontinuity at $x = 0$*

**To Prove:** $f(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ is not differentiable at $x = 0$.

**Solution:**

Let us test continuity of $f(x)$ at $x = 0$.

Simplifying $f(x)$ for non-zero $x$:

$$f(x) = \begin{cases} \frac{x}{x} = 1, & x > 0 \\ \frac{x}{-x} = -1, & x < 0 \\ 1, & x = 0 \end{cases}$$

1. Left-Hand Limit at $x = 0$:

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (-1) = -1$$

2. Right-Hand Limit at $x = 0$:

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (1) = 1$$

Since $\lim_{x\to 0^-} f(x) = -1 \neq 1 = \lim_{x\to 0^+} f(x)$, the limit $\lim_{x\to 0} f(x)$ does not exist.

Hence, $f(x)$ is discontinuous at $x = 0$.

Since continuity is a necessary condition for differentiability, a discontinuous function cannot be differentiable.

**Hence Proved**

Since $f(x)$ is discontinuous at $x = 0$, it is not differentiable at $x = 0$.

## Question 17

If $f(x) = \begin{cases} x^2, & \text{if } x \geq 1 \\ x, & \text{if } x < 1 \end{cases}$, then show that $f$ is not differentiable at $x = 1$.

**TYPE 15** **Differentiability of Piecewise Functions**

**To Prove:** $f(x) = \begin{cases} x^2, & x \geq 1 \\ x, & x < 1 \end{cases}$ is not differentiable at $x = 1$.

**Concept / Formula Used:** $Lf'(c) = Rf'(c)$

**Solution:**

Compute $f(1) = 1^2 = 1$.

1. Left-Hand Derivative at $x = 1$:

$$\begin{aligned} Lf'(1) &= \lim_{h \to 0^-} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \to 0^-} \frac{(1+h) - 1}{h} \\ &= \lim_{h \to 0^-} \frac{h}{h} \\ &= 1 \end{aligned}$$

2. Right-Hand Derivative at $x = 1$:

$$\begin{aligned} Rf'(1) &= \lim_{h \to 0^+} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \to 0^+} \frac{(1+h)^2 - 1}{h} \\ &= \lim_{h \to 0^+} \frac{1 + 2h + h^2 - 1}{h} \\ &= \lim_{h \to 0^+} \frac{2h + h^2}{h} \\ &= \lim_{h \to 0^+} (2 + h) \\ &= 2 \end{aligned}$$

Since $Lf'(1) = 1 \neq 2 = Rf'(1)$, $f(x)$ is not differentiable at $x = 1$.

**Hence Proved**

$Lf'(1) = 1 \neq 2 = Rf'(1)$, hence $f$ is not differentiable at $x = 1$.

## Question 18

If $f(x) = \begin{cases} ax+b, & 0 < x \leq 1 \\ 2x^2 - x, & 1 < x < 2 \end{cases}$ is a differentiable function in $(0,2)$, then find the values of $a$ and $b$.

**TYPE 17** **Finding Constants for Differentiability**

★ **Likely Exam Question**

**Given:** $f(x) = \begin{cases} ax+b, & 0 < x \leq 1 \\ 2x^2 - x, & 1 < x < 2 \end{cases}$ is differentiable in $(0,2)$.

**To Find:** The values of $a$ and $b$.

**Concept / Formula Used:** $x = c$ LHL = RHL = $f(c)$ LHD = RHD $x = c$

**Solution:**

Since $f(x)$ is differentiable in $(0,2)$, it must be continuous and differentiable at $x = 1$.

**Step 1: Applying Continuity at $x = 1$**

$$\begin{aligned} \text{LHL} &= \lim_{x \to 1^-} (ax+b) = a(1) + b = a + b \\ \text{RHL} &= \lim_{x \to 1^+} (2x^2 - x) = 2(1)^2 - 1 = 1 \\ f(1) &= a(1) + b = a + b \end{aligned}$$

Equating LHL and RHL:

$$a + b = 1 \tag{1}$$

**Step 2: Applying Differentiability at $x = 1$**

$$\begin{aligned} Lf'(1) &= \lim_{h \to 0^-} \frac{f(1+h) - f(1)}{h} = \lim_{h \to 0^-} \frac{[a(1+h)+b] - (a+b)}{h} = \lim_{h \to 0^-} \frac{ah}{h} = a \\ Rf'(1) &= \lim_{h \to 0^+} \frac{f(1+h) - f(1)}{h} = \lim_{h \to 0^+} \frac{[2(1+h)^2 - (1+h)] - 1}{h} \\ &= \lim_{h \to 0^+} \frac{2(1 + 2h + h^2) - 1 - h - 1}{h} \\ &= \lim_{h \to 0^+} \frac{2 + 4h + 2h^2 - 2 - h}{h} \\ &= \lim_{h \to 0^+} \frac{3h + 2h^2}{h} = 3 \end{aligned}$$

Equating $Lf'(1)$ and $Rf'(1)$:

$$a = 3 \tag{2}$$

**Step 3: Solving for $b$**

Substitute $a = 3$ from Equation (2) into Equation (1):

$$\begin{aligned} 3 + b &= 1 \\ b &= 1 - 3 \\ b &= -2 \end{aligned}$$

**Final Answer**

$a = 3, \quad b = -2$

## Question 19

If the function $f(x) = \begin{cases} -\ln(4-x), & x \leq 3 \\ p - qx, & x > 3 \end{cases}$ is differentiable at $x = 3$, find the values of $p$ and $q$.

**TYPE 17** **Finding Constants for Differentiability**

**Given:** Function $f(x) = \begin{cases} -\ln(4-x), & x \leq 3 \\ p - qx, & x > 3 \end{cases}$ is differentiable at $x = 3$.

**To Find:** The values of $p$ and $q$.

**Concept / Formula Used:** $x = c \implies \quad \text{LHL} = \text{RHL} \quad \text{LHD} = \text{RHD}$

**Solution:**

Since $f(x)$ is differentiable at $x = 3$, it must be continuous and differentiable at $x = 3$.

**Step 1: Applying Continuity at $x = 3$**

$$\begin{aligned} \text{LHL} &= \lim_{x \to 3^-} [-\ln(4-x)] = -\ln(4-3) = -\ln(1) = 0 \\ \text{RHL} &= \lim_{x \to 3^+} (p - qx) = p - 3q \\ f(3) &= -\ln(4-3) = 0 \end{aligned}$$

Equating LHL and RHL:

$$p - 3q = 0 \implies p = 3q \tag{3}$$

**Step 2: Applying Differentiability at $x = 3$**

$$\begin{aligned} Lf'(3) &= \lim_{x \to 3^-} \frac{d}{dx}[-\ln(4-x)] = -\frac{1}{4-x} \cdot (-1)\Big|_{x=3} = \frac{1}{4-3} = 1 \\ Rf'(3) &= \lim_{x \to 3^+} \frac{d}{dx}(p - qx) = -q \end{aligned}$$

Equating $Lf'(3)$ and $Rf'(3)$:

$$-q = 1 \implies q = -1 \tag{4}$$

**Step 3: Solving for $p$**

Substitute $q = -1$ into Equation (3):

$$\begin{aligned} p &= 3(-1) \\ p &= -3 \end{aligned}$$

**Final Answer**

$p = -3, \quad q = -1$

## Question 20 (i)

Examine for continuity and differentiability:

$$f(x) = \begin{cases} x \sin \dfrac{1}{x}, & x < 0 \\ 0, & x \geq 0 \end{cases} \quad \text{at } x = 0$$

**TYPE 18** **Squeeze Theorem for Differentiability**

**Given:** Function $f(x) = \begin{cases} x \sin \dfrac{1}{x}, & x < 0 \\ 0, & x \geq 0 \end{cases}$ at $x = 0$.

**To Find:** Continuity and differentiability at $x = 0$.

**Concept / Formula Used:** $-|x| \leq x \sin \dfrac{1}{x} \leq |x|$

**Solution:**

**Part 1: Continuity at $x = 0$**

$$\text{LHL} = \lim_{x \to 0^-} x \sin \frac{1}{x}$$

Since $-1 \leq \sin \frac{1}{x} \leq 1$ for all $x \neq 0$:

$$\begin{aligned} \lim_{x \to 0^-} \left| x \sin \frac{1}{x} \right| \leq \lim_{x \to 0^-} |x| \cdot 1 = 0 \implies \text{LHL} &= 0 \\ \text{RHL} &= \lim_{x \to 0^+} 0 = 0 \\ f(0) &= 0 \end{aligned}$$

Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

**Part 2: Differentiability at $x = 0$**

$$\begin{aligned} Lf'(0) &= \lim_{h\to 0^-} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h\to 0^-} \frac{h\sin\frac{1}{h} - 0}{h} \\ &= \lim_{h\to 0^-} \sin\frac{1}{h} \end{aligned}$$

As $h \to 0^-$, $\sin\frac{1}{h}$ oscillates endlessly between $-1$ and $1$, so this limit does not exist.

**Final Answer**

$f(x)$ is continuous at $x = 0$, but not differentiable at $x = 0$.

## Question 20 (ii)

Examine for continuity and differentiability:

$$f(x) = \begin{cases} |x|\sin\dfrac{1}{x}, & x > 0 \\ 0, & x \le 0 \end{cases} \quad \text{at } x = 0$$

**TYPE 18** **Squeeze Theorem for Differentiability**

**Given:** Function $f(x) = \begin{cases} |x|\sin\dfrac{1}{x}, & x > 0 \\ 0, & x \le 0 \end{cases}$ at $x = 0$.

**To Find:** Continuity and differentiability at $x = 0$.

**Concept / Formula Used:** $x > 0 \qquad |x| = x$

**Solution:**

For $x > 0$, $f(x) = x\sin\dfrac{1}{x}$.

**Part 1: Continuity at $x = 0$**

$$\begin{aligned} \text{RHL} &= \lim_{x\to 0^+} x\sin\frac{1}{x} = 0 \quad \text{(by Squeeze Theorem, since } -x \le x\sin\frac{1}{x} \le x) \\ \text{LHL} &= \lim_{x\to 0^-} 0 = 0 \\ f(0) &= 0 \end{aligned}$$

Since $\text{LHL} = \text{RHL} = f(0) = 0$, $f(x)$ is continuous at $x = 0$.

**Part 2: Differentiability at $x = 0$**

$$\begin{aligned} Rf'(0) &= \lim_{h\to 0^+} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h\to 0^+} \frac{h \sin \frac{1}{h} - 0}{h} \\ &= \lim_{h\to 0^+} \sin \frac{1}{h} \end{aligned}$$

As $h \to 0^+$, $\sin \frac{1}{h}$ oscillates infinitely between $-1$ and $1$, so $Rf'(0)$ does not exist.

**Final Answer**

$f(x)$ is continuous at $x = 0$, but not differentiable at $x = 0$.

## Question 20 (iii)

Examine for continuity and differentiability:

$$f(x) = \begin{cases} (x-c)^2 \cos \dfrac{1}{x-c}, & x \neq c \\ 0, & x = c \end{cases} \quad \text{at } x = c$$

**TYPE 18** **Squeeze Theorem for Differentiability**

★ **Likely Exam Question**

**Given:** Function $f(x) = \begin{cases} (x-c)^2 \cos \dfrac{1}{x-c}, & x \neq c \\ 0, & x = c \end{cases}$ at $x = c$.

**To Find:** Continuity and differentiability at $x = c$.

**Concept / Formula Used:** $(x-c)^2 \cos \dfrac{1}{x-c}$

**Solution:**

**Part 1: Continuity at $x = c$**

$$\lim_{x\to c} f(x) = \lim_{x\to c} (x-c)^2 \cos \frac{1}{x-c}$$

Since $-1 \leq \cos \frac{1}{x-c} \leq 1$:

$$-(x-c)^2 \leq (x-c)^2 \cos \frac{1}{x-c} \leq (x-c)^2$$

Taking limits as $x \to c$:

$$\lim_{x\to c} [-(x-c)^2] = 0 \quad \text{and} \quad \lim_{x\to c} (x-c)^2 = 0$$

By the Squeeze Theorem:

$$\lim_{x \to c} f(x) = 0 = f(c)$$

Hence, $f(x)$ is continuous at $x = c$.

**Part 2: Differentiability at $x = c$**

$$\begin{aligned} f'(c) &= \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \\ &= \lim_{h \to 0} \frac{(c+h-c)^2 \cos \frac{1}{c+h-c} - 0}{h} \\ &= \lim_{h \to 0} \frac{h^2 \cos \frac{1}{h}}{h} \\ &= \lim_{h \to 0} h \cos \frac{1}{h} \end{aligned}$$

Using Squeeze Theorem again:

$$-|h| \leq h \cos \frac{1}{h} \leq |h| \implies \lim_{h \to 0} h \cos \frac{1}{h} = 0$$

Since $f'(c) = 0$ exists finitely, $f(x)$ is differentiable at $x = c$.

**Final Answer**

$f(x)$ is both continuous and differentiable at $x = c$, with $f'(c) = 0$.

## Question 21

Find the values of $a$ and $b$ so that the following function is differentiable for all values of $x$:

$$f(x) = \begin{cases} ax + b, & x > -1 \\ bx^2 - 3, & x \leq -1 \end{cases}$$

**TYPE 17** **Finding Constants for Differentiability**

**★ Likely Exam Question**

**Given:** Function $f(x) = \begin{cases} ax + b, & x > -1 \\ bx^2 - 3, & x \leq -1 \end{cases}$ is differentiable for all $x \in \mathbb{R}$.

**To Find:** The values of $a$ and $b$.

**Concept / Formula Used:** $x = -1 \implies \qquad x = -1 \qquad Lf'(-1) = Rf'(-1)$

**Solution:**

The function is differentiable for all $x$, so it must be continuous and differentiable at the transition point $x = -1$.

**Step 1: Applying Continuity at $x = -1$**

$$\begin{aligned}
\text{LHL} &= \lim_{x \to -1^-} (bx^2 - 3) = b(-1)^2 - 3 = b - 3 \\
\text{RHL} &= \lim_{x \to -1^+} (ax + b) = a(-1) + b = -a + b \\
f(-1) &= b(-1)^2 - 3 = b - 3
\end{aligned}$$

Equating LHL and RHL:

$$\begin{aligned}
b - 3 &= -a + b \\
-3 &= -a \\
a &= 3
\end{aligned}$$

**Step 2: Applying Differentiability at $x = -1$**

$$\begin{aligned}
Lf'(-1) &= \frac{d}{dx}(bx^2 - 3)\bigg|_{x=-1} = 2bx|_{x=-1} = -2b \\
Rf'(-1) &= \frac{d}{dx}(ax + b)\bigg|_{x=-1} = a
\end{aligned}$$

Equating $Lf'(-1)$ and $Rf'(-1)$:

$$-2b = a$$

Substitute $a = 3$:

$$\begin{aligned}
-2b &= 3 \\
b &= -\frac{3}{2}
\end{aligned}$$

**Final Answer**

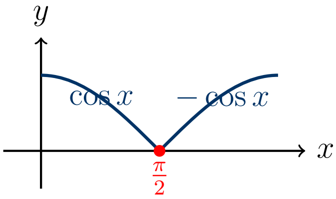
$a = 3, \quad b = -\dfrac{3}{2}$

## Question 22

Check the differentiability of $f(x) = |\cos x|$ at $x = \dfrac{\pi}{2}$.

**TYPE 12** **Differentiability of Absolute Value Functions**

**★ Likely Exam Question**




*Graph of $f(x) = |\cos x|$ showing sharp corner at $x = \pi/2$*

**Given:** Function $f(x) = |\cos x|$ at $x = \dfrac{\pi}{2}$.

**To Find:** Differentiability of $f(x)$ at $x = \dfrac{\pi}{2}$.

**Concept / Formula Used:** $\cos x$ $x = \dfrac{\pi}{2}$ $\cos x > 0$ $x < \dfrac{\pi}{2}$ $\cos x <$ $0$ $x > \dfrac{\pi}{2}$

### Solution:

Redefine $f(x)$ in the neighborhood of $x = \dfrac{\pi}{2}$:

$$f(x) = \begin{cases} \cos x, & \text{for } x \le \dfrac{\pi}{2} \\ -\cos x, & \text{for } x > \dfrac{\pi}{2} \end{cases}$$

Value at point $x = \dfrac{\pi}{2}$: $f\left(\frac{\pi}{2}\right) = \left|\cos \frac{\pi}{2}\right| = 0$.

1. Left-Hand Derivative at $x = \dfrac{\pi}{2}$:

$$\begin{aligned} Lf'\left(\frac{\pi}{2}\right) &= \lim_{h \to 0^-} \frac{f\left(\frac{\pi}{2} + h\right) - f\left(\frac{\pi}{2}\right)}{h} \\ &= \lim_{h \to 0^-} \frac{\cos\left(\frac{\pi}{2} + h\right) - 0}{h} \\ &= \lim_{h \to 0^-} \frac{-\sin h}{h} \\ &= -1 \end{aligned}$$

2. Right-Hand Derivative at $x = \dfrac{\pi}{2}$:

$$\begin{aligned}
Rf'\left(\frac{\pi}{2}\right) &= \lim_{h\to 0^+} \frac{f\left(\frac{\pi}{2}+h\right) - f\left(\frac{\pi}{2}\right)}{h} \\
&= \lim_{h\to 0^+} \frac{-\cos\left(\frac{\pi}{2}+h\right) - 0}{h} \\
&= \lim_{h\to 0^+} \frac{-(-\sin h)}{h} \\
&= \lim_{h\to 0^+} \frac{\sin h}{h} \\
&= 1
\end{aligned}$$

Since $Lf'\left(\frac{\pi}{2}\right) = -1 \neq 1 = Rf'\left(\frac{\pi}{2}\right)$, $f(x)$ is not differentiable at $x = \dfrac{\pi}{2}$.

**Final Answer**

$f(x) = |\cos x|$ is not differentiable at $x = \dfrac{\pi}{2}$.

## 5.5 – Derivatives of Standard and Absolute Value Functions

### Question 1 (i)

If $f(x) = \sqrt{4-x^2}, x \in (-2, 2)$, find $f'(x)$.

**TYPE 19 Chain Rule for Differentiation**

**Given:** $f(x) = \sqrt{4-x^2}$ for $x \in (-2, 2)$

**To Find:** $f'(x)$

**Concept / Formula Used:** $\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}}\frac{du}{dx}$

**Solution:**

Expressing the square root as a fractional power:

$$f(x) = (4-x^2)^{1/2}$$

Differentiating with respect to $x$ using the chain rule:

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left[(4-x^2)^{1/2}\right] \\ &= \frac{1}{2}(4-x^2)^{1/2-1}\cdot\frac{d}{dx}(4-x^2) \\ &= \frac{1}{2}(4-x^2)^{-1/2}\cdot(0-2x) \\ &= \frac{-2x}{2\sqrt{4-x^2}} \\ &= -\frac{x}{\sqrt{4-x^2}} \end{aligned}$$

**Final Answer**

$f'(x) = -\frac{x}{\sqrt{4-x^2}}$

## Question 1 (ii)

If $f(x) = \sqrt{1-x^2}, x \in (0,1)$, find $f'(x)$.

**TYPE 19** **Chain Rule for Differentiation**

**Given:** $f(x) = \sqrt{1-x^2}$ for $x \in (0,1)$

**To Find:** $f'(x)$

**Concept / Formula Used:** $\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}}\frac{du}{dx}$

**Solution:**

Expressing $f(x)$ as a power function:

$$f(x) = (1-x^2)^{1/2}$$

Differentiating with respect to $x$ using the chain rule:

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left[(1-x^2)^{1/2}\right] \\ &= \frac{1}{2}(1-x^2)^{-1/2}\cdot\frac{d}{dx}(1-x^2) \\ &= \frac{1}{2\sqrt{1-x^2}}\cdot(-2x) \\ &= -\frac{x}{\sqrt{1-x^2}} \end{aligned}$$

**Final Answer**

$f'(x) = -\frac{x}{\sqrt{1-x^2}}$

## Question 2 (i)

Differentiate $\sqrt{2x-1}$ w.r.t. $x$.

**TYPE 19** **Chain Rule for Differentiation**

**Given:** $y = \sqrt{2x-1}$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}}\frac{du}{dx}$

**Solution:**

Let $y = (2x-1)^{1/2}$. Differentiating with respect to $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2}(2x-1)^{-1/2} \cdot \frac{d}{dx}(2x-1) \\ &= \frac{1}{2\sqrt{2x-1}} \cdot (2) \\ &= \frac{1}{\sqrt{2x-1}}\end{aligned}$$

**Final Answer**

$\frac{1}{\sqrt{2x-1}}$

## Question 2 (ii)

Differentiate $(3x^2-9x+5)^9$ w.r.t. $x$.

**TYPE 19** **Chain Rule for Differentiation**

**Given:** $y = (3x^2-9x+5)^9$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}(u^n) = nu^{n-1}\frac{du}{dx}$

**Solution:**

Differentiating with respect to $x$ using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= 9(3x^2-9x+5)^8 \cdot \frac{d}{dx}(3x^2-9x+5) \\ &= 9(3x^2-9x+5)^8 \cdot (6x-9) \\ &= 9 \cdot 3(2x-3)(3x^2-9x+5)^8 \\ &= 27(2x-3)(3x^2-9x+5)^8\end{aligned}$$

**Final Answer**

$27(2x-3)(3x^2-9x+5)^8$

## Question 3 (i)

Differentiate $\sqrt{3x+2}+\frac{1}{\sqrt{2x^2+4}}$ w.r.t. $x$.

**TYPE 20** **Sum and Product Rules**

**Given:** $y=\sqrt{3x+2}+\frac{1}{\sqrt{2x^2+4}}$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}(u+v)=\frac{du}{dx}+\frac{dv}{dx}$

**Solution:**

Rewriting $y$ using fractional exponents:

$$y=(3x+2)^{1/2}+(2x^2+4)^{-1/2}$$

Differentiating term by term with respect to $x$:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx}\left[(3x+2)^{1/2}\right]+\frac{d}{dx}\left[(2x^2+4)^{-1/2}\right] \\
&= \frac{1}{2}(3x+2)^{-1/2}\cdot\frac{d}{dx}(3x+2)+\left(-\frac{1}{2}\right)(2x^2+4)^{-3/2}\cdot\frac{d}{dx}(2x^2+4) \\
&= \frac{1}{2\sqrt{3x+2}}\cdot(3)-\frac{1}{2(2x^2+4)^{3/2}}\cdot(4x) \\
&= \frac{3}{2\sqrt{3x+2}}-\frac{2x}{(2x^2+4)^{3/2}}
\end{aligned}$$

**Final Answer**

$\frac{3}{2\sqrt{3x+2}}-\frac{2x}{(2x^2+4)^{3/2}}$

## Question 3 (ii)

Differentiate $\sin^3 x+\cos^6 x$ w.r.t. $x$.

**TYPE 20** **Sum and Product Rules**

**Given:** $y=\sin^3 x+\cos^6 x$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}[f(x)]^n=n[f(x)]^{n-1}f'(x)$

**Solution:**

Differentiating term by term using the power rule for trigonometric functions:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(\sin^3 x) + \frac{d}{dx}(\cos^6 x)\\ &= 3\sin^2 x \cdot \frac{d}{dx}(\sin x) + 6\cos^5 x \cdot \frac{d}{dx}(\cos x)\\ &= 3\sin^2 x \cos x + 6\cos^5 x(-\sin x)\\ &= 3\sin^2 x \cos x - 6\sin x \cos^5 x\\ &= 3\sin x \cos x(\sin x - 2\cos^4 x)\end{aligned}$$

**Final Answer**

$3\sin x \cos x(\sin x - 2\cos^4 x)$

## Question 4 (i)

Differentiate $\sin(x^2)$ w.r.t. $x$.

**TYPE 21** **Chain Rule for Trigonometric Functions**

**Given:** $y = \sin(x^2)$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\sin(u) = \cos(u)\frac{du}{dx}$

**Solution:**

Differentiating with respect to $x$ using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \cos(x^2) \cdot \frac{d}{dx}(x^2)\\ &= \cos(x^2) \cdot (2x)\\ &= 2x\cos(x^2)\end{aligned}$$

**Final Answer**

$2x\cos(x^2)$

## Question 4 (ii)

Differentiate $\cos\sqrt{x}$ w.r.t. $x$.

**TYPE 21** **Chain Rule for Trigonometric Functions**

**Given:** $y = \cos\sqrt{x}$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\cos(u) = -\sin(u)\frac{du}{dx}$

**Solution:**

Differentiating with respect to $x$ using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= -\sin\sqrt{x}\cdot\frac{d}{dx}(\sqrt{x}) \\ &= -\sin\sqrt{x}\cdot\left(\frac{1}{2\sqrt{x}}\right) \\ &= -\frac{\sin\sqrt{x}}{2\sqrt{x}}\end{aligned}$$

**Final Answer**

$-\frac{\sin\sqrt{x}}{2\sqrt{x}}$

## Question 5 (i)

Differentiate $\sin(ax+b)$ w.r.t. $x$.

**TYPE 19** **Chain Rule for Differentiation**

**Given:** $y = \sin(ax+b)$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\sin(u) = \cos(u)\frac{du}{dx}$

**Solution:**

Differentiating with respect to $x$ using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \cos(ax+b)\cdot\frac{d}{dx}(ax+b) \\ &= \cos(ax+b)\cdot a \\ &= a\cos(ax+b)\end{aligned}$$

**Final Answer**

$a\cos(ax+b)$

## Question 5 (ii)

Differentiate $\tan(2x+3)$ w.r.t. $x$.

**TYPE 19** **Chain Rule for Differentiation**

**Given:** $y = \tan(2x + 3)$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\tan(u) = \sec^2(u)\frac{du}{dx}$

**Solution:**

Differentiating with respect to $x$ using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \sec^2(2x+3)\cdot\frac{d}{dx}(2x+3)\\ &= \sec^2(2x+3)\cdot 2\\ &= 2\sec^2(2x+3)\end{aligned}$$

**Final Answer**

$2\sec^2(2x + 3)$

## Question 6 (i)

Differentiate $\cos(\sin x)$ w.r.t. $x$.

**TYPE 21** **Chain Rule for Trigonometric Functions**

**Given:** $y = \cos(\sin x)$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\cos(u) = -\sin(u)\frac{du}{dx}$

**Solution:**

Differentiating with respect to $x$ using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= -\sin(\sin x)\cdot\frac{d}{dx}(\sin x)\\ &= -\sin(\sin x)\cdot\cos x\\ &= -\cos x\sin(\sin x)\end{aligned}$$

**Final Answer**

$-\cos x\sin(\sin x)$

## Question 6 (ii)

Differentiate $\sin(\cos(x^2))$ w.r.t. $x$.

**TYPE 22** **Repeated Chain Rule**

**Given:** $y = \sin(\cos(x^2))$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\sin(u) = \cos(u)\frac{du}{dx}$

**Solution:**

Applying chain rule repeatedly:

$$\begin{aligned}\frac{dy}{dx} &= \cos(\cos(x^2)) \cdot \frac{d}{dx}(\cos(x^2)) \\ &= \cos(\cos(x^2)) \cdot \left[-\sin(x^2) \cdot \frac{d}{dx}(x^2)\right] \\ &= \cos(\cos(x^2)) \cdot \left[-\sin(x^2) \cdot (2x)\right] \\ &= -2x\sin(x^2)\cos(\cos(x^2))\end{aligned}$$

**Final Answer**

$-2x\sin(x^2)\cos(\cos(x^2))$

## Question 7 (i)

Differentiate $\cos(\tan\sqrt{x+1})$ w.r.t. $x$.

**TYPE 22** Repeated Chain Rule

**Given:** $y = \cos(\tan\sqrt{x+1})$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\cos(u) = -\sin(u)\frac{du}{dx}$, $\quad \frac{d}{dx}\tan(v) = \sec^2(v)\frac{dv}{dx}$

**Solution:**

Applying chain rule step-by-step:

$$\begin{aligned}\frac{dy}{dx} &= -\sin(\tan\sqrt{x+1}) \cdot \frac{d}{dx}(\tan\sqrt{x+1}) \\ &= -\sin(\tan\sqrt{x+1}) \cdot \sec^2(\sqrt{x+1}) \cdot \frac{d}{dx}(\sqrt{x+1}) \\ &= -\sin(\tan\sqrt{x+1}) \cdot \sec^2(\sqrt{x+1}) \cdot \frac{1}{2\sqrt{x+1}} \\ &= -\frac{\sec^2(\sqrt{x+1})\sin(\tan\sqrt{x+1})}{2\sqrt{x+1}}\end{aligned}$$

**Final Answer**

$-\frac{\sec^2(\sqrt{x+1})\sin(\tan\sqrt{x+1})}{2\sqrt{x+1}}$

## Question 7 (ii)

Differentiate $\sqrt{\tan\sqrt{x}}$ w.r.t. $x$.

**TYPE 22 Repeated Chain Rule**

**Given:** $y = \sqrt{\tan\sqrt{x}}$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\sqrt{u} = \frac{1}{2\sqrt{u}}\frac{du}{dx}$, $\quad \frac{d}{dx}\tan(v) = \sec^2(v)\frac{dv}{dx}$

**Solution:**

Applying chain rule step-by-step:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{\tan\sqrt{x}}} \cdot \frac{d}{dx}(\tan\sqrt{x}) \\ &= \frac{1}{2\sqrt{\tan\sqrt{x}}} \cdot \sec^2\sqrt{x} \cdot \frac{d}{dx}(\sqrt{x}) \\ &= \frac{1}{2\sqrt{\tan\sqrt{x}}} \cdot \sec^2\sqrt{x} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{\sec^2\sqrt{x}}{4\sqrt{x}\sqrt{\tan\sqrt{x}}}\end{aligned}$$

**Final Answer**

$\frac{\sec^2\sqrt{x}}{4\sqrt{x}\sqrt{\tan\sqrt{x}}}$

## Question 8 (i)

If $f(x) = x - 1$, find $\frac{d}{dx}((f \circ g)(x))$.

**TYPE 23 Derivative of Composite Functions**

**Given:** $f(x) = x - 1$

**To Find:** $\frac{d}{dx}((f \circ g)(x))$

**Concept / Formula Used:** $(f \circ g)(x) = f(g(x))$

**Solution:**

Evaluating the composite function $(f \circ g)(x)$:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= g(x) - 1\end{aligned}$$

Differentiating with respect to $x$:

$$\begin{aligned}\frac{d}{dx}((f \circ g)(x)) &= \frac{d}{dx}(g(x) - 1)\\ &= g'(x) - 0\\ &= g'(x)\end{aligned}$$

**Final Answer**

$g'(x)$

## Question 8 (ii)

If $f(x) = x + 7$ and $g(x) = x - 7$, find $\frac{d}{dx}((f \circ f)(x))$.

**TYPE 23** **Derivative of Composite Functions**

**Given:** $f(x) = x + 7, \quad g(x) = x - 7$

**To Find:** $\frac{d}{dx}((f \circ f)(x))$

**Concept / Formula Used:** $(f \circ f)(x) = f(f(x))$

**Solution:**

Evaluating the composite function $(f \circ f)(x)$:

$$\begin{aligned}(f \circ f)(x) &= f(f(x))\\ &= f(x + 7)\\ &= (x + 7) + 7\\ &= x + 14\end{aligned}$$

Differentiating with respect to $x$:

$$\begin{aligned}\frac{d}{dx}((f \circ f)(x)) &= \frac{d}{dx}(x + 14)\\ &= 1 + 0\\ &= 1\end{aligned}$$

**Final Answer**

1

## Question 9

If $f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \frac{x^{98}}{98} + \cdots + x + 1$, show that $f'(1) = 100\,f'(0)$.

**★ Likely Exam Question**

**TYPE 24** **Term-by-Term Polynomial Differentiation**

**Given:** $f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \frac{x^{98}}{98} + \cdots + x + 1$

**To Prove:** $f'(1) = 100\, f'(0)$

**Concept / Formula Used:** $\frac{d}{dx}(x^n) = nx^{n-1}$

**Solution:**

Differentiating $f(x)$ term by term with respect to $x$:

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left(\frac{x^{100}}{100} + \frac{x^{99}}{99} + \cdots + x + 1\right) \\ &= \frac{100x^{99}}{100} + \frac{99x^{98}}{99} + \frac{98x^{97}}{98} + \cdots + 1 + 0 \\ &= x^{99} + x^{98} + x^{97} + \cdots + x + 1 \end{aligned}$$

Evaluating $f'(1)$ by substituting $x = 1$:

$$\begin{aligned} f'(1) &= 1^{99} + 1^{98} + 1^{97} + \cdots + 1 + 1 \\ &= \underbrace{1 + 1 + 1 + \cdots + 1}_{\text{100 terms}} \\ &= 100 \end{aligned}$$

Evaluating $f'(0)$ by substituting $x = 0$:

$$\begin{aligned} f'(0) &= 0^{99} + 0^{98} + \cdots + 0 + 1 \\ &= 1 \end{aligned}$$

Evaluating $100\, f'(0)$:

$$\begin{aligned} 100\, f'(0) &= 100 \times 1 \\ &= 100 \end{aligned}$$

Since $f'(1) = 100$ and $100\, f'(0) = 100$, we have $f'(1) = 100\, f'(0)$.

**Hence Proved**

$f'(1) = 100\, f'(0)$, hence proved.

## Question 10

Find the derivative of $|2x^2 - 3|$ with respect to $x$.

**TYPE 25** **Derivative of Absolute Value Functions**

**Given:** $y = |2x^2 - 3|$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}|u| = \frac{u}{|u|}\frac{du}{dx} \qquad u \neq 0$

**Solution:**

Using the property $|u| = \sqrt{u^2}$, we express $y$ as:

$$y = \sqrt{(2x^2-3)^2}$$

Differentiating with respect to $x$ using chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{(2x^2-3)^2}} \cdot \frac{d}{dx}\left[(2x^2-3)^2\right] \\ &= \frac{1}{2|2x^2-3|} \cdot 2(2x^2-3) \cdot \frac{d}{dx}(2x^2-3) \\ &= \frac{2x^2-3}{|2x^2-3|} \cdot (4x) \\ &= \frac{4x(2x^2-3)}{|2x^2-3|}, \quad \text{for } x \neq \pm\sqrt{\frac{3}{2}}\end{aligned}$$

**Final Answer**

$\frac{4x(2x^2-3)}{|2x^2-3|}, \quad x \neq \pm\sqrt{\frac{3}{2}}$

## Question 11

If $f(x) = |\cos x|$, find $f'\left(\frac{3\pi}{4}\right)$.

**★ Likely Exam Question**

**TYPE 26 Derivative of Absolute Value Functions at a Point**

**Given:** $f(x) = |\cos x|$, point $x = \frac{3\pi}{4}$

**To Find:** $f'\left(\frac{3\pi}{4}\right)$

**Concept / Formula Used:** $\cos x < 0 \qquad |\cos x| = -\cos x$

**Solution:**

Evaluating $\cos x$ at $x = \frac{3\pi}{4}$:

$$\cos\left(\frac{3\pi}{4}\right) = -\frac{1}{\sqrt{2}} < 0$$

Since $\cos x < 0$ in the second quadrant, in a neighborhood around $x = \frac{3\pi}{4}$:

$$f(x) = -\cos x$$

Differentiating with respect to $x$:

$$\begin{aligned} f'(x) &= \frac{d}{dx}(-\cos x) \\ &= -(-\sin x) \\ &= \sin x \end{aligned}$$

Substituting $x = \frac{3\pi}{4}$:

$$\begin{aligned} f'\left(\frac{3\pi}{4}\right) &= \sin\left(\frac{3\pi}{4}\right) \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

**Final Answer**

$\frac{1}{\sqrt{2}}$

## Question 12

The table below shows some of the values of differentiable functions $u$ and $v$ and their derivatives.

| $x$ | $u(x)$ | $u'(x)$ | $v(x)$ | $v'(x)$ |
|---|---|---|---|---|
| 1 | 0 | 2 | $-1$ | 5 |
| 2 | 4 | 5 | 3 | 2 |
| 3 | 2 | $-1$ | 3 | 0 |

If $r(x) = u(v(x))$, find $r'(2)$.

★ **Likely Exam Question**

**TYPE 27 Chain Rule with Tabular Values**

**Given:** Values of $u(x), u'(x), v(x), v'(x)$ in table; $r(x) = u(v(x))$

**To Find:** $r'(2)$

**Concept / Formula Used:** $r'(x) = u'(v(x)) \cdot v'(x)$

### Solution:

Applying the chain rule to $r(x) = u(v(x))$:

$$r'(x) = u'(v(x)) \cdot v'(x)$$

Substituting $x = 2$:

$$r'(2) = u'(v(2)) \cdot v'(2)$$

Reading values from the table at $x = 2$:

$$\begin{aligned} v(2) &= 3 \\ v'(2) &= 2 \end{aligned}$$

Substituting $v(2) = 3$ into the expression:

$$r'(2) = u'(3) \cdot 2$$

Reading $u'(3)$ from the table at $x = 3$:

$$u'(3) = -1$$

Substituting $u'(3) = -1$:

$$\begin{aligned} r'(2) &= (-1) \cdot 2 \\ &= -2 \end{aligned}$$

**Final Answer**

$-2$

## Question 13 (i)

Differentiate $\sqrt{\frac{a^2-x^2}{a^2+x^2}}$ w.r.t. $x$.

**TYPE 28 Quotient Rule for Differentiation**

**Given:** $y = \sqrt{\frac{a^2-x^2}{a^2+x^2}}$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu'-uv'}{v^2}$

**Solution:**

Expressing $y$ with fractional exponent:

$$y = \left(\frac{a^2 - x^2}{a^2 + x^2}\right)^{1/2}$$

Differentiating using chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2}\left(\frac{a^2 - x^2}{a^2 + x^2}\right)^{-1/2} \cdot \frac{d}{dx}\left(\frac{a^2 - x^2}{a^2 + x^2}\right) \\ &= \frac{1}{2}\sqrt{\frac{a^2 + x^2}{a^2 - x^2}} \cdot \frac{(a^2 + x^2)\frac{d}{dx}(a^2 - x^2) - (a^2 - x^2)\frac{d}{dx}(a^2 + x^2)}{(a^2 + x^2)^2} \\ &= \frac{1}{2}\sqrt{\frac{a^2 + x^2}{a^2 - x^2}} \cdot \frac{(a^2 + x^2)(-2x) - (a^2 - x^2)(2x)}{(a^2 + x^2)^2} \\ &= \frac{1}{2}\sqrt{\frac{a^2 + x^2}{a^2 - x^2}} \cdot \frac{-2a^2x - 2x^3 - 2a^2x + 2x^3}{(a^2 + x^2)^2} \\ &= \frac{1}{2}\sqrt{\frac{a^2 + x^2}{a^2 - x^2}} \cdot \frac{-4a^2x}{(a^2 + x^2)^2} \\ &= \frac{-2a^2x}{\sqrt{a^2 - x^2}(a^2 + x^2)^{3/2}} \end{aligned}$$

**Final Answer**

$-\frac{2a^2x}{\sqrt{a^2-x^2}(a^2+x^2)^{3/2}}$

## Question 13 (ii)

Differentiate $\sin\sqrt{x} + \cos^2\sqrt{x}$ w.r.t. $x$.

**TYPE 20** **Sum and Product Rules**

**Given:** $y = \sin\sqrt{x} + \cos^2\sqrt{x}$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\sin(u) = \cos(u)\frac{du}{dx}, \quad \frac{d}{dx}(u^2) = 2u\frac{du}{dx}$

**Solution:**

Differentiating term by term with respect to $x$:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx}(\sin\sqrt{x}) + \frac{d}{dx}(\cos^2\sqrt{x}) \\
&= \cos\sqrt{x}\cdot\frac{d}{dx}(\sqrt{x}) + 2\cos\sqrt{x}\cdot\frac{d}{dx}(\cos\sqrt{x}) \\
&= \cos\sqrt{x}\cdot\frac{1}{2\sqrt{x}} + 2\cos\sqrt{x}\cdot\left(-\sin\sqrt{x}\cdot\frac{1}{2\sqrt{x}}\right) \\
&= \frac{\cos\sqrt{x}}{2\sqrt{x}} - \frac{2\sin\sqrt{x}\cos\sqrt{x}}{2\sqrt{x}} \\
&= \frac{\cos\sqrt{x}(1-2\sin\sqrt{x})}{2\sqrt{x}}
\end{aligned}$$

**Final Answer**

$\frac{\cos\sqrt{x}(1-2\sin\sqrt{x})}{2\sqrt{x}}$

## Question 14 (i)

Differentiate $\frac{\sin(ax+b)}{\cos(cx+d)}$ w.r.t. $x$.

**TYPE 28** **Quotient Rule for Differentiation**

**Given:** $y = \frac{\sin(ax+b)}{\cos(cx+d)}$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu'-uv'}{v^2}$

**Solution:**

Let $u = \sin(ax+b)$ and $v = \cos(cx+d)$. Differentiating $u$ and $v$:

$$u' = a\cos(ax+b)$$
$$v' = -c\sin(cx+d)$$

Applying the quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\cos(cx+d)\cdot[a\cos(ax+b)] - \sin(ax+b)\cdot[-c\sin(cx+d)]}{\cos^2(cx+d)} \\ &= \frac{a\cos(ax+b)\cos(cx+d) + c\sin(ax+b)\sin(cx+d)}{\cos^2(cx+d)} \\ &= \frac{a\cos(ax+b)\cos(cx+d)}{\cos^2(cx+d)} + \frac{c\sin(ax+b)\sin(cx+d)}{\cos^2(cx+d)} \\ &= a\cos(ax+b)\sec(cx+d) + c\sin(ax+b)\tan(cx+d)\sec(cx+d)\end{aligned}$$

**Final Answer**

$a\cos(ax+b)\sec(cx+d) + c\sin(ax+b)\tan(cx+d)\sec(cx+d)$

## Question 14 (ii)

Differentiate $\frac{\sin x + x^2}{\cot 2x}$ w.r.t. $x$.

**TYPE 28 Quotient Rule for Differentiation**

**Given:** $y = \frac{\sin x + x^2}{\cot 2x}$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{1}{\cot 2x} = \tan 2x, \quad \frac{d}{dx}(uv) = u'v + uv'$

**Solution:**

Rewriting $y$ using trigonometric reciprocal identity $\frac{1}{\cot 2x} = \tan 2x$:

$$y = (\sin x + x^2)\tan 2x$$

Applying the product rule with $u = \sin x + x^2$ and $v = \tan 2x$:

$$u' = \cos x + 2x$$
$$v' = 2\sec^2 2x$$

Differentiating:

$$\begin{aligned}\frac{dy}{dx} &= u'v + uv' \\ &= (\cos x + 2x)\tan 2x + (\sin x + x^2)(2\sec^2 2x) \\ &= (\cos x + 2x)\tan 2x + 2(\sin x + x^2)\sec^2 2x\end{aligned}$$

**Final Answer**

$(\cos x + 2x)\tan 2x + 2(\sin x + x^2)\sec^2 2x$

## Question 15 (i)

Differentiate $\sin^m x \cos^n x$ w.r.t. $x$.

TYPE 20 **Sum and Product Rules**

**Given:** $y = \sin^m x \cos^n x$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}(uv) = u'v + uv'$

**Solution:**

Let $u = \sin^m x$ and $v = \cos^n x$. Differentiating $u$ and $v$ using chain rule:

$$u' = m\sin^{m-1} x \cos x$$
$$v' = n\cos^{n-1} x(-\sin x) = -n\cos^{n-1} x \sin x$$

Applying the product rule:

$$\begin{aligned}\frac{dy}{dx} &= (m\sin^{m-1} x\cos x)(\cos^n x) + (\sin^m x)(-n\cos^{n-1} x\sin x)\\ &= m\sin^{m-1} x\cos^{n+1} x - n\sin^{m+1} x\cos^{n-1} x\\ &= \sin^{m-1} x\cos^{n-1} x\left(m\cos^2 x - n\sin^2 x\right)\end{aligned}$$

**Final Answer**

$\sin^{m-1} x\cos^{n-1} x(m\cos^2 x - n\sin^2 x)$

## Question 15 (ii)

Differentiate $\sin^n(ax^2 + bx + c)$ w.r.t. $x$.

TYPE 22 **Repeated Chain Rule**

**Given:** $y = \sin^n(ax^2 + bx + c)$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\frac{d}{dx}[f(u)]^n = n[f(u)]^{n-1} f'(u)\frac{du}{dx}$

**Solution:**

Differentiating using repeated chain rule applications:

$$\begin{aligned}
\frac{dy}{dx} &= n\sin^{n-1}(ax^2+bx+c)\cdot\frac{d}{dx}[\sin(ax^2+bx+c)] \\
&= n\sin^{n-1}(ax^2+bx+c)\cdot\cos(ax^2+bx+c)\cdot\frac{d}{dx}(ax^2+bx+c) \\
&= n\sin^{n-1}(ax^2+bx+c)\cdot\cos(ax^2+bx+c)\cdot(2ax+b) \\
&= n(2ax+b)\sin^{n-1}(ax^2+bx+c)\cos(ax^2+bx+c)
\end{aligned}$$

**Final Answer**

$n(2ax+b)\sin^{n-1}(ax^2+bx+c)\cos(ax^2+bx+c)$

## Question 16 (i)

Differentiate $\frac{\sqrt{a+x}-\sqrt{a-x}}{\sqrt{a+x}+\sqrt{a-x}}$ w.r.t. $x$.

**TYPE 29** **Rationalization before Differentiation**

**Given:** $y = \frac{\sqrt{a+x}-\sqrt{a-x}}{\sqrt{a+x}+\sqrt{a-x}}$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $(\sqrt{a+x}-\sqrt{a-x})$

**Solution:**

Rationalizing the denominator:

$$\begin{aligned}
y &= \frac{(\sqrt{a+x}-\sqrt{a-x})^2}{(\sqrt{a+x}+\sqrt{a-x})(\sqrt{a+x}-\sqrt{a-x})} \\
&= \frac{(a+x)+(a-x)-2\sqrt{(a+x)(a-x)}}{(a+x)-(a-x)} \\
&= \frac{2a-2\sqrt{a^2-x^2}}{2x} \\
&= \frac{a-\sqrt{a^2-x^2}}{x}
\end{aligned}$$

Differentiating $y = \frac{a-\sqrt{a^2-x^2}}{x}$ using quotient rule:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{x \cdot \frac{d}{dx}\left(a - \sqrt{a^2-x^2}\right) - \left(a - \sqrt{a^2-x^2}\right) \cdot \frac{d}{dx}(x)}{x^2} \\
&= \frac{x \cdot \left(-\frac{-2x}{2\sqrt{a^2-x^2}}\right) - \left(a - \sqrt{a^2-x^2}\right) \cdot 1}{x^2} \\
&= \frac{\frac{x^2}{\sqrt{a^2-x^2}} - a + \sqrt{a^2-x^2}}{x^2} \\
&= \frac{x^2 - a\sqrt{a^2-x^2} + (a^2-x^2)}{x^2\sqrt{a^2-x^2}} \\
&= \frac{a^2 - a\sqrt{a^2-x^2}}{x^2\sqrt{a^2-x^2}} \\
&= \frac{a(a-\sqrt{a^2-x^2})}{x^2\sqrt{a^2-x^2}}
\end{aligned}$$

**Final Answer**

$\frac{a(a-\sqrt{a^2-x^2})}{x^2\sqrt{a^2-x^2}}$

## Question 16 (ii)

Differentiate $\frac{\sqrt{x^2+1}+\sqrt{x^2-1}}{\sqrt{x^2+1}-\sqrt{x^2-1}}$ w.r.t. $x$.

**TYPE 29** **Rationalization before Differentiation**

**Given:** $y = \frac{\sqrt{x^2+1}+\sqrt{x^2-1}}{\sqrt{x^2+1}-\sqrt{x^2-1}}$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $(\sqrt{x^2+1} + \sqrt{x^2-1})$

**Solution:**

Rationalizing the denominator:

$$\begin{aligned}
y &= \frac{(\sqrt{x^2+1}+\sqrt{x^2-1})^2}{(\sqrt{x^2+1}-\sqrt{x^2-1})(\sqrt{x^2+1}+\sqrt{x^2-1})} \\
&= \frac{(x^2+1)+(x^2-1)+2\sqrt{(x^2+1)(x^2-1)}}{(x^2+1)-(x^2-1)} \\
&= \frac{2x^2+2\sqrt{x^4-1}}{2} \\
&= x^2 + \sqrt{x^4-1}
\end{aligned}$$

Differentiating with respect to $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^2) + \frac{d}{dx}\left(\sqrt{x^4-1}\right)\\ &= 2x + \frac{1}{2\sqrt{x^4-1}} \cdot \frac{d}{dx}(x^4-1)\\ &= 2x + \frac{4x^3}{2\sqrt{x^4-1}}\\ &= 2x + \frac{2x^3}{\sqrt{x^4-1}}\end{aligned}$$

**Final Answer**

$2x + \frac{2x^3}{\sqrt{x^4-1}}$

## Question 17

If $f(x) = \sqrt{\frac{\sec x - 1}{\sec x + 1}}$, find $f'(x)$. Also find $f'\left(\frac{\pi}{3}\right)$.

**TYPE 30** **Trigonometric Simplification before Differentiation**

**Given:** $f(x) = \sqrt{\frac{\sec x-1}{\sec x+1}}$

**To Find:** $f'(x)$ and $f'\left(\frac{\pi}{3}\right)$

**Concept / Formula Used:** $\sec x = \frac{1}{\cos x}$, $\quad \frac{1-\cos x}{1+\cos x} = \tan^2\left(\frac{x}{2}\right)$

**Solution:**

Simplifying $f(x)$ using trigonometric identities:

$$\begin{aligned}f(x) &= \sqrt{\frac{\frac{1}{\cos x}-1}{\frac{1}{\cos x}+1}}\\ &= \sqrt{\frac{1-\cos x}{1+\cos x}}\\ &= \sqrt{\tan^2\left(\frac{x}{2}\right)}\\ &= \tan\left(\frac{x}{2}\right) \quad \text{for } x \in \left(0, \frac{\pi}{2}\right)\end{aligned}$$

Differentiating $f(x) = \tan\left(\frac{x}{2}\right)$ with respect to $x$:

$$\begin{aligned}f'(x) &= \sec^2\left(\frac{x}{2}\right) \cdot \frac{d}{dx}\left(\frac{x}{2}\right)\\ &= \frac{1}{2}\sec^2\left(\frac{x}{2}\right)\end{aligned}$$

Evaluating at $x = \frac{\pi}{3}$:

$$\begin{aligned} f'\left(\frac{\pi}{3}\right) &= \frac{1}{2}\sec^2\left(\frac{\pi}{6}\right) \\ &= \frac{1}{2}\cdot\left(\frac{2}{\sqrt{3}}\right)^2 \\ &= \frac{1}{2}\cdot\frac{4}{3} \\ &= \frac{2}{3} \end{aligned}$$

**Final Answer**

$f'(x) = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)$,
$f'\left(\frac{\pi}{3}\right) = \frac{2}{3}$

## Question 18 (i)

Find the derivative of $\cos\left(2x + \frac{\pi}{3}\right)$ at $x = \frac{\pi}{3}$.

**TYPE 31 Chain Rule and Evaluation**

**Given:** $f(x) = \cos\left(2x + \frac{\pi}{3}\right)$, point $x = \frac{\pi}{3}$

**To Find:** $f'\left(\frac{\pi}{3}\right)$

**Concept / Formula Used:** $\frac{d}{dx}\cos(u) = -\sin(u)\frac{du}{dx}$

**Solution:**

Differentiating $f(x)$ with respect to $x$:

$$\begin{aligned} f'(x) &= -\sin\left(2x + \frac{\pi}{3}\right)\cdot\frac{d}{dx}\left(2x + \frac{\pi}{3}\right) \\ &= -2\sin\left(2x + \frac{\pi}{3}\right) \end{aligned}$$

Substituting $x = \frac{\pi}{3}$:

$$\begin{aligned} f'\left(\frac{\pi}{3}\right) &= -2\sin\left(2\left(\frac{\pi}{3}\right) + \frac{\pi}{3}\right) \\ &= -2\sin\left(\frac{2\pi}{3} + \frac{\pi}{3}\right) \\ &= -2\sin(\pi) \\ &= -2(0) \\ &= 0 \end{aligned}$$

**Final Answer**

0

## Question 18 (ii)

Find the derivative of $\frac{1-\sin x}{1+\cos x}$ at $x = \frac{\pi}{2}$.

**TYPE 28** Quotient Rule for Differentiation

**Given:** $f(x) = \frac{1-\sin x}{1+\cos x}$, point $x = \frac{\pi}{2}$

**To Find:** $f'\left(\frac{\pi}{2}\right)$

**Concept / Formula Used:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu'-uv'}{v^2}$

**Solution:**

Let $u = 1 - \sin x$ and $v = 1 + \cos x$. Differentiating $u$ and $v$:

$$u' = -\cos x$$
$$v' = -\sin x$$

Applying quotient rule:

$$\begin{aligned} f'(x) &= \frac{(1+\cos x)(-\cos x) - (1-\sin x)(-\sin x)}{(1+\cos x)^2} \\ &= \frac{-\cos x - \cos^2 x + \sin x - \sin^2 x}{(1+\cos x)^2} \\ &= \frac{\sin x - \cos x - (\sin^2 x + \cos^2 x)}{(1+\cos x)^2} \\ &= \frac{\sin x - \cos x - 1}{(1+\cos x)^2} \end{aligned}$$

Evaluating at $x = \frac{\pi}{2}$ using $\sin\left(\frac{\pi}{2}\right) = 1$ and $\cos\left(\frac{\pi}{2}\right) = 0$:

$$\begin{aligned} f'\left(\frac{\pi}{2}\right) &= \frac{1-0-1}{(1+0)^2} \\ &= \frac{0}{1} \\ &= 0 \end{aligned}$$

**Final Answer**

0

## Question 19

If $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, then show that $2x\frac{dy}{dx} + y = 2\sqrt{x}$.

**TYPE 32** Verifying Differential Relations

**Given:** $y = x^{1/2} + x^{-1/2}$

**To Prove:** $2x\frac{dy}{dx} + y = 2\sqrt{x}$

**Concept / Formula Used:** $\frac{d}{dx}(x^n) = nx^{n-1}$

**Solution:**

Differentiating $y = x^{1/2} + x^{-1/2}$ with respect to $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2}x^{-1/2} - \frac{1}{2}x^{-3/2}\\ &= \frac{1}{2\sqrt{x}} - \frac{1}{2x\sqrt{x}}\end{aligned}$$

Multiplying $\frac{dy}{dx}$ by $2x$:

$$\begin{aligned}2x\frac{dy}{dx} &= 2x\left(\frac{1}{2\sqrt{x}} - \frac{1}{2x\sqrt{x}}\right)\\ &= \sqrt{x} - \frac{1}{\sqrt{x}}\end{aligned}$$

Substituting into the left-hand side LHS $= 2x\frac{dy}{dx} + y$:

$$\begin{aligned}\text{LHS} &= \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right) + \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)\\ &= \sqrt{x} + \sqrt{x}\\ &= 2\sqrt{x}\\ &= \text{RHS}\end{aligned}$$

**Hence Proved**

LHS = RHS, hence proved.

## Question 20

If $y = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$, prove that $\frac{dy}{dx} + \sec^2\left(\frac{\pi}{4} - x\right) = 0$.

**TYPE 30** Trigonometric Simplification before Differentiation

**Given:** $y = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$

**To Prove:** $\frac{dy}{dx} + \sec^2\left(\frac{\pi}{4} - x\right) = 0$

**Concept / Formula Used:** $1-\sin 2x = (\cos x - \sin x)^2, \quad 1+\sin 2x = (\cos x + \sin x)^2$

**Solution:**

Expressing $1 \pm \sin 2x$ as perfect squares:

$$1 - \sin 2x = \cos^2 x + \sin^2 x - 2\sin x \cos x = (\cos x - \sin x)^2$$
$$1 + \sin 2x = \cos^2 x + \sin^2 x + 2\sin x \cos x = (\cos x + \sin x)^2$$

Substituting these into $y$:

$$y = \sqrt{\frac{(\cos x - \sin x)^2}{(\cos x + \sin x)^2}}$$
$$= \frac{\cos x - \sin x}{\cos x + \sin x}$$

Dividing numerator and denominator by $\cos x$:

$$y = \frac{1 - \tan x}{1 + \tan x}$$
$$= \tan\left(\frac{\pi}{4} - x\right)$$

Differentiating $y$ with respect to $x$:

$$\frac{dy}{dx} = \sec^2\left(\frac{\pi}{4} - x\right) \cdot \frac{d}{dx}\left(\frac{\pi}{4} - x\right)$$
$$= \sec^2\left(\frac{\pi}{4} - x\right) \cdot (-1)$$
$$= -\sec^2\left(\frac{\pi}{4} - x\right)$$

Transposing terms:

$$\frac{dy}{dx} + \sec^2\left(\frac{\pi}{4} - x\right) = 0$$

**Hence Proved**

$\frac{dy}{dx} + \sec^2\left(\frac{\pi}{4} - x\right) = 0$, hence proved.

## Question 21

Differentiate $|\cos x|$ w.r.t. $x$. Is this function differentiable? What can you say about the differentiability of $\cos|x|$?

**TYPE 25 Derivative of Absolute Value Functions**

**Given:** Functions $f(x) = |\cos x|$ and $g(x) = \cos|x|$

**To Find:** Derivative of $|\cos x|$ and differentiability analysis of both functions

**Concept / Formula Used:** $\frac{d}{dx}|u| = \frac{u}{|u|}u' \qquad u \neq 0$

**Solution:**

First, differentiating $f(x) = |\cos x|$:

$$\begin{aligned}\frac{d}{dx}|\cos x| &= \frac{\cos x}{|\cos x|} \cdot \frac{d}{dx}(\cos x)\\ &= \frac{\cos x}{|\cos x|} \cdot (-\sin x)\\ &= -\frac{\sin x \cos x}{|\cos x|}, \quad \text{for } \cos x \neq 0\end{aligned}$$

Second, analyzing differentiability of $f(x) = |\cos x|$: The function $|\cos x|$ is not differentiable at points where $\cos x = 0$, i.e., at $x = (2n+1)\frac{\pi}{2}$ for $n \in \mathbb{Z}$, because the absolute value creates sharp corners at the roots of $\cos x$.

Third, analyzing differentiability of $g(x) = \cos|x|$: Since $\cos(-\theta) = \cos\theta$ for all $\theta \in \mathbb{R}$, we have:

$$g(x) = \cos|x| = \cos x \quad \text{for all } x \in \mathbb{R}$$

Since $\cos x$ is differentiable everywhere on $\mathbb{R}$, $\cos|x|$ is differentiable everywhere on $\mathbb{R}$.

**Final Answer**

$\frac{d}{dx}|\cos x| = -\frac{\sin x \cos x}{|\cos x|}$ for $x \neq (2n+1)\frac{\pi}{2}$;
$|\cos x|$ is not differentiable at $x = (2n+1)\frac{\pi}{2}$;
$\cos|x|$ is differentiable everywhere on $\mathbb{R}$.

## Question 22

Prove that the derivative of an odd function is always an even function.

TYPE 33 **Proofs Using Chain Rule**

**Given:** An odd function $f(x)$ satisfying $f(-x) = -f(x)$ for all $x$

**To Prove:** $f'(x)$ is an even function, i.e., $f'(-x) = f'(x)$

**Concept / Formula Used:** $f(-x) = -f(x)$

**Solution:**

Since $f(x)$ is an odd function, by definition:

$$f(-x) = -f(x)$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}[f(-x)] = \frac{d}{dx}[-f(x)]$$

Applying the chain rule to the left-hand side:

$$\begin{aligned} f'(-x) \cdot \frac{d}{dx}(-x) &= -f'(x) \\ f'(-x) \cdot (-1) &= -f'(x) \\ -f'(-x) &= -f'(x) \end{aligned}$$

Multiplying both sides by $-1$:

$$f'(-x) = f'(x)$$

By definition, $f'(x)$ is an even function.

**Hence Proved**

$f'(-x) = f'(x)$, hence the derivative of an odd function is an even function.

## Question 23

If $2f(x) + 3f(-x) = x^2 + x + 1$, find $f'(1)$.

**TYPE 34** Solving Functional Equations for Derivatives

**Given:** $2f(x) + 3f(-x) = x^2 + x + 1$

**To Find:** $f'(1)$

**Concept / Formula Used:** $f(x)$ $-x$ $x$

### Solution:

Given functional equation:

$$2f(x) + 3f(-x) = x^2 + x + 1 \tag{1}$$

Replacing $x$ with $-x$ in equation (1):

$$\begin{aligned} 2f(-x) + 3f(x) &= (-x)^2 + (-x) + 1 \\ 3f(x) + 2f(-x) &= x^2 - x + 1 \end{aligned} \tag{2}$$

Multiplying equation (1) by 2:

$$4f(x) + 6f(-x) = 2x^2 + 2x + 2 \tag{3}$$

Multiplying equation (2) by 3:

$$9f(x) + 6f(-x) = 3x^2 - 3x + 3 \tag{4}$$

Subtracting equation (3) from equation (4):

$$\begin{aligned} [9f(x) + 6f(-x)] - [4f(x) + 6f(-x)] &= (3x^2 - 3x + 3) - (2x^2 + 2x + 2) \\ 5f(x) &= x^2 - 5x + 1 \\ f(x) &= \frac{1}{5}x^2 - x + \frac{1}{5} \end{aligned}$$

Differentiating $f(x)$ with respect to $x$:

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left(\frac{1}{5}x^2 - x + \frac{1}{5}\right) \\ &= \frac{2}{5}x - 1 \end{aligned}$$

Substituting $x = 1$:

$$\begin{aligned} f'(1) &= \frac{2}{5}(1) - 1 \\ &= \frac{2}{5} - 1 \\ &= -\frac{3}{5} \end{aligned}$$

**Final Answer**

$f'(1) = -\frac{3}{5}$

## Question 24

If $f(x) = |\tan 2x|$, then find the value of $f'(x)$ at $x = \frac{\pi}{3}$.

**TYPE 26 Derivative of Absolute Value Functions at a Point**

**Given:** $f(x) = |\tan 2x|$, point $x = \frac{\pi}{3}$

**To Find:** $f'\left(\frac{\pi}{3}\right)$

**Concept / Formula Used:** $\tan\left(\frac{2\pi}{3}\right) < 0 \implies |\tan 2x| = -\tan 2x$

**Solution:**

Evaluating $\tan 2x$ at $x = \frac{\pi}{3}$:

$$\begin{aligned} 2x &= \frac{2\pi}{3} \\ \tan\left(\frac{2\pi}{3}\right) &= -\sqrt{3} < 0 \end{aligned}$$

Since $\tan 2x < 0$ in a neighborhood of $x = \frac{\pi}{3}$:

$$f(x) = -\tan 2x$$

Differentiating $f(x)$ with respect to $x$:

$$\begin{aligned} f'(x) &= \frac{d}{dx}(-\tan 2x) \\ &= -\sec^2 2x \cdot \frac{d}{dx}(2x) \\ &= -2\sec^2 2x \end{aligned}$$

Substituting $x = \frac{\pi}{3}$:

$$\begin{aligned} f'\left(\frac{\pi}{3}\right) &= -2\sec^2\left(\frac{2\pi}{3}\right) \\ &= -2 \cdot (-2)^2 \\ &= -2 \cdot 4 \\ &= -8 \end{aligned}$$

**Final Answer**

$-8$

## Question 25

If $y = \cos^3(\sec^2 2t)$, find $\frac{dy}{dt}$.

**TYPE 22** **Repeated Chain Rule**

**Given:** $y = \cos^3(\sec^2 2t)$

**To Find:** $\frac{dy}{dt}$

**Concept / Formula Used:** $\frac{d}{dt}[u^n] = nu^{n-1}u'$, $\frac{d}{du}\cos u = -\sin u$, $\frac{d}{dv}\sec^2 v = 2\sec^2 v\tan v$

**Solution:**

Differentiating $y = [\cos(\sec^2 2t)]^3$ with respect to $t$ using the chain rule:

$$\frac{dy}{dt} = 3\cos^2(\sec^2 2t) \cdot \frac{d}{dt}\left[\cos(\sec^2 2t)\right]$$

Differentiating the inner function $\cos(\sec^2 2t)$:

$$\frac{d}{dt}\left[\cos(\sec^2 2t)\right] = -\sin(\sec^2 2t) \cdot \frac{d}{dt}(\sec^2 2t)$$

Differentiating $\sec^2 2t = (\sec 2t)^2$:

$$\begin{aligned} \frac{d}{dt}(\sec^2 2t) &= 2\sec 2t \cdot \frac{d}{dt}(\sec 2t) \\ &= 2\sec 2t \cdot (\sec 2t \tan 2t \cdot 2) \\ &= 4\sec^2 2t \tan 2t \end{aligned}$$

Combining all the results:

$$\begin{aligned} \frac{dy}{dt} &= 3\cos^2(\sec^2 2t) \cdot \left[-\sin(\sec^2 2t)\right] \cdot \left[4\sec^2 2t \tan 2t\right] \\ &= -12\sec^2 2t \tan 2t \cos^2(\sec^2 2t)\sin(\sec^2 2t) \end{aligned}$$

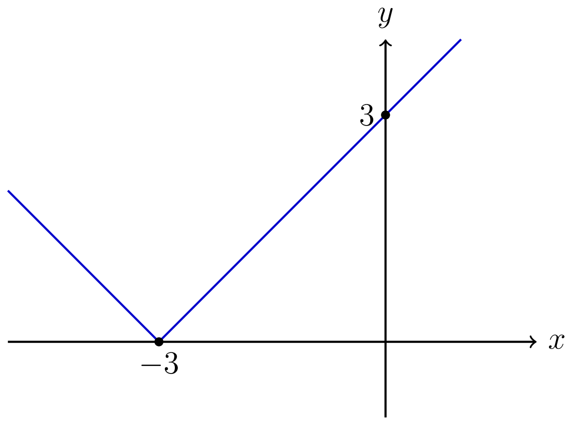
**Final Answer**

$-12\sec^2 2t \tan 2t \cos^2(\sec^2 2t)\sin(\sec^2 2t)$

## Question 26

The figure given below shows the graph of a function $y = f(x)$. What is the derivative of the function?

**TYPE 25** **Derivative of Absolute Value Functions**




*V-shaped graph of* $y = f(x)$ *with corner at* $(-3, 0)$

**Given:** V-shaped graph passing through $(-3, 0)$, $(0, 3)$, and $(-6, 3)$

**To Find:** The derivative $f'(x)$

**Concept / Formula Used:** $f'(x) \quad y = f(x)$

**Solution:**

From the given graph, the function is represented by two linear rays meeting at the vertex $(-3, 0)$:

For $x > -3$, the line passes through $(-3, 0)$ and $(0, 3)$. Its slope is:

$$\begin{aligned} m_1 &= \frac{3 - 0}{0 - (-3)} \\ &= \frac{3}{3} \\ &= 1 \end{aligned}$$

For $x < -3$, the line passes through $(-3, 0)$ and $(-6, 3)$. Its slope is:

$$\begin{aligned} m_2 &= \frac{3 - 0}{-6 - (-3)} \\ &= \frac{3}{-3} \\ &= -1 \end{aligned}$$

Thus, $f(x) = |x+3|$. Differentiating $f(x)$ for $x \neq -3$:

$$f'(x) = \begin{cases} 1, & \text{if } x > -3 \\ -1, & \text{if } x < -3 \end{cases}$$

Equivalently, in absolute value notation:

$$f'(x) = \frac{x+3}{|x+3|}, \quad \text{for } x \neq -3$$

At $x = -3$, $f'(x)$ does not exist due to a sharp corner.

**Final Answer**

$f'(x) = \frac{x+3}{|x+3|}, \quad x \neq -3$

## 5.6 – Derivatives of Inverse Trigonometric Functions

### Question 1

Find the domain of differentiability of the following functions:

(i) $\sin^{-1} x$
(ii) $\cos^{-1} x$
(iii) $\sec^{-1} x$
(iv) $\csc^{-1} x$
(v) $\tan^{-1} x$
(vi) $\cot^{-1} x$

**TYPE 35** **Domain of Differentiability**

**To Find:** The domain of differentiability for each given inverse trigonometric function.

**Concept / Formula Used:** $f(x) \qquad f'(x)$

**Solution:**

(i) Let $f(x) = \sin^{-1} x$. Using the standard derivative formula:

$$f'(x) = \frac{1}{\sqrt{1-x^2}}$$

For $f'(x)$ to be defined and real, we require $1 - x^2 > 0$, which gives $x^2 < 1$, i.e., $x \in (-1, 1)$.

(ii) Let $f(x) = \cos^{-1} x$. Using the standard derivative formula:

$$f'(x) = -\frac{1}{\sqrt{1-x^2}}$$

For $f'(x)$ to be defined and real, we require $1 - x^2 > 0$, which gives $x \in (-1, 1)$.

(iii) Let $f(x) = \sec^{-1} x$. Using the standard derivative formula:

$$f'(x) = \frac{1}{|x|\sqrt{x^2-1}}$$

For $f'(x)$ to be defined and real, we require $x \neq 0$ and $x^2 - 1 > 0$, which gives $|x| > 1$, i.e., $x \in (-\infty, -1) \cup (1, \infty)$.

(iv) Let $f(x) = \csc^{-1} x$. Using the standard derivative formula:

$$f'(x) = -\frac{1}{|x|\sqrt{x^2-1}}$$

For $f'(x)$ to be defined and real, we require $x \neq 0$ and $x^2 - 1 > 0$, which gives $x \in (-\infty, -1) \cup (1, \infty)$.

(v) Let $f(x) = \tan^{-1} x$. Using the standard derivative formula:

$$f'(x) = \frac{1}{1+x^2}$$

Since $1 + x^2 \geq 1 > 0$ for all real numbers $x$, $f'(x)$ exists for all $x \in \mathbb{R}$.

(vi) Let $f(x) = \cot^{-1} x$. Using the standard derivative formula:

$$f'(x) = -\frac{1}{1+x^2}$$

Since $1 + x^2 \geq 1 > 0$ for all real numbers $x$, $f'(x)$ exists for all $x \in \mathbb{R}$.

**Final Answer**

(i) $(-1, 1)$
(ii) $(-1, 1)$
(iii) $(-\infty, -1) \cup (1, \infty)$
(iv) $(-\infty, -1) \cup (1, \infty)$
(v) $\mathbb{R}$ or $(-\infty, \infty)$
(vi) $\mathbb{R}$ or $(-\infty, \infty)$

## Question 2

Find the derivatives of the following functions:

(i) $\tan^{-1}(x\sqrt{x})$
(ii) $\cos^{-1}(3x)$
(iii) $\sqrt{\sin^{-1}(2x)}$
(iv) $\sin(\tan^{-1} x)$
(v) $\cot^{-1} \sqrt{x}$

(vi) $(\tan^{-1} x)^2$

**TYPE 36** **Chain Rule for Inverse Trigonometric Functions**

**To Find:** The derivative $\frac{dy}{dx}$ for each function.

**Concept / Formula Used:** $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

**Solution:**

(i) Let $y = \tan^{-1}(x\sqrt{x}) = \tan^{-1}(x^{3/2})$.
Let $u = x^{3/2}$.
Differentiating $u$ w.r.t. $x$:

$$\frac{du}{dx} = \frac{3}{2}x^{1/2} = \frac{3}{2}\sqrt{x}$$

Using the standard result $\frac{d}{du}\tan^{-1} u = \frac{1}{1+u^2}$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{1+u^2} \cdot \frac{du}{dx} \\ &= \frac{1}{1+(x^{3/2})^2} \cdot \left(\frac{3}{2}\sqrt{x}\right) \\ &= \frac{3\sqrt{x}}{2(1+x^3)}\end{aligned}$$

(ii) Let $y = \cos^{-1}(3x)$.
Let $u = 3x$.
Differentiating $u$ w.r.t. $x$:

$$\frac{du}{dx} = 3$$

Using the standard result $\frac{d}{du}\cos^{-1} u = -\frac{1}{\sqrt{1-u^2}}$:

$$\begin{aligned}\frac{dy}{dx} &= -\frac{1}{\sqrt{1-u^2}} \cdot \frac{du}{dx} \\ &= -\frac{1}{\sqrt{1-(3x)^2}} \cdot 3 \\ &= -\frac{3}{\sqrt{1-9x^2}}\end{aligned}$$

(iii) Let $y = \sqrt{\sin^{-1}(2x)}$.
Let $u = \sin^{-1}(2x)$.

Let $v = 2x \implies \frac{dv}{dx} = 2$.
Using the standard result $\frac{d}{dv}\sin^{-1} v = \frac{1}{\sqrt{1-v^2}}$:

$$\frac{du}{dx} = \frac{1}{\sqrt{1-(2x)^2}} \cdot 2 = \frac{2}{\sqrt{1-4x^2}}$$

Now for $y = \sqrt{u}$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{u}} \cdot \frac{du}{dx} \\ &= \frac{1}{2\sqrt{\sin^{-1}(2x)}} \cdot \frac{2}{\sqrt{1-4x^2}} \\ &= \frac{1}{\sqrt{1-4x^2}\sqrt{\sin^{-1}(2x)}}\end{aligned}$$

(iv) Let $y = \sin(\tan^{-1} x)$.
Let $\tan^{-1} x = \theta \implies \tan\theta = x$.
Using the trigonometric identity $\sin\theta = \frac{\tan\theta}{\sqrt{1+\tan^2\theta}}$:

$$y = \frac{x}{\sqrt{1+x^2}}$$

Differentiating $y$ w.r.t. $x$ using the Quotient Rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\sqrt{1+x^2} \cdot \frac{d}{dx}(x) - x \cdot \frac{d}{dx}(\sqrt{1+x^2})}{(\sqrt{1+x^2})^2} \\ &= \frac{\sqrt{1+x^2}(1) - x \cdot \left(\frac{2x}{2\sqrt{1+x^2}}\right)}{1+x^2} \\ &= \frac{\sqrt{1+x^2} - \frac{x^2}{\sqrt{1+x^2}}}{1+x^2} \\ &= \frac{1+x^2-x^2}{(1+x^2)^{3/2}} \\ &= \frac{1}{(1+x^2)^{3/2}}\end{aligned}$$

(v) Let $y = \cot^{-1}\sqrt{x}$.
Let $u = \sqrt{x}$.
Differentiating $u$ w.r.t. $x$:

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$$

Using the standard result $\frac{d}{du}\cot^{-1} u = -\frac{1}{1+u^2}$:

$$\begin{aligned}\frac{dy}{dx} &= -\frac{1}{1+u^2} \cdot \frac{du}{dx} \\ &= -\frac{1}{1+(\sqrt{x})^2} \cdot \frac{1}{2\sqrt{x}} \\ &= -\frac{1}{2\sqrt{x}(1+x)}\end{aligned}$$

(vi) Let $y = (\tan^{-1} x)^2$.
Using the Power Rule combined with Chain Rule:

$$\begin{aligned}\frac{dy}{dx} &= 2(\tan^{-1} x) \cdot \frac{d}{dx}(\tan^{-1} x) \\ &= 2(\tan^{-1} x) \cdot \frac{1}{1+x^2} \\ &= \frac{2\tan^{-1} x}{1+x^2}\end{aligned}$$

**Final Answer**

(i) $\frac{3\sqrt{x}}{2(1+x^3)}$

(ii) $-\frac{3}{\sqrt{1-9x^2}}$

(iii) $\frac{1}{\sqrt{1-4x^2}\sqrt{\sin^{-1}(2x)}}$

(iv) $\frac{1}{(1+x^2)^{3/2}}$

(v) $-\frac{1}{2\sqrt{x}(1+x)}$

(vi) $\frac{2\tan^{-1} x}{1+x^2}$

## Question 3

Find the derivatives of the following functions:

(i) $\sec^{-1}(\csc x)$

(ii) $\tan^{-1}(\cot x)$

**TYPE 37 Inverse Trigonometric Simplification before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each given expression.

**Concept / Formula Used:** $\csc x = \sec\left(\frac{\pi}{2} - x\right)$ $\quad \cot x = \tan\left(\frac{\pi}{2} - x\right)$

**Solution:**

(i) Let $y = \sec^{-1}(\csc x)$.
Using the identity $\csc x = \sec\left(\frac{\pi}{2} - x\right)$:

$$\begin{aligned}y &= \sec^{-1}\left(\sec\left(\frac{\pi}{2} - x\right)\right) \\ &= \frac{\pi}{2} - x\end{aligned}$$

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\left(\frac{\pi}{2}\right) - \frac{d}{dx}(x) \\ &= 0 - 1 \\ &= -1\end{aligned}$$

(ii) Let $y = \tan^{-1}(\cot x)$.
Using the identity $\cot x = \tan\left(\frac{\pi}{2} - x\right)$:

$$\begin{aligned} y &= \tan^{-1}\left(\tan\left(\frac{\pi}{2} - x\right)\right) \\ &= \frac{\pi}{2} - x \end{aligned}$$

Differentiating w.r.t. $x$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(\frac{\pi}{2}\right) - \frac{d}{dx}(x) \\ &= 0 - 1 \\ &= -1 \end{aligned}$$

**Final Answer**

(i) $-1$

(ii) $-1$

## Question 4

If $f(x) = \cot^{-1}(\tan x)$, show that $f'(x) = -1$.

**TYPE 37 Inverse Trigonometric Simplification before Differentiation**

**To Prove:** $f'(x) = -1$ where $f(x) = \cot^{-1}(\tan x)$.

**Concept / Formula Used:** $\tan x = \cot\left(\frac{\pi}{2} - x\right)$

**Solution:**

Given function:

$$f(x) = \cot^{-1}(\tan x)$$

Using the complementary angle identity $\tan x = \cot\left(\frac{\pi}{2} - x\right)$:

$$\begin{aligned} f(x) &= \cot^{-1}\left(\cot\left(\frac{\pi}{2} - x\right)\right) \\ &= \frac{\pi}{2} - x \end{aligned}$$

Differentiating $f(x)$ w.r.t. $x$:

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left(\frac{\pi}{2} - x\right) \\ &= 0 - 1 \\ &= -1 \end{aligned}$$

**Hence Proved**

LHS $= f'(x) = -1$, hence proved.

## Question 5

If $y = \sin^{-1} x + \cos^{-1} x$, find $\frac{dy}{dx}$.

**TYPE 37** **Inverse Trigonometric Simplification before Differentiation**

**Given:** $y = \sin^{-1} x + \cos^{-1} x$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \qquad x \in [-1, 1]$

**Solution:**

Given function:

$$y = \sin^{-1} x + \cos^{-1} x$$

Using the standard inverse trigonometric identity $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$:

$$y = \frac{\pi}{2}$$

Differentiating w.r.t. $x$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(\frac{\pi}{2}\right) \\ &= 0 \end{aligned}$$

**Final Answer**

$\frac{dy}{dx} = 0$

## Question 6

If $f(x) = \sin^{-1} x + \sec^{-1} \frac{1}{x}$, find $f'(x)$.

**TYPE 37** **Inverse Trigonometric Simplification before Differentiation**

**Given:** $f(x) = \sin^{-1} x + \sec^{-1} \frac{1}{x}$

**To Find:** $f'(x)$

**Concept / Formula Used:** $\sec^{-1}\left(\frac{1}{x}\right) = \cos^{-1} x \qquad \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

**Solution:**

Given function:

$$f(x) = \sin^{-1} x + \sec^{-1}\left(\frac{1}{x}\right)$$

Using the reciprocal property $\sec^{-1}\left(\frac{1}{x}\right) = \cos^{-1} x$:

$$f(x) = \sin^{-1} x + \cos^{-1} x$$

Using the identity $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$:

$$f(x) = \frac{\pi}{2}$$

Differentiating w.r.t. $x$:

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left(\frac{\pi}{2}\right) \\ &= 0 \end{aligned}$$

**Final Answer**

$f'(x) = 0$

## Question 7

If $y = \tan^{-1} x + \cot^{-1} \frac{1}{x}$, $x > 0$, find $\frac{dy}{dx}$.

**TYPE 37 Inverse Trigonometric Simplification before Differentiation**

**Given:** $y = \tan^{-1} x + \cot^{-1} \frac{1}{x}$ for $x > 0$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $x > 0 \qquad \cot^{-1}\left(\frac{1}{x}\right) = \tan^{-1} x$

**Solution:**

Given function for $x > 0$:

$$y = \tan^{-1} x + \cot^{-1}\left(\frac{1}{x}\right)$$

Using the identity $\cot^{-1}\left(\frac{1}{x}\right) = \tan^{-1} x$ for $x > 0$:

$$\begin{aligned} y &= \tan^{-1} x + \tan^{-1} x \\ &= 2\tan^{-1} x \end{aligned}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = 2 \cdot \frac{d}{dx}(\tan^{-1} x)$$

Using the standard result $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$:

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

**Final Answer**

$\frac{dy}{dx} = \frac{2}{1+x^2}$

## Question 8

If $y = \sin^{-1}(\cos x) + 3\csc^{-1}(\sec x)$, find $\frac{dy}{dx}$.

**TYPE 37 Inverse Trigonometric Simplification before Differentiation**

**Given:** $y = \sin^{-1}(\cos x) + 3\csc^{-1}(\sec x)$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\cos x = \sin\left(\frac{\pi}{2} - x\right) \qquad \sec x = \csc\left(\frac{\pi}{2} - x\right)$

**Solution:**

Given function:

$$y = \sin^{-1}(\cos x) + 3\csc^{-1}(\sec x)$$

Using complementary angle transformations:

$$\begin{aligned} \cos x = \sin\left(\frac{\pi}{2} - x\right) &\implies \sin^{-1}(\cos x) = \frac{\pi}{2} - x \\ \sec x = \csc\left(\frac{\pi}{2} - x\right) &\implies \csc^{-1}(\sec x) = \frac{\pi}{2} - x \end{aligned}$$

Substituting these into $y$:

$$\begin{aligned} y &= \left(\frac{\pi}{2} - x\right) + 3\left(\frac{\pi}{2} - x\right) \\ &= 4\left(\frac{\pi}{2} - x\right) \\ &= 2\pi - 4x \end{aligned}$$

Differentiating w.r.t. $x$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(2\pi - 4x) \\ &= 0 - 4 \\ &= -4 \end{aligned}$$

**Final Answer**

$\frac{dy}{dx} = -4$

## Question 9

If $y = 2\cos^{-1}(\sin x) + 3\cot^{-1}(\tan x)$, find $\frac{dy}{dx}$.

**TYPE 37 Inverse Trigonometric Simplification before Differentiation**

**Given:** $y = 2\cos^{-1}(\sin x) + 3\cot^{-1}(\tan x)$

**To Find:** $\frac{dy}{dx}$

**Concept / Formula Used:** $\sin x = \cos\left(\frac{\pi}{2} - x\right) \qquad \tan x = \cot\left(\frac{\pi}{2} - x\right)$

**Solution:**

Given function:

$$y = 2\cos^{-1}(\sin x) + 3\cot^{-1}(\tan x)$$

Using complementary angle transformations:

$$\begin{aligned} \sin x = \cos\left(\frac{\pi}{2} - x\right) &\implies \cos^{-1}(\sin x) = \frac{\pi}{2} - x \\ \tan x = \cot\left(\frac{\pi}{2} - x\right) &\implies \cot^{-1}(\tan x) = \frac{\pi}{2} - x \end{aligned}$$

Substituting these into $y$:

$$\begin{aligned} y &= 2\left(\frac{\pi}{2} - x\right) + 3\left(\frac{\pi}{2} - x\right) \\ &= 5\left(\frac{\pi}{2} - x\right) \\ &= \frac{5\pi}{2} - 5x \end{aligned}$$

Differentiating w.r.t. $x$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(\frac{5\pi}{2} - 5x\right) \\ &= 0 - 5 \\ &= -5 \end{aligned}$$

**Final Answer**

$\frac{dy}{dx} = -5$

## Question 10

Differentiate the following functions w.r.t. $x$:

(i) $x\tan^{-1} x$

(ii) $x\cos^{-1}\sqrt{x}$

(iii) $\frac{\sin^{-1} x}{x}$

**TYPE 38 Product and Quotient Rules with Inverse Trigonometric Functions**

**To Find:** The derivative $\frac{dy}{dx}$ for each given expression.

**Concept / Formula Used:** $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx} \qquad \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

**Solution:**

(i) Let $y = x \tan^{-1} x$.
Using Product Rule with $u = x$ and $v = \tan^{-1} x$:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot \frac{d}{dx}(\tan^{-1} x) + \tan^{-1} x \cdot \frac{d}{dx}(x) \\ &= x \cdot \frac{1}{1+x^2} + (\tan^{-1} x) \cdot 1 \\ &= \frac{x}{1+x^2} + \tan^{-1} x\end{aligned}$$

(ii) Let $y = x \cos^{-1} \sqrt{x}$.
Using Product Rule with $u = x$ and $v = \cos^{-1} \sqrt{x}$:

$$\frac{dy}{dx} = x \cdot \frac{d}{dx}\left(\cos^{-1} \sqrt{x}\right) + \cos^{-1} \sqrt{x} \cdot \frac{d}{dx}(x)$$

To find $\frac{d}{dx}\left(\cos^{-1} \sqrt{x}\right)$, let $w = \sqrt{x} \implies \frac{dw}{dx} = \frac{1}{2\sqrt{x}}$:

$$\begin{aligned}\frac{d}{dx}\left(\cos^{-1} \sqrt{x}\right) &= -\frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}} \\ &= -\frac{1}{2\sqrt{x}\sqrt{1-x}}\end{aligned}$$

Substituting back:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot \left(-\frac{1}{2\sqrt{x}\sqrt{1-x}}\right) + \cos^{-1} \sqrt{x} \\ &= -\frac{\sqrt{x}}{2\sqrt{1-x}} + \cos^{-1} \sqrt{x}\end{aligned}$$

(iii) Let $y = \frac{\sin^{-1} x}{x}$.
Using Quotient Rule with $u = \sin^{-1} x$ and $v = x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{x \cdot \frac{d}{dx}(\sin^{-1} x) - \sin^{-1} x \cdot \frac{d}{dx}(x)}{x^2} \\ &= \frac{x \cdot \frac{1}{\sqrt{1-x^2}} - \sin^{-1} x \cdot 1}{x^2} \\ &= \frac{\frac{x}{\sqrt{1-x^2}} - \sin^{-1} x}{x^2} \\ &= \frac{x - \sqrt{1-x^2}\sin^{-1} x}{x^2\sqrt{1-x^2}}\end{aligned}$$

**Final Answer**

(i) $\frac{x}{1+x^2} + \tan^{-1} x$

(ii) $\cos^{-1} \sqrt{x} - \frac{\sqrt{x}}{2\sqrt{1-x}}$

(iii) $\frac{x-\sqrt{1-x^2}\sin^{-1} x}{x^2\sqrt{1-x^2}}$

## Question 11

Differentiate the following functions w.r.t. $x$:

(i) $\cos^{-1}\left(\frac{1-x}{1+x}\right)$

(ii) $\sec^{-1}\left(\frac{1}{x-1}\right)$

(iii) $\sin^{-1}\left(\frac{1}{\sqrt{x+1}}\right)$

★ **Likely Exam Question**

**TYPE 39 Trigonometric Substitution before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each function.

**Concept / Formula Used:** $\cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}$ $\quad \sec^{-1}\left(\frac{1}{u}\right) = \cos^{-1} u$

**Solution:**

(i) Let $y = \cos^{-1}\left(\frac{1-x}{1+x}\right)$.
Let $x = \tan^2\theta \implies \sqrt{x} = \tan\theta$.
Differentiating $\tan\theta = \sqrt{x}$:

$$\theta = \tan^{-1}\sqrt{x}$$

Using double-angle identity:

$$\frac{1-x}{1+x} = \frac{1-\tan^2\theta}{1+\tan^2\theta} = \cos 2\theta$$

Substituting this into $y$:

$$y = \cos^{-1}(\cos 2\theta) = 2\theta = 2\tan^{-1}\sqrt{x}$$

Differentiating $y$ w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= 2 \cdot \frac{1}{1+(\sqrt{x})^2} \cdot \frac{d}{dx}(\sqrt{x}) \\ &= \frac{2}{1+x} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{1}{\sqrt{x}(1+x)}\end{aligned}$$

(ii) Let $y = \sec^{-1}\left(\frac{1}{x-1}\right)$.
Using identity $\sec^{-1}\left(\frac{1}{u}\right) = \cos^{-1} u$:

$$y = \cos^{-1}(x-1)$$

Differentiating w.r.t. $x$: Using standard result $\frac{d}{du}\cos^{-1}u = -\frac{1}{\sqrt{1-u^2}}$ with $u = x-1$:

$$\begin{aligned}\frac{dy}{dx} &= -\frac{1}{\sqrt{1-(x-1)^2}} \cdot \frac{d}{dx}(x-1) \\ &= -\frac{1}{\sqrt{1-(x^2-2x+1)}} \cdot 1 \\ &= -\frac{1}{\sqrt{2x-x^2}}\end{aligned}$$

(iii) Let $y = \sin^{-1}\left(\frac{1}{\sqrt{x+1}}\right)$.
Let $x = \cot^2\theta \implies \sqrt{x} = \cot\theta \implies \theta = \cot^{-1}\sqrt{x}$.
Then:

$$\begin{aligned}x+1 &= \cot^2\theta + 1 = \csc^2\theta \\ \sqrt{x+1} &= \csc\theta \\ \frac{1}{\sqrt{x+1}} &= \frac{1}{\csc\theta} = \sin\theta\end{aligned}$$

Substituting this into $y$:

$$y = \sin^{-1}(\sin\theta) = \theta = \cot^{-1}\sqrt{x}$$

Differentiating $y$ w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= -\frac{1}{1+(\sqrt{x})^2} \cdot \frac{d}{dx}(\sqrt{x}) \\ &= -\frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}} \\ &= -\frac{1}{2\sqrt{x}(1+x)}\end{aligned}$$

**Final Answer**

(i) $\frac{1}{\sqrt{x}(1+x)}$

(ii) $-\frac{1}{\sqrt{2x-x^2}}$

(iii) $-\frac{1}{2\sqrt{x}(1+x)}$

## Question 12

Differentiate the following functions w.r.t. $x$:
(i) $\sin^{-1}x + \sin^{-1}\sqrt{1-x^2}$
(ii) $x(\sin^{-1}x)^2 + 2\sqrt{1-x^2}\sin^{-1}x - 2x$

**TYPE 37** **Inverse Trigonometric Simplification before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each given expression.

**Concept / Formula Used:** $\sin^{-1}\sqrt{1-x^2} = \cos^{-1} x \qquad x \in [0,1]$

**Solution:**

(i) Let $y = \sin^{-1} x + \sin^{-1}\sqrt{1-x^2}$.
For $x \in [0,1]$, let $x = \sin\theta \implies \theta = \sin^{-1} x$.
Then $\sqrt{1-x^2} = \cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$.
Thus $\sin^{-1}\sqrt{1-x^2} = \frac{\pi}{2} - \theta = \frac{\pi}{2} - \sin^{-1} x$.
Substituting into $y$:

$$\begin{aligned} y &= \sin^{-1} x + \left(\frac{\pi}{2} - \sin^{-1} x\right) \\ &= \frac{\pi}{2} \end{aligned}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = 0$$

(ii) Let $y = x(\sin^{-1} x)^2 + 2\sqrt{1-x^2}\sin^{-1} x - 2x$.
Differentiating term by term w.r.t. $x$:

$$\begin{aligned} \frac{d}{dx}\left[x(\sin^{-1} x)^2\right] &= 1 \cdot (\sin^{-1} x)^2 + x \cdot 2(\sin^{-1} x) \cdot \frac{1}{\sqrt{1-x^2}} \\ &= (\sin^{-1} x)^2 + \frac{2x\sin^{-1} x}{\sqrt{1-x^2}} \end{aligned}$$

$$\begin{aligned} \frac{d}{dx}\left[2\sqrt{1-x^2}\sin^{-1} x\right] &= 2\left[\frac{-2x}{2\sqrt{1-x^2}}\sin^{-1} x + \sqrt{1-x^2} \cdot \frac{1}{\sqrt{1-x^2}}\right] \\ &= -\frac{2x\sin^{-1} x}{\sqrt{1-x^2}} + 2 \end{aligned}$$

$$\frac{d}{dx}[-2x] = -2$$

Summing all three derivatives:

$$\begin{aligned} \frac{dy}{dx} &= (\sin^{-1} x)^2 + \frac{2x\sin^{-1} x}{\sqrt{1-x^2}} - \frac{2x\sin^{-1} x}{\sqrt{1-x^2}} + 2 - 2 \\ &= (\sin^{-1} x)^2 \end{aligned}$$

**Final Answer**

(i) 0

(ii) $(\sin^{-1} x)^2$

## Question 13

Differentiate the following functions w.r.t. $x$:

(i) $\tan^{-1}(\sin x + \cos x)$
(ii) $\tan^{-1}\left(\frac{\sin x}{1+\cos x}\right)$
(iii) $\tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)$
(iv) $\tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$

**TYPE 37 Inverse Trigonometric Simplification before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each given expression.

**Concept / Formula Used:** $\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}$ $\quad 1 + \cos x = 2\cos^2\frac{x}{2}$

**Solution:**

(i) Let $y = \tan^{-1}(\sin x + \cos x)$.
Let $u = \sin x + \cos x \implies \frac{du}{dx} = \cos x - \sin x$.
Using standard result $\frac{d}{du}\tan^{-1} u = \frac{1}{1+u^2}$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{1+(\sin x+\cos x)^2}\cdot(\cos x - \sin x)\\ &= \frac{\cos x - \sin x}{1+\sin^2 x+\cos^2 x + 2\sin x\cos x}\\ &= \frac{\cos x - \sin x}{1+1+\sin 2x}\\ &= \frac{\cos x - \sin x}{2+\sin 2x}\end{aligned}$$

(ii) Let $y = \tan^{-1}\left(\frac{\sin x}{1+\cos x}\right)$.
Using half-angle identities $\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}$ and $1+\cos x = 2\cos^2\frac{x}{2}$:

$$\frac{\sin x}{1+\cos x} = \frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}} = \tan\frac{x}{2}$$

$$y = \tan^{-1}\left(\tan\frac{x}{2}\right) = \frac{x}{2}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{1}{2}$$

(iii) Let $y = \tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)$.
Using half-angle identities:

$$\frac{1+\cos x}{\sin x} = \frac{2\cos^2\frac{x}{2}}{2\sin\frac{x}{2}\cos\frac{x}{2}} = \cot\frac{x}{2} = \tan\left(\frac{\pi}{2}-\frac{x}{2}\right)$$

$$y = \tan^{-1}\left(\tan\left(\frac{\pi}{2}-\frac{x}{2}\right)\right) = \frac{\pi}{2}-\frac{x}{2}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = 0 - \frac{1}{2} = -\frac{1}{2}$$

(iv) Let $y = \tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$.
Converting sine and cosine to complementary angles:

$$\cos x = \sin\left(\frac{\pi}{2} - x\right) = 2\sin\left(\frac{\pi}{4} - \frac{x}{2}\right)\cos\left(\frac{\pi}{4} - \frac{x}{2}\right)$$
$$1 + \sin x = 1 + \cos\left(\frac{\pi}{2} - x\right) = 2\cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right)$$
$$\frac{\cos x}{1+\sin x} = \frac{2\sin\left(\frac{\pi}{4} - \frac{x}{2}\right)\cos\left(\frac{\pi}{4} - \frac{x}{2}\right)}{2\cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right)} = \tan\left(\frac{\pi}{4} - \frac{x}{2}\right)$$
$$y = \tan^{-1}\left(\tan\left(\frac{\pi}{4} - \frac{x}{2}\right)\right) = \frac{\pi}{4} - \frac{x}{2}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -\frac{1}{2}$$

**Final Answer**

(i) $\frac{\cos x - \sin x}{2+\sin 2x}$
(ii) $\frac{1}{2}$
(iii) $-\frac{1}{2}$
(iv) $-\frac{1}{2}$

## Question 14

Differentiate the following functions w.r.t. $x$:
(i) $\cot^{-1}(\csc x + \cot x)$
(ii) $\tan^{-1}\left(\sqrt{\frac{1-\cos x}{1+\cos x}}\right)$
(iii) $\cot^{-1}\left(\sqrt{\frac{1-\sin x}{1+\sin x}}\right)$
(iv) $\tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$

**TYPE 37** **Inverse Trigonometric Simplification before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each given expression.

**Concept / Formula Used:** $1 - \cos x = 2\sin^2\frac{x}{2}$ $\qquad 1 + \cos x = 2\cos^2\frac{x}{2}$

**Solution:**

(i) Let $y = \cot^{-1}(\csc x + \cot x)$.
Expressing inside terms in sine and cosine:

$$\begin{aligned} \csc x + \cot x &= \frac{1}{\sin x} + \frac{\cos x}{\sin x} = \frac{1+\cos x}{\sin x} \\ &= \frac{2\cos^2\frac{x}{2}}{2\sin\frac{x}{2}\cos\frac{x}{2}} = \cot\frac{x}{2} \\ y &= \cot^{-1}\left(\cot\frac{x}{2}\right) = \frac{x}{2} \end{aligned}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{1}{2}$$

(ii) Let $y = \tan^{-1}\left(\sqrt{\frac{1-\cos x}{1+\cos x}}\right)$.
Using standard identities $1 - \cos x = 2\sin^2\frac{x}{2}$ and $1 + \cos x = 2\cos^2\frac{x}{2}$:

$$\begin{aligned} \sqrt{\frac{1-\cos x}{1+\cos x}} &= \sqrt{\frac{2\sin^2\frac{x}{2}}{2\cos^2\frac{x}{2}}} = \sqrt{\tan^2\frac{x}{2}} = \tan\frac{x}{2} \\ y &= \tan^{-1}\left(\tan\frac{x}{2}\right) = \frac{x}{2} \end{aligned}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{1}{2}$$

(iii) Let $y = \cot^{-1}\left(\sqrt{\frac{1-\sin x}{1+\sin x}}\right)$.
Using $1 - \sin x = \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)^2$ and $1 + \sin x = \left(\cos\frac{x}{2} + \sin\frac{x}{2}\right)^2$:

$$\begin{aligned} \sqrt{\frac{1-\sin x}{1+\sin x}} &= \frac{\cos\frac{x}{2} - \sin\frac{x}{2}}{\cos\frac{x}{2} + \sin\frac{x}{2}} = \frac{1-\tan\frac{x}{2}}{1+\tan\frac{x}{2}} = \tan\left(\frac{\pi}{4} - \frac{x}{2}\right) \\ y &= \cot^{-1}\left(\tan\left(\frac{\pi}{4} - \frac{x}{2}\right)\right) \\ &= \cot^{-1}\left(\cot\left(\frac{\pi}{2} - \left(\frac{\pi}{4} - \frac{x}{2}\right)\right)\right) = \frac{\pi}{4} + \frac{x}{2} \end{aligned}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{1}{2}$$

(iv) Let $y = \tan^{-1}\left(\frac{\cos x - \sin x}{\cos x + \sin x}\right)$.
Dividing numerator and denominator by $\cos x$:

$$\begin{aligned} \frac{\cos x - \sin x}{\cos x + \sin x} &= \frac{1-\tan x}{1+\tan x} = \tan\left(\frac{\pi}{4} - x\right) \\ y &= \tan^{-1}\left(\tan\left(\frac{\pi}{4} - x\right)\right) = \frac{\pi}{4} - x \end{aligned}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -1$$

**Final Answer**

(i) $\frac{1}{2}$
(ii) $\frac{1}{2}$
(iii) $\frac{1}{2}$
(iv) $-1$

## Question 15

(i) If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, prove that $(1 - x^2)\frac{dy}{dx} - xy = 1$.
(ii) If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, prove that $(1 + x^2)\frac{dy}{dx} = 2$.

★ **Likely Exam Question**

**TYPE 32 Verifying Differential Relations**

**To Prove:** (i) $(1 - x^2)\frac{dy}{dx} - xy = 1$
(ii) $(1 + x^2)\frac{dy}{dx} = 2$

**Concept / Formula Used:** $\sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta}$

**Solution:**

(i) Given $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$.
Cross-multiplying:

$$y\sqrt{1-x^2} = \sin^{-1} x$$

Differentiating both sides w.r.t. $x$ using Product Rule on LHS:

$$\begin{aligned} \frac{d}{dx}\left[y\sqrt{1-x^2}\right] &= \frac{d}{dx}(\sin^{-1} x) \\ y \cdot \frac{d}{dx}\left(\sqrt{1-x^2}\right) + \sqrt{1-x^2} \cdot \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}} \\ y \cdot \left(\frac{-2x}{2\sqrt{1-x^2}}\right) + \sqrt{1-x^2}\frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}} \\ -\frac{xy}{\sqrt{1-x^2}} + \sqrt{1-x^2}\frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}} \end{aligned}$$

Multiplying the entire equation by $\sqrt{1-x^2}$:

$$-xy + (1-x^2)\frac{dy}{dx} = 1$$

$$(1-x^2)\frac{dy}{dx} - xy = 1$$

**Hence Proved**

$LHS = RHS, hence proved.$

(ii) Given $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$.
Let $x = \tan\theta \implies \theta = \tan^{-1}x$.
Using identity $\sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta}$:

$$\begin{aligned} y &= \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right) \\ &= \sin^{-1}(\sin 2\theta) \\ &= 2\theta \\ &= 2\tan^{-1}x \end{aligned}$$

Differentiating w.r.t. $x$:

$$\begin{aligned} \frac{dy}{dx} &= 2 \cdot \frac{d}{dx}(\tan^{-1}x) \\ \frac{dy}{dx} &= 2 \cdot \frac{1}{1+x^2} \end{aligned}$$

Multiplying both sides by $(1+x^2)$:

$$(1+x^2)\frac{dy}{dx} = 2$$

**Hence Proved**

$LHS = RHS, hence proved.$

## Question 16

Differentiate the following functions w.r.t. $x$:

(i) $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$

(ii) $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

(iii) $\cos^{-1}(1-2x^2)$

(iv) $\sin^{-1}(2x\sqrt{1-x^2})$

★ **Likely Exam Question**

**TYPE 39 Trigonometric Substitution before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each given expression.

**Concept / Formula Used:** $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$ $\cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta} = 1 - 2\sin^2\theta$ $\sin 2\theta = 2\sin\theta\cos\theta$

**Solution:**

(i) Let $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$.
Let $x = \tan\theta \implies \theta = \tan^{-1}x$.
Using identity $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$:

$$y = \tan^{-1}(\tan 2\theta) = 2\theta = 2\tan^{-1}x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = 2 \cdot \frac{1}{1+x^2} = \frac{2}{1+x^2}$$

(ii) Let $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$.
Let $x = \tan\theta \implies \theta = \tan^{-1} x$.
Using identity $\cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}$:

$$y = \cos^{-1}(\cos 2\theta) = 2\theta = 2\tan^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

(iii) Let $y = \cos^{-1}(1-2x^2)$.
Let $x = \sin\theta \implies \theta = \sin^{-1} x$.
Using identity $\cos 2\theta = 1 - 2\sin^2\theta$:

$$y = \cos^{-1}(\cos 2\theta) = 2\theta = 2\sin^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = 2 \cdot \frac{1}{\sqrt{1-x^2}} = \frac{2}{\sqrt{1-x^2}}$$

(iv) Let $y = \sin^{-1}(2x\sqrt{1-x^2})$.
Let $x = \sin\theta \implies \theta = \sin^{-1} x$.
Then $\sqrt{1-x^2} = \cos\theta$.
Using identity $\sin 2\theta = 2\sin\theta\cos\theta$:

$$y = \sin^{-1}(\sin 2\theta) = 2\theta = 2\sin^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$$

**Final Answer**

(i) $\frac{2}{1+x^2}$

(ii) $\frac{2}{1+x^2}$

(iii) $\frac{2}{\sqrt{1-x^2}}$

(iv) $\frac{2}{\sqrt{1-x^2}}$

## Question 17

Differentiate the following functions w.r.t. $x$:

(i) $\sin^{-1}(3x - 4x^3)$

(ii) $\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$

(iii) $\cos^{-1}\left(\frac{2x}{1+x^2}\right)$

(iv) $\sec^{-1}\left(\frac{1}{4x^3-3x}\right)$

**★ Likely Exam Question**

**TYPE 39 Trigonometric Substitution before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each function.

**Concept / Formula Used:** $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ $\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}$ $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

**Solution:**

(i) Let $y = \sin^{-1}(3x - 4x^3)$.
Let $x = \sin\theta \implies \theta = \sin^{-1} x$.
Using triple-angle identity $3\sin\theta - 4\sin^3\theta = \sin 3\theta$:

$$y = \sin^{-1}(\sin 3\theta) = 3\theta = 3\sin^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = 3 \cdot \frac{1}{\sqrt{1-x^2}} = \frac{3}{\sqrt{1-x^2}}$$

(ii) Let $y = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$.
Let $x = \tan\theta \implies \theta = \tan^{-1} x$.
Using triple-angle identity $\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta} = \tan 3\theta$:

$$y = \tan^{-1}(\tan 3\theta) = 3\theta = 3\tan^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = 3 \cdot \frac{1}{1+x^2} = \frac{3}{1+x^2}$$

(iii) Let $y = \cos^{-1}\left(\frac{2x}{1+x^2}\right)$.
Let $x = \tan\theta \implies \theta = \tan^{-1} x$.
Since $\frac{2\tan\theta}{1+\tan^2\theta} = \sin 2\theta = \cos\left(\frac{\pi}{2} - 2\theta\right)$:

$$y = \cos^{-1}\left(\cos\left(\frac{\pi}{2} - 2\theta\right)\right) = \frac{\pi}{2} - 2\theta = \frac{\pi}{2} - 2\tan^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -\frac{2}{1+x^2}$$

(iv) Let $y = \sec^{-1}\left(\frac{1}{4x^3-3x}\right)$.
Using identity $\sec^{-1}\left(\frac{1}{u}\right) = \cos^{-1} u$:

$$y = \cos^{-1}(4x^3 - 3x)$$

Let $x = \cos\theta \implies \theta = \cos^{-1} x$.
Using triple-angle identity $4\cos^3\theta - 3\cos\theta = \cos 3\theta$:

$$y = \cos^{-1}(\cos 3\theta) = 3\theta = 3\cos^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = 3 \cdot \left(-\frac{1}{\sqrt{1-x^2}}\right) = -\frac{3}{\sqrt{1-x^2}}$$

**Final Answer**

(i) $\frac{3}{\sqrt{1-x^2}}$
(ii) $\frac{3}{1+x^2}$
(iii) $-\frac{2}{1+x^2}$
(iv) $-\frac{3}{\sqrt{1-x^2}}$

## Question 18

Differentiate the following functions w.r.t. $x$:
(i) $\sec^{-1}\left(\frac{x^2+1}{x^2-1}\right)$
(ii) $\sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$
(iii) $\tan^{-1}(\sqrt{1+x^2} - x)$
(iv) $\cot^{-1}(\sqrt{1+x^2} + x)$

**TYPE 39** **Trigonometric Substitution before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each given function.

**Concept / Formula Used:** $\csc\theta - \cot\theta = \tan\frac{\theta}{2}$ $\quad$ $\csc\theta + \cot\theta = \cot\frac{\theta}{2}$

**Solution:**

(i) Let $y = \sec^{-1}\left(\frac{x^2+1}{x^2-1}\right)$.
Using reciprocal identity $\sec^{-1} u = \cos^{-1}\left(\frac{1}{u}\right)$:

$$y = \cos^{-1}\left(\frac{x^2-1}{x^2+1}\right) = \cos^{-1}\left(-\frac{1-x^2}{1+x^2}\right)$$
$$= \pi - \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

Let $x = \tan\theta \implies \theta = \tan^{-1} x$.
Using $\frac{1-\tan^2\theta}{1+\tan^2\theta} = \cos 2\theta$:

$$y = \pi - \cos^{-1}(\cos 2\theta) = \pi - 2\theta = \pi - 2\tan^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -\frac{2}{1+x^2}$$

(ii) Let $y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$.
Let $x = \tan\theta \implies \theta = \tan^{-1} x$.
Since $\frac{1-\tan^2\theta}{1+\tan^2\theta} = \cos 2\theta = \sin\left(\frac{\pi}{2} - 2\theta\right)$:

$$y = \sin^{-1}\left(\sin\left(\frac{\pi}{2} - 2\theta\right)\right) = \frac{\pi}{2} - 2\theta = \frac{\pi}{2} - 2\tan^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -\frac{2}{1+x^2}$$

(iii) Let $y = \tan^{-1}(\sqrt{1+x^2} - x)$.
Let $x = \cot\theta \implies \theta = \cot^{-1} x$.
Then $\sqrt{1+x^2} = \sqrt{1+\cot^2\theta} = \csc\theta$.
Expression inside inverse tan:

$$\csc\theta - \cot\theta = \frac{1-\cos\theta}{\sin\theta} = \frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = \tan\frac{\theta}{2}$$
$$y = \tan^{-1}\left(\tan\frac{\theta}{2}\right) = \frac{\theta}{2} = \frac{1}{2}\cot^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{1}{2}\left(-\frac{1}{1+x^2}\right) = -\frac{1}{2(1+x^2)}$$

(iv) Let $y = \cot^{-1}(\sqrt{1+x^2} + x)$.
Let $x = \cot\theta \implies \theta = \cot^{-1} x$.
Then $\sqrt{1+x^2} = \csc\theta$.
Expression inside inverse cot:

$$\csc\theta + \cot\theta = \frac{1+\cos\theta}{\sin\theta} = \frac{2\cos^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = \cot\frac{\theta}{2}$$
$$y = \cot^{-1}\left(\cot\frac{\theta}{2}\right) = \frac{\theta}{2} = \frac{1}{2}\cot^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -\frac{1}{2(1+x^2)}$$

**Final Answer**

(i) $-\frac{2}{1+x^2}$

(ii) $-\frac{2}{1+x^2}$

(iii) $-\frac{1}{2(1+x^2)}$

(iv) $-\frac{1}{2(1+x^2)}$

## Question 19

Differentiate the following functions w.r.t. $x$:

(i) $\tan^{-1}\left(\frac{x}{\sqrt{a^2-x^2}}\right)$

(ii) $\sin^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$

(iii) $\cos^{-1}\left(\sqrt{\frac{1+x}{2}}\right)$

(iv) $\sin^{-1}\left(\frac{x}{\sqrt{x^2+a^2}}\right)$

**TYPE 39** **Trigonometric Substitution before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each expression.

**Concept / Formula Used:** $\frac{d}{dx}\sin^{-1}\left(\frac{x}{a}\right) = \frac{1}{\sqrt{a^2-x^2}}$ $\quad \frac{d}{dx}\tan^{-1}\left(\frac{x}{a}\right) = \frac{a}{a^2+x^2}$

**Solution:**

(i) Let $y = \tan^{-1}\left(\frac{x}{\sqrt{a^2-x^2}}\right)$.
Let $x = a\sin\theta \implies \sin\theta = \frac{x}{a} \implies \theta = \sin^{-1}\left(\frac{x}{a}\right)$.
Then $\sqrt{a^2-x^2} = \sqrt{a^2 - a^2\sin^2\theta} = a\cos\theta$.
Substituting into $y$:

$$y = \tan^{-1}\left(\frac{a\sin\theta}{a\cos\theta}\right) = \tan^{-1}(\tan\theta) = \theta = \sin^{-1}\left(\frac{x}{a}\right)$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(x/a)^2}} \cdot \frac{1}{a} = \frac{1}{\sqrt{a^2-x^2}}$$

(ii) Let $y = \sin^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$.
Let $x = \cot\theta \implies \theta = \cot^{-1}x$.
Then $\sqrt{1+x^2} = \csc\theta \implies \frac{1}{\sqrt{1+x^2}} = \sin\theta$.
Substituting into $y$:

$$y = \sin^{-1}(\sin\theta) = \theta = \cot^{-1}x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -\frac{1}{1+x^2}$$

(iii) Let $y = \cos^{-1}\left(\sqrt{\frac{1+x}{2}}\right)$.
Let $x = \cos\theta \implies \theta = \cos^{-1} x$.
Using identity $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$:

$$\sqrt{\frac{1+\cos\theta}{2}} = \sqrt{\cos^2\frac{\theta}{2}} = \cos\frac{\theta}{2}$$
$$y = \cos^{-1}\left(\cos\frac{\theta}{2}\right) = \frac{\theta}{2} = \frac{1}{2}\cos^{-1} x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{1}{2}\left(-\frac{1}{\sqrt{1-x^2}}\right) = -\frac{1}{2\sqrt{1-x^2}}$$

(iv) Let $y = \sin^{-1}\left(\frac{x}{\sqrt{x^2+a^2}}\right)$.
Let $x = a\tan\theta \implies \tan\theta = \frac{x}{a} \implies \theta = \tan^{-1}\left(\frac{x}{a}\right)$.
Then $\sqrt{x^2+a^2} = a\sec\theta \implies \frac{x}{\sqrt{x^2+a^2}} = \frac{a\tan\theta}{a\sec\theta} = \sin\theta$.
Substituting into $y$:

$$y = \sin^{-1}(\sin\theta) = \theta = \tan^{-1}\left(\frac{x}{a}\right)$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{1}{1+(x/a)^2}\cdot\frac{1}{a} = \frac{a}{a^2+x^2}$$

**Final Answer**

(i) $\frac{1}{\sqrt{a^2-x^2}}$
(ii) $-\frac{1}{1+x^2}$
(iii) $-\frac{1}{2\sqrt{1-x^2}}$
(iv) $\frac{a}{a^2+x^2}$

## Question 20

Differentiate the following functions w.r.t. $x$:

(i) $\sin^{-1}\left(\sqrt{\frac{1+x^2}{2}}\right)$
(ii) $\tan^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$
(iii) $\sin^2\left(\cot^{-1}\sqrt{\frac{1+x}{1-x}}\right)$
(iv) $\sin^{-1}\left(\frac{2\sec x}{1+\sec^2 x}\right)$

**TYPE 39 Trigonometric Substitution before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each given expression.

**Concept / Formula Used:** $1 + \cos 2\theta = 2\cos^2\theta \qquad 1 - \cos 2\theta = 2\sin^2\theta$

**Solution:**

(i) Let $y = \sin^{-1}\left(\sqrt{\frac{1+x^2}{2}}\right)$.
Let $x^2 = \cos 2\theta \implies 2\theta = \cos^{-1}(x^2) \implies \theta = \frac{1}{2}\cos^{-1}(x^2)$.
Then $\frac{1+x^2}{2} = \frac{1+\cos 2\theta}{2} = \cos^2\theta \implies \sqrt{\frac{1+x^2}{2}} = \cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$.

$$y = \sin^{-1}\left(\sin\left(\frac{\pi}{2} - \theta\right)\right) = \frac{\pi}{2} - \theta = \frac{\pi}{2} - \frac{1}{2}\cos^{-1}(x^2)$$

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= -\frac{1}{2}\cdot\left(-\frac{1}{\sqrt{1-(x^2)^2}}\right)\cdot\frac{d}{dx}(x^2)\\ &= \frac{1}{2\sqrt{1-x^4}}\cdot 2x\\ &= \frac{x}{\sqrt{1-x^4}}\end{aligned}$$

(ii) Let $y = \tan^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$.
Let $x = \cos 2\theta \implies \theta = \frac{1}{2}\cos^{-1}x$.
Then $\sqrt{1+x} = \sqrt{2}\cos\theta$ and $\sqrt{1-x} = \sqrt{2}\sin\theta$.

$$\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}} = \frac{\sqrt{2}\cos\theta - \sqrt{2}\sin\theta}{\sqrt{2}\cos\theta + \sqrt{2}\sin\theta} = \frac{1-\tan\theta}{1+\tan\theta} = \tan\left(\frac{\pi}{4} - \theta\right)$$
$$y = \tan^{-1}\left(\tan\left(\frac{\pi}{4} - \theta\right)\right) = \frac{\pi}{4} - \theta = \frac{\pi}{4} - \frac{1}{2}\cos^{-1}x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -\frac{1}{2}\cdot\left(-\frac{1}{\sqrt{1-x^2}}\right) = \frac{1}{2\sqrt{1-x^2}}$$

(iii) Let $y = \sin^2\left(\cot^{-1}\sqrt{\frac{1+x}{1-x}}\right)$.
Let $x = \cos 2\theta \implies 2\theta = \cos^{-1}x$.
Then $\sqrt{\frac{1+x}{1-x}} = \sqrt{\frac{2\cos^2\theta}{2\sin^2\theta}} = \cot\theta \implies \cot^{-1}(\cot\theta) = \theta$.

$$y = \sin^2\theta = \frac{1-\cos 2\theta}{2} = \frac{1-x}{2} = \frac{1}{2} - \frac{x}{2}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -\frac{1}{2}$$

(iv) Let $y = \sin^{-1}\left(\frac{2\sec x}{1+\sec^2 x}\right)$.
Let $\sec x = \tan\theta \implies \theta = \tan^{-1}(\sec x)$.
Using identity $\frac{2\tan\theta}{1+\tan^2\theta} = \sin 2\theta$:

$$y = \sin^{-1}(\sin 2\theta) = 2\theta = 2\tan^{-1}(\sec x)$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = 2 \cdot \frac{1}{1+\sec^2 x} \cdot \frac{d}{dx}(\sec x)$$
$$= \frac{2\sec x \tan x}{1+\sec^2 x}$$

**Final Answer**

(i) $\frac{x}{\sqrt{1-x^4}}$
(ii) $\frac{1}{2\sqrt{1-x^2}}$
(iii) $-\frac{1}{2}$
(iv) $\frac{2\sec x\tan x}{1+\sec^2 x}$

## Question 21

Differentiate the following functions w.r.t. $x$:
(i) $\cos^{-1}\left(\frac{\sin x+\cos x}{\sqrt{2}}\right), -\frac{\pi}{4} < x < \frac{\pi}{4}$
(ii) $\cos^{-1}\left(\frac{3\cos x-4\sin x}{5}\right)$
(iii) $\sin^{-1}\left(\frac{x+\sqrt{1-x^2}}{\sqrt{2}}\right)$
(iv) $\cos^{-1}\left(\frac{3x+4\sqrt{1-x^2}}{5}\right)$

★ **Likely Exam Question**

**TYPE 39 Trigonometric Substitution before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each expression.

**Concept / Formula Used:** $A\cos\theta + B\sin\theta = R\cos(\theta-\alpha) \qquad R = \sqrt{A^2+B^2}$

**Solution:**

(i) Let $y = \cos^{-1}\left(\frac{\sin x+\cos x}{\sqrt{2}}\right)$.
Expanding inside terms:

$$\frac{\cos x+\sin x}{\sqrt{2}} = \cos x\cos\frac{\pi}{4} + \sin x\sin\frac{\pi}{4} = \cos\left(x-\frac{\pi}{4}\right)$$

For $-\frac{\pi}{4} < x < \frac{\pi}{4}$, we have $-\frac{\pi}{2} < x-\frac{\pi}{4} < 0$.
Since $\cos(-\theta) = \cos\theta$, we write $\cos\left(x-\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}-x\right)$ where $0 < \frac{\pi}{4}-x < \frac{\pi}{2}$.

$$y = \cos^{-1}\left(\cos\left(\frac{\pi}{4}-x\right)\right) = \frac{\pi}{4} - x$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -1$$

(ii) Let $y = \cos^{-1}\left(\frac{3\cos x - 4\sin x}{5}\right)$.
Let $\cos\alpha = \frac{3}{5}$ and $\sin\alpha = \frac{4}{5}$, so $\alpha = \tan^{-1}\frac{4}{3}$ is a constant.

$$\frac{3\cos x - 4\sin x}{5} = \cos x\cos\alpha - \sin x\sin\alpha = \cos(x+\alpha)$$
$$y = \cos^{-1}(\cos(x+\alpha)) = x+\alpha$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = 1 + 0 = 1$$

(iii) Let $y = \sin^{-1}\left(\frac{x+\sqrt{1-x^2}}{\sqrt{2}}\right)$.
Let $x = \sin\theta \implies \theta = \sin^{-1}x \implies \sqrt{1-x^2} = \cos\theta$.

$$\frac{x+\sqrt{1-x^2}}{\sqrt{2}} = \frac{\sin\theta + \cos\theta}{\sqrt{2}} = \sin\theta\cos\frac{\pi}{4} + \cos\theta\sin\frac{\pi}{4} = \sin\left(\theta + \frac{\pi}{4}\right)$$
$$y = \sin^{-1}\left(\sin\left(\theta+\frac{\pi}{4}\right)\right) = \theta + \frac{\pi}{4} = \sin^{-1}x + \frac{\pi}{4}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

(iv) Let $y = \cos^{-1}\left(\frac{3x+4\sqrt{1-x^2}}{5}\right)$.
Let $x = \cos\theta \implies \theta = \cos^{-1}x \implies \sqrt{1-x^2} = \sin\theta$.
Let $\cos\alpha = \frac{3}{5}$ and $\sin\alpha = \frac{4}{5} \implies \alpha = \cos^{-1}\frac{3}{5}$ (constant).

$$\frac{3x+4\sqrt{1-x^2}}{5} = \cos\alpha\cos\theta + \sin\alpha\sin\theta = \cos(\theta-\alpha)$$
$$y = \cos^{-1}(\cos(\theta-\alpha)) = \theta - \alpha = \cos^{-1}x - \alpha$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

**Final Answer**

(i) $-1$

(ii) $1$

(iii) $\frac{1}{\sqrt{1-x^2}}$

(iv) $-\frac{1}{\sqrt{1-x^2}}$

## Question 22

Differentiate the following functions w.r.t. $x$:

(i) $\tan^{-1}\left(\frac{4\sqrt{x}}{1-4x}\right)$

(ii) $\tan^{-1}\left(\frac{x^{1/3}+a^{1/3}}{1-x^{1/3}a^{1/3}}\right)$

(iii) $\sin(2\sin^{-1}x)$

(iv) $\cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right)$

**TYPE 39 Trigonometric Substitution before Differentiation**

**To Find:** The derivative $\frac{dy}{dx}$ for each expression.

**Concept / Formula Used:** $\tan^{-1}\left(\frac{u+v}{1-uv}\right) = \tan^{-1}u + \tan^{-1}v$ $\sin(2\theta) = 2\sin\theta\cos\theta$

**Solution:**

(i) Let $y = \tan^{-1}\left(\frac{4\sqrt{x}}{1-4x}\right)$.
Rewriting: $4\sqrt{x} = 2(2\sqrt{x})$ and $4x = (2\sqrt{x})^2$.
Let $u = 2\sqrt{x}$.

$$y = \tan^{-1}\left(\frac{2u}{1-u^2}\right) = 2\tan^{-1}u = 2\tan^{-1}(2\sqrt{x})$$

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= 2\cdot\frac{1}{1+(2\sqrt{x})^2}\cdot\frac{d}{dx}(2\sqrt{x})\\ &= \frac{2}{1+4x}\cdot\left(2\cdot\frac{1}{2\sqrt{x}}\right)\\ &= \frac{2}{\sqrt{x}(1+4x)}\end{aligned}$$

(ii) Let $y = \tan^{-1}\left(\frac{x^{1/3}+a^{1/3}}{1-x^{1/3}a^{1/3}}\right)$.
Using identity $\tan^{-1}\left(\frac{u+v}{1-uv}\right) = \tan^{-1}u + \tan^{-1}v$:

$$y = \tan^{-1}(x^{1/3}) + \tan^{-1}(a^{1/3})$$

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{1+(x^{1/3})^2}\cdot\frac{d}{dx}(x^{1/3}) + 0\\ &= \frac{1}{1+x^{2/3}}\cdot\frac{1}{3}x^{-2/3}\\ &= \frac{1}{3x^{2/3}(1+x^{2/3})}\end{aligned}$$

(iii) Let $y = \sin(2\sin^{-1} x)$.
Let $\sin^{-1} x = \theta \implies \sin\theta = x$ and $\cos\theta = \sqrt{1-x^2}$.

$$y = \sin(2\theta) = 2\sin\theta\cos\theta = 2x\sqrt{1-x^2}$$

Differentiating w.r.t. $x$ using Product Rule:

$$\begin{aligned}
\frac{dy}{dx} &= 2 \cdot \left[\frac{d}{dx}(x) \cdot \sqrt{1-x^2} + x \cdot \frac{d}{dx}\left(\sqrt{1-x^2}\right)\right] \\
&= 2\left[1 \cdot \sqrt{1-x^2} + x \cdot \frac{-2x}{2\sqrt{1-x^2}}\right] \\
&= 2\left[\sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}}\right] \\
&= 2\left[\frac{1-x^2-x^2}{\sqrt{1-x^2}}\right] \\
&= \frac{2(1-2x^2)}{\sqrt{1-x^2}}
\end{aligned}$$

(iv) Let $y = \cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right)$.
Using $1+\sin x = \left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)^2$ and $1-\sin x = \left(\cos\frac{x}{2}-\sin\frac{x}{2}\right)^2$:

$$\begin{aligned}
\sqrt{1+\sin x} &= \cos\frac{x}{2} + \sin\frac{x}{2} \\
\sqrt{1-\sin x} &= \cos\frac{x}{2} - \sin\frac{x}{2} \\
\text{Numerator} &= 2\cos\frac{x}{2} \\
\text{Denominator} &= 2\sin\frac{x}{2} \\
\frac{\text{Numerator}}{\text{Denominator}} &= \frac{2\cos\frac{x}{2}}{2\sin\frac{x}{2}} = \cot\frac{x}{2} \\
y &= \cot^{-1}\left(\cot\frac{x}{2}\right) = \frac{x}{2}
\end{aligned}$$

Differentiating w.r.t. $x$:

$$\frac{dy}{dx} = \frac{1}{2}$$

**Final Answer**

(i) $\frac{2}{\sqrt{x}(1+4x)}$

(ii) $\frac{1}{3x^{2/3}(1+x^{2/3})}$

(iii) $\frac{2(1-2x^2)}{\sqrt{1-x^2}}$

(iv) $\frac{1}{2}$

## Question 23

If $y = \sin(\tan^{-1} e^x)$, then find $\frac{dy}{dx}$ at $x = 0$.

**TYPE 31** **Chain Rule and Evaluation**

**Given:** $y = \sin(\tan^{-1} e^x)$

**To Find:** $\frac{dy}{dx}\big|_{x=0}$

**Concept / Formula Used:** $\sin(\tan^{-1} u) = \frac{u}{\sqrt{1+u^2}}$ $\quad \frac{d}{dx}(e^x) = e^x$

**Solution:**

Let $u = e^x$.
Using trigonometric simplification $\sin(\tan^{-1} u) = \frac{u}{\sqrt{1+u^2}}$:

$$y = \frac{e^x}{\sqrt{1+(e^x)^2}} = \frac{e^x}{\sqrt{1+e^{2x}}}$$

Differentiating $y$ w.r.t. $x$ using Quotient Rule:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{\sqrt{1+e^{2x}} \cdot \frac{d}{dx}(e^x) - e^x \cdot \frac{d}{dx}\left(\sqrt{1+e^{2x}}\right)}{1+e^{2x}} \\
&= \frac{\sqrt{1+e^{2x}} \cdot e^x - e^x \cdot \left(\frac{2e^{2x}}{2\sqrt{1+e^{2x}}}\right)}{1+e^{2x}} \\
&= \frac{\frac{e^x(1+e^{2x}) - e^{3x}}{\sqrt{1+e^{2x}}}}{1+e^{2x}} \\
&= \frac{e^x + e^{3x} - e^{3x}}{(1+e^{2x})^{3/2}} \\
&= \frac{e^x}{(1+e^{2x})^{3/2}}
\end{aligned}$$

Now evaluating at $x = 0$:

$$\begin{aligned}
e^0 &= 1 \\
\frac{dy}{dx}\bigg|_{x=0} &= \frac{1}{(1+1^2)^{3/2}} \\
&= \frac{1}{2^{3/2}} \\
&= \frac{1}{2\sqrt{2}}
\end{aligned}$$

**Final Answer**

$\frac{dy}{dx}\big|_{x=0} = \frac{1}{2\sqrt{2}}$

## Question 24

If $y = \csc(\cot^{-1} x)$, then prove that $\sqrt{1+x^2}\frac{dy}{dx} - x = 0$.

**TYPE 32** **Verifying Differential Relations**

**Given:** $y = \csc(\cot^{-1} x)$

**To Prove:** $\sqrt{1+x^2}\frac{dy}{dx} - x = 0$

**Concept / Formula Used:** $\csc(\cot^{-1} x) = \sqrt{1+x^2}$

**Solution:**

Given function:

$$y = \csc(\cot^{-1} x)$$

Let $\theta = \cot^{-1} x \implies \cot\theta = x$.
Using the identity $\csc\theta = \sqrt{1+\cot^2\theta}$:

$$y = \sqrt{1+x^2}$$

Differentiating both sides w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\left(\sqrt{1+x^2}\right)\\ &= \frac{1}{2\sqrt{1+x^2}} \cdot \frac{d}{dx}(1+x^2)\\ &= \frac{2x}{2\sqrt{1+x^2}}\\ &= \frac{x}{\sqrt{1+x^2}}\end{aligned}$$

Multiplying both sides by $\sqrt{1+x^2}$:

$$\sqrt{1+x^2}\frac{dy}{dx} = x$$
$$\sqrt{1+x^2}\frac{dy}{dx} - x = 0$$

**Hence Proved**

LHS $= \sqrt{1+x^2}\frac{dy}{dx} - x = 0$, hence proved.

## 5.7 – Implicit Differentiation

## Question 1 (i)

Find $\frac{dy}{dx}$ in $x - y = \pi$.

**TYPE 40** **Implicit Differentiation**

**Given:** The implicit equation $x - y = \pi$.

**Concept / Formula Used:** $\frac{d}{dx}(x) = 1 \qquad \frac{d}{dx}(y) = \frac{dy}{dx} \qquad \frac{d}{dx}(c) = 0 \qquad c$

**Solution:**

Given equation:

$$x - y = \pi$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(x - y) = \frac{d}{dx}(\pi)$$
$$\frac{d}{dx}(x) - \frac{d}{dx}(y) = 0$$
$$1 - \frac{dy}{dx} = 0$$

Rearranging to solve for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = 1$$

**Final Answer**

$\frac{dy}{dx} = 1$

## Question 1 (ii)

Find $\frac{dy}{dx}$ in $x^2 + y^2 = 25$.

**TYPE 40** **Implicit Differentiation**

**Given:** The equation of circle $x^2 + y^2 = 25$.

**Concept / Formula Used:** $\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$

**Solution:**

Given equation:

$$x^2 + y^2 = 25$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$$
$$2x + 2y\frac{dy}{dx} = 0$$

Subtracting $2x$ from both sides:

$$2y\frac{dy}{dx} = -2x$$

Dividing both sides by $2y$:

$$\frac{dy}{dx} = -\frac{2x}{2y}$$
$$\frac{dy}{dx} = -\frac{x}{y}$$

**Final Answer**

$\frac{dy}{dx} = -\frac{x}{y}$

## Question 2 (i)

Find $\frac{dy}{dx}$ in $xy = c^2$.

**TYPE 40 Implicit Differentiation**

**Given:** The hyperbola equation $xy = c^2$.

**Concept / Formula Used:** $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$

**Solution:**

Given equation:

$$xy = c^2$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(xy) = \frac{d}{dx}(c^2)$$
$$x\frac{d}{dx}(y) + y\frac{d}{dx}(x) = 0$$
$$x\frac{dy}{dx} + y(1) = 0$$
$$x\frac{dy}{dx} + y = 0$$

Subtracting $y$ from both sides:

$$x\frac{dy}{dx} = -y$$

Dividing both sides by $x$:

$$\frac{dy}{dx} = -\frac{y}{x}$$

**Final Answer**

$\frac{dy}{dx} = -\frac{y}{x}$

## Question 2 (ii)

Find $\frac{dy}{dx}$ in $\frac{x^2}{a^2} + \frac{y^2}{b^2} = c^2$.

**TYPE 40** **Implicit Differentiation**

**Given:** The equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = c^2$.

**Concept / Formula Used:** $\frac{d}{dx}(x^2) = 2x \qquad \frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$

**Solution:**

Given equation:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = c^2$$

Differentiating both sides with respect to $x$:

$$\frac{1}{a^2}\frac{d}{dx}(x^2) + \frac{1}{b^2}\frac{d}{dx}(y^2) = \frac{d}{dx}(c^2)$$
$$\frac{1}{a^2}(2x) + \frac{1}{b^2}\left(2y\frac{dy}{dx}\right) = 0$$
$$\frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} = 0$$

Subtracting $\frac{2x}{a^2}$ from both sides:

$$\frac{2y}{b^2}\frac{dy}{dx} = -\frac{2x}{a^2}$$

Multiplying both sides by $\frac{b^2}{2y}$:

$$\frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y}$$
$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

**Final Answer**

$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$

## Question 3 (i)

Find $\dfrac{dy}{dx}$ in $\sqrt{x}+\sqrt{y}=\sqrt{c}$.

**TYPE 40** **Implicit Differentiation**

**Given:** The equation $\sqrt{x}+\sqrt{y}=\sqrt{c}$.

**Concept / Formula Used:** $\dfrac{d}{dx}(\sqrt{x})=\dfrac{1}{2\sqrt{x}}$ $\quad\dfrac{d}{dx}(\sqrt{y})=\dfrac{1}{2\sqrt{y}}\dfrac{dy}{dx}$

**Solution:**

Given equation:

$$\sqrt{x}+\sqrt{y}=\sqrt{c}$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(\sqrt{x})+\frac{d}{dx}(\sqrt{y})=\frac{d}{dx}(\sqrt{c})$$
$$\frac{1}{2\sqrt{x}}+\frac{1}{2\sqrt{y}}\frac{dy}{dx}=0$$

Subtracting $\dfrac{1}{2\sqrt{x}}$ from both sides:

$$\frac{1}{2\sqrt{y}}\frac{dy}{dx}=-\frac{1}{2\sqrt{x}}$$

Multiplying both sides by $2\sqrt{y}$:

$$\frac{dy}{dx}=-\frac{2\sqrt{y}}{2\sqrt{x}}$$
$$\frac{dy}{dx}=-\sqrt{\frac{y}{x}}$$

**Final Answer**

$\dfrac{dy}{dx}=-\sqrt{\dfrac{y}{x}}$

## Question 3 (ii)

Find $\dfrac{dy}{dx}$ in $x^3+y^3=3axy$.

**TYPE 40** **Implicit Differentiation**

**★ Likely Exam Question**

**Given:** The folium equation $x^3 + y^3 = 3axy$.

**Concept / Formula Used:** $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y \qquad \frac{d}{dx}(y^3) = 3y^2\frac{dy}{dx}$

**Solution:**

Given equation:

$$x^3 + y^3 = 3axy$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}\frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) &= 3a\frac{d}{dx}(xy)\\ 3x^2 + 3y^2\frac{dy}{dx} &= 3a\left(x\frac{dy}{dx} + y(1)\right)\\ 3x^2 + 3y^2\frac{dy}{dx} &= 3ax\frac{dy}{dx} + 3ay\end{aligned}$$

Dividing both sides by 3:

$$x^2 + y^2\frac{dy}{dx} = ax\frac{dy}{dx} + ay$$

Rearranging terms with $\frac{dy}{dx}$ to the left side:

$$y^2\frac{dy}{dx} - ax\frac{dy}{dx} = ay - x^2$$

Factoring out $\frac{dy}{dx}$:

$$(y^2 - ax)\frac{dy}{dx} = ay - x^2$$

Dividing by $(y^2 - ax)$:

$$\frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}$$

**Final Answer**

$\frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}$

## Question 4 (i)

Find $\frac{dy}{dx}$ in $2x + 3y = \sin x$.

**TYPE 40 Implicit Differentiation**

**Given:** The equation $2x + 3y = \sin x$.

**Concept / Formula Used:** $\frac{d}{dx}(\sin x) = \cos x$

**Solution:**

Given equation:

$$2x + 3y = \sin x$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(2x + 3y) = \frac{d}{dx}(\sin x)$$
$$2(1) + 3\frac{dy}{dx} = \cos x$$
$$2 + 3\frac{dy}{dx} = \cos x$$

Subtracting 2 from both sides:

$$3\frac{dy}{dx} = \cos x - 2$$

Dividing by 3:

$$\frac{dy}{dx} = \frac{\cos x - 2}{3}$$

**Final Answer**

$\dfrac{dy}{dx} = \dfrac{\cos x - 2}{3}$

## Question 4 (ii)

Find $\dfrac{dy}{dx}$ in $2x + 3y = \sin y$.

**TYPE 40 Implicit Differentiation**

**Given:** The equation $2x + 3y = \sin y$.

**Concept / Formula Used:** $\dfrac{d}{dx}(\sin y) = \cos y\dfrac{dy}{dx}$

**Solution:**

Given equation:

$$2x + 3y = \sin y$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(2x + 3y) = \frac{d}{dx}(\sin y)$$
$$2 + 3\frac{dy}{dx} = \cos y\frac{dy}{dx}$$

Rearranging terms containing $\dfrac{dy}{dx}$ to the right side:

$$2 = \cos y\frac{dy}{dx} - 3\frac{dy}{dx}$$
$$2 = (\cos y - 3)\frac{dy}{dx}$$

Dividing both sides by $(\cos y - 3)$:

$$\frac{dy}{dx} = \frac{2}{\cos y - 3}$$

**Final Answer**

$$\frac{dy}{dx} = \frac{2}{\cos y - 3}$$

## Question 5 (i)

Find $\frac{dy}{dx}$ when $(x^2 + y^2)^2 = xy$.

**TYPE 40** **Implicit Differentiation**

**Given:** The equation $(x^2 + y^2)^2 = xy$.

**Concept / Formula Used:** $\frac{d}{dx}(u^2) = 2u\frac{du}{dx} \qquad \frac{d}{dx}(xy) = x\frac{dy}{dx} + y$

**Solution:**

Given equation:

$$(x^2 + y^2)^2 = xy$$

Differentiating both sides with respect to $x$:

$$\begin{aligned} \frac{d}{dx}\left((x^2 + y^2)^2\right) &= \frac{d}{dx}(xy) \\ 2(x^2 + y^2)\frac{d}{dx}(x^2 + y^2) &= x\frac{dy}{dx} + y(1) \\ 2(x^2 + y^2)\left(2x + 2y\frac{dy}{dx}\right) &= x\frac{dy}{dx} + y \end{aligned}$$

Expanding the left-hand side:

$$\begin{aligned} 4x(x^2 + y^2) + 4y(x^2 + y^2)\frac{dy}{dx} &= x\frac{dy}{dx} + y \\ (4x^3 + 4xy^2) + (4x^2y + 4y^3)\frac{dy}{dx} &= x\frac{dy}{dx} + y \end{aligned}$$

Grouping all terms containing $\frac{dy}{dx}$ on the left-hand side:

$$\begin{aligned} (4x^2y + 4y^3)\frac{dy}{dx} - x\frac{dy}{dx} &= y - (4x^3 + 4xy^2) \\ (4x^2y + 4y^3 - x)\frac{dy}{dx} &= y - 4x^3 - 4xy^2 \end{aligned}$$

Dividing both sides by $(4x^2y + 4y^3 - x)$:

$$\frac{dy}{dx} = \frac{y - 4x^3 - 4xy^2}{4x^2y + 4y^3 - x}$$

**Final Answer**

$$\frac{dy}{dx} = \frac{y - 4x^3 - 4xy^2}{4x^2y + 4y^3 - x}$$

## Question 5 (ii)

Find $\dfrac{dy}{dx}$ when $\sin(x + y) = \dfrac{2}{3}$.

**TYPE 40** **Implicit Differentiation**

**Given:** The equation $\sin(x + y) = \dfrac{2}{3}$.

**Concept / Formula Used:** $\dfrac{d}{dx}(\sin u) = \cos u \dfrac{du}{dx}$

**Solution:**

Given equation:

$$\sin(x + y) = \frac{2}{3}$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}\frac{d}{dx}(\sin(x + y)) &= \frac{d}{dx}\left(\frac{2}{3}\right)\\ \cos(x + y) \cdot \frac{d}{dx}(x + y) &= 0\\ \cos(x + y)\left(1 + \frac{dy}{dx}\right) &= 0\end{aligned}$$

Since $\sin(x + y) = \dfrac{2}{3}$, $\cos(x + y) = \pm\sqrt{1 - \left(\dfrac{2}{3}\right)^2} = \pm\dfrac{\sqrt{5}}{3} \neq 0$. Dividing both sides by $\cos(x + y)$:

$$\begin{aligned}1 + \frac{dy}{dx} &= 0\\ \frac{dy}{dx} &= -1\end{aligned}$$

**Final Answer**

$$\frac{dy}{dx} = -1$$

## Question 6

If $x^{2/3} + y^{2/3} = 2$, find $\dfrac{dy}{dx}$ at $(1, 1)$.

**TYPE 41** **Implicit Differentiation at a Point**

## ★ Likely Exam Question

**Given:** The astroid equation $x^{2/3} + y^{2/3} = 2$ and the point $(x, y) = (1, 1)$.

**Concept / Formula Used:** $\frac{d}{dx}(x^n) = nx^{n-1}$

**Solution:**

Given equation:

$$x^{2/3} + y^{2/3} = 2$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}\frac{d}{dx}\left(x^{2/3}\right) + \frac{d}{dx}\left(y^{2/3}\right) &= \frac{d}{dx}(2)\\ \frac{2}{3}x^{2/3-1} + \frac{2}{3}y^{2/3-1}\frac{dy}{dx} &= 0\\ \frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}\frac{dy}{dx} &= 0\end{aligned}$$

Multiplying through by $\frac{3}{2}$:

$$x^{-1/3} + y^{-1/3}\frac{dy}{dx} = 0$$

Isolating $\frac{dy}{dx}$:

$$\begin{aligned}y^{-1/3}\frac{dy}{dx} &= -x^{-1/3}\\ \frac{dy}{dx} &= -\frac{x^{-1/3}}{y^{-1/3}}\\ \frac{dy}{dx} &= -\left(\frac{y}{x}\right)^{1/3}\end{aligned}$$

Substituting $(x, y) = (1, 1)$:

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{(1,1)} &= -\left(\frac{1}{1}\right)^{1/3}\\ &= -1\end{aligned}$$

**Final Answer**

$$\left.\frac{dy}{dx}\right|_{(1,1)} = -1$$

## Question 7 (i)

Use implicit differentiation to verify that $\dfrac{dy}{dx} \cdot \dfrac{dx}{dy} = 1$, when $y^2 = 4ax$.

**TYPE 42** **Verifying Reciprocal Derivative Relation**

**To Prove:** $\dfrac{dy}{dx} \cdot \dfrac{dx}{dy} = 1$ for the parabola $y^2 = 4ax$.

**Concept / Formula Used:** $x$ $y$

**Solution:**

Given equation:

$$y^2 = 4ax$$

Differentiating with respect to $x$:

$$\begin{aligned} \frac{d}{dx}(y^2) &= \frac{d}{dx}(4ax) \\ 2y\frac{dy}{dx} &= 4a \\ \frac{dy}{dx} &= \frac{4a}{2y} = \frac{2a}{y} \quad \text{— (Equation 1)} \end{aligned}$$

Differentiating $y^2 = 4ax$ with respect to $y$:

$$\begin{aligned} \frac{d}{dy}(y^2) &= \frac{d}{dy}(4ax) \\ 2y &= 4a\frac{dx}{dy} \\ \frac{dx}{dy} &= \frac{2y}{4a} = \frac{y}{2a} \quad \text{— (Equation 2)} \end{aligned}$$

Multiplying Equation 1 and Equation 2:

$$\begin{aligned} \frac{dy}{dx} \cdot \frac{dx}{dy} &= \left(\frac{2a}{y}\right) \cdot \left(\frac{y}{2a}\right) \\ &= 1 \end{aligned}$$

**Hence Proved**

$\dfrac{dy}{dx} \cdot \dfrac{dx}{dy} = 1$ is verified.

## Question 7 (ii)

Use implicit differentiation to verify that $\dfrac{dy}{dx} \cdot \dfrac{dx}{dy} = 1$, when $x^4 + y^4 = 3x^2y^2$.

**TYPE 42** **Verifying Reciprocal Derivative Relation**

**To Prove:** $\dfrac{dy}{dx} \cdot \dfrac{dx}{dy} = 1$ for $x^4 + y^4 = 3x^2y^2$.

**Concept / Formula Used:** $x$ $y$

**Solution:**

Given equation:

$$x^4 + y^4 = 3x^2y^2$$

First, differentiating both sides with respect to $x$:

$$\begin{aligned}
\frac{d}{dx}(x^4 + y^4) &= \frac{d}{dx}(3x^2y^2) \\
4x^3 + 4y^3\frac{dy}{dx} &= 3\left(x^2 \cdot 2y\frac{dy}{dx} + y^2 \cdot 2x\right) \\
4x^3 + 4y^3\frac{dy}{dx} &= 6x^2y\frac{dy}{dx} + 6xy^2
\end{aligned}$$

Grouping terms with $\dfrac{dy}{dx}$:

$$\begin{aligned}
(4y^3 - 6x^2y)\frac{dy}{dx} &= 6xy^2 - 4x^3 \\
2y(2y^2 - 3x^2)\frac{dy}{dx} &= 2x(3y^2 - 2x^2) \\
\frac{dy}{dx} &= \frac{x(3y^2 - 2x^2)}{y(2y^2 - 3x^2)} \quad \text{— (Equation 1)}
\end{aligned}$$

Next, differentiating $x^4 + y^4 = 3x^2y^2$ with respect to $y$:

$$\begin{aligned}
\frac{d}{dy}(x^4 + y^4) &= \frac{d}{dy}(3x^2y^2) \\
4x^3\frac{dx}{dy} + 4y^3 &= 3\left(x^2 \cdot 2y + y^2 \cdot 2x\frac{dx}{dy}\right) \\
4x^3\frac{dx}{dy} + 4y^3 &= 6x^2y + 6xy^2\frac{dx}{dy}
\end{aligned}$$

Grouping terms with $\dfrac{dx}{dy}$:

$$\begin{aligned}
(4x^3 - 6xy^2)\frac{dx}{dy} &= 6x^2y - 4y^3 \\
2x(2x^2 - 3y^2)\frac{dx}{dy} &= 2y(3x^2 - 2y^2) \\
\frac{dx}{dy} &= \frac{y(3x^2 - 2y^2)}{x(2x^2 - 3y^2)} \quad \text{— (Equation 2)}
\end{aligned}$$

Note that $(3x^2 - 2y^2) = -(2y^2 - 3x^2)$ and $(2x^2 - 3y^2) = -(3y^2 - 2x^2)$. Thus:

$$\frac{dx}{dy} = \frac{y(2y^2 - 3x^2)}{x(3y^2 - 2x^2)}$$

Multiplying Equation 1 and Equation 2:

$$\begin{aligned} \frac{dy}{dx} \cdot \frac{dx}{dy} &= \left[\frac{x(3y^2 - 2x^2)}{y(2y^2 - 3x^2)}\right] \cdot \left[\frac{y(2y^2 - 3x^2)}{x(3y^2 - 2x^2)}\right] \\ &= 1 \end{aligned}$$

**Hence Proved**

$\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ is verified.

## Question 8 (i)

Find $\frac{dy}{dx}$, when $y + \sin y = \cos x$.

**TYPE 40** **Implicit Differentiation**

**Given:** The equation $y + \sin y = \cos x$.

**Concept / Formula Used:** $\frac{d}{dx}(\sin y) = \cos y \frac{dy}{dx} \qquad \frac{d}{dx}(\cos x) = -\sin x$

**Solution:**

Given equation:

$$y + \sin y = \cos x$$

Differentiating both sides with respect to $x$:

$$\begin{aligned} \frac{d}{dx}(y) + \frac{d}{dx}(\sin y) &= \frac{d}{dx}(\cos x) \\ \frac{dy}{dx} + \cos y \frac{dy}{dx} &= -\sin x \end{aligned}$$

Factoring out $\frac{dy}{dx}$:

$$(1 + \cos y)\frac{dy}{dx} = -\sin x$$

Dividing both sides by $(1 + \cos y)$:

$$\frac{dy}{dx} = -\frac{\sin x}{1 + \cos y}$$

**Final Answer**

$\frac{dy}{dx} = -\frac{\sin x}{1 + \cos y}$

## Question 8 (ii)

Find $\dfrac{dy}{dx}$, when $ax + by^2 = \cos y$.

**TYPE 40** Implicit Differentiation

**Given:** The equation $ax + by^2 = \cos y$.

**Concept / Formula Used:** $\dfrac{d}{dx}(by^2) = 2by\dfrac{dy}{dx}$ $\quad \dfrac{d}{dx}(\cos y) = -\sin y\dfrac{dy}{dx}$

**Solution:**

Given equation:

$$ax + by^2 = \cos y$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}\frac{d}{dx}(ax + by^2) &= \frac{d}{dx}(\cos y)\\ a(1) + 2by\frac{dy}{dx} &= -\sin y\frac{dy}{dx}\\ a + 2by\frac{dy}{dx} &= -\sin y\frac{dy}{dx}\end{aligned}$$

Rearranging terms with $\dfrac{dy}{dx}$ to one side:

$$\begin{aligned}2by\frac{dy}{dx} + \sin y\frac{dy}{dx} &= -a\\ (2by + \sin y)\frac{dy}{dx} &= -a\end{aligned}$$

Dividing both sides by $(2by + \sin y)$:

$$\frac{dy}{dx} = -\frac{a}{2by + \sin y} = \frac{a}{-\sin y - 2by}$$

**Final Answer**

$\dfrac{dy}{dx} = \dfrac{a}{-\sin y - 2by}$

## Question 8 (iii)

Find $\dfrac{dy}{dx}$, when $y\sec x + \tan x + x^2y = 0$.

**TYPE 40** Implicit Differentiation

**Given:** The equation $y\sec x + \tan x + x^2y = 0$.

**Concept / Formula Used:** $\frac{d}{dx}(\sec x) = \sec x \tan x \qquad \frac{d}{dx}(\tan x) = \sec^2 x$

### Solution:

Given equation:

$$y \sec x + \tan x + x^2 y = 0$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(y \sec x) + \frac{d}{dx}(\tan x) + \frac{d}{dx}(x^2 y) = 0$$
$$\left(y \cdot \sec x \tan x + \sec x \frac{dy}{dx}\right) + \sec^2 x + \left(x^2 \frac{dy}{dx} + y \cdot 2x\right) = 0$$

Grouping terms containing $\frac{dy}{dx}$:

$$(\sec x + x^2)\frac{dy}{dx} + y \sec x \tan x + \sec^2 x + 2xy = 0$$

Transferring non-$\frac{dy}{dx}$ terms to the right side:

$$(\sec x + x^2)\frac{dy}{dx} = -(2xy + \sec^2 x + y \sec x \tan x)$$

Dividing both sides by $(\sec x + x^2)$:

$$\frac{dy}{dx} = -\frac{2xy + \sec^2 x + y \sec x \tan x}{x^2 + \sec x}$$

**Final Answer**

$$\frac{dy}{dx} = -\frac{2xy + \sec^2 x + y \sec x \tan x}{x^2 + \sec x}$$

## Question 8 (iv)

Find $\frac{dy}{dx}$, when $\sin^2 x + \cos^2 y = 1$.

**TYPE 40** **Implicit Differentiation**

**Given:** The equation $\sin^2 x + \cos^2 y = 1$.

**Concept / Formula Used:** $\sin 2\theta = 2 \sin\theta \cos\theta$

### Solution:

Given equation:

$$\sin^2 x + \cos^2 y = 1$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(\sin^2 x) + \frac{d}{dx}(\cos^2 y) = \frac{d}{dx}(1)$$
$$2\sin x \frac{d}{dx}(\sin x) + 2\cos y \frac{d}{dx}(\cos y) = 0$$
$$2\sin x \cos x + 2\cos y \left(-\sin y \frac{dy}{dx}\right) = 0$$
$$2\sin x \cos x - 2\sin y \cos y \frac{dy}{dx} = 0$$

Applying $\sin 2\theta = 2\sin\theta\cos\theta$:

$$\sin 2x - \sin 2y \frac{dy}{dx} = 0$$

Isolating $\frac{dy}{dx}$:

$$\sin 2y \frac{dy}{dx} = \sin 2x$$
$$\frac{dy}{dx} = \frac{\sin 2x}{\sin 2y}$$

**Final Answer**

$\frac{dy}{dx} = \frac{\sin 2x}{\sin 2y}$

## Question 8 (v)

Find $\frac{dy}{dx}$, when $xy + y^2 = \tan x + y$.

**TYPE 40** **Implicit Differentiation**

**Given:** The equation $xy + y^2 = \tan x + y$.

**Concept / Formula Used:** $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y \qquad \frac{d}{dx}(\tan x) = \sec^2 x$

**Solution:**

Given equation:

$$xy + y^2 = \tan x + y$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(xy) + \frac{d}{dx}(y^2) = \frac{d}{dx}(\tan x) + \frac{d}{dx}(y)$$
$$\left(x\frac{dy}{dx} + y(1)\right) + 2y\frac{dy}{dx} = \sec^2 x + \frac{dy}{dx}$$
$$x\frac{dy}{dx} + y + 2y\frac{dy}{dx} = \sec^2 x + \frac{dy}{dx}$$

Grouping all terms containing $\frac{dy}{dx}$ on the left side:

$$x\frac{dy}{dx} + 2y\frac{dy}{dx} - \frac{dy}{dx} = \sec^2 x - y$$
$$(x + 2y - 1)\frac{dy}{dx} = \sec^2 x - y$$

Dividing both sides by $(x + 2y - 1)$:

$$\frac{dy}{dx} = \frac{\sec^2 x - y}{x + 2y - 1}$$

**Final Answer**

$\frac{dy}{dx} = \frac{\sec^2 x - y}{x + 2y - 1}$

## Question 8 (vi)

Find $\frac{dy}{dx}$, when $y = \tan(x + y)$.

**TYPE 40 Implicit Differentiation**

**Given:** The equation $y = \tan(x + y)$.

**Concept / Formula Used:** $\frac{d}{dx}(\tan u) = \sec^2 u\frac{du}{dx}$ $\quad 1 - \sec^2 u = -\tan^2 u$

**Solution:**

Given equation:

$$y = \tan(x + y)$$

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = \frac{d}{dx}(\tan(x + y))$$
$$\frac{dy}{dx} = \sec^2(x + y) \cdot \frac{d}{dx}(x + y)$$
$$\frac{dy}{dx} = \sec^2(x + y)\left(1 + \frac{dy}{dx}\right)$$
$$\frac{dy}{dx} = \sec^2(x + y) + \sec^2(x + y)\frac{dy}{dx}$$

Grouping terms containing $\frac{dy}{dx}$ on the left side:

$$\frac{dy}{dx} - \sec^2(x + y)\frac{dy}{dx} = \sec^2(x + y)$$
$$\left(1 - \sec^2(x + y)\right)\frac{dy}{dx} = \sec^2(x + y)$$

Using $1-\sec^2(x+y) = -\tan^2(x+y)$:

$$-\tan^2(x+y)\frac{dy}{dx} = \sec^2(x+y)$$

Dividing both sides by $-\tan^2(x+y)$:

$$\begin{aligned}
\frac{dy}{dx} &= -\frac{\sec^2(x+y)}{\tan^2(x+y)} \\
&= -\frac{\frac{1}{\cos^2(x+y)}}{\frac{\sin^2(x+y)}{\cos^2(x+y)}} \\
&= -\frac{1}{\sin^2(x+y)} \\
&= -\csc^2(x+y)
\end{aligned}$$

**Final Answer**

$\dfrac{dy}{dx} = -\csc^2(x+y)$

## Question 8 (vii)

Find $\dfrac{dy}{dx}$, when $\sin(xy) + \dfrac{x}{y} = x^2 - y$.

**TYPE 40** **Implicit Differentiation**

**Given:** The equation $\sin(xy) + \dfrac{x}{y} = x^2 - y$.

**Concept / Formula Used:** $\dfrac{d}{dx}\left(\dfrac{x}{y}\right) = \dfrac{y(1) - x\dfrac{dy}{dx}}{y^2}$

**Solution:**

Given equation:

$$\sin(xy) + \frac{x}{y} = x^2 - y$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}
\frac{d}{dx}(\sin(xy)) + \frac{d}{dx}\left(\frac{x}{y}\right) &= \frac{d}{dx}(x^2) - \frac{d}{dx}(y) \\
\cos(xy)\left(x\frac{dy}{dx} + y\right) + \frac{y(1) - x\frac{dy}{dx}}{y^2} &= 2x - \frac{dy}{dx} \\
x\cos(xy)\frac{dy}{dx} + y\cos(xy) + \frac{1}{y} - \frac{x}{y^2}\frac{dy}{dx} &= 2x - \frac{dy}{dx}
\end{aligned}$$

Multiplying the entire equation by $y^2$:

$$xy^2\cos(xy)\frac{dy}{dx} + y^3\cos(xy) + y - x\frac{dy}{dx} = 2xy^2 - y^2\frac{dy}{dx}$$

Grouping terms containing $\frac{dy}{dx}$ on the left-hand side:

$$\left(xy^2\cos(xy) - x + y^2\right)\frac{dy}{dx} = 2xy^2 - y - y^3\cos(xy)$$

Dividing by $(xy^2\cos(xy) - x + y^2)$:

$$\frac{dy}{dx} = \frac{2xy^2 - y - y^3\cos(xy)}{xy^2\cos(xy) - x + y^2}$$

**Final Answer**

$\frac{dy}{dx} = \frac{2xy^2 - y - y^3\cos(xy)}{xy^2\cos(xy) - x + y^2}$

## Question 8 (viii)

Find $\frac{dy}{dx}$, when $\sec(x + y) = xy$.

**TYPE 40 Implicit Differentiation**

**Given:** The equation $\sec(x + y) = xy$.

**Concept / Formula Used:** $\frac{d}{dx}(\sec u) = \sec u \tan u \frac{du}{dx}$

**Solution:**

Given equation:

$$\sec(x + y) = xy$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(\sec(x + y)) = \frac{d}{dx}(xy)$$

$$\sec(x + y)\tan(x + y) \cdot \left(1 + \frac{dy}{dx}\right) = x\frac{dy}{dx} + y(1)$$

$$\sec(x + y)\tan(x + y) + \sec(x + y)\tan(x + y)\frac{dy}{dx} = x\frac{dy}{dx} + y$$

Grouping terms containing $\frac{dy}{dx}$ on the left-hand side:

$$(\sec(x + y)\tan(x + y) - x)\frac{dy}{dx} = y - \sec(x + y)\tan(x + y)$$

Dividing both sides by $(\sec(x + y)\tan(x + y) - x)$:

$$\frac{dy}{dx} = \frac{y - \sec(x + y)\tan(x + y)}{\sec(x + y)\tan(x + y) - x}$$

**Final Answer**

$\frac{dy}{dx} = \frac{y - \sec(x + y)\tan(x + y)}{\sec(x + y)\tan(x + y) - x}$

## Question 9

If $\sec\left(\frac{x+y}{x-y}\right) = a$, prove that $\frac{dy}{dx} = \frac{y}{x}$.

**TYPE 43** **Simplification before Implicit Differentiation**

★ **Likely Exam Question**

**To Prove:** $\frac{dy}{dx} = \frac{y}{x}$ given $\sec\left(\frac{x+y}{x-y}\right) = a$.

**Concept / Formula Used:** $f(x, y) = \text{constant}$ $\quad \frac{x+y}{x-y} = \text{constant}$

**Solution:**

Given equation:

$$\sec\left(\frac{x+y}{x-y}\right) = a$$

Taking inverse secant on both sides:

$$\frac{x+y}{x-y} = \sec^{-1}(a)$$

Since $a$ is a constant, $\sec^{-1}(a)$ is also a constant. Let $\sec^{-1}(a) = k$.

$$\frac{x+y}{x-y} = k$$

Cross-multiplying:

$$x + y = k(x - y)$$
$$x + y = kx - ky$$

Rearranging to isolate $y$ terms on one side:

$$y + ky = kx - x$$
$$y(1 + k) = x(k - 1)$$
$$\frac{y}{x} = \frac{k-1}{k+1}$$

Let the constant $\frac{k-1}{k+1} = m$. Thus:

$$y = mx$$

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = m$$

Since $m = \frac{y}{x}$:

$$\frac{dy}{dx} = \frac{y}{x}$$

**Hence Proved**

Hence proved.

## Question 10 (i)

If $y = x\sin(a+y)$, prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin(a+y) - y\cos(a+y)}$.

**TYPE 44** **Differentiation by Isolating Variables**

**★ Likely Exam Question**

**To Prove:** $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin(a+y) - y\cos(a+y)}$ given $y = x\sin(a+y)$.

**Concept / Formula Used:** $\frac{d}{dy}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$ $\qquad \frac{dy}{dx} = \frac{1}{dx/dy}$

**Solution:**

Given relation:

$$y = x\sin(a+y)$$

Isolating $x$ in terms of $y$:

$$x = \frac{y}{\sin(a+y)}$$

Differentiating $x$ with respect to $y$ using the Quotient Rule:

$$\frac{dx}{dy} = \frac{\sin(a+y)\cdot\frac{d}{dy}(y) - y\cdot\frac{d}{dy}(\sin(a+y))}{\sin^2(a+y)}$$

$$\frac{dx}{dy} = \frac{\sin(a+y)\cdot 1 - y\cdot\cos(a+y)}{\sin^2(a+y)}$$

$$\frac{dx}{dy} = \frac{\sin(a+y) - y\cos(a+y)}{\sin^2(a+y)}$$

Taking the reciprocal to find $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{\sin^2(a+y)}{\sin(a+y) - y\cos(a+y)}$$

**Hence Proved**

Hence proved.

## Question 10 (ii)

If $\sin x = y \sin(x+a)$, prove that $\dfrac{dy}{dx} = \dfrac{\sin a}{\sin^2(x+a)}$.

**TYPE 44** **Differentiation by Isolating Variables**

★ **Likely Exam Question**

**To Prove:** $\dfrac{dy}{dx} = \dfrac{\sin a}{\sin^2(x+a)}$ given $\sin x = y \sin(x+a)$.

**Concept / Formula Used:** $\sin(A-B) = \sin A \cos B - \cos A \sin B$

**Solution:**

Given equation:

$$\sin x = y \sin(x+a)$$

Isolating $y$ in terms of $x$:

$$y = \frac{\sin x}{\sin(x+a)}$$

Differentiating $y$ with respect to $x$ using the Quotient Rule:

$$\frac{dy}{dx} = \frac{\sin(x+a) \cdot \frac{d}{dx}(\sin x) - \sin x \cdot \frac{d}{dx}(\sin(x+a))}{\sin^2(x+a)}$$

$$\frac{dy}{dx} = \frac{\sin(x+a)\cos x - \sin x \cos(x+a)}{\sin^2(x+a)}$$

Applying $\sin(A-B) = \sin A \cos B - \cos A \sin B$ with $A = x+a$ and $B = x$:

$$\begin{aligned} \sin(x+a)\cos x - \sin x \cos(x+a) &= \sin((x+a)-x) \\ &= \sin a \end{aligned}$$

Substituting this into the numerator:

$$\frac{dy}{dx} = \frac{\sin a}{\sin^2(x+a)}$$

**Hence Proved**

Hence proved.

## Question 10 (iii)

If $\sin y = x \sin(a+y)$, prove that $\dfrac{dy}{dx} = \dfrac{\sin^2(a+y)}{\sin a}$.

**TYPE 44** **Differentiation by Isolating Variables**

### ★ Likely Exam Question

**To Prove:** $\dfrac{dy}{dx} = \dfrac{\sin^2(a+y)}{\sin a}$ given $\sin y = x\sin(a+y)$.

**Concept / Formula Used:** $\sin(A-B) = \sin A\cos B - \cos A\sin B$

**Solution:**

Given equation:

$$\sin y = x\sin(a+y)$$

Isolating $x$ in terms of $y$:

$$x = \frac{\sin y}{\sin(a+y)}$$

Differentiating $x$ with respect to $y$ using the Quotient Rule:

$$\frac{dx}{dy} = \frac{\sin(a+y)\cdot\frac{d}{dy}(\sin y) - \sin y\cdot\frac{d}{dy}(\sin(a+y))}{\sin^2(a+y)}$$
$$\frac{dx}{dy} = \frac{\sin(a+y)\cos y - \sin y\cos(a+y)}{\sin^2(a+y)}$$

Applying $\sin(A-B) = \sin A\cos B - \cos A\sin B$ with $A = a+y$ and $B = y$:

$$\begin{aligned}\sin(a+y)\cos y - \cos(a+y)\sin y &= \sin((a+y)-y)\\ &= \sin a\end{aligned}$$

Thus:

$$\frac{dx}{dy} = \frac{\sin a}{\sin^2(a+y)}$$

Taking the reciprocal to get $\dfrac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{\sin^2(a+y)}{\sin a}$$

**Hence Proved**

Hence proved.

## Question 11

If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3(x^3-y^3)$, prove that $\dfrac{dy}{dx} = \dfrac{x^2}{y^2}\sqrt{\dfrac{1-y^6}{1-x^6}}$.

**TYPE 39** Trigonometric Substitution before Differentiation

### ★ Likely Exam Question

**To Prove:** $\dfrac{dy}{dx} = \dfrac{x^2}{y^2}\sqrt{\dfrac{1-y^6}{1-x^6}}$ given $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3(x^3 - y^3)$.

**Concept / Formula Used:** $x^3 = \sin\theta$ $y^3 = \sin\phi$ $\cos\theta + \cos\phi = 2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right)$

**Solution:**

Let $x^3 = \sin\theta \implies \theta = \sin^{-1}(x^3)$
and $y^3 = \sin\phi \implies \phi = \sin^{-1}(y^3)$.

Substituting these into the given equation:

$$\begin{aligned}\sqrt{1-\sin^2\theta} + \sqrt{1-\sin^2\phi} &= a^3(\sin\theta - \sin\phi) \\ \cos\theta + \cos\phi &= a^3(\sin\theta - \sin\phi)\end{aligned}$$

Applying the sum-to-product formulas:

$$2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = a^3 \cdot 2\cos\left(\frac{\theta+\phi}{2}\right)\sin\left(\frac{\theta-\phi}{2}\right)$$

Cancelling $2\cos\left(\dfrac{\theta+\phi}{2}\right)$ on both sides:

$$\begin{aligned}\cos\left(\frac{\theta-\phi}{2}\right) &= a^3\sin\left(\frac{\theta-\phi}{2}\right) \\ \cot\left(\frac{\theta-\phi}{2}\right) &= a^3 \\ \frac{\theta-\phi}{2} &= \cot^{-1}(a^3) \\ \theta - \phi &= 2\cot^{-1}(a^3)\end{aligned}$$

Substituting back $\theta = \sin^{-1}(x^3)$ and $\phi = \sin^{-1}(y^3)$:

$$\sin^{-1}(x^3) - \sin^{-1}(y^3) = 2\cot^{-1}(a^3)$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}\frac{d}{dx}\left(\sin^{-1}(x^3)\right) - \frac{d}{dx}\left(\sin^{-1}(y^3)\right) &= \frac{d}{dx}\left(2\cot^{-1}(a^3)\right) \\ \frac{1}{\sqrt{1-(x^3)^2}} \cdot 3x^2 - \frac{1}{\sqrt{1-(y^3)^2}} \cdot 3y^2\frac{dy}{dx} &= 0 \\ \frac{3x^2}{\sqrt{1-x^6}} - \frac{3y^2}{\sqrt{1-y^6}}\frac{dy}{dx} &= 0\end{aligned}$$

Isolating $\frac{dy}{dx}$:

$$\frac{3y^2}{\sqrt{1-y^6}}\frac{dy}{dx} = \frac{3x^2}{\sqrt{1-x^6}}$$
$$\frac{dy}{dx} = \frac{3x^2}{\sqrt{1-x^6}} \cdot \frac{\sqrt{1-y^6}}{3y^2}$$
$$\frac{dy}{dx} = \frac{x^2}{y^2}\sqrt{\frac{1-y^6}{1-x^6}}$$

**Hence Proved**

Hence proved.

## Question 12 (i)

If $y = \sqrt{x + \sqrt{x + \sqrt{x + \ldots \infty}}}$, prove that $(2y-1)\frac{dy}{dx} = 1$.

**TYPE 45** **Differentiation of Infinite Recursive Functions**

**★ Likely Exam Question**

**To Prove:** $(2y-1)\frac{dy}{dx} = 1$ given $y = \sqrt{x + \sqrt{x + \ldots \infty}}$.

**Concept / Formula Used:** $y = \sqrt{x+y}$

**Solution:**

Given infinite radical series:

$$y = \sqrt{x + \sqrt{x + \sqrt{x + \ldots \infty}}}$$

Using self-similarity, rewrite as:

$$y = \sqrt{x+y}$$

Squaring both sides:

$$y^2 = x + y$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(y^2) = \frac{d}{dx}(x+y)$$
$$2y\frac{dy}{dx} = 1 + \frac{dy}{dx}$$

Subtracting $\frac{dy}{dx}$ from both sides:

$$2y\frac{dy}{dx} - \frac{dy}{dx} = 1$$
$$(2y-1)\frac{dy}{dx} = 1$$

**Hence Proved**

Hence proved.

## Question 12 (ii)

If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \ldots \infty}}}$, prove that $(2y-1)\dfrac{dy}{dx} = \cos x$.

TYPE 45 **Differentiation of Infinite Recursive Functions**

★ **Likely Exam Question**

**To Prove:** $(2y-1)\dfrac{dy}{dx} = \cos x$ given $y = \sqrt{\sin x + \sqrt{\sin x + \ldots \infty}}$.

**Concept / Formula Used:** $y = \sqrt{\sin x + y}$

### Solution:

Given infinite radical series:

$$y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \ldots \infty}}}$$

Using self-similarity, rewrite as:

$$y = \sqrt{\sin x + y}$$

Squaring both sides:

$$y^2 = \sin x + y$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(y^2) = \frac{d}{dx}(\sin x + y)$$
$$2y\frac{dy}{dx} = \cos x + \frac{dy}{dx}$$

Subtracting $\dfrac{dy}{dx}$ from both sides:

$$2y\frac{dy}{dx} - \frac{dy}{dx} = \cos x$$
$$(2y-1)\frac{dy}{dx} = \cos x$$

**Hence Proved**

Hence proved.

## Question 12 (iii)

If $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \ldots \infty}}}$, prove that $(2y-1)\dfrac{dy}{dx} = \sec^2 x$.

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**To Prove:** $(2y-1)\dfrac{dy}{dx} = \sec^2 x$ given $y = \sqrt{\tan x + \sqrt{\tan x + \ldots \infty}}$.

**Concept / Formula Used:** $y = \sqrt{\tan x + y}$

**Solution:**

Given infinite radical series:

$$y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \ldots \infty}}}$$

Using self-similarity, rewrite as:

$$y = \sqrt{\tan x + y}$$

Squaring both sides:

$$y^2 = \tan x + y$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}(y^2) = \frac{d}{dx}(\tan x + y)$$
$$2y\frac{dy}{dx} = \sec^2 x + \frac{dy}{dx}$$

Subtracting $\dfrac{dy}{dx}$ from both sides:

$$2y\frac{dy}{dx} - \frac{dy}{dx} = \sec^2 x$$
$$(2y-1)\frac{dy}{dx} = \sec^2 x$$

**Hence Proved**

Hence proved.

## Question 13

If $y = x + \cfrac{1}{x + \cfrac{1}{x + \ldots \text{ to } \infty}}$, prove that $\dfrac{dy}{dx} = \dfrac{y^2}{1+y^2}$.

**TYPE 45** Differentiation of Infinite Recursive Functions

★ **Likely Exam Question**

**To Prove:** $\frac{dy}{dx} = \frac{y^2}{1+y^2}$ given $y = x + \cfrac{1}{x + \cfrac{1}{x + \dots \text{ to } \infty}}$.

**Concept / Formula Used:** $y = x + \frac{1}{y}$

**Solution:**

Given infinite continued fraction:

$$y = x + \frac{1}{x + \frac{1}{x + \dots \text{ to } \infty}}$$

Using self-similarity, rewrite as:

$$y = x + \frac{1}{y}$$

Isolating $x$ in terms of $y$:

$$x = y - \frac{1}{y}$$

Differentiating $x$ with respect to $y$:

$$\frac{dx}{dy} = \frac{d}{dy}\left(y - y^{-1}\right)$$

$$\frac{dx}{dy} = 1 - (-1)y^{-2}$$

$$\frac{dx}{dy} = 1 + \frac{1}{y^2}$$

$$\frac{dx}{dy} = \frac{y^2+1}{y^2}$$

Taking the reciprocal to evaluate $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{y^2}{1+y^2}$$

**Hence Proved**

Hence proved.

## 5.8 – Differentiation of Exponential and Logarithmic Functions

## Question 1

Differentiate the following functions w.r.t. $x$:

(i) $\log x + \dfrac{1}{x}$

(ii) $10^x + \dfrac{1}{3}e^x - 2\log x$

**TYPE 20** **Sum and Product Rules**

**To Find:** The derivative of (i) $y = \log x + \dfrac{1}{x}$ and (ii) $y = 10^x + \dfrac{1}{3}e^x - 2\log x$ with respect to $x$.

**Concept / Formula Used:** $\dfrac{d}{dx}(x^n) = nx^{n-1}$ $\quad \dfrac{d}{dx}(\log x) = \dfrac{1}{x}$ $\quad \dfrac{d}{dx}(a^x) = a^x \log a$ $\quad \dfrac{d}{dx}(e^x) = e^x$

**Solution:**

**Part (i):** Let $y = \log x + \dfrac{1}{x} = \log x + x^{-1}$.

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(\log x) + \frac{d}{dx}(x^{-1}) \\ &= \frac{1}{x} + (-1)x^{-2} \\ &= \frac{1}{x} - \frac{1}{x^2}\end{aligned}$$

**Part (ii):** Let $y = 10^x + \dfrac{1}{3}e^x - 2\log x$.

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(10^x) + \frac{1}{3}\frac{d}{dx}(e^x) - 2\frac{d}{dx}(\log x) \\ &= 10^x \log 10 + \frac{1}{3}e^x - 2 \cdot \frac{1}{x} \\ &= 10^x \log 10 + \frac{1}{3}e^x - \frac{2}{x}\end{aligned}$$

### Final Answer

**(i)** $\dfrac{dy}{dx} = \dfrac{1}{x} - \dfrac{1}{x^2}$

**(ii)** $\dfrac{dy}{dx} = 10^x \log 10 + \dfrac{1}{3}e^x - \dfrac{2}{x}$

## Question 2

Differentiate the following functions w.r.t. $x$:
(i) $e^{-x}$
(ii) $\sin(\log x), \quad x > 0$

**TYPE 19** **Chain Rule for Differentiation**

**To Find:** The derivative of (i) $y = e^{-x}$ and (ii) $y = \sin(\log x)$ w.r.t. $x$.

**Concept / Formula Used:** $\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$

**Solution:**

**Part (i):** Let $y = e^{-x}$.

Differentiating w.r.t. $x$ using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= e^{-x} \cdot \frac{d}{dx}(-x) \\ &= e^{-x} \cdot (-1) \\ &= -e^{-x}\end{aligned}$$

**Part (ii):** Let $y = \sin(\log x)$.

Differentiating w.r.t. $x$ using the chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \cos(\log x) \cdot \frac{d}{dx}(\log x) \\ &= \cos(\log x) \cdot \frac{1}{x} \\ &= \frac{\cos(\log x)}{x}\end{aligned}$$

**Final Answer**

**(i)** $\frac{dy}{dx} = -e^{-x}$

**(ii)** $\frac{dy}{dx} = \frac{\cos(\log x)}{x}$

## Question 3

Differentiate the following functions w.r.t. $x$:
(i) $e^{\cos x}$
(ii) $e^{\sin^{-1} x}$

**TYPE 19** **Chain Rule for Differentiation**

**To Find:** The derivative of (i) $y = e^{\cos x}$ and (ii) $y = e^{\sin^{-1} x}$ w.r.t. $x$.

**Concept / Formula Used:** $\frac{d}{dx}(e^u) = e^u \frac{du}{dx} \qquad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

**Solution:**

**Part (i):** Let $y = e^{\cos x}$.

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= e^{\cos x} \cdot \frac{d}{dx}(\cos x) \\ &= e^{\cos x} \cdot (-\sin x) \\ &= -\sin x \cdot e^{\cos x}\end{aligned}$$

**Part (ii):** Let $y = e^{\sin^{-1} x}$.

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= e^{\sin^{-1} x} \cdot \frac{d}{dx}(\sin^{-1} x) \\ &= e^{\sin^{-1} x} \cdot \frac{1}{\sqrt{1-x^2}} \\ &= \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}\end{aligned}$$

**Final Answer**

**(i)** $\frac{dy}{dx} = -\sin x \cdot e^{\cos x}$

**(ii)** $\frac{dy}{dx} = \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}$

## Question 4

Differentiate the following functions w.r.t. $x$:

(i) $e^{x^3}$

(ii) $\sqrt{e^{\sqrt{x}}}, \quad x > 0$

**TYPE 22** **Repeated Chain Rule**

**To Find:** The derivative of (i) $y = e^{x^3}$ and (ii) $y = \sqrt{e^{\sqrt{x}}}$ w.r.t. $x$.

**Concept / Formula Used:** $\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}}\frac{du}{dx}$

**Solution:**

**Part (i):** Let $y = e^{x^3}$.

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= e^{x^3} \cdot \frac{d}{dx}(x^3) \\ &= e^{x^3} \cdot (3x^2) \\ &= 3x^2 e^{x^3}\end{aligned}$$

**Part (ii):** Let $y = \sqrt{e^{\sqrt{x}}}$.
Differentiating w.r.t. $x$ using chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2\sqrt{e^{\sqrt{x}}}} \cdot \frac{d}{dx}\left(e^{\sqrt{x}}\right) \\ &= \frac{1}{2\sqrt{e^{\sqrt{x}}}} \cdot e^{\sqrt{x}} \cdot \frac{d}{dx}\left(\sqrt{x}\right) \\ &= \frac{e^{\sqrt{x}}}{2\sqrt{e^{\sqrt{x}}}} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{\sqrt{e^{\sqrt{x}}}}{4\sqrt{x}}\end{aligned}$$

**Final Answer**

**(i)** $\frac{dy}{dx} = 3x^2 e^{x^3}$

**(ii)** $\frac{dy}{dx} = \frac{\sqrt{e^{\sqrt{x}}}}{4\sqrt{x}}$

## Question 5

Differentiate the following functions w.r.t. $x$:
(i) $\log(\log x), \quad x > 1$
(ii) $\log_7(\log x), \quad x > 1$

**TYPE 19** **Chain Rule for Differentiation**

**To Find:** The derivative of (i) $y = \log(\log x)$ and (ii) $y = \log_7(\log x)$ w.r.t. $x$.

**Concept / Formula Used:** $\log_a b = \frac{\log b}{\log a}$

**Solution:**

**Part (i):** Let $y = \log(\log x)$.

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\log x}\cdot\frac{d}{dx}(\log x)\\ &= \frac{1}{\log x}\cdot\frac{1}{x}\\ &= \frac{1}{x\log x}\end{aligned}$$

**Part (ii):** Let $y = \log_7(\log x)$.
Applying change of base formula to natural logarithm:

$$y = \frac{\log(\log x)}{\log 7}$$

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\log 7}\cdot\frac{d}{dx}(\log(\log x))\\ &= \frac{1}{\log 7}\cdot\frac{1}{\log x}\cdot\frac{1}{x}\\ &= \frac{1}{x\log 7\log x}\end{aligned}$$

**Final Answer**

**(i)** $\frac{dy}{dx} = \frac{1}{x\log x}$

**(ii)** $\frac{dy}{dx} = \frac{1}{x\log 7\log x}$

## Question 6

Differentiate the following functions w.r.t. $x$:
(i) $\log(\cos e^x)$
(ii) $\tan^{-1}(e^{\sin x})$

**TYPE 22** **Repeated Chain Rule**

**To Find:** The derivative of (i) $y = \log(\cos e^x)$ and (ii) $y = \tan^{-1}(e^{\sin x})$ w.r.t. $x$.

**Concept / Formula Used:** $\frac{d}{dx}(\tan^{-1}u) = \frac{1}{1+u^2}\frac{du}{dx}$

**Solution:**

**Part (i):** Let $y = \log(\cos e^x)$.

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\cos e^x} \cdot \frac{d}{dx}(\cos e^x)\\ &= \frac{1}{\cos e^x} \cdot (-\sin e^x) \cdot \frac{d}{dx}(e^x)\\ &= -\frac{\sin e^x}{\cos e^x} \cdot e^x\\ &= -e^x \tan(e^x)\end{aligned}$$

**Part (ii):** Let $y = \tan^{-1}(e^{\sin x})$.
Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{1+(e^{\sin x})^2} \cdot \frac{d}{dx}(e^{\sin x})\\ &= \frac{1}{1+e^{2\sin x}} \cdot e^{\sin x} \cdot \frac{d}{dx}(\sin x)\\ &= \frac{e^{\sin x}\cos x}{1+e^{2\sin x}}\end{aligned}$$

**Final Answer**

**(i)** $\dfrac{dy}{dx} = -e^x \tan(e^x)$

**(ii)** $\dfrac{dy}{dx} = \dfrac{e^{\sin x}\cos x}{1+e^{2\sin x}}$

## Question 7

Is the function $f(x) = \log(2x-1)$ derivable at $x = 0$?

**TYPE 35 Domain of Differentiability**

**To Find:** Whether $f(x) = \log(2x-1)$ is derivable at $x = 0$.

**Concept / Formula Used:** $\log u \qquad u > 0$

**Solution:**

For $f(x) = \log(2x-1)$, the argument of the logarithm must be strictly positive:

$$\begin{aligned}2x - 1 &> 0\\ 2x &> 1\\ x &> \frac{1}{2}\end{aligned}$$

Thus, the domain of $f(x)$ is $\left(\frac{1}{2}, \infty\right)$.
Evaluating the argument at $x = 0$:

$$2(0) - 1 = -1 \leq 0$$

Since $f(0) = \log(-1)$ is undefined in real numbers, $f(x)$ is not defined in any neighbourhood of $x = 0$.

Therefore, $f(x)$ is not derivable at $x = 0$.

**Final Answer**

No, the function $f(x) = \log(2x - 1)$ is not derivable at $x = 0$ because it is not defined at $x = 0$.

## Question 8

If $f(x) = \log(\log x)$, find $f'(e)$.

**TYPE 31** **Chain Rule and Evaluation**

**Given:** $f(x) = \log(\log x)$

**To Find:** $f'(e)$

**Concept / Formula Used:** $\dfrac{d}{dx}(\log u) = \dfrac{1}{u}\dfrac{du}{dx}$ $\qquad \log e = 1$

**Solution:**

Given function:

$$f(x) = \log(\log x)$$

Differentiating $f(x)$ with respect to $x$:

$$\begin{aligned} f'(x) &= \frac{1}{\log x} \cdot \frac{d}{dx}(\log x) \\ &= \frac{1}{\log x} \cdot \frac{1}{x} \\ &= \frac{1}{x \log x} \end{aligned}$$

Evaluating at $x = e$:

$$f'(e) = \frac{1}{e \log e}$$

Since $\log e = 1$:

$$\begin{aligned} f'(e) &= \frac{1}{e \cdot 1} \\ &= \frac{1}{e} \end{aligned}$$

**Final Answer**

$f'(e) = \dfrac{1}{e}$

## Question 9

If $y = \log(\tan x)$, show that $\dfrac{dy}{dx} = 2\csc 2x$.

**TYPE 33** **Proofs Using Chain Rule**

**To Prove:** $\dfrac{dy}{dx} = 2\csc 2x$

**Concept / Formula Used:** $\dfrac{d}{dx}(\tan x) = \sec^2 x$ $\quad \sin 2x = 2\sin x\cos x$ $\quad \csc\theta = \dfrac{1}{\sin\theta}$

**Solution:**

Given:

$$y = \log(\tan x)$$

Differentiating $y$ w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{\tan x}\cdot\frac{d}{dx}(\tan x)\\ &= \frac{1}{\tan x}\cdot\sec^2 x\end{aligned}$$

Expressing in terms of $\sin x$ and $\cos x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\cos x}{\sin x}\cdot\frac{1}{\cos^2 x}\\ &= \frac{1}{\sin x\cos x}\end{aligned}$$

Multiplying numerator and denominator by 2:

$$\frac{dy}{dx} = \frac{2}{2\sin x\cos x}$$

Using the identity $\sin 2x = 2\sin x\cos x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{2}{\sin 2x}\\ &= 2\csc 2x\end{aligned}$$

**Hence Proved**

$\dfrac{dy}{dx} = 2\csc 2x$, hence proved.

## Question 10

If $f(1) = 4$ and $f'(1) = 2$, find the value of the derivative of $\log f(e^x)$ w.r.t. $x$ at $x = 0$.

**TYPE 31 Chain Rule and Evaluation**

**Given:** $f(1) = 4$ and $f'(1) = 2$

**To Find:** $\frac{d}{dx}[\log f(e^x)]\Big|_{x=0}$

**Concept / Formula Used:** $\frac{d}{dx}[g(h(x))] = g'(h(x)) \cdot h'(x)$

**Solution:**

Let $y = \log f(e^x)$.

Differentiating $y$ w.r.t. $x$ using chain rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{f(e^x)} \cdot \frac{d}{dx}\left[f(e^x)\right] \\ &= \frac{1}{f(e^x)} \cdot f'(e^x) \cdot \frac{d}{dx}(e^x) \\ &= \frac{f'(e^x) \cdot e^x}{f(e^x)}\end{aligned}$$

Evaluating at $x = 0$:

$$\frac{dy}{dx}\Big|_{x=0} = \frac{f'(e^0) \cdot e^0}{f(e^0)}$$

Since $e^0 = 1$:

$$\begin{aligned}\frac{dy}{dx}\Big|_{x=0} &= \frac{f'(1) \cdot 1}{f(1)} \\ &= \frac{f'(1)}{f(1)}\end{aligned}$$

Substituting the given values $f(1) = 4$ and $f'(1) = 2$:

$$\begin{aligned}\frac{dy}{dx}\Big|_{x=0} &= \frac{2}{4} \\ &= \frac{1}{2}\end{aligned}$$

**Final Answer**

$$\frac{d}{dx}[\log f(e^x)]\Big|_{x=0} = \frac{1}{2}$$

## Question 11

If $f(x) = e^{xg(x)}$, $g(0) = 2$ and $g'(0) = 1$, then find $f'(0)$.

**TYPE 31** **Chain Rule and Evaluation**

**Given:** $f(x) = e^{xg(x)}$, $g(0) = 2$, $g'(0) = 1$

**To Find:** $f'(0)$

**Concept / Formula Used:** $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$

**Solution:**

Given function:

$$f(x) = e^{xg(x)}$$

Differentiating $f(x)$ w.r.t. $x$ using chain rule:

$$f'(x) = e^{xg(x)} \cdot \frac{d}{dx}\left[xg(x)\right]$$

Applying product rule to $\frac{d}{dx}[xg(x)]$:

$$\begin{aligned}\frac{d}{dx}[xg(x)] &= x \cdot g'(x) + g(x) \cdot \frac{d}{dx}(x) \\ &= xg'(x) + g(x) \cdot 1 \\ &= xg'(x) + g(x)\end{aligned}$$

So,

$$f'(x) = e^{xg(x)}\left[xg'(x) + g(x)\right]$$

Evaluating at $x = 0$:

$$\begin{aligned}f'(0) &= e^{0\cdot g(0)}\left[0 \cdot g'(0) + g(0)\right] \\ &= e^{0}\left[0 + g(0)\right] \\ &= 1 \cdot g(0)\end{aligned}$$

Substituting $g(0) = 2$:

$$f'(0) = 2$$

**Final Answer**

$f'(0) = 2$

## Question 12

Differentiate the following functions w.r.t. $x$:

(i) $\dfrac{e^x}{\sin x}$

(ii) $\dfrac{e^x}{1+\sin x}$

**TYPE 28** **Quotient Rule for Differentiation**

**To Find:** The derivative of (i) $y = \dfrac{e^x}{\sin x}$ and (ii) $y = \dfrac{e^x}{1+\sin x}$ w.r.t. $x$.

**Concept / Formula Used:** $\dfrac{d}{dx}\left(\dfrac{u}{v}\right) = \dfrac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

**Solution:**

**Part (i):** Let $y = \dfrac{e^x}{\sin x}$.
Using quotient rule with $u = e^x$ and $v = \sin x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\sin x \cdot \frac{d}{dx}(e^x) - e^x \cdot \frac{d}{dx}(\sin x)}{(\sin x)^2} \\ &= \frac{\sin x \cdot e^x - e^x \cdot \cos x}{\sin^2 x} \\ &= \frac{e^x(\sin x - \cos x)}{\sin^2 x}\end{aligned}$$

**Part (ii):** Let $y = \dfrac{e^x}{1+\sin x}$.
Using quotient rule with $u = e^x$ and $v = 1 + \sin x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{(1+\sin x) \cdot \frac{d}{dx}(e^x) - e^x \cdot \frac{d}{dx}(1+\sin x)}{(1+\sin x)^2} \\ &= \frac{(1+\sin x) \cdot e^x - e^x \cdot (0+\cos x)}{(1+\sin x)^2} \\ &= \frac{e^x(1+\sin x - \cos x)}{(1+\sin x)^2}\end{aligned}$$

**Final Answer**

**(i)** $\dfrac{dy}{dx} = \dfrac{e^x(\sin x - \cos x)}{\sin^2 x}$

**(ii)** $\dfrac{dy}{dx} = \dfrac{e^x(1+\sin x - \cos x)}{(1+\sin x)^2}$

## Question 13

Differentiate the following functions w.r.t. $x$:

(i) $2x\tan^{-1}x - \log(1+x^2)$

(ii) $\dfrac{\cos x}{\log x}, \quad x > 0$

**TYPE 38 Product and Quotient Rules with Inverse Trigonometric Functions**

**To Find:** The derivative of (i) $y = 2x\tan^{-1}x - \log(1+x^2)$ and (ii) $y = \dfrac{\cos x}{\log x}$ w.r.t. $x$.

**Concept / Formula Used:** $\dfrac{d}{dx}(\tan^{-1}x) = \dfrac{1}{1+x^2}$

**Solution:**

**Part (i):** Let $y = 2x\tan^{-1}x - \log(1+x^2)$.

Differentiating $2x\tan^{-1}x$ using product rule:

$$\begin{aligned}\frac{d}{dx}(2x\tan^{-1}x) &= 2x\cdot\frac{d}{dx}(\tan^{-1}x) + \tan^{-1}x\cdot\frac{d}{dx}(2x)\\ &= 2x\cdot\frac{1}{1+x^2} + \tan^{-1}x\cdot 2\\ &= \frac{2x}{1+x^2} + 2\tan^{-1}x\end{aligned}$$

Differentiating $\log(1+x^2)$ using chain rule:

$$\begin{aligned}\frac{d}{dx}(\log(1+x^2)) &= \frac{1}{1+x^2}\cdot\frac{d}{dx}(1+x^2)\\ &= \frac{1}{1+x^2}\cdot 2x\\ &= \frac{2x}{1+x^2}\end{aligned}$$

Combining both terms:

$$\begin{aligned}\frac{dy}{dx} &= \left(\frac{2x}{1+x^2} + 2\tan^{-1}x\right) - \frac{2x}{1+x^2}\\ &= 2\tan^{-1}x\end{aligned}$$

**Part (ii):** Let $y = \dfrac{\cos x}{\log x}$.

Using quotient rule with $u = \cos x$ and $v = \log x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\log x \cdot \frac{d}{dx}(\cos x) - \cos x \cdot \frac{d}{dx}(\log x)}{(\log x)^2} \\ &= \frac{\log x \cdot (-\sin x) - \cos x \cdot \frac{1}{x}}{(\log x)^2} \\ &= \frac{-\sin x \log x - \frac{\cos x}{x}}{(\log x)^2} \\ &= \frac{-(x \sin x \log x + \cos x)}{x(\log x)^2} \\ &= -\frac{x \sin x \log x + \cos x}{x(\log x)^2}\end{aligned}$$

**Final Answer**

**(i)** $\dfrac{dy}{dx} = 2\tan^{-1} x$

**(ii)** $\dfrac{dy}{dx} = -\dfrac{x \sin x \log x + \cos x}{x(\log x)^2}$

## Question 14

Differentiate the following functions w.r.t. $x$:

(i) $e^{\sec^2 x} + 3\cos^{-1} x$

(ii) $e^x + e^{x^2} + \cdots + e^{x^5}$

**TYPE 20** **Sum and Product Rules**

**To Find:** The derivative of (i) $y = e^{\sec^2 x} + 3\cos^{-1} x$ and (ii) $y = e^x + e^{x^2} + \cdots + e^{x^5}$ w.r.t. $x$.

**Concept / Formula Used:** $\dfrac{d}{dx}(e^u) = e^u \dfrac{du}{dx}$ $\qquad \dfrac{d}{dx}(\cos^{-1} x) = -\dfrac{1}{\sqrt{1-x^2}}$

**Solution:**

**Part (i):** Let $y = e^{\sec^2 x} + 3\cos^{-1} x$.

Differentiating first term:

$$\begin{aligned}\frac{d}{dx}(e^{\sec^2 x}) &= e^{\sec^2 x} \cdot \frac{d}{dx}(\sec^2 x) \\ &= e^{\sec^2 x} \cdot 2\sec x \cdot \frac{d}{dx}(\sec x) \\ &= e^{\sec^2 x} \cdot 2\sec x \cdot (\sec x \tan x) \\ &= 2\sec^2 x \tan x \, e^{\sec^2 x}\end{aligned}$$

Differentiating second term:

$$\frac{d}{dx}(3\cos^{-1} x) = 3 \cdot \left(-\frac{1}{\sqrt{1-x^2}}\right) = -\frac{3}{\sqrt{1-x^2}}$$

Thus,

$$\frac{dy}{dx} = 2\sec^2 x \tan x \, e^{\sec^2 x} - \frac{3}{\sqrt{1-x^2}}$$

**Part (ii):** Let $y = e^x + e^{x^2} + e^{x^3} + e^{x^4} + e^{x^5}$.
Differentiating term by term using chain rule:

$$\begin{aligned}
\frac{d}{dx}(e^x) &= e^x \\
\frac{d}{dx}(e^{x^2}) &= e^{x^2} \cdot (2x) = 2xe^{x^2} \\
\frac{d}{dx}(e^{x^3}) &= e^{x^3} \cdot (3x^2) = 3x^2e^{x^3} \\
\frac{d}{dx}(e^{x^4}) &= e^{x^4} \cdot (4x^3) = 4x^3e^{x^4} \\
\frac{d}{dx}(e^{x^5}) &= e^{x^5} \cdot (5x^4) = 5x^4e^{x^5}
\end{aligned}$$

Summing all terms:

$$\frac{dy}{dx} = e^x + 2xe^{x^2} + 3x^2e^{x^3} + 4x^3e^{x^4} + 5x^4e^{x^5}$$

**Final Answer**

**(i)** $\dfrac{dy}{dx} = 2\sec^2 x \tan x \, e^{\sec^2 x} - \dfrac{3}{\sqrt{1-x^2}}$

**(ii)** $\dfrac{dy}{dx} = e^x + 2xe^{x^2} + 3x^2e^{x^3} + 4x^3e^{x^4} + 5x^4e^{x^5}$

## Question 15

Differentiate the following functions w.r.t. $x$:

(i) $\dfrac{5^x \log x}{x^2+1}$

(ii) $\cos(\log x + e^x)$

**TYPE 28** **Quotient Rule for Differentiation**

**To Find:** The derivative of (i) $y = \dfrac{5^x \log x}{x^2+1}$ and (ii) $y = \cos(\log x + e^x)$ w.r.t. $x$.

**Concept / Formula Used:** $\dfrac{d}{dx}(5^x) = 5^x \log 5$

### Solution:

**Part (i):** Let $y = \dfrac{5^x \log x}{x^2+1}$.

Let numerator $u = 5^x \log x$ and denominator $v = x^2 + 1$.

Differentiating $u$ using product rule:

$$\begin{aligned}\frac{du}{dx} &= 5^x \cdot \frac{d}{dx}(\log x) + \log x \cdot \frac{d}{dx}(5^x)\\ &= 5^x \cdot \frac{1}{x} + \log x \cdot (5^x \log 5)\\ &= 5^x \left(\frac{1}{x} + \log 5 \log x\right)\\ &= 5^x \left(\frac{1 + x\log 5 \log x}{x}\right)\end{aligned}$$

Applying quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}\\ &= \frac{(x^2+1)\cdot 5^x \left(\frac{1+x\log 5\log x}{x}\right) - (5^x \log x)\cdot(2x)}{(x^2+1)^2}\\ &= \frac{5^x\left[(x^2+1)(1+x\log 5\log x) - 2x^2\log x\right]}{x(x^2+1)^2}\end{aligned}$$

**Part (ii):** Let $y = \cos(\log x + e^x)$.

Differentiating w.r.t. $x$ using chain rule:

$$\begin{aligned}\frac{dy}{dx} &= -\sin(\log x + e^x) \cdot \frac{d}{dx}(\log x + e^x)\\ &= -\sin(\log x + e^x)\cdot\left(\frac{1}{x} + e^x\right)\\ &= -\left(\frac{1}{x} + e^x\right)\sin(\log x + e^x)\end{aligned}$$

**Final Answer**

**(i)** $\dfrac{dy}{dx} = \dfrac{5^x\left[(x^2+1)(1+x\log 5\log x) - 2x^2\log x\right]}{x(x^2+1)^2}$

**(ii)** $\dfrac{dy}{dx} = -\left(\dfrac{1}{x} + e^x\right)\sin(\log x + e^x)$

## Question 16

Differentiate the following functions w.r.t. $x$:

(i) $\log\left(\dfrac{x+\sqrt{x^2-a^2}}{x-\sqrt{x^2-a^2}}\right)$

(ii) $\log\sqrt{\dfrac{1-\cos x}{1+\cos x}}$

**TYPE 30** **Trigonometric Simplification before Differentiation**

**To Find:** The derivative of (i) $y = \log\left(\dfrac{x+\sqrt{x^2-a^2}}{x-\sqrt{x^2-a^2}}\right)$ and (ii) $y = \log\sqrt{\dfrac{1-\cos x}{1+\cos x}}$ w.r.t. $x$.

**Concept / Formula Used:** $\tan^2\left(\frac{x}{2}\right) = \frac{1-\cos x}{1+\cos x}$ $\qquad \log(u^2) = 2\log u$

**Solution:**

**Part (i):** Let $y = \log\left(\frac{x+\sqrt{x^2-a^2}}{x-\sqrt{x^2-a^2}}\right)$.

Rationalizing the denominator inside the logarithm:

$$\begin{aligned}\frac{x+\sqrt{x^2-a^2}}{x-\sqrt{x^2-a^2}} &= \frac{(x+\sqrt{x^2-a^2})^2}{(x)^2-(\sqrt{x^2-a^2})^2}\\ &= \frac{(x+\sqrt{x^2-a^2})^2}{x^2-(x^2-a^2)}\\ &= \frac{(x+\sqrt{x^2-a^2})^2}{a^2}\end{aligned}$$

Substitute back into $y$:

$$\begin{aligned}y &= \log\left(\frac{(x+\sqrt{x^2-a^2})^2}{a^2}\right)\\ &= 2\log(x+\sqrt{x^2-a^2}) - \log(a^2)\end{aligned}$$

Differentiating w.r.t. $x$:

$$\begin{aligned}\frac{dy}{dx} &= 2\cdot\frac{1}{x+\sqrt{x^2-a^2}}\cdot\frac{d}{dx}\left(x+\sqrt{x^2-a^2}\right) - 0\\ &= \frac{2}{x+\sqrt{x^2-a^2}}\cdot\left(1+\frac{1}{2\sqrt{x^2-a^2}}\cdot 2x\right)\\ &= \frac{2}{x+\sqrt{x^2-a^2}}\cdot\left(1+\frac{x}{\sqrt{x^2-a^2}}\right)\\ &= \frac{2}{x+\sqrt{x^2-a^2}}\cdot\left(\frac{\sqrt{x^2-a^2}+x}{\sqrt{x^2-a^2}}\right)\\ &= \frac{2}{\sqrt{x^2-a^2}}\end{aligned}$$

**Part (ii):** Let $y = \log\sqrt{\frac{1-\cos x}{1+\cos x}}$.

Using trigonometric identity $\frac{1-\cos x}{1+\cos x} = \tan^2\left(\frac{x}{2}\right)$:

$$\begin{aligned}y &= \log\sqrt{\tan^2\left(\frac{x}{2}\right)}\\ &= \log\tan\left(\frac{x}{2}\right)\end{aligned}$$

Differentiating w.r.t. $x$:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{1}{\tan(x/2)} \cdot \frac{d}{dx}\left[\tan\left(\frac{x}{2}\right)\right] \\
&= \frac{1}{\tan(x/2)} \cdot \sec^2\left(\frac{x}{2}\right) \cdot \frac{1}{2} \\
&= \frac{\cos(x/2)}{\sin(x/2)} \cdot \frac{1}{\cos^2(x/2)} \cdot \frac{1}{2} \\
&= \frac{1}{2\sin(x/2)\cos(x/2)} \\
&= \frac{1}{\sin x} \\
&= \csc x
\end{aligned}$$

**Final Answer**

**(i)** $\frac{dy}{dx} = \frac{2}{\sqrt{x^2 - a^2}}$

**(ii)** $\frac{dy}{dx} = \csc x$

## Question 17

Differentiate the following functions w.r.t. $x$:

(i) $e^{\cot^{-1} x^2}$

(ii) $x\sqrt{1+x^2} + \log(x + \sqrt{x^2+1})$

**TYPE 36** **Chain Rule for Inverse Trigonometric Functions**

**To Find:** The derivative of (i) $y = e^{\cot^{-1} x^2}$ and (ii) $y = x\sqrt{1+x^2} + \log(x + \sqrt{x^2+1})$ w.r.t. $x$.

**Concept / Formula Used:** $\frac{d}{dx}(\cot^{-1} u) = -\frac{1}{1+u^2}\frac{du}{dx}$ $\quad \frac{d}{dx}\left[\log(x + \sqrt{x^2+a^2})\right] = \frac{1}{\sqrt{x^2+a^2}}$

**Solution:**

**Part (i):** Let $y = e^{\cot^{-1} x^2}$.

Differentiating w.r.t. $x$ using chain rule:

$$\begin{aligned}
\frac{dy}{dx} &= e^{\cot^{-1}x^2} \cdot \frac{d}{dx}\left(\cot^{-1}x^2\right) \\
&= e^{\cot^{-1}x^2} \cdot \left(-\frac{1}{1+(x^2)^2}\right) \cdot \frac{d}{dx}(x^2) \\
&= e^{\cot^{-1}x^2} \cdot \left(-\frac{1}{1+x^4}\right) \cdot (2x) \\
&= -\frac{2xe^{\cot^{-1}x^2}}{1+x^4}
\end{aligned}$$

**Part (ii):** Let $y = x\sqrt{1+x^2} + \log(x+\sqrt{x^2+1})$.

Differentiating the first term using product rule:

$$\begin{aligned}
\frac{d}{dx}\left(x\sqrt{1+x^2}\right) &= x \cdot \frac{d}{dx}\left(\sqrt{1+x^2}\right) + \sqrt{1+x^2} \cdot \frac{d}{dx}(x) \\
&= x \cdot \frac{1}{2\sqrt{1+x^2}} \cdot (2x) + \sqrt{1+x^2} \cdot 1 \\
&= \frac{x^2}{\sqrt{1+x^2}} + \sqrt{1+x^2} \\
&= \frac{x^2+(1+x^2)}{\sqrt{1+x^2}} \\
&= \frac{2x^2+1}{\sqrt{1+x^2}}
\end{aligned}$$

Differentiating the second term:

$$\begin{aligned}
\frac{d}{dx}\left[\log(x+\sqrt{x^2+1})\right] &= \frac{1}{x+\sqrt{x^2+1}} \cdot \left(1+\frac{2x}{2\sqrt{x^2+1}}\right) \\
&= \frac{1}{x+\sqrt{x^2+1}} \cdot \left(\frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}}\right) \\
&= \frac{1}{\sqrt{x^2+1}}
\end{aligned}$$

Adding both derivatives:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{2x^2+1}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+x^2}} \\
&= \frac{2x^2+2}{\sqrt{1+x^2}} \\
&= \frac{2(x^2+1)}{\sqrt{x^2+1}} \\
&= 2\sqrt{x^2+1}
\end{aligned}$$

**Final Answer**

**(i)** $\frac{dy}{dx} = -\frac{2xe^{\cot^{-1}x^2}}{1+x^4}$

**(ii)** $\frac{dy}{dx} = 2\sqrt{x^2+1}$

## Question 18

(i) If $y = \dfrac{\log(x+\sqrt{x^2+1})}{\sqrt{x^2+1}}$, prove that $(x^2+1)\dfrac{dy}{dx} + xy = 1$.

(ii) If $y = e^{2\log x + 3x}$, prove that $\dfrac{dy}{dx} = x(2+3x)e^{3x}$.

**TYPE 32** **Verifying Differential Relations**

### ★ Likely Exam Question

**To Prove:** (i) $(x^2+1)\dfrac{dy}{dx} + xy = 1$

(ii) $\dfrac{dy}{dx} = x(2+3x)e^{3x}$

**Concept / Formula Used:** $e^{a\log b} = b^a$

**Solution:**

**Part (i):** Given:

$$y = \frac{\log(x+\sqrt{x^2+1})}{\sqrt{x^2+1}}$$

Cross-multiplying:

$$y\sqrt{x^2+1} = \log(x+\sqrt{x^2+1})$$

Differentiating both sides w.r.t. $x$:

$$\frac{d}{dx}\left(y\sqrt{x^2+1}\right) = \frac{d}{dx}\left[\log(x+\sqrt{x^2+1})\right]$$

Using product rule on LHS:

$$\begin{aligned}
\sqrt{x^2+1}\frac{dy}{dx} + y\cdot\frac{d}{dx}\left(\sqrt{x^2+1}\right) &= \frac{1}{x+\sqrt{x^2+1}}\cdot\left(1+\frac{2x}{2\sqrt{x^2+1}}\right) \\
\sqrt{x^2+1}\frac{dy}{dx} + y\cdot\frac{2x}{2\sqrt{x^2+1}} &= \frac{1}{x+\sqrt{x^2+1}}\cdot\left(\frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}}\right) \\
\sqrt{x^2+1}\frac{dy}{dx} + \frac{xy}{\sqrt{x^2+1}} &= \frac{1}{\sqrt{x^2+1}}
\end{aligned}$$

Multiplying the entire equation by $\sqrt{x^2+1}$:

$$(x^2+1)\frac{dy}{dx} + xy = 1$$

**Part (ii):** Given:

$$y = e^{2\log x + 3x}$$

Simplifying using logarithm properties:

$$\begin{aligned} y &= e^{\log(x^2)+3x} \\ &= e^{\log(x^2)} \cdot e^{3x} \\ &= x^2 e^{3x} \end{aligned}$$

Differentiating $y$ w.r.t. $x$ using product rule:

$$\begin{aligned} \frac{dy}{dx} &= x^2 \cdot \frac{d}{dx}(e^{3x}) + e^{3x} \cdot \frac{d}{dx}(x^2) \\ &= x^2 \cdot (3e^{3x}) + e^{3x} \cdot (2x) \\ &= 3x^2 e^{3x} + 2xe^{3x} \\ &= x(2+3x)e^{3x} \end{aligned}$$

**Hence Proved**

Both identities (i) and (ii) are proved.

## Question 19

Differentiate $\tan^{-1}\left(\frac{2^{x+1}}{1-4^x}\right)$ w.r.t. $x$.

**TYPE 39** **Trigonometric Substitution before Differentiation**

**To Find:** The derivative of $y = \tan^{-1}\left(\frac{2^{x+1}}{1-4^x}\right)$ w.r.t. $x$.

**Concept / Formula Used:** $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$ $\quad \frac{d}{dx}(a^x) = a^x \log a$

**Solution:**

Let $y = \tan^{-1}\left(\frac{2^{x+1}}{1-4^x}\right)$.

Rewrite the terms inside the argument:

$$\begin{aligned} 2^{x+1} &= 2 \cdot 2^x \\ 4^x &= (2^2)^x = (2^x)^2 \end{aligned}$$

Substituting these into $y$:

$$y = \tan^{-1}\left(\frac{2 \cdot 2^x}{1-(2^x)^2}\right)$$

Let $2^x = \tan\theta$, which gives $\theta = \tan^{-1}(2^x)$.

Substituting $2^x = \tan\theta$:

$$y = \tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)$$

Using standard identity $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$:

$$\begin{aligned} y &= \tan^{-1}(\tan 2\theta) \\ &= 2\theta \end{aligned}$$

Back-substituting $\theta = \tan^{-1}(2^x)$:

$$y = 2\tan^{-1}(2^x)$$

Differentiating $y$ w.r.t. $x$ using chain rule:

$$\begin{aligned} \frac{dy}{dx} &= 2 \cdot \frac{1}{1+(2^x)^2} \cdot \frac{d}{dx}(2^x) \\ &= \frac{2}{1+4^x} \cdot (2^x \log 2) \\ &= \frac{2 \cdot 2^x \log 2}{1+4^x} \\ &= \frac{2^{x+1}\log 2}{1+4^x} \end{aligned}$$

**Final Answer**

$$\frac{dy}{dx} = \frac{2^{x+1}\log 2}{1+4^x}$$

## Question 20

Find $\dfrac{dy}{dx}$ when

(i) $xy + xe^{-y} + ye^x = x^2$

(ii) $e^{x-y} = \log\left(\dfrac{x}{y}\right)$

(iii) $5^x + 5^y = 5^{x+y}$

**TYPE 40** **Implicit Differentiation**

**To Find:** $\dfrac{dy}{dx}$ for each given implicit equation.

**Concept / Formula Used:** $x$ $\dfrac{dy}{dx}$

### Solution:

**Part (i):** Given: $xy + xe^{-y} + ye^x = x^2$.

Differentiating term-by-term w.r.t. $x$:

$$\begin{aligned} \frac{d}{dx}(xy) + \frac{d}{dx}(xe^{-y}) + \frac{d}{dx}(ye^x) &= \frac{d}{dx}(x^2) \\ \left(x\frac{dy}{dx} + y \cdot 1\right) + \left(x\left(-e^{-y}\frac{dy}{dx}\right) + e^{-y} \cdot 1\right) + \left(ye^x + e^x\frac{dy}{dx}\right) &= 2x \\ x\frac{dy}{dx} + y - xe^{-y}\frac{dy}{dx} + e^{-y} + ye^x + e^x\frac{dy}{dx} &= 2x \end{aligned}$$

Grouping terms containing $\frac{dy}{dx}$ on LHS:

$$\left(x - xe^{-y} + e^x\right)\frac{dy}{dx} = 2x - y - e^{-y} - ye^x$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{2x - y - e^{-y} - ye^x}{x - xe^{-y} + e^x}$$

**Part (ii):** Given: $e^{x-y} = \log\left(\frac{x}{y}\right) = \log x - \log y$.

Differentiating both sides w.r.t. $x$:

$$\begin{aligned}
\frac{d}{dx}\left(e^{x-y}\right) &= \frac{d}{dx}(\log x - \log y) \\
e^{x-y}\left(1 - \frac{dy}{dx}\right) &= \frac{1}{x} - \frac{1}{y}\frac{dy}{dx} \\
e^{x-y} - e^{x-y}\frac{dy}{dx} &= \frac{1}{x} - \frac{1}{y}\frac{dy}{dx}
\end{aligned}$$

Rearranging terms with $\frac{dy}{dx}$:

$$\begin{aligned}
\left(\frac{1}{y} - e^{x-y}\right)\frac{dy}{dx} &= \frac{1}{x} - e^{x-y} \\
\left(\frac{1 - ye^{x-y}}{y}\right)\frac{dy}{dx} &= \frac{1 - xe^{x-y}}{x}
\end{aligned}$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{y(1 - xe^{x-y})}{x(1 - ye^{x-y})}$$

**Part (iii):** Given: $5^x + 5^y = 5^{x+y}$.

Differentiating both sides w.r.t. $x$:

$$5^x \log 5 + 5^y \log 5\frac{dy}{dx} = 5^{x+y}\log 5\left(1 + \frac{dy}{dx}\right)$$

Dividing both sides by $\log 5$:

$$5^x + 5^y\frac{dy}{dx} = 5^{x+y} + 5^{x+y}\frac{dy}{dx}$$

Rearranging terms with $\frac{dy}{dx}$:

$$\begin{aligned}
\left(5^y - 5^{x+y}\right)\frac{dy}{dx} &= 5^{x+y} - 5^x \\
\frac{dy}{dx} &= \frac{5^{x+y} - 5^x}{5^y - 5^{x+y}}
\end{aligned}$$

Substituting $5^{x+y} = 5^x + 5^y$ into the expression:

$$\begin{aligned}\frac{dy}{dx} &= \frac{(5^x + 5^y) - 5^x}{5^y - (5^x + 5^y)} \\ &= \frac{5^y}{-5^x} \\ &= -5^{y-x}\end{aligned}$$

**Final Answer**

**(i)** $\dfrac{dy}{dx} = \dfrac{2x - y - e^{-y} - ye^x}{x - xe^{-y} + e^x}$

**(ii)** $\dfrac{dy}{dx} = \dfrac{y(1 - xe^{x-y})}{x(1 - ye^{x-y})}$

**(iii)** $\dfrac{dy}{dx} = -5^{y-x}$

## Question 21

If $\log(x^2 + y^2) = 2\tan^{-1}\left(\frac{y}{x}\right)$, show that $\dfrac{dy}{dx} = \dfrac{x+y}{x-y}$.

**TYPE 40 Implicit Differentiation**

★ **Likely Exam Question**

**To Prove:** $\dfrac{dy}{dx} = \dfrac{x+y}{x-y}$

**Concept / Formula Used:** $\dfrac{d}{dx}(\tan^{-1} u) = \dfrac{1}{1+u^2}\dfrac{du}{dx}$

**Solution:**

Given equation:

$$\log(x^2 + y^2) = 2\tan^{-1}\left(\frac{y}{x}\right)$$

Differentiating both sides w.r.t. $x$:

$$\frac{d}{dx}\left[\log(x^2 + y^2)\right] = \frac{d}{dx}\left[2\tan^{-1}\left(\frac{y}{x}\right)\right]$$

Differentiating LHS:

$$\frac{1}{x^2 + y^2} \cdot \left(2x + 2y\frac{dy}{dx}\right) = \frac{2(x + y\frac{dy}{dx})}{x^2 + y^2}$$

Differentiating RHS using chain rule and quotient rule:

$$2 \cdot \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \frac{d}{dx}\left(\frac{y}{x}\right) = 2 \cdot \frac{1}{\frac{x^2+y^2}{x^2}} \cdot \left(\frac{x\frac{dy}{dx} - y}{x^2}\right)$$
$$= \frac{2x^2}{x^2+y^2} \cdot \left(\frac{x\frac{dy}{dx} - y}{x^2}\right)$$
$$= \frac{2\left(x\frac{dy}{dx} - y\right)}{x^2+y^2}$$

Equating LHS and RHS:

$$\frac{2(x + y\frac{dy}{dx})}{x^2+y^2} = \frac{2(x\frac{dy}{dx} - y)}{x^2+y^2}$$

Multiplying both sides by $\frac{x^2+y^2}{2}$:

$$x + y\frac{dy}{dx} = x\frac{dy}{dx} - y$$

Rearranging terms:

$$x + y = x\frac{dy}{dx} - y\frac{dy}{dx}$$
$$x + y = (x - y)\frac{dy}{dx}$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

**Hence Proved**

$\frac{dy}{dx} = \frac{x+y}{x-y}$, hence proved.

## Question 22

If $y \log x = x - y$, prove that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$.

**TYPE 44** **Differentiation by Isolating Variables**

**★ Likely Exam Question**

**To Prove:** $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$

**Concept / Formula Used:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

**Solution:**

Given equation:

$$y \log x = x - y$$

Grouping terms involving $y$ on the LHS:

$$y \log x + y = x$$
$$y(1 + \log x) = x$$
$$y = \frac{x}{1 + \log x}$$

Differentiating $y$ w.r.t. $x$ using the quotient rule with $u = x$ and $v = 1 + \log x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{(1 + \log x) \cdot \frac{d}{dx}(x) - x \cdot \frac{d}{dx}(1 + \log x)}{(1 + \log x)^2} \\ &= \frac{(1 + \log x) \cdot 1 - x \cdot \left(0 + \frac{1}{x}\right)}{(1 + \log x)^2} \\ &= \frac{1 + \log x - 1}{(1 + \log x)^2} \\ &= \frac{\log x}{(1 + \log x)^2}\end{aligned}$$

**Hence Proved**

$\frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}$, hence proved.

## Question 23

Differentiate the following functions w.r.t. $x$:

(i) $\log_x(2x - 3)$

(ii) $\log_{\cos x} \sin x$

**TYPE 28** **Quotient Rule for Differentiation**

**To Find:** The derivative of (i) $y = \log_x(2x - 3)$ and (ii) $y = \log_{\cos x} \sin x$ w.r.t. $x$.

**Concept / Formula Used:** $\log_a b = \frac{\log b}{\log a}$

**Solution:**

**Part (i):** Let $y = \log_x(2x - 3)$.

Applying change of base formula to express $y$ using natural logarithms:

$$y = \frac{\log(2x - 3)}{\log x}$$

Differentiating $y$ w.r.t. $x$ using quotient rule:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{\log x \cdot \frac{d}{dx}\left[\log(2x-3)\right] - \log(2x-3) \cdot \frac{d}{dx}(\log x)}{(\log x)^2} \\
&= \frac{\log x \cdot \left(\frac{2}{2x-3}\right) - \log(2x-3) \cdot \left(\frac{1}{x}\right)}{(\log x)^2} \\
&= \frac{\frac{2\log x}{2x-3} - \frac{\log(2x-3)}{x}}{(\log x)^2} \\
&= \frac{2x\log x - (2x-3)\log(2x-3)}{x(2x-3)(\log x)^2}
\end{aligned}$$

**Part (ii):** Let $y = \log_{\cos x} \sin x$.
Applying change of base formula to express $y$ using natural logarithms:

$$y = \frac{\log \sin x}{\log \cos x}$$

Differentiating $y$ w.r.t. $x$ using quotient rule:

$$\frac{dy}{dx} = \frac{\log \cos x \cdot \frac{d}{dx}(\log \sin x) - \log \sin x \cdot \frac{d}{dx}(\log \cos x)}{(\log \cos x)^2}$$

Evaluating the individual derivatives:

$$\begin{aligned}
\frac{d}{dx}(\log \sin x) &= \frac{1}{\sin x} \cdot \cos x = \cot x \\
\frac{d}{dx}(\log \cos x) &= \frac{1}{\cos x} \cdot (-\sin x) = -\tan x
\end{aligned}$$

Substituting these back into $\frac{dy}{dx}$:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{\log \cos x \cdot (\cot x) - \log \sin x \cdot (-\tan x)}{(\log \cos x)^2} \\
&= \frac{\cot x \log \cos x + \tan x \log \sin x}{(\log \cos x)^2}
\end{aligned}$$

**Final Answer**

**(i)** $\dfrac{dy}{dx} = \dfrac{2x\log x - (2x-3)\log(2x-3)}{x(2x-3)(\log x)^2}$

**(ii)** $\dfrac{dy}{dx} = \dfrac{\cot x \log \cos x + \tan x \log \sin x}{(\log \cos x)^2}$

## 5.9 – Logarithmic Differentiation

## Question 1 (i)

Differentiate $(x+3)^2(x+4)^3(x+5)^4$ w.r.t. $x$.

**TYPE 46 Logarithmic Differentiation of Products**

**Given:** Let $y = (x+3)^2(x+4)^3(x+5)^4$

**Concept / Formula Used:** $\log(abc) = \log a + \log b + \log c \qquad \log(u^k) = k \log u$

**Solution:**

Taking natural logarithm on both sides:

$$\begin{aligned}\log y &= \log\left((x+3)^2(x+4)^3(x+5)^4\right)\\ &= \log((x+3)^2) + \log((x+4)^3) + \log((x+5)^4)\\ &= 2\log(x+3) + 3\log(x+4) + 4\log(x+5)\end{aligned}$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}\frac{1}{y}\frac{dy}{dx} &= 2 \cdot \frac{1}{x+3} + 3 \cdot \frac{1}{x+4} + 4 \cdot \frac{1}{x+5}\\ \frac{dy}{dx} &= y\left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5}\right]\end{aligned}$$

Substituting $y = (x+3)^2(x+4)^3(x+5)^4$:

$$\frac{dy}{dx} = (x+3)^2(x+4)^3(x+5)^4\left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5}\right]$$

**Final Answer**

$$\frac{dy}{dx} = (x+3)^2(x+4)^3(x+5)^4\left[\frac{2}{x+3} + \frac{3}{x+4} + \frac{4}{x+5}\right]$$

## Question 1 (ii)

Differentiate $\cos x \cos 2x \cos 3x$ w.r.t. $x$.

**TYPE 46 Logarithmic Differentiation of Products**

**★ Likely Exam Question**

**Given:** Let $y = \cos x \cos 2x \cos 3x$

**Concept / Formula Used:** $\log(uvw) = \log u + \log v + \log w$

**Solution:**

Taking natural logarithm on both sides:

$$\begin{aligned}\log y &= \log(\cos x \cos 2x \cos 3x)\\ &= \log(\cos x) + \log(\cos 2x) + \log(\cos 3x)\end{aligned}$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}\frac{1}{y}\frac{dy}{dx} &= \frac{1}{\cos x}\cdot(-\sin x) + \frac{1}{\cos 2x}\cdot(-\sin 2x)\cdot 2 + \frac{1}{\cos 3x}\cdot(-\sin 3x)\cdot 3\\ \frac{1}{y}\frac{dy}{dx} &= -\tan x - 2\tan 2x - 3\tan 3x\\ \frac{dy}{dx} &= -y(\tan x + 2\tan 2x + 3\tan 3x)\end{aligned}$$

Substituting $y = \cos x \cos 2x \cos 3x$:

$$\frac{dy}{dx} = -\cos x \cos 2x \cos 3x(\tan x + 2\tan 2x + 3\tan 3x)$$

**Final Answer**

$\frac{dy}{dx} = -\cos x \cos 2x \cos 3x(\tan x + 2\tan 2x + 3\tan 3x)$

## Question 2 (i)

Differentiate $e^x \cos^3 x \sin^2 x$ w.r.t. $x$.

**TYPE 46** **Logarithmic Differentiation of Products**

**Given:** Let $y = e^x \cos^3 x \sin^2 x$

**Concept / Formula Used:** $\log(abc) = \log a + \log b + \log c$

**Solution:**

Taking natural logarithm on both sides:

$$\begin{aligned}\log y &= \log(e^x \cos^3 x \sin^2 x)\\ &= \log(e^x) + \log(\cos^3 x) + \log(\sin^2 x)\\ &= x + 3\log(\cos x) + 2\log(\sin x)\end{aligned}$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}\frac{1}{y}\frac{dy}{dx} &= 1 + 3\cdot\frac{1}{\cos x}\cdot(-\sin x) + 2\cdot\frac{1}{\sin x}\cdot(\cos x)\\ \frac{1}{y}\frac{dy}{dx} &= 1 - 3\tan x + 2\cot x\\ \frac{dy}{dx} &= y(1 - 3\tan x + 2\cot x)\end{aligned}$$

Substituting $y = e^x \cos^3 x \sin^2 x$:

$$\frac{dy}{dx} = e^x \cos^3 x \sin^2 x(1 - 3\tan x + 2\cot x)$$

**Final Answer**

$$\frac{dy}{dx} = e^x \cos^3 x \sin^2 x(1 - 3\tan x + 2\cot x)$$

## Question 2 (ii)

Differentiate $\dfrac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$ w.r.t. $x$.

**TYPE 46** **Logarithmic Differentiation of Products**

**Given:** Let $y = \dfrac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$

**Concept / Formula Used:** $\log\left(\dfrac{uv}{w}\right) = \log u + \log v - \log w$

**Solution:**

Taking natural logarithm on both sides:

$$\begin{aligned}\log y &= \log\left(\frac{e^{x^2} \tan^{-1} x}{(1+x^2)^{1/2}}\right) \\ &= \log(e^{x^2}) + \log(\tan^{-1} x) - \log((1+x^2)^{1/2}) \\ &= x^2 + \log(\tan^{-1} x) - \frac{1}{2}\log(1+x^2)\end{aligned}$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}\frac{1}{y}\frac{dy}{dx} &= 2x + \frac{1}{\tan^{-1} x} \cdot \frac{1}{1+x^2} - \frac{1}{2} \cdot \frac{1}{1+x^2} \cdot 2x \\ \frac{1}{y}\frac{dy}{dx} &= 2x + \frac{1}{(1+x^2)\tan^{-1} x} - \frac{x}{1+x^2} \\ \frac{dy}{dx} &= y\left[2x + \frac{1}{(1+x^2)\tan^{-1} x} - \frac{x}{1+x^2}\right]\end{aligned}$$

Substituting $y = \dfrac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$:

$$\frac{dy}{dx} = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}\left[2x + \frac{1}{(1+x^2)\tan^{-1} x} - \frac{x}{1+x^2}\right]$$

**Final Answer**

$$\frac{dy}{dx} = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}\left[2x + \frac{1}{(1+x^2)\tan^{-1} x} - \frac{x}{1+x^2}\right]$$

## Question 3 (i)

Differentiate $(\sin x)^{\cos x}, 0 < x < \pi$ w.r.t. $x$.

**TYPE 47** **Logarithmic Differentiation of Variable Exponents**

**Given:** Let $y = (\sin x)^{\cos x}$, where $0 < x < \pi$

**Concept / Formula Used:** $\log(u^v) = v \log u$

**Solution:**

Taking natural logarithm on both sides:

$$\log y = \cos x \cdot \log(\sin x)$$

Differentiating both sides with respect to $x$ using the product rule:

$$\begin{aligned}
\frac{1}{y}\frac{dy}{dx} &= \cos x \cdot \frac{d}{dx}[\log(\sin x)] + \log(\sin x) \cdot \frac{d}{dx}[\cos x] \\
&= \cos x \cdot \left(\frac{1}{\sin x} \cdot \cos x\right) + \log(\sin x) \cdot (-\sin x) \\
&= \cos x \cot x - \sin x \log(\sin x) \\
\frac{dy}{dx} &= y\,[\cos x \cot x - \sin x \log(\sin x)]
\end{aligned}$$

Substituting $y = (\sin x)^{\cos x}$:

$$\frac{dy}{dx} = (\sin x)^{\cos x}[\cos x \cot x - \sin x \log(\sin x)]$$

**Final Answer**

$\frac{dy}{dx} = (\sin x)^{\cos x}[\cos x \cot x - \sin x \log(\sin x)]$

## Question 3 (ii)

Differentiate $(\sin x)^{\sin x}, 0 < x < \pi$ w.r.t. $x$.

**TYPE 47** **Logarithmic Differentiation of Variable Exponents**

★ **Likely Exam Question**

**Given:** Let $y = (\sin x)^{\sin x}$, where $0 < x < \pi$

**Concept / Formula Used:** $\log(u^v) = v \log u$

**Solution:**

Taking natural logarithm on both sides:

$$\log y = \sin x \cdot \log(\sin x)$$

Differentiating both sides with respect to $x$ using the product rule:

$$\begin{aligned}
\frac{1}{y}\frac{dy}{dx} &= \sin x \cdot \frac{d}{dx}[\log(\sin x)] + \log(\sin x) \cdot \frac{d}{dx}[\sin x] \\
&= \sin x \cdot \left(\frac{1}{\sin x} \cdot \cos x\right) + \log(\sin x) \cdot \cos x \\
&= \cos x + \cos x \log(\sin x) \\
&= \cos x(1 + \log(\sin x)) \\
\frac{dy}{dx} &= y \cdot \cos x(1 + \log(\sin x))
\end{aligned}$$

Substituting $y = (\sin x)^{\sin x}$:

$$\frac{dy}{dx} = (\sin x)^{\sin x} \cos x(1 + \log(\sin x))$$

**Final Answer**

$\frac{dy}{dx} = (\sin x)^{\sin x}(1 + \log(\sin x)) \cos x$

## Question 3 (iii)

Differentiate $(\tan x)^{\tan x}, 0 < x < \frac{\pi}{2}$ w.r.t. $x$.

**TYPE 47 Logarithmic Differentiation of Variable Exponents**

**Given:** Let $y = (\tan x)^{\tan x}$, where $0 < x < \frac{\pi}{2}$

**Concept / Formula Used:** $\log(u^v) = v \log u$

**Solution:**

Taking natural logarithm on both sides:

$$\log y = \tan x \cdot \log(\tan x)$$

Differentiating both sides with respect to $x$ using the product rule:

$$\begin{aligned}
\frac{1}{y}\frac{dy}{dx} &= \tan x \cdot \frac{d}{dx}[\log(\tan x)] + \log(\tan x) \cdot \frac{d}{dx}[\tan x] \\
&= \tan x \cdot \left(\frac{1}{\tan x} \cdot \sec^2 x\right) + \log(\tan x) \cdot \sec^2 x \\
&= \sec^2 x + \sec^2 x \log(\tan x) \\
&= \sec^2 x(1 + \log(\tan x)) \\
\frac{dy}{dx} &= y \cdot \sec^2 x(1 + \log(\tan x))
\end{aligned}$$

Substituting $y = (\tan x)^{\tan x}$:

$$\frac{dy}{dx} = (\tan x)^{\tan x}(1 + \log(\tan x))\sec^2 x$$

**Final Answer**

$\dfrac{dy}{dx} = (\tan x)^{\tan x}(1 + \log(\tan x))\sec^2 x$

## Question 4 (i)

Differentiate $x^{\sin x}, x > 0$ w.r.t. $x$.

**TYPE 47 Logarithmic Differentiation of Variable Exponents**

**★ Likely Exam Question**

**Given:** Let $y = x^{\sin x}$, where $x > 0$

**Concept / Formula Used:** $\log(u^v) = v \log u$

**Solution:**

Taking natural logarithm on both sides:

$$\log y = \sin x \cdot \log x$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}
\frac{1}{y}\frac{dy}{dx} &= \sin x \cdot \frac{d}{dx}[\log x] + \log x \cdot \frac{d}{dx}[\sin x] \\
&= \sin x \cdot \frac{1}{x} + \log x \cdot \cos x \\
&= \frac{\sin x}{x} + \cos x \log x \\
\frac{dy}{dx} &= y\left[\frac{\sin x}{x} + \cos x \log x\right]
\end{aligned}$$

Substituting $y = x^{\sin x}$:

$$\frac{dy}{dx} = x^{\sin x}\left[\frac{\sin x}{x} + \cos x \log x\right]$$

**Final Answer**

$\dfrac{dy}{dx} = x^{\sin x}\left[\dfrac{\sin x}{x} + \cos x \log x\right]$

## Question 4 (ii)

Differentiate $(2x+3)^{x-5}, x > -\frac{3}{2}$ w.r.t. $x$.

**TYPE 47** **Logarithmic Differentiation of Variable Exponents**

**Given:** Let $y = (2x+3)^{x-5}$, where $x > -\frac{3}{2}$

**Concept / Formula Used:** $\log(u^v) = v \log u$

**Solution:**

Taking natural logarithm on both sides:

$$\log y = (x-5)\log(2x+3)$$

Differentiating both sides with respect to $x$ using the product rule:

$$\begin{aligned}
\frac{1}{y}\frac{dy}{dx} &= (x-5)\cdot\frac{d}{dx}[\log(2x+3)] + \log(2x+3)\cdot\frac{d}{dx}[x-5] \\
&= (x-5)\cdot\left(\frac{1}{2x+3}\cdot 2\right) + \log(2x+3)\cdot 1 \\
&= \frac{2(x-5)}{2x+3} + \log(2x+3) \\
\frac{dy}{dx} &= y\left[\frac{2(x-5)}{2x+3} + \log(2x+3)\right]
\end{aligned}$$

Substituting $y = (2x+3)^{x-5}$:

$$\frac{dy}{dx} = (2x+3)^{x-5}\left[\frac{2(x-5)}{2x+3} + \log(2x+3)\right]$$

**Final Answer**

$$\frac{dy}{dx} = (2x+3)^{x-5}\left[\frac{2(x-5)}{2x+3} + \log(2x+3)\right]$$

## Question 5 (i)

Differentiate $(\log x)^{\cos x}, x > 1$ w.r.t. $x$.

**TYPE 47** **Logarithmic Differentiation of Variable Exponents**

**★ Likely Exam Question**

**Given:** Let $y = (\log x)^{\cos x}$, where $x > 1$

**Concept / Formula Used:** $\log(u^v) = v \log u$

**Solution:**

Taking natural logarithm on both sides:

$$\log y = \cos x \cdot \log(\log x)$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}
\frac{1}{y}\frac{dy}{dx} &= \cos x \cdot \frac{d}{dx}[\log(\log x)] + \log(\log x) \cdot \frac{d}{dx}[\cos x] \\
&= \cos x \cdot \left(\frac{1}{\log x} \cdot \frac{1}{x}\right) + \log(\log x) \cdot (-\sin x) \\
&= \frac{\cos x}{x \log x} - \sin x \log(\log x) \\
\frac{dy}{dx} &= y\left[\frac{\cos x}{x \log x} - \sin x \log(\log x)\right]
\end{aligned}$$

Substituting $y = (\log x)^{\cos x}$:

$$\frac{dy}{dx} = (\log x)^{\cos x}\left[\frac{\cos x}{x \log x} - \sin x \log(\log x)\right]$$

**Final Answer**

$\frac{dy}{dx} = (\log x)^{\cos x}\left[\frac{\cos x}{x \log x} - \sin x \log(\log x)\right]$

## Question 5 (ii)

Differentiate $(\log x)^{\log x}, x > 1$ w.r.t. $x$.

**TYPE 47 Logarithmic Differentiation of Variable Exponents**

★ **Likely Exam Question**

**Given:** Let $y = (\log x)^{\log x}$, where $x > 1$

**Concept / Formula Used:** $\log(u^v) = v \log u$

**Solution:**

Taking natural logarithm on both sides:

$$\log y = \log x \cdot \log(\log x)$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}
\frac{1}{y}\frac{dy}{dx} &= \log x \cdot \frac{d}{dx}[\log(\log x)] + \log(\log x) \cdot \frac{d}{dx}[\log x] \\
&= \log x \cdot \left(\frac{1}{\log x} \cdot \frac{1}{x}\right) + \log(\log x) \cdot \frac{1}{x} \\
&= \frac{1}{x} + \frac{\log(\log x)}{x} \\
&= \frac{1}{x}(1 + \log(\log x)) \\
\frac{dy}{dx} &= y \cdot \frac{1}{x}(1 + \log(\log x))
\end{aligned}$$

Substituting $y = (\log x)^{\log x}$:

$$\frac{dy}{dx} = (\log x)^{\log x} \cdot \frac{1}{x}(1 + \log(\log x))$$

**Final Answer**

$\frac{dy}{dx} = (\log x)^{\log x} \cdot \frac{1}{x}(1 + \log(\log x))$

## Question 6 (i)

If $y = x^y$, prove that $x\frac{dy}{dx} = \frac{y^2}{1 - y\log x}$.

**TYPE 48** **Implicit Logarithmic Differentiation**

**★ Likely Exam Question**

**To Prove:** $x\frac{dy}{dx} = \frac{y^2}{1 - y\log x}$

**Concept / Formula Used:** $\log(u^v) = v\log u$

**Solution:**

Given:

$$y = x^y$$

Taking natural logarithm on both sides:

$$\begin{aligned}
\log y &= \log(x^y) \\
\log y &= y\log x
\end{aligned}$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}
\frac{1}{y}\frac{dy}{dx} &= y \cdot \frac{d}{dx}[\log x] + \log x \cdot \frac{dy}{dx} \\
\frac{1}{y}\frac{dy}{dx} &= y \cdot \frac{1}{x} + \log x\frac{dy}{dx}
\end{aligned}$$

Grouping terms containing $\frac{dy}{dx}$ on the left-hand side:

$$\frac{1}{y}\frac{dy}{dx} - \log x \frac{dy}{dx} = \frac{y}{x}$$
$$\left(\frac{1}{y} - \log x\right)\frac{dy}{dx} = \frac{y}{x}$$
$$\left(\frac{1 - y\log x}{y}\right)\frac{dy}{dx} = \frac{y}{x}$$

Multiplying both sides by $x$:

$$x\left(\frac{1 - y\log x}{y}\right)\frac{dy}{dx} = y$$
$$x\frac{dy}{dx} = \frac{y^2}{1 - y\log x}$$

**Hence Proved**

$x\frac{dy}{dx} = \frac{y^2}{1 - y\log x}$, hence proved.

## Question 6 (ii)

If $x = e^{x/y}$, prove that $\frac{dy}{dx} = \frac{x - y}{x\log x}$.

**TYPE 48** **Implicit Logarithmic Differentiation**

**★ Likely Exam Question**

**To Prove:** $\frac{dy}{dx} = \frac{x - y}{x\log x}$

**Concept / Formula Used:** $\log(e^z) = z$

**Solution:**

Given:

$$x = e^{x/y}$$

Taking natural logarithm on both sides:

$$\log x = \log(e^{x/y})$$
$$\log x = \frac{x}{y}$$

From this, we can write:

$$y\log x = x$$
$$y = \frac{x}{\log x}$$

Differentiating $y$ with respect to $x$ using the quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\log x \cdot \frac{d}{dx}[x] - x \cdot \frac{d}{dx}[\log x]}{(\log x)^2} \\ &= \frac{\log x \cdot 1 - x \cdot \frac{1}{x}}{(\log x)^2} \\ &= \frac{\log x - 1}{(\log x)^2}\end{aligned}$$

Since $\log x = \dfrac{x}{y}$, we substitute $\log x = \dfrac{x}{y}$ into the numerator:

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{x}{y} - 1}{\frac{x}{y} \cdot \log x} \\ &= \frac{\frac{x-y}{y}}{\frac{x \log x}{y}} \\ &= \frac{x-y}{x \log x}\end{aligned}$$

**Hence Proved**

$\dfrac{dy}{dx} = \dfrac{x-y}{x \log x}$, hence proved.

## Question 6 (iii)

If $x^y = e^{x-y}$, prove that $\dfrac{dy}{dx} = \dfrac{\log x}{(\log xe)^2}$.

**TYPE 48** **Implicit Logarithmic Differentiation**

★ **Likely Exam Question**

**To Prove:** $\dfrac{dy}{dx} = \dfrac{\log x}{(\log xe)^2}$

**Concept / Formula Used:** $\log(ab) = \log a + \log b \qquad \log e = 1$

**Solution:**

Given equation:

$$x^y = e^{x-y}$$

Taking natural logarithm on both sides:

$$\begin{aligned}\log(x^y) &= \log(e^{x-y}) \\ y \log x &= (x-y) \log e \\ y \log x &= x - y \quad (\text{since } \log e = 1)\end{aligned}$$

Rearranging to express $y$ explicitly in terms of $x$:

$$\begin{aligned} y\log x + y &= x \\ y(1+\log x) &= x \\ y &= \frac{x}{1+\log x} \end{aligned}$$

Differentiating $y$ with respect to $x$ using the quotient rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{(1+\log x)\cdot \frac{d}{dx}[x] - x\cdot \frac{d}{dx}[1+\log x]}{(1+\log x)^2} \\ &= \frac{(1+\log x)\cdot 1 - x\cdot \frac{1}{x}}{(1+\log x)^2} \\ &= \frac{1+\log x - 1}{(1+\log x)^2} \\ &= \frac{\log x}{(1+\log x)^2} \end{aligned}$$

Since $1+\log x = \log e + \log x = \log(xe)$:

$$\frac{dy}{dx} = \frac{\log x}{(\log xe)^2}$$

**Hence Proved**

$\dfrac{dy}{dx} = \dfrac{\log x}{(\log xe)^2}$, hence proved.

## Question 6 (iv)

If $x^{16}y^9 = (x^2+y)^{17}$, prove that $\dfrac{dy}{dx} = \dfrac{2y}{x}$.

**TYPE 48 Implicit Logarithmic Differentiation**

**★ Likely Exam Question**

**To Prove:** $\dfrac{dy}{dx} = \dfrac{2y}{x}$

**Concept / Formula Used:** $\log(u\cdot v) = \log u + \log v \qquad \log(u^k) = k\log u$

**Solution:**

Given equation:

$$x^{16}y^9 = (x^2+y)^{17}$$

Taking natural logarithm on both sides:

$$\begin{aligned} \log(x^{16}y^9) &= \log((x^2+y)^{17}) \\ \log(x^{16}) + \log(y^9) &= 17\log(x^2+y) \\ 16\log x + 9\log y &= 17\log(x^2+y) \end{aligned}$$

Differentiating both sides with respect to $x$:

$$\frac{16}{x} + \frac{9}{y}\frac{dy}{dx} = \frac{17}{x^2+y} \cdot \left(2x + \frac{dy}{dx}\right)$$
$$\frac{16}{x} + \frac{9}{y}\frac{dy}{dx} = \frac{34x}{x^2+y} + \frac{17}{x^2+y}\frac{dy}{dx}$$

Grouping terms involving $\frac{dy}{dx}$ on one side:

$$\left(\frac{9}{y} - \frac{17}{x^2+y}\right)\frac{dy}{dx} = \frac{34x}{x^2+y} - \frac{16}{x}$$
$$\left(\frac{9(x^2+y) - 17y}{y(x^2+y)}\right)\frac{dy}{dx} = \frac{34x^2 - 16(x^2+y)}{x(x^2+y)}$$
$$\left(\frac{9x^2+9y-17y}{y(x^2+y)}\right)\frac{dy}{dx} = \frac{34x^2 - 16x^2 - 16y}{x(x^2+y)}$$
$$\left(\frac{9x^2-8y}{y(x^2+y)}\right)\frac{dy}{dx} = \frac{18x^2 - 16y}{x(x^2+y)}$$
$$\left(\frac{9x^2-8y}{y(x^2+y)}\right)\frac{dy}{dx} = \frac{2(9x^2 - 8y)}{x(x^2+y)}$$

Cancelling $(x^2+y)$ from both denominators and $(9x^2-8y)$ from both numerators:

$$\frac{1}{y}\frac{dy}{dx} = \frac{2}{x}$$
$$\frac{dy}{dx} = \frac{2y}{x}$$

**Hence Proved**

$\frac{dy}{dx} = \frac{2y}{x}$, hence proved.

## Question 7

Find the derivative of $x^x + a^x + x^a + a^a$ for some fixed $a > 0, x > 0$.

**TYPE 49 Logarithmic Differentiation of Sums**

**★ Likely Exam Question**

**Given:** Let $y = x^x + a^x + x^a + a^a$, where $a > 0, x > 0$

**Concept / Formula Used:** $\frac{d}{dx}(a^x) = a^x \log a$ $\quad \frac{d}{dx}(x^a) = ax^{a-1}$ $\quad \frac{d}{dx}(a^a) = 0$ $\quad \frac{d}{dx}(x^x) = x^x(1 + \log x)$

**Solution:**

Let $y = u + v + w + z$, where:

$$u = x^x, \quad v = a^x, \quad w = x^a, \quad z = a^a$$

To find $\dfrac{du}{dx}$, take logarithm on both sides of $u = x^x$:

$$\begin{aligned}\log u &= x \log x \\ \frac{1}{u}\frac{du}{dx} &= x \cdot \frac{1}{x} + \log x \cdot 1 \\ \frac{du}{dx} &= u(1 + \log x) = x^x(1 + \log x)\end{aligned}$$

For standard terms:

$$\begin{aligned}\frac{dv}{dx} &= \frac{d}{dx}(a^x) = a^x \log a \\ \frac{dw}{dx} &= \frac{d}{dx}(x^a) = ax^{a-1} \\ \frac{dz}{dx} &= \frac{d}{dx}(a^a) = 0 \quad \text{(since } a^a \text{ is constant)}\end{aligned}$$

Therefore, differentiating $y$ with respect to $x$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx} + \frac{dz}{dx} \\ &= x^x(1 + \log x) + a^x \log a + ax^{a-1} + 0\end{aligned}$$

**Final Answer**

$\dfrac{dy}{dx} = x^x(1 + \log x) + a^x \log a + ax^{a-1}$

## Question 8 (i)

Differentiate $x^{\log x} + (\log x)^x$ w.r.t. $x$.

**TYPE 49** **Logarithmic Differentiation of Sums**

**Given:** Let $y = u + v$, where $u = x^{\log x}$ and $v = (\log x)^x$

**Concept / Formula Used:** $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

**Solution:**

For $u = x^{\log x}$:

$$\begin{aligned}\log u &= \log x \cdot \log x = (\log x)^2 \\ \frac{1}{u}\frac{du}{dx} &= 2 \log x \cdot \frac{1}{x} \\ \frac{du}{dx} &= u \cdot \frac{2 \log x}{x} = x^{\log x} \cdot \frac{2 \log x}{x}\end{aligned}$$

For $v = (\log x)^x$:

$$\log v = x \log(\log x)$$
$$\frac{1}{v}\frac{dv}{dx} = x \cdot \left(\frac{1}{\log x} \cdot \frac{1}{x}\right) + \log(\log x) \cdot 1$$
$$\frac{1}{v}\frac{dv}{dx} = \frac{1}{\log x} + \log(\log x)$$
$$\frac{dv}{dx} = v\left[\frac{1}{\log x} + \log(\log x)\right] = (\log x)^x \left[\frac{1}{\log x} + \log(\log x)\right]$$

Combining both derivatives:

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$
$$= x^{\log x} \cdot \frac{2\log x}{x} + (\log x)^x \left[\frac{1}{\log x} + \log(\log x)\right]$$

**Final Answer**

$$\frac{dy}{dx} = x^{\log x} \cdot \frac{2\log x}{x} + (\log x)^x \left[\frac{1}{\log x} + \log(\log x)\right]$$

## Question 8 (ii)

Differentiate $(\sin x)^{\cos x} + x^{\sin x}$ w.r.t. $x$.

**TYPE 49 Logarithmic Differentiation of Sums**

**Given:** Let $y = u + v$, where $u = (\sin x)^{\cos x}$ and $v = x^{\sin x}$

**Concept / Formula Used:** $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

### Solution:

For $u = (\sin x)^{\cos x}$:

$$\log u = \cos x \log(\sin x)$$
$$\frac{1}{u}\frac{du}{dx} = \cos x \cdot \frac{\cos x}{\sin x} + \log(\sin x) \cdot (-\sin x)$$
$$= \cos x \cot x - \sin x \log(\sin x)$$
$$\frac{du}{dx} = (\sin x)^{\cos x}[\cos x \cot x - \sin x \log(\sin x)]$$

For $v = x^{\sin x}$:

$$\log v = \sin x \log x$$
$$\frac{1}{v}\frac{dv}{dx} = \sin x \cdot \frac{1}{x} + \log x \cdot \cos x$$
$$\frac{dv}{dx} = x^{\sin x}\left[\frac{\sin x}{x} + \cos x \log x\right]$$

Combining the results:

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$
$$= (\sin x)^{\cos x}[\cos x \cot x - \sin x \log(\sin x)] + x^{\sin x}\left[\frac{\sin x}{x} + \cos x \log x\right]$$

**Final Answer**

$$\frac{dy}{dx} = (\sin x)^{\cos x}[\cos x \cot x - \sin x \log(\sin x)] + x^{\sin x}\left[\frac{\sin x}{x} + \cos x \log x\right]$$

## Question 8 (iii)

Differentiate $x^{\cos x} + (\cos x)^x$ w.r.t. $x$.

**TYPE 49 Logarithmic Differentiation of Sums**

**Given:** Let $y = u + v$, where $u = x^{\cos x}$ and $v = (\cos x)^x$

**Concept / Formula Used:** $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

**Solution:**

For $u = x^{\cos x}$:

$$\log u = \cos x \log x$$
$$\frac{1}{u}\frac{du}{dx} = \cos x \cdot \frac{1}{x} + \log x \cdot (-\sin x)$$
$$= \frac{\cos x}{x} - \sin x \log x$$
$$\frac{du}{dx} = x^{\cos x}\left[\frac{\cos x}{x} - \sin x \log x\right]$$

For $v = (\cos x)^x$:

$$\log v = x \log(\cos x)$$
$$\frac{1}{v}\frac{dv}{dx} = x \cdot \frac{1}{\cos x}(-\sin x) + \log(\cos x) \cdot 1$$
$$= -x \tan x + \log(\cos x)$$
$$\frac{dv}{dx} = (\cos x)^x[\log(\cos x) - x \tan x]$$

Combining the two derivatives:

$$\frac{dy}{dx} = x^{\cos x}\left[\frac{\cos x}{x} - \sin x \log x\right] + (\cos x)^x[\log(\cos x) - x \tan x]$$

**Final Answer**

$$\frac{dy}{dx} = x^{\cos x}\left[\frac{\cos x}{x} - \sin x \log x\right] + (\cos x)^x[\log(\cos x) - x \tan x]$$

## Question 8 (iv)

Differentiate $x^{\cos x} + (\sin x)^{\tan x}$ w.r.t. $x$.

**TYPE 49** **Logarithmic Differentiation of Sums**

**Given:** Let $y = u + v$, where $u = x^{\cos x}$ and $v = (\sin x)^{\tan x}$

**Concept / Formula Used:** $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

**Solution:**

For $u = x^{\cos x}$:

$$\begin{aligned}\log u &= \cos x \log x\\ \frac{1}{u}\frac{du}{dx} &= \cos x \cdot \frac{1}{x} - \sin x \log x\\ \frac{du}{dx} &= x^{\cos x}\left[\frac{\cos x}{x} - \sin x \log x\right]\end{aligned}$$

For $v = (\sin x)^{\tan x}$:

$$\begin{aligned}\log v &= \tan x \log(\sin x)\\ \frac{1}{v}\frac{dv}{dx} &= \tan x \cdot \frac{\cos x}{\sin x} + \log(\sin x) \cdot \sec^2 x\\ &= \frac{\sin x}{\cos x} \cdot \frac{\cos x}{\sin x} + \sec^2 x \log(\sin x)\\ &= 1 + \sec^2 x \log(\sin x)\\ \frac{dv}{dx} &= (\sin x)^{\tan x}[1 + \sec^2 x \log(\sin x)]\end{aligned}$$

Combining both:

$$\frac{dy}{dx} = x^{\cos x}\left[\frac{\cos x}{x} - \sin x \log x\right] + (\sin x)^{\tan x}[1 + \sec^2 x \log(\sin x)]$$

**Final Answer**

$$\frac{dy}{dx} = x^{\cos x}\left[\frac{\cos x}{x} - \sin x \log x\right] + (\sin x)^{\tan x}[1 + \sec^2 x \log(\sin x)]$$

## Question 8 (v)

Differentiate $(\sin 2x)^x + \sin^{-1}\sqrt{3x}$ w.r.t. $x$.

**TYPE 49** **Logarithmic Differentiation of Sums**

**Given:** Let $y = u + v$, where $u = (\sin 2x)^x$ and $v = \sin^{-1}\sqrt{3x}$

**Concept / Formula Used:** $\frac{d}{dx}(\sin^{-1} z) = \frac{1}{\sqrt{1-z^2}} \cdot \frac{dz}{dx}$

**Solution:**

For $u = (\sin 2x)^x$:

$$\begin{aligned}\log u &= x \log(\sin 2x) \\ \frac{1}{u}\frac{du}{dx} &= x \cdot \frac{1}{\sin 2x} \cdot (2\cos 2x) + \log(\sin 2x) \cdot 1 \\ &= 2x \cot 2x + \log(\sin 2x) \\ \frac{du}{dx} &= (\sin 2x)^x[2x \cot 2x + \log(\sin 2x)]\end{aligned}$$

For $v = \sin^{-1}\sqrt{3x}$:

$$\begin{aligned}\frac{dv}{dx} &= \frac{1}{\sqrt{1-(\sqrt{3x})^2}} \cdot \frac{d}{dx}(\sqrt{3x}) \\ &= \frac{1}{\sqrt{1-3x}} \cdot \frac{\sqrt{3}}{2\sqrt{x}} \\ &= \frac{\sqrt{3}}{2\sqrt{x(1-3x)}} \\ &= \frac{3}{2\sqrt{3x-9x^2}}\end{aligned}$$

Combining both parts:

$$\frac{dy}{dx} = (\sin 2x)^x[2x \cot 2x + \log(\sin 2x)] + \frac{3}{2\sqrt{3x-9x^2}}$$

**Final Answer**

$$\frac{dy}{dx} = (\sin 2x)^x[2x \cot 2x + \log(\sin 2x)] + \frac{3}{2\sqrt{3x-9x^2}}$$

## Question 8 (vi)

Differentiate $e^{\sin x} + (\tan x)^x$ w.r.t. $x$.

**TYPE 49** **Logarithmic Differentiation of Sums**

**Given:** Let $y = u + v$, where $u = e^{\sin x}$ and $v = (\tan x)^x$

**Concept / Formula Used:** $\frac{d}{dx}(e^{g(x)}) = e^{g(x)}g'(x)$

**Solution:**

For $u = e^{\sin x}$:

$$\frac{du}{dx} = e^{\sin x} \cdot \frac{d}{dx}(\sin x) = e^{\sin x}\cos x$$

For $v = (\tan x)^x$:

$$\begin{aligned}
\log v &= x \log(\tan x) \\
\frac{1}{v}\frac{dv}{dx} &= x \cdot \frac{1}{\tan x} \cdot \sec^2 x + \log(\tan x) \cdot 1 \\
&= x \cdot \frac{\cos x}{\sin x} \cdot \frac{1}{\cos^2 x} + \log(\tan x) \\
&= \frac{x}{\sin x \cos x} + \log(\tan x) \\
&= \frac{2x}{2 \sin x \cos x} + \log(\tan x) \\
&= 2x \csc 2x + \log(\tan x) \\
\frac{dv}{dx} &= (\tan x)^x [2x \csc 2x + \log(\tan x)]
\end{aligned}$$

Combining both:

$$\frac{dy}{dx} = e^{\sin x} \cos x + (\tan x)^x [2x \csc 2x + \log(\tan x)]$$

**Final Answer**

$\frac{dy}{dx} = e^{\sin x} \cos x + (\tan x)^x [2x \csc 2x + \log(\tan x)]$

## Question 8 (vii)

Differentiate $x^x - 2^{\sin x}$ w.r.t. $x$.

**TYPE 49 Logarithmic Differentiation of Sums**

### ★ Likely Exam Question

**Given:** Let $y = u - v$, where $u = x^x$ and $v = 2^{\sin x}$

**Concept / Formula Used:** $\frac{d}{dx}(a^{g(x)}) = a^{g(x)} \log a \cdot g'(x)$

**Solution:**

For $u = x^x$:

$$\begin{aligned}
\log u &= x \log x \\
\frac{1}{u}\frac{du}{dx} &= x \cdot \frac{1}{x} + \log x \cdot 1 = 1 + \log x \\
\frac{du}{dx} &= x^x (1 + \log x)
\end{aligned}$$

For $v = 2^{\sin x}$:

$$\begin{aligned}
\frac{dv}{dx} &= 2^{\sin x} \cdot \log 2 \cdot \frac{d}{dx}(\sin x) \\
&= 2^{\sin x} \cdot \log 2 \cdot \cos x
\end{aligned}$$

Combining the results:

$$\frac{dy}{dx} = \frac{du}{dx} - \frac{dv}{dx}$$
$$= x^x(1 + \log x) - 2^{\sin x} \cos x \log 2$$

**Final Answer**

$$\frac{dy}{dx} = x^x(1 + \log x) - 2^{\sin x} \cos x \log 2$$

## Question 8 (viii)

Differentiate $x^{x \cos x} + \dfrac{x^2 + 1}{x^2 - 1}$ w.r.t. $x$.

**TYPE 49** **Logarithmic Differentiation of Sums**

**Given:** Let $y = u + v$, where $u = x^{x \cos x}$ and $v = \dfrac{x^2 + 1}{x^2 - 1}$

**Concept / Formula Used:** $\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{gf' - fg'}{g^2}$

**Solution:**

For $u = x^{x \cos x}$:

$$\log u = x \cos x \log x$$

Differentiating w.r.t. $x$ using the product rule for three terms:

$$\frac{1}{u}\frac{du}{dx} = \frac{d}{dx}(x) \cdot \cos x \log x + x \cdot \frac{d}{dx}(\cos x) \cdot \log x + x \cos x \cdot \frac{d}{dx}(\log x)$$
$$= 1 \cdot \cos x \log x + x(-\sin x) \log x + x \cos x \cdot \frac{1}{x}$$
$$= \cos x \log x - x \sin x \log x + \cos x$$
$$= (1 + \log x) \cos x - x \sin x \log x$$
$$\frac{du}{dx} = x^{x \cos x}[(1 + \log x) \cos x - x \sin x \log x]$$

For $v = \dfrac{x^2 + 1}{x^2 - 1}$:

$$\frac{dv}{dx} = \frac{(x^2 - 1)\frac{d}{dx}(x^2 + 1) - (x^2 + 1)\frac{d}{dx}(x^2 - 1)}{(x^2 - 1)^2}$$
$$= \frac{(x^2 - 1)(2x) - (x^2 + 1)(2x)}{(x^2 - 1)^2}$$
$$= \frac{2x^3 - 2x - 2x^3 - 2x}{(x^2 - 1)^2}$$
$$= \frac{-4x}{(x^2 - 1)^2}$$

Combining both derivatives:

$$\frac{dy}{dx} = x^{x\cos x}[(1+\log x)\cos x - x\sin x\log x] - \frac{4x}{(x^2-1)^2}$$

**Final Answer**

$$\frac{dy}{dx} = x^{x\cos x}[(1+\log x)\cos x - x\sin x\log x] - \frac{4x}{(x^2-1)^2}$$

## Question 9 (i)

If $y = (\log x)^{\cos x} + \dfrac{x^2+1}{x^2-1}$, find $\dfrac{dy}{dx}$.

**TYPE 49 Logarithmic Differentiation of Sums**

**Given:** Let $y = u + v$, where $u = (\log x)^{\cos x}$ and $v = \dfrac{x^2+1}{x^2-1}$

**To Find:** $\dfrac{dy}{dx}$

**Concept / Formula Used:** $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

**Solution:**

For $u = (\log x)^{\cos x}$:

$$\begin{aligned}
\log u &= \cos x \log(\log x) \\
\frac{1}{u}\frac{du}{dx} &= \cos x \cdot \frac{1}{\log x} \cdot \frac{1}{x} + \log(\log x) \cdot (-\sin x) \\
&= \frac{\cos x}{x\log x} - \sin x \log(\log x) \\
\frac{du}{dx} &= (\log x)^{\cos x}\left[\frac{\cos x}{x\log x} - \sin x\log(\log x)\right]
\end{aligned}$$

For $v = \dfrac{x^2+1}{x^2-1}$:

$$\begin{aligned}
\frac{dv}{dx} &= \frac{(x^2-1)(2x) - (x^2+1)(2x)}{(x^2-1)^2} \\
&= \frac{-4x}{(x^2-1)^2}
\end{aligned}$$

Combining the two derivatives:

$$\begin{aligned}
\frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} \\
&= (\log x)^{\cos x}\left[\frac{\cos x}{x\log x} - \sin x\log(\log x)\right] - \frac{4x}{(x^2-1)^2}
\end{aligned}$$

**Final Answer**

$$\frac{dy}{dx} = (\log x)^{\cos x}\left[\frac{\cos x}{x\log x} - \sin x\log(\log x)\right] - \frac{4x}{(x^2-1)^2}$$

## Question 9 (ii)

If $y = e^{x^2\cos x} + (\cos x)^x$, find $\dfrac{dy}{dx}$.

**TYPE 49** **Logarithmic Differentiation of Sums**

**Given:** Let $y = u + v$, where $u = e^{x^2\cos x}$ and $v = (\cos x)^x$

**To Find:** $\dfrac{dy}{dx}$

**Concept / Formula Used:** $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

**Solution:**

For $u = e^{x^2\cos x}$:

$$\begin{aligned}\frac{du}{dx} &= e^{x^2\cos x}\cdot\frac{d}{dx}(x^2\cos x)\\ &= e^{x^2\cos x}[x^2(-\sin x) + \cos x(2x)]\\ &= e^{x^2\cos x}(2x\cos x - x^2\sin x)\end{aligned}$$

For $v = (\cos x)^x$:

$$\begin{aligned}\log v &= x\log(\cos x)\\ \frac{1}{v}\frac{dv}{dx} &= x\cdot\frac{-\sin x}{\cos x} + \log(\cos x)\cdot 1\\ &= -x\tan x + \log(\cos x)\\ \frac{dv}{dx} &= (\cos x)^x[\log(\cos x) - x\tan x]\end{aligned}$$

Combining both parts:

$$\frac{dy}{dx} = e^{x^2\cos x}(2x\cos x - x^2\sin x) + (\cos x)^x[\log(\cos x) - x\tan x]$$

**Final Answer**

$$\frac{dy}{dx} = e^{x^2\cos x}(2x\cos x - x^2\sin x) + (\cos x)^x[\log(\cos x) - x\tan x]$$

## Question 10 (i)

Find $\dfrac{dy}{dx}$ when $x^y y^x = a^b$.

**TYPE 48** **Implicit Logarithmic Differentiation**

**Given:** $x^y y^x = a^b$, where $a, b$ are constants

**To Find:** $\dfrac{dy}{dx}$

**Concept / Formula Used:** $\log(uv) = \log u + \log v$

**Solution:**

Taking natural logarithm on both sides:

$$\log(x^y y^x) = \log(a^b)$$
$$\log(x^y) + \log(y^x) = \log(a^b)$$
$$y \log x + x \log y = b \log a$$

Differentiating both sides with respect to $x$:

$$\frac{d}{dx}[y \log x] + \frac{d}{dx}[x \log y] = \frac{d}{dx}[b \log a]$$
$$\left(y \cdot \frac{1}{x} + \log x \cdot \frac{dy}{dx}\right) + \left(x \cdot \frac{1}{y}\frac{dy}{dx} + \log y \cdot 1\right) = 0$$
$$\frac{y}{x} + \log x \frac{dy}{dx} + \frac{x}{y}\frac{dy}{dx} + \log y = 0$$

Grouping terms containing $\dfrac{dy}{dx}$:

$$\left(\log x + \frac{x}{y}\right)\frac{dy}{dx} = -\left(\log y + \frac{y}{x}\right)$$
$$\left(\frac{y \log x + x}{y}\right)\frac{dy}{dx} = -\left(\frac{x \log y + y}{x}\right)$$
$$\frac{dy}{dx} = -\frac{y}{x} \cdot \frac{x \log y + y}{y \log x + x}$$

**Final Answer**

$$\frac{dy}{dx} = -\frac{y(y + x \log y)}{x(x + y \log x)}$$

## Question 10 (ii)

Find $\dfrac{dy}{dx}$ when $x^y + y^x = a^b$.

**TYPE 49** **Logarithmic Differentiation of Sums**

★ **Likely Exam Question**

**Given:** $x^y + y^x = a^b$, where $a, b$ are constants

**To Find:** $\dfrac{dy}{dx}$

**Concept / Formula Used:** $u = x^y \qquad v = y^x \qquad \frac{du}{dx} + \frac{dv}{dx} = 0$

**Solution:**

Let $u = x^y$ and $v = y^x$. Then $u + v = a^b$. Differentiating with respect to $x$:

$$\frac{du}{dx} + \frac{dv}{dx} = 0$$

For $u = x^y$:

$$\begin{aligned}
\log u &= y \log x \\
\frac{1}{u}\frac{du}{dx} &= y \cdot \frac{1}{x} + \log x \frac{dy}{dx} \\
\frac{du}{dx} &= x^y \left( \frac{y}{x} + \log x \frac{dy}{dx} \right) = yx^{y-1} + x^y \log x \frac{dy}{dx}
\end{aligned}$$

For $v = y^x$:

$$\begin{aligned}
\log v &= x \log y \\
\frac{1}{v}\frac{dv}{dx} &= x \cdot \frac{1}{y}\frac{dy}{dx} + \log y \cdot 1 \\
\frac{dv}{dx} &= y^x \left( \frac{x}{y}\frac{dy}{dx} + \log y \right) = xy^{x-1}\frac{dy}{dx} + y^x \log y
\end{aligned}$$

Substituting $\dfrac{du}{dx}$ and $\dfrac{dv}{dx}$ into $\dfrac{du}{dx} + \dfrac{dv}{dx} = 0$:

$$\begin{aligned}
yx^{y-1} + x^y \log x \frac{dy}{dx} + xy^{x-1}\frac{dy}{dx} + y^x \log y = 0 \\
(x^y \log x + xy^{x-1})\frac{dy}{dx} &= -(yx^{y-1} + y^x \log y) \\
\frac{dy}{dx} &= -\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}
\end{aligned}$$

**Final Answer**

$$\frac{dy}{dx} = -\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$$

## Question 10 (iii)

Find $\dfrac{dy}{dx}$ when $x^x + y^x = 1$.

**TYPE 49** **Logarithmic Differentiation of Sums**

**Given:** $x^x + y^x = 1$

**To Find:** $\dfrac{dy}{dx}$

**Concept / Formula Used:** $u = x^x \qquad v = y^x \qquad \frac{du}{dx} + \frac{dv}{dx} = 0$

**Solution:**

Let $u = x^x$ and $v = y^x$. Then $u + v = 1$. Differentiating with respect to $x$:

$$\frac{du}{dx} + \frac{dv}{dx} = 0$$

For $u = x^x$:

$$\begin{aligned}
\log u &= x \log x \\
\frac{1}{u}\frac{du}{dx} &= x \cdot \frac{1}{x} + \log x \cdot 1 = 1 + \log x \\
\frac{du}{dx} &= x^x(1 + \log x)
\end{aligned}$$

For $v = y^x$:

$$\begin{aligned}
\log v &= x \log y \\
\frac{1}{v}\frac{dv}{dx} &= x \cdot \frac{1}{y}\frac{dy}{dx} + \log y \cdot 1 \\
\frac{dv}{dx} &= y^x\left(\frac{x}{y}\frac{dy}{dx} + \log y\right) = xy^{x-1}\frac{dy}{dx} + y^x \log y
\end{aligned}$$

Substituting into $\dfrac{du}{dx} + \dfrac{dv}{dx} = 0$:

$$\begin{aligned}
x^x(1 + \log x) + xy^{x-1}\frac{dy}{dx} + y^x \log y = 0 \\
xy^{x-1}\frac{dy}{dx} &= -[x^x(1 + \log x) + y^x \log y] \\
\frac{dy}{dx} &= -\frac{x^x(1 + \log x) + y^x \log y}{xy^{x-1}}
\end{aligned}$$

**Final Answer**

$$\frac{dy}{dx} = -\frac{x^x(1 + \log x) + y^x \log y}{xy^{x-1}}$$

## Question 10 (iv)

Find $\dfrac{dy}{dx}$ when $(\sin x)^y = x + y$.

**TYPE 48** **Implicit Logarithmic Differentiation**

**Given:** $(\sin x)^y = x + y$

**To Find:** $\dfrac{dy}{dx}$

**Concept / Formula Used:** $\log(u^v) = v \log u$

**Solution:**

Taking natural logarithm on both sides:

$$\begin{aligned}\log\left((\sin x)^y\right) &= \log(x+y)\\ y\log(\sin x) &= \log(x+y)\end{aligned}$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}y\cdot\frac{d}{dx}[\log(\sin x)] + \log(\sin x)\cdot\frac{dy}{dx} &= \frac{d}{dx}[\log(x+y)]\\ y\cdot\left(\frac{1}{\sin x}\cdot\cos x\right) + \log(\sin x)\frac{dy}{dx} &= \frac{1}{x+y}\left(1+\frac{dy}{dx}\right)\\ y\cot x + \log(\sin x)\frac{dy}{dx} &= \frac{1}{x+y} + \frac{1}{x+y}\frac{dy}{dx}\end{aligned}$$

Grouping terms containing $\dfrac{dy}{dx}$ on the left side:

$$\begin{aligned}\left(\log(\sin x) - \frac{1}{x+y}\right)\frac{dy}{dx} &= \frac{1}{x+y} - y\cot x\\ \left(\frac{(x+y)\log(\sin x)-1}{x+y}\right)\frac{dy}{dx} &= \frac{1-y(x+y)\cot x}{x+y}\\ ((x+y)\log(\sin x)-1)\frac{dy}{dx} &= 1-y(x+y)\cot x\\ \frac{dy}{dx} &= \frac{1-y(x+y)\cot x}{(x+y)\log(\sin x)-1}\end{aligned}$$

**Final Answer**

$$\frac{dy}{dx} = \frac{1-y(x+y)\cot x}{(x+y)\log(\sin x)-1}$$

## Question 10 (v)

Find $\dfrac{dy}{dx}$ when $xy = e^{x-y}$.

TYPE 48 **Implicit Logarithmic Differentiation**

**★ Likely Exam Question**

**Given:** $xy = e^{x-y}$

**To Find:** $\dfrac{dy}{dx}$

**Concept / Formula Used:** $\log(xy) = \log x + \log y \qquad \log(e^{x-y}) = x - y$

**Solution:**

Taking natural logarithm on both sides:

$$\log(xy) = \log(e^{x-y})$$
$$\log x + \log y = x - y$$

Differentiating both sides with respect to $x$:

$$\frac{1}{x} + \frac{1}{y}\frac{dy}{dx} = 1 - \frac{dy}{dx}$$

Grouping terms involving $\dfrac{dy}{dx}$ on one side:

$$\frac{1}{y}\frac{dy}{dx} + \frac{dy}{dx} = 1 - \frac{1}{x}$$
$$\left(\frac{1}{y} + 1\right)\frac{dy}{dx} = \frac{x-1}{x}$$
$$\left(\frac{1+y}{y}\right)\frac{dy}{dx} = \frac{x-1}{x}$$
$$\frac{dy}{dx} = \frac{y(x-1)}{x(1+y)}$$

**Final Answer**

$$\frac{dy}{dx} = \frac{y(x-1)}{x(y+1)}$$

## Question 10 (vi)

Find $\dfrac{dy}{dx}$ when $(\cos x)^y = (\cos y)^x$.

**TYPE 48** **Implicit Logarithmic Differentiation**

**Given:** $(\cos x)^y = (\cos y)^x$

**To Find:** $\dfrac{dy}{dx}$

**Concept / Formula Used:** $\log(u^v) = v \log u$

**Solution:**

Taking natural logarithm on both sides:

$$\log((\cos x)^y) = \log((\cos y)^x)$$
$$y\log(\cos x) = x\log(\cos y)$$

Differentiating both sides with respect to $x$ using the product rule:

$$y \cdot \frac{d}{dx}[\log(\cos x)] + \log(\cos x) \cdot \frac{dy}{dx} = x \cdot \frac{d}{dx}[\log(\cos y)] + \log(\cos y) \cdot \frac{d}{dx}[x]$$
$$y \cdot \left(\frac{-\sin x}{\cos x}\right) + \log(\cos x)\frac{dy}{dx} = x \cdot \left(\frac{-\sin y}{\cos y}\frac{dy}{dx}\right) + \log(\cos y) \cdot 1$$
$$-y\tan x + \log(\cos x)\frac{dy}{dx} = -x\tan y\frac{dy}{dx} + \log(\cos y)$$

Grouping $\frac{dy}{dx}$ terms:

$$(\log(\cos x) + x\tan y)\frac{dy}{dx} = \log(\cos y) + y\tan x$$
$$\frac{dy}{dx} = \frac{\log(\cos y) + y\tan x}{\log(\cos x) + x\tan y}$$

**Final Answer**

$$\frac{dy}{dx} = \frac{\log(\cos y) + y\tan x}{\log(\cos x) + x\tan y}$$

## Question 11

If $y = x^{x^{x^{\cdot^{\cdot^{\infty}}}}}$, prove that $\dfrac{dy}{dx} = \dfrac{y^2}{x(1 - y\log x)}$.

**TYPE 45 Differentiation of Infinite Recursive Functions**

★ **Likely Exam Question**

**To Prove:** $\dfrac{dy}{dx} = \dfrac{y^2}{x(1 - y\log x)}$

**Concept / Formula Used:** $y = x^y$

**Solution:**

Given the infinite power tower:

$$y = x^{x^{x^{\cdot^{\cdot^{\infty}}}}}$$

Since the exponent tower is infinite, we can rewrite the relation as:

$$y = x^y$$

Taking natural logarithm on both sides:

$$\log y = \log(x^y)$$
$$\log y = y \log x$$

Differentiating both sides with respect to $x$:

$$\frac{1}{y}\frac{dy}{dx} = y \cdot \frac{d}{dx}[\log x] + \log x \cdot \frac{dy}{dx}$$
$$\frac{1}{y}\frac{dy}{dx} = \frac{y}{x} + \log x \frac{dy}{dx}$$

Collecting terms involving $\frac{dy}{dx}$ on the left-hand side:

$$\left(\frac{1}{y} - \log x\right)\frac{dy}{dx} = \frac{y}{x}$$
$$\left(\frac{1 - y\log x}{y}\right)\frac{dy}{dx} = \frac{y}{x}$$
$$\frac{dy}{dx} = \frac{y}{x} \cdot \frac{y}{1 - y\log x}$$
$$\frac{dy}{dx} = \frac{y^2}{x(1 - y\log x)}$$

**Hence Proved**

$\frac{dy}{dx} = \frac{y^2}{x(1 - y\log x)}$, hence proved.

## Question 12

If $y = x^x$, then find $\frac{dy}{dx}$ at $x = 1$.

**TYPE 50** **Logarithmic Differentiation with Evaluation**

**Given:** $y = x^x$

**To Find:** The value of $\frac{dy}{dx}$ at $x = 1$

**Concept / Formula Used:** $\log(u^v) = v \log u$

**Solution:**

Given function:

$$y = x^x$$

Taking natural logarithm on both sides:

$$\log y = x \log x$$

Differentiating both sides with respect to $x$:

$$\frac{1}{y}\frac{dy}{dx} = x \cdot \frac{1}{x} + \log x \cdot 1$$
$$\frac{1}{y}\frac{dy}{dx} = 1 + \log x$$
$$\frac{dy}{dx} = y(1 + \log x)$$

Substituting back $y = x^x$:

$$\frac{dy}{dx} = x^x(1 + \log x)$$

Now, evaluating $\frac{dy}{dx}$ at $x = 1$:

$$\left.\frac{dy}{dx}\right|_{x=1} = (1)^1 \cdot (1 + \log 1)$$

Since $\log 1 = 0$:

$$\left.\frac{dy}{dx}\right|_{x=1} = 1 \cdot (1 + 0) = 1$$

**Final Answer**

$\left.\frac{dy}{dx}\right|_{x=1} = 1$

## Question 9(i)

Find $\frac{dy}{dx}$ if $y = (\log x)^{\cos x}$.

**TYPE 61** Logarithmic Differentiation

**Given:** $y = (\log x)^{\cos x}$

**Concept / Formula Used:** $\frac{d}{dx}[u^v] = u^v\left(v' \ln u + v\frac{u'}{u}\right)$ log

**Solution:**

Taking natural logarithm on both sides of the given equation:

$$\log y = \log\left((\log x)^{\cos x}\right)$$

Using the logarithmic property $\log(a^b) = b \log a$:

$$\log y = \cos x \cdot \log(\log x)$$

Differentiating both sides with respect to $x$ using the chain rule on the left and the product rule on the right:

$$\frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}[\cos x] \cdot \log(\log x) + \cos x \cdot \frac{d}{dx}[\log(\log x)]$$

Evaluating the derivatives on the right-hand side:

$$\frac{1}{y}\frac{dy}{dx} = (-\sin x) \cdot \log(\log x) + \cos x \cdot \frac{1}{\log x} \cdot \frac{d}{dx}[\log x]$$

$$\frac{1}{y}\frac{dy}{dx} = -\sin x \log(\log x) + \cos x \cdot \frac{1}{\log x} \cdot \frac{1}{x}$$

Multiplying both sides by $y$ to isolate $\frac{dy}{dx}$:

$$\frac{dy}{dx} = y\left(-\sin x \log(\log x) + \frac{\cos x}{x \log x}\right)$$

Back-substituting $y = (\log x)^{\cos x}$:

$$\frac{dy}{dx} = (\log x)^{\cos x}\left(\frac{\cos x}{x \log x} - \sin x \log(\log x)\right)$$

**Final Answer**

$$\frac{dy}{dx} = (\log x)^{\cos x}\left[\frac{\cos x}{x \log x} - \sin x \log(\log x)\right]$$

**Common Mistake:** Forgetting to apply the chain rule when differentiating $\log(\log x)$ with respect to $x$.

## Question 9(ii)

Find $\frac{dy}{dx}$ if $y = x^{e^{x^2} \cos x}$.

**TYPE 61** **Logarithmic Differentiation**

**Given:** $y = x^{e^{x^2} \cos x}$

**Concept / Formula Used:** $\frac{d}{dx}[u^v] = u^v\left(v' \ln u + v\frac{u'}{u}\right)$

**Solution:**

Taking natural logarithm on both sides:

$$\log y = \log\left(x^{e^{x^2} \cos x}\right)$$

Using the property $\log(a^b) = b \log a$:

$$\log y = e^{x^2} \cos x \cdot \log x$$

Differentiating both sides with respect to $x$ using the product rule for three functions on the right-hand side:

$$\frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}[e^{x^2}] \cdot \cos x \cdot \log x + e^{x^2} \cdot \frac{d}{dx}[\cos x] \cdot \log x + e^{x^2} \cdot \cos x \cdot \frac{d}{dx}[\log x]$$

Evaluating each derivative term:

$$\begin{aligned}\frac{1}{y}\frac{dy}{dx} &= (e^{x^2} \cdot 2x) \cdot \cos x \cdot \log x + e^{x^2} \cdot (-\sin x) \cdot \log x + e^{x^2} \cdot \cos x \cdot \frac{1}{x} \\ &= e^{x^2}\left[2x \cos x \log x - \sin x \log x + \frac{\cos x}{x}\right]\end{aligned}$$

Multiplying both sides by $y = x^{e^{x^2} \cos x}$:

$$\frac{dy}{dx} = x^{e^{x^2} \cos x} \cdot e^{x^2}\left(2x \cos x \log x - \sin x \log x + \frac{\cos x}{x}\right)$$

**Final Answer**

$$\frac{dy}{dx} = x^{e^{x^2} \cos x} e^{x^2}\left(2x \cos x \log x - \sin x \log x + \frac{\cos x}{x}\right)$$

**Common Mistake:** Treating the exponent as a simple function and failing to apply the product rule correctly for the product of three terms.

## Question 10(i)

Find $\dfrac{dy}{dx}$ if $x^y + y^x = a^b$. (Implicit differentiation / Logarithmic differentiation)

**TYPE 62 Implicit Differentiation with Exponential/Logarithmic Terms**

**Given:** $x^y + y^x = a^b$

**Concept / Formula Used:** $u = x^y$ $\quad v = y^x$ $\quad u + v = a^b$ $\quad \dfrac{du}{dx} + \dfrac{dv}{dx} = 0$

**Solution:**

Let $u = x^y$ and $v = y^x$. Then the given equation becomes:

$$u + v = a^b$$

Differentiating implicitly with respect to $x$:

$$\frac{du}{dx} + \frac{dv}{dx} = 0$$

Now, considering $u = x^y$:

$$\log u = y \log x$$

Differentiating both sides with respect to $x$:

$$\frac{1}{u}\frac{du}{dx} = \frac{dy}{dx}\log x + y \cdot \frac{1}{x}$$

$$\frac{du}{dx} = x^y\left(\log x\frac{dy}{dx} + \frac{y}{x}\right)$$

Next, considering $v = y^x$:

$$\log v = x \log y$$

Differentiating both sides with respect to $x$:

$$\frac{1}{v}\frac{dv}{dx} = 1 \cdot \log y + x \cdot \frac{1}{y}\frac{dy}{dx}$$

$$\frac{dv}{dx} = y^x\left(\log y + \frac{x}{y}\frac{dy}{dx}\right)$$

Substituting $\frac{du}{dx}$ and $\frac{dv}{dx}$ into $\frac{du}{dx} + \frac{dv}{dx} = 0$:

$$x^y\left(\log x\frac{dy}{dx} + \frac{y}{x}\right) + y^x\left(\log y + \frac{x}{y}\frac{dy}{dx}\right) = 0$$

Grouping the terms containing $\frac{dy}{dx}$:

$$\frac{dy}{dx}\left(x^y \log x + y^x\frac{x}{y}\right) = -\left(x^y\frac{y}{x} + y^x \log y\right)$$

Isolating $\frac{dy}{dx}$:

$$\frac{dy}{dx} = -\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$$

**Final Answer**

$$\frac{dy}{dx} = -\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$$

**Common Mistake:** Incorrectly differentiating $x^y$ directly as $yx^{y-1}$ without accounting for the variable exponent using logarithmic differentiation.

## Question 10(ii)

Find $\frac{dy}{dx}$ if $x^{y-1} + y^{x-1} = 1$.

**TYPE 62 Implicit Differentiation with Exponential/Logarithmic Terms**

**Given:** $x^{y-1} + y^{x-1} = 1$

**Concept / Formula Used:** $u(x, y) + v(x, y) = C \qquad \frac{d}{dx}[u] + \frac{d}{dx}[v] = 0$

**Solution:**

Let $u = x^{y-1}$ and $v = y^{x-1}$. The equation is $u + v = 1$, so:

$$\frac{du}{dx} + \frac{dv}{dx} = 0$$

For $u = x^{y-1}$:

$$\log u = (y-1)\log x$$

Differentiating with respect to $x$:

$$\frac{1}{u}\frac{du}{dx} = \frac{dy}{dx}\log x + (y-1)\frac{1}{x}$$

$$\frac{du}{dx} = x^{y-1}\left(\log x\frac{dy}{dx} + \frac{y-1}{x}\right)$$

For $v = y^{x-1}$:

$$\log v = (x-1)\log y$$

Differentiating with respect to $x$:

$$\frac{1}{v}\frac{dv}{dx} = 1 \cdot \log y + (x-1)\frac{1}{y}\frac{dy}{dx}$$

$$\frac{dv}{dx} = y^{x-1}\left(\log y + \frac{x-1}{y}\frac{dy}{dx}\right)$$

Substituting $\frac{du}{dx}$ and $\frac{dv}{dx}$ into the sum derivative equation:

$$x^{y-1}\left(\log x\frac{dy}{dx} + \frac{y-1}{x}\right) + y^{x-1}\left(\log y + \frac{x-1}{y}\frac{dy}{dx}\right) = 0$$

Rearranging terms to collect coefficients of $\frac{dy}{dx}$:

$$\frac{dy}{dx}\left[x^{y-1}\log x + y^{x-1}\frac{x-1}{y}\right] = -\left[x^{y-1}\frac{y-1}{x} + y^{x-1}\log y\right]$$

Solving for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = -\frac{y^{x-1}\log y + x^{y-2}(y-1)}{x^{y-1}\log x + y^{x-2}(x-1)}$$

**Final Answer**

$$\frac{dy}{dx} = -\frac{y^{x-1}\log y + x^{y-1}\left(\frac{y-1}{x}\right)}{x^{y-1}\log x + y^{x-1}\left(\frac{x-1}{y}\right)}$$

**Common Mistake:** Algebraic errors when combining fractional powers of $x$ and $y$ after differentiation.

## Question 10(iii)

Find $\dfrac{dy}{dx}$ if $x^x + y^x = xy$.

**TYPE 62** **Implicit Differentiation with Exponential/Logarithmic Terms**

**Given:** $x^x + y^x = xy$

**Concept / Formula Used:** $x^x \quad x^x(1 + \ln x) \quad y^x$

**Solution:**

Let $u = x^x$ and $v = y^x$. The equation can be written as $u + v = xy$. Differentiating both sides with respect to $x$:

$$\frac{du}{dx} + \frac{dv}{dx} = x\frac{dy}{dx} + y \cdot 1$$

For $u = x^x$:

$$\log u = x \log x \implies \frac{1}{u}\frac{du}{dx} = 1 \cdot \log x + x \cdot \frac{1}{x} = \log x + 1$$

$$\frac{du}{dx} = x^x(1 + \log x)$$

For $v = y^x$:

$$\log v = x \log y \implies \frac{1}{v}\frac{dv}{dx} = 1 \cdot \log y + x \cdot \frac{1}{y}\frac{dy}{dx}$$

$$\frac{dv}{dx} = y^x\left(\log y + \frac{x}{y}\frac{dy}{dx}\right)$$

Substituting $\dfrac{du}{dx}$ and $\dfrac{dv}{dx}$ into the derivative relation:

$$x^x(1 + \log x) + y^x\left(\log y + \frac{x}{y}\frac{dy}{dx}\right) = x\frac{dy}{dx} + y$$

Collecting all terms involving $\dfrac{dy}{dx}$ on one side:

$$y^x\frac{x}{y}\frac{dy}{dx} - x\frac{dy}{dx} = y - x^x(1 + \log x) - y^x \log y$$

$$\frac{dy}{dx}\left(xy^{x-1} - x\right) = y - x^x(1 + \log x) - y^x \log y$$

Isolating $\dfrac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{y - x^x(1 + \log x) - y^x \log y}{x(y^{x-1} - 1)}$$

**Final Answer**

$$\frac{dy}{dx} = \frac{y - x^x(1 + \log x) - y^x \log y}{xy^{x-1} - x}$$

**Common Mistake:** Dropping the product rule terms on the right-hand side when differentiating $xy$.

## Question 10(iv)

Find $\dfrac{dy}{dx}$ if $(x+y)^y = \log(\sin x)$.

TYPE 61 **Logarithmic Differentiation**

**Given:** $(x+y)^y = \log(\sin x)$

**Concept / Formula Used:** $\dfrac{d}{dx}[f(x)^{g(x)}]$

**Solution:**

Taking natural logarithm on both sides:

$$\log\left((x+y)^y\right) = \log(\log(\sin x))$$

Using the power property of logarithms:

$$y\log(x+y) = \log(\log(\sin x))$$

Differentiating both sides with respect to $x$ using the product rule on the left-hand side and the chain rule on the right-hand side:

$$\frac{d}{dx}[y]\cdot\log(x+y) + y\cdot\frac{d}{dx}[\log(x+y)] = \frac{1}{\log(\sin x)}\cdot\frac{d}{dx}[\log(\sin x)]$$

Evaluating the derivatives:

$$\frac{dy}{dx}\log(x+y) + y\cdot\frac{1}{x+y}\left(1+\frac{dy}{dx}\right) = \frac{1}{\log(\sin x)}\cdot\frac{1}{\sin x}\cdot\cos x$$

Expanding the left-hand side:

$$\frac{dy}{dx}\log(x+y) + \frac{y}{x+y} + \frac{y}{x+y}\frac{dy}{dx} = \frac{\cot x}{\log(\sin x)}$$

Grouping terms with $\dfrac{dy}{dx}$:

$$\frac{dy}{dx}\left[\log(x+y) + \frac{y}{x+y}\right] = \frac{\cot x}{\log(\sin x)} - \frac{y}{x+y}$$

Writing over common denominators and solving for $\dfrac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{\frac{\cot x}{\log(\sin x)} - \frac{y}{x+y}}{\log(x+y) + \frac{y}{x+y}}$$

Multiplying numerator and denominator by $(x+y)\log(\sin x)$:

$$\frac{dy}{dx} = \frac{(x+y)\cot x - y\log(\sin x)}{(x+y)\log(\sin x)\left[\log(x+y) + \frac{y}{x+y}\right]}$$

$$\frac{dy}{dx} = \frac{(x+y)\cot x - y\log(\sin x)}{(x+y)\log(\sin x)\log(x+y) + y\log(\sin x)}$$

**Final Answer**

$$\frac{dy}{dx} = \frac{(x+y)\cot x - y\log(\sin x)}{\log(\sin x)\left[(x+y)\log(x+y) + y\right]}$$

**Common Mistake:** Incorrectly applying the chain rule to the composite logarithmic term $\log(\sin x)$.

## Question 10(v)

Find $\frac{dy}{dx}$ if $y^x = x^y$.

**TYPE 61 Logarithmic Differentiation**

**Given:** $y^x = x^y$

**Concept / Formula Used:** $\ln(a^b) = b\ln a$

### Solution:

Taking natural logarithm on both sides of the equation:

$$\log(y^x) = \log(x^y)$$

Using the power property of logarithms:

$$x\log y = y\log x$$

Differentiating both sides with respect to $x$ using the product rule on both sides:

$$\frac{d}{dx}[x]\cdot\log y + x\cdot\frac{d}{dx}[\log y] = \frac{d}{dx}[y]\cdot\log x + y\cdot\frac{d}{dx}[\log x]$$

Evaluating the derivatives:

$$1\cdot\log y + x\cdot\frac{1}{y}\frac{dy}{dx} = \frac{dy}{dx}\log x + y\cdot\frac{1}{x}$$

$$\log y + \frac{x}{y}\frac{dy}{dx} = \log x\frac{dy}{dx} + \frac{y}{x}$$

Rearranging terms to collect $\frac{dy}{dx}$ on one side:

$$\frac{x}{y}\frac{dy}{dx} - \log x\frac{dy}{dx} = \frac{y}{x} - \log y$$

$$\frac{dy}{dx}\left(\frac{x}{y}-\log x\right)=\frac{y}{x}-\log y$$

$$\frac{dy}{dx}\left(\frac{x-y\log x}{y}\right)=\frac{y-x\log y}{x}$$

Isolating $\frac{dy}{dx}$:

$$\frac{dy}{dx}=\frac{y}{x}\left(\frac{y-x\log y}{x-y\log x}\right)$$

**Final Answer**

$$\frac{dy}{dx}=\frac{y(y-x\log y)}{x(x-y\log x)}$$

**Common Mistake:** Forgetting to apply the product rule to both sides of the logarithmically simplified equation.

## Question 10(vi)

Find $\frac{dy}{dx}$ if $y=\frac{y\tan x+\log\cos y}{x\tan y+\log\cos x}$. (Implicit / Logarithmic structure)

**TYPE 62** **Implicit Differentiation with Exponential/Logarithmic Terms**

**Given:** $y=\frac{y\tan x+\log\cos y}{x\tan y+\log\cos x}$

**Solution:**

Clearing the denominator by cross-multiplication:

$$y(x\tan y+\log\cos x)=y\tan x+\log\cos y$$

$$xy\tan y+y\log\cos x=y\tan x+\log\cos y$$

Differentiating both sides with respect to $x$: For the first term $xy\tan y$, use product rule on $xy$ and chain rule on $\tan y$:

$$\frac{d}{dx}[xy\tan y]=\left(x\frac{dy}{dx}+y\right)\tan y+xy\sec^2 y\frac{dy}{dx}$$

For the second term $y\log\cos x$:

$$\frac{d}{dx}[y\log\cos x]=\frac{dy}{dx}\log\cos x+y\cdot\frac{1}{\cos x}(-\sin x)=\frac{dy}{dx}\log\cos x-y\tan x$$

For the right-hand side $y\tan x+\log\cos y$:

$$\frac{d}{dx}[y\tan x]=\frac{dy}{dx}\tan x+y\sec^2 x$$

$$\frac{d}{dx}[\log\cos y]=\frac{1}{\cos y}(-\sin y)\frac{dy}{dx}=-\tan y\frac{dy}{dx}$$

Combining all derivatives:

$$\left(x\tan y\frac{dy}{dx}+y\tan y+xy\sec^2 y\frac{dy}{dx}\right)+\left(\log\cos x\frac{dy}{dx}-y\tan x\right)=\left(\tan x\frac{dy}{dx}+y\sec^2 x\right)-\tan y\frac{dy}{dx}$$

Grouping all terms containing $\frac{dy}{dx}$ on the left-hand side:

$$\frac{dy}{dx}\left(x\tan y+xy\sec^2 y+\log\cos x-\tan x+\tan y\right)=y\sec^2 x+y\tan x-y\tan y$$

Isolating $\frac{dy}{dx}$:

$$\frac{dy}{dx}=\frac{y\sec^2 x+y\tan x-y\tan y}{x\tan y+xy\sec^2 y+\log\cos x-\tan x+\tan y}$$

**Final Answer**

$$\frac{dy}{dx}=\frac{y(\sec^2 x+\tan x-\tan y)}{x\tan y+xy\sec^2 y+\log\cos x-\tan x+\tan y}$$

**Common Mistake:** Algebraic omission of terms when differentiating products involving trigonometric and logarithmic functions simultaneously.

## 5.10 – Parametric Differentiation

### Question 1 (i)

Find $\frac{dy}{dx}$ when $x=at^2$ and $y=2at$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x=at^2$ and $y=2at$

**Concept / Formula Used:** $x=f(t)\qquad y=g(t)\qquad \frac{dy}{dx}=\frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x=at^2$$
$$y=2at$$

Differentiating $x$ with respect to $t$:

$$\frac{dx}{dt} = \frac{d}{dt}(at^2)$$
$$\frac{dx}{dt} = a \cdot (2t)$$
$$\frac{dx}{dt} = 2at$$

Differentiating $y$ with respect to $t$:

$$\frac{dy}{dt} = \frac{d}{dt}(2at)$$
$$\frac{dy}{dt} = 2a \cdot (1)$$
$$\frac{dy}{dt} = 2a$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{2a}{2at}$$
$$\frac{dy}{dx} = \frac{1}{t}$$

**Final Answer**

$\frac{dy}{dx} = \frac{1}{t}$

## Question 1 (ii)

Find $\frac{dy}{dx}$ when $x = 2at^2$ and $y = at^4$.

**TYPE 51 Parametric Differentiation**

**Given:** $x = 2at^2$ and $y = at^4$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = 2at^2$$
$$y = at^4$$

Differentiating $x$ with respect to $t$:

$$\frac{dx}{dt} = \frac{d}{dt}(2at^2)$$
$$\frac{dx}{dt} = 2a \cdot (2t)$$
$$\frac{dx}{dt} = 4at$$

Differentiating $y$ with respect to $t$:

$$\frac{dy}{dt} = \frac{d}{dt}(at^4)$$
$$\frac{dy}{dt} = a \cdot (4t^3)$$
$$\frac{dy}{dt} = 4at^3$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{4at^3}{4at}$$
$$\frac{dy}{dx} = t^2$$

**Final Answer**

$\frac{dy}{dx} = t^2$

## Question 2 (i)

Find $\frac{dy}{dx}$ when $x = a\cos\theta$ and $y = b\cos\theta$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = a\cos\theta$ and $y = b\cos\theta$

**Concept / Formula Used:** $x = f(\theta) \qquad y = g(\theta) \qquad \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

**Solution:**

We are given the parametric equations:

$$x = a\cos\theta$$
$$y = b\cos\theta$$

Differentiating $x$ with respect to $\theta$:

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(a\cos\theta)$$
$$\frac{dx}{d\theta} = -a\sin\theta$$

Differentiating $y$ with respect to $\theta$:

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(b\cos\theta)$$
$$\frac{dy}{d\theta} = -b\sin\theta$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$
$$\frac{dy}{dx} = \frac{-b\sin\theta}{-a\sin\theta}$$
$$\frac{dy}{dx} = \frac{b}{a}$$

**Final Answer**

$\frac{dy}{dx} = \frac{b}{a}$

## Question 2 (ii)

Find $\frac{dy}{dx}$ when $x = a\cos\theta$ and $y = a\sin\theta$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = a\cos\theta$ and $y = a\sin\theta$

**Concept / Formula Used:** $x = f(\theta) \qquad y = g(\theta) \qquad \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

**Solution:**

We are given the parametric equations:

$$x = a\cos\theta$$
$$y = a\sin\theta$$

Differentiating $x$ with respect to $\theta$:

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(a\cos\theta)$$
$$\frac{dx}{d\theta} = -a\sin\theta$$

Differentiating $y$ with respect to $\theta$:

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(a\sin\theta)$$
$$\frac{dy}{d\theta} = a\cos\theta$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$
$$\frac{dy}{dx} = \frac{a\cos\theta}{-a\sin\theta}$$
$$\frac{dy}{dx} = -\cot\theta$$

**Final Answer**

$\frac{dy}{dx} = -\cot\theta$

## Question 3 (i)

Find $\frac{dy}{dx}$ when $x = a\sec\theta$ and $y = b\tan\theta$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = a\sec\theta$ and $y = b\tan\theta$

**Concept / Formula Used:** $x = f(\theta) \qquad y = g(\theta) \qquad \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

**Solution:**

We are given the parametric equations:

$$x = a\sec\theta$$
$$y = b\tan\theta$$

Differentiating $x$ with respect to $\theta$:

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(a\sec\theta)$$
$$\frac{dx}{d\theta} = a\sec\theta\tan\theta$$

Differentiating $y$ with respect to $\theta$:

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(b\tan\theta)$$
$$\frac{dy}{d\theta} = b\sec^2\theta$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$
$$\frac{dy}{dx} = \frac{b\sec^2\theta}{a\sec\theta\tan\theta}$$
$$\frac{dy}{dx} = \frac{b\sec\theta}{a\tan\theta}$$

Expressing in terms of sine and cosine:

$$\frac{dy}{dx} = \frac{b\left(\frac{1}{\cos\theta}\right)}{a\left(\frac{sin\theta}{\cos\theta}\right)}$$
$$\frac{dy}{dx} = \frac{b}{a\sin\theta}$$
$$\frac{dy}{dx} = \frac{b}{a}\csc\theta$$

**Final Answer**

$\frac{dy}{dx} = \frac{b}{a}\csc\theta$

## Question 3 (ii)

Find $\frac{dy}{dx}$ when $x = \sin t$ and $y = \cos 2t$.

**TYPE 51 Parametric Differentiation**

**Given:** $x = \sin t$ and $y = \cos 2t$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = \sin t$$
$$y = \cos 2t$$

Differentiating $x$ with respect to $t$:

$$\frac{dx}{dt} = \frac{d}{dt}(\sin t)$$
$$\frac{dx}{dt} = \cos t$$

Differentiating $y$ with respect to $t$:

$$\frac{dy}{dt} = \frac{d}{dt}(\cos 2t)$$
$$\frac{dy}{dt} = -\sin 2t \cdot \frac{d}{dt}(2t)$$
$$\frac{dy}{dt} = -2\sin 2t$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{-2\sin 2t}{\cos t}$$

Using the standard double-angle identity $\sin 2t = 2\sin t\cos t$:

$$\frac{dy}{dx} = \frac{-2(2\sin t\cos t)}{\cos t}$$
$$\frac{dy}{dx} = -4\sin t$$

**Final Answer**

$\frac{dy}{dx} = -4\sin t$

## Question 4 (i)

Find $\frac{dy}{dx}$ when $x = 4t$ and $y = \frac{4}{t}$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = 4t$ and $y = \frac{4}{t}$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = 4t$$
$$y = 4t^{-1}$$

Differentiating $x$ with respect to $t$:

$$\frac{dx}{dt} = \frac{d}{dt}(4t)$$
$$\frac{dx}{dt} = 4$$

Differentiating $y$ with respect to $t$:

$$\frac{dy}{dt} = \frac{d}{dt}(4t^{-1})$$
$$\frac{dy}{dt} = -4t^{-2}$$
$$\frac{dy}{dt} = -\frac{4}{t^2}$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{-\frac{4}{t^2}}{4}$$
$$\frac{dy}{dx} = -\frac{1}{t^2}$$

**Final Answer**

$\frac{dy}{dx} = -\frac{1}{t^2}$

## Question 4 (ii)

Find $\frac{dy}{dx}$ when $x = t + \frac{1}{t}$ and $y = t - \frac{1}{t}$.

**TYPE 51 Parametric Differentiation**

**Given:** $x = t + \frac{1}{t}$ and $y = t - \frac{1}{t}$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = t + t^{-1}$$
$$y = t - t^{-1}$$

Differentiating $x$ with respect to $t$:

$$\frac{dx}{dt} = \frac{d}{dt}(t + t^{-1})$$
$$\frac{dx}{dt} = 1 - t^{-2}$$
$$\frac{dx}{dt} = 1 - \frac{1}{t^2}$$
$$\frac{dx}{dt} = \frac{t^2 - 1}{t^2}$$

Differentiating $y$ with respect to $t$:

$$\frac{dy}{dt} = \frac{d}{dt}(t - t^{-1})$$
$$\frac{dy}{dt} = 1 - (-t^{-2})$$
$$\frac{dy}{dt} = 1 + \frac{1}{t^2}$$
$$\frac{dy}{dt} = \frac{t^2 + 1}{t^2}$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{\frac{t^2+1}{t^2}}{\frac{t^2-1}{t^2}}$$
$$\frac{dy}{dx} = \frac{t^2 + 1}{t^2 - 1}$$

**Final Answer**

$\frac{dy}{dx} = \frac{t^2 + 1}{t^2 - 1}$

## Question 4 (iii)

Find $\frac{dy}{dx}$ when $x = a(t - \sin t)$ and $y = a(1 + \cos t)$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = a(t - \sin t)$ and $y = a(1 + \cos t)$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = a(t - \sin t)$$
$$y = a(1 + \cos t)$$

Differentiating $x$ with respect to $t$:

$$\frac{dx}{dt} = a(1 - \cos t)$$

Differentiating $y$ with respect to $t$:

$$\frac{dy}{dt} = a(0 - \sin t)$$
$$\frac{dy}{dt} = -a \sin t$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{-a \sin t}{a(1 - \cos t)}$$
$$\frac{dy}{dx} = -\frac{\sin t}{1 - \cos t}$$

Using the standard half-angle identities $\sin t = 2 \sin \frac{t}{2} \cos \frac{t}{2}$ and $1 - \cos t = 2 \sin^2 \frac{t}{2}$:

$$\frac{dy}{dx} = -\frac{2 \sin \frac{t}{2} \cos \frac{t}{2}}{2 \sin^2 \frac{t}{2}}$$
$$\frac{dy}{dx} = -\frac{\cos \frac{t}{2}}{\sin \frac{t}{2}}$$
$$\frac{dy}{dx} = -\cot \frac{t}{2}$$

**Final Answer**

$$\frac{dy}{dx} = -\cot \frac{t}{2}$$

**Common Mistake:** The textbook answer key lists $-\tan \frac{t}{2}$ for this question. This is a typographical error in the textbook's key; the correct mathematical derivative is indeed $-\cot \frac{t}{2}$.

## Question 5 (i)

Find $\dfrac{dy}{dx}$ when $x = a\cos^3 t$ and $y = b\sin^3 t$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = a\cos^3 t$ and $y = b\sin^3 t$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \dfrac{dy}{dx} = \dfrac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = a\cos^3 t$$
$$y = b\sin^3 t$$

Differentiating $x$ with respect to $t$ using the chain rule:

$$\frac{dx}{dt} = a \cdot 3\cos^2 t \cdot \frac{d}{dt}(\cos t)$$
$$\frac{dx}{dt} = 3a\cos^2 t \cdot (-\sin t)$$
$$\frac{dx}{dt} = -3a\cos^2 t\sin t$$

Differentiating $y$ with respect to $t$ using the chain rule:

$$\frac{dy}{dt} = b \cdot 3\sin^2 t \cdot \frac{d}{dt}(\sin t)$$
$$\frac{dy}{dt} = 3b\sin^2 t \cdot (\cos t)$$
$$\frac{dy}{dt} = 3b\sin^2 t\cos t$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{3b\sin^2 t\cos t}{-3a\cos^2 t\sin t}$$
$$\frac{dy}{dx} = -\frac{b\sin t}{a\cos t}$$
$$\frac{dy}{dx} = -\frac{b}{a}\tan t$$

**Final Answer**

$$\frac{dy}{dx} = -\frac{b}{a}\tan t$$

## Question 5 (ii)

Find $\dfrac{dy}{dx}$ when $x = a(1 - \sin t)$ and $y = a(1 + \cos t)$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = a(1 - \sin t)$ and $y = a(1 + \cos t)$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \dfrac{dy}{dx} = \dfrac{\dfrac{dy}{dt}}{\dfrac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = a(1 - \sin t)$$
$$y = a(1 + \cos t)$$

Differentiating $x$ with respect to $t$:

$$\frac{dx}{dt} = a(0 - \cos t)$$
$$\frac{dx}{dt} = -a\cos t$$

Differentiating $y$ with respect to $t$:

$$\frac{dy}{dt} = a(0 - \sin t)$$
$$\frac{dy}{dt} = -a\sin t$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{-a\sin t}{-a\cos t}$$
$$\frac{dy}{dx} = \tan t$$

**Final Answer**

$\dfrac{dy}{dx} = \tan t$

**Common Mistake:** The textbook answer key lists $-\cot\dfrac{t}{2}$ for this question. This indicates that the question in the textbook was likely intended to be $x = a(t - \sin t)$ instead of $x = a(1 - \sin t)$. If $x = a(t - \sin t)$, then $\dfrac{dx}{dt} = a(1 - \cos t)$, which yields $\dfrac{dy}{dx} = -\cot\dfrac{t}{2}$.

## Question 6 (i)

Find $\dfrac{dy}{dx}$ when $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$

**Concept / Formula Used:** $x = f(\theta) \qquad y = g(\theta) \qquad \dfrac{dy}{dx} = \dfrac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

### Solution:

We are given the parametric equations:

$$x = a(\theta + \sin\theta)$$
$$y = a(1 - \cos\theta)$$

Differentiating $x$ with respect to $\theta$:

$$\frac{dx}{d\theta} = a(1 + \cos\theta)$$

Differentiating $y$ with respect to $\theta$:

$$\frac{dy}{d\theta} = a(0 - (-\sin\theta))$$
$$\frac{dy}{d\theta} = a\sin\theta$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$
$$\frac{dy}{dx} = \frac{a\sin\theta}{a(1 + \cos\theta)}$$
$$\frac{dy}{dx} = \frac{\sin\theta}{1 + \cos\theta}$$

Using the standard half-angle identities $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$ and $1+\cos\theta = 2\cos^2\frac{\theta}{2}$:

$$\frac{dy}{dx} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}}$$
$$\frac{dy}{dx} = \frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}$$
$$\frac{dy}{dx} = \tan\frac{\theta}{2}$$

**Final Answer**

$\frac{dy}{dx} = \tan\frac{\theta}{2}$

## Question 6 (ii)

Find $\frac{dy}{dx}$ when $x = a(1-\cos\theta)$ and $y = a(\theta+\sin\theta)$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = a(1-\cos\theta)$ and $y = a(\theta+\sin\theta)$

**Concept / Formula Used:** $x = f(\theta) \qquad y = g(\theta) \qquad \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

**Solution:**

We are given the parametric equations:

$$x = a(1-\cos\theta)$$
$$y = a(\theta+\sin\theta)$$

Differentiating $x$ with respect to $\theta$:

$$\frac{dx}{d\theta} = a(0-(-\sin\theta))$$
$$\frac{dx}{d\theta} = a\sin\theta$$

Differentiating $y$ with respect to $\theta$:

$$\frac{dy}{d\theta} = a(1+\cos\theta)$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$
$$\frac{dy}{dx} = \frac{a(1+\cos\theta)}{a\sin\theta}$$
$$\frac{dy}{dx} = \frac{1+\cos\theta}{\sin\theta}$$

Using the standard half-angle identities $1+\cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$:

$$\frac{dy}{dx} = \frac{2\cos^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}$$
$$\frac{dy}{dx} = \frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}}$$
$$\frac{dy}{dx} = \cot\frac{\theta}{2}$$

**Final Answer**

$\dfrac{dy}{dx} = \cot\dfrac{\theta}{2}$

## Question 7 (i)

Find $\dfrac{dy}{dx}$ when $x = e^t(\sin t + \cos t)$ and $y = e^t(\sin t - \cos t)$.

**TYPE 51 Parametric Differentiation**

**Given:** $x = e^t(\sin t + \cos t)$ and $y = e^t(\sin t - \cos t)$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \dfrac{dy}{dx} = \dfrac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = e^t(\sin t + \cos t)$$
$$y = e^t(\sin t - \cos t)$$

Differentiating $x$ with respect to $t$ using the product rule:

$$\frac{dx}{dt} = e^t \frac{d}{dt}(\sin t + \cos t) + (\sin t + \cos t)\frac{d}{dt}(e^t)$$
$$\frac{dx}{dt} = e^t(\cos t - \sin t) + e^t(\sin t + \cos t)$$
$$\frac{dx}{dt} = e^t(\cos t - \sin t + \sin t + \cos t)$$
$$\frac{dx}{dt} = e^t(2\cos t)$$
$$\frac{dx}{dt} = 2e^t \cos t$$

Differentiating $y$ with respect to $t$ using the product rule:

$$\frac{dy}{dt} = e^t \frac{d}{dt}(\sin t - \cos t) + (\sin t - \cos t)\frac{d}{dt}(e^t)$$
$$\frac{dy}{dt} = e^t(\cos t - (-\sin t)) + e^t(\sin t - \cos t)$$
$$\frac{dy}{dt} = e^t(\cos t + \sin t) + e^t(\sin t - \cos t)$$
$$\frac{dy}{dt} = e^t(\cos t + \sin t + \sin t - \cos t)$$
$$\frac{dy}{dt} = e^t(2\sin t)$$
$$\frac{dy}{dt} = 2e^t \sin t$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{2e^t \sin t}{2e^t \cos t}$$
$$\frac{dy}{dx} = \tan t$$

**Final Answer**

$\frac{dy}{dx} = \tan t$

## Question 7 (ii)

Find $\frac{dy}{dx}$ when $x = \cos\theta - \cos 2\theta$ and $y = \sin\theta - \sin 2\theta$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = \cos\theta - \cos 2\theta$ and $y = \sin\theta - \sin 2\theta$

**Concept / Formula Used:** $x = f(\theta) \qquad y = g(\theta) \qquad \dfrac{dy}{dx} = \dfrac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

**Solution:**

We are given the parametric equations:

$$x = \cos\theta - \cos 2\theta$$
$$y = \sin\theta - \sin 2\theta$$

Differentiating $x$ with respect to $\theta$:

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(\cos\theta - \cos 2\theta)$$
$$\frac{dx}{d\theta} = -\sin\theta - (-2\sin 2\theta)$$
$$\frac{dx}{d\theta} = 2\sin 2\theta - \sin\theta$$

Differentiating $y$ with respect to $\theta$:

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(\sin\theta - \sin 2\theta)$$
$$\frac{dy}{d\theta} = \cos\theta - 2\cos 2\theta$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$
$$\frac{dy}{dx} = \frac{\cos\theta - 2\cos 2\theta}{2\sin 2\theta - \sin\theta}$$

**Final Answer**

$$\frac{dy}{dx} = \frac{\cos\theta - 2\cos 2\theta}{2\sin 2\theta - \sin\theta}$$

## Question 8 (i)

Find $\dfrac{dy}{dx}$ when $x = a(\cos\theta + \cos 2\theta)$ and $y = b(\sin\theta + \sin 2\theta)$.

**TYPE 51 Parametric Differentiation**

**Given:** $x = a(\cos\theta + \cos 2\theta)$ and $y = b(\sin\theta + \sin 2\theta)$

**Concept / Formula Used:** $x = f(\theta) \qquad y = g(\theta) \qquad \dfrac{dy}{dx} = \dfrac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

**Solution:**

We are given the parametric equations:

$$x = a(\cos\theta + \cos 2\theta)$$
$$y = b(\sin\theta + \sin 2\theta)$$

Differentiating $x$ with respect to $\theta$:

$$\frac{dx}{d\theta} = a(-\sin\theta - 2\sin 2\theta)$$
$$\frac{dx}{d\theta} = -a(\sin\theta + 2\sin 2\theta)$$

Differentiating $y$ with respect to $\theta$:

$$\frac{dy}{d\theta} = b(\cos\theta + 2\cos 2\theta)$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$
$$\frac{dy}{dx} = \frac{b(\cos\theta + 2\cos 2\theta)}{-a(\sin\theta + 2\sin 2\theta)}$$
$$\frac{dy}{dx} = -\frac{b}{a}\left(\frac{\cos\theta + 2\cos 2\theta}{\sin\theta + 2\sin 2\theta}\right)$$

**Final Answer**

$$\frac{dy}{dx} = -\frac{b}{a}\left(\frac{\cos\theta + 2\cos 2\theta}{\sin\theta + 2\sin 2\theta}\right)$$

## Question 8 (ii)

Find $\frac{dy}{dx}$ when $x = \frac{1+\log t}{t^2}$ and $y = \frac{3+2\log t}{t}$.

**TYPE 51 Parametric Differentiation**

**Given:** $x = \frac{1+\log t}{t^2}$ and $y = \frac{3+2\log t}{t}$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = \frac{1+\log t}{t^2}$$
$$y = \frac{3+2\log t}{t}$$

Differentiating $x$ with respect to $t$ using the quotient rule:

$$\frac{dx}{dt} = \frac{t^2 \frac{d}{dt}(1 + \log t) - (1 + \log t)\frac{d}{dt}(t^2)}{(t^2)^2}$$
$$\frac{dx}{dt} = \frac{t^2\left(\frac{1}{t}\right) - (1 + \log t)(2t)}{t^4}$$
$$\frac{dx}{dt} = \frac{t - 2t(1 + \log t)}{t^4}$$
$$\frac{dx}{dt} = \frac{t(1 - 2 - 2\log t)}{t^4}$$
$$\frac{dx}{dt} = \frac{-1 - 2\log t}{t^3}$$
$$\frac{dx}{dt} = -\frac{1 + 2\log t}{t^3}$$

Differentiating $y$ with respect to $t$ using the quotient rule:

$$\frac{dy}{dt} = \frac{t\frac{d}{dt}(3 + 2\log t) - (3 + 2\log t)\frac{d}{dt}(t)}{t^2}$$
$$\frac{dy}{dt} = \frac{t\left(\frac{2}{t}\right) - (3 + 2\log t)(1)}{t^2}$$
$$\frac{dy}{dt} = \frac{2 - 3 - 2\log t}{t^2}$$
$$\frac{dy}{dt} = \frac{-1 - 2\log t}{t^2}$$
$$\frac{dy}{dt} = -\frac{1 + 2\log t}{t^2}$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{-\frac{1+2\log t}{t^2}}{-\frac{1+2\log t}{t^3}}$$
$$\frac{dy}{dx} = \frac{t^3}{t^2}$$
$$\frac{dy}{dx} = t$$

**Final Answer**

$$\frac{dy}{dx} = t$$

## Question 9

Find $\dfrac{dy}{dx}$ when $x = 3\cos\theta - 2\cos^3\theta$ and $y = 3\sin\theta - 2\sin^3\theta$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = 3\cos\theta - 2\cos^3\theta$ and $y = 3\sin\theta - 2\sin^3\theta$

**Concept / Formula Used:** $x = f(\theta) \qquad y = g(\theta) \qquad \dfrac{dy}{dx} = \dfrac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

**Solution:**

We are given the parametric equations:

$$x = 3\cos\theta - 2\cos^3\theta$$
$$y = 3\sin\theta - 2\sin^3\theta$$

Differentiating $x$ with respect to $\theta$:

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(3\cos\theta - 2\cos^3\theta)$$
$$\frac{dx}{d\theta} = -3\sin\theta - 2\cdot 3\cos^2\theta\cdot(-\sin\theta)$$
$$\frac{dx}{d\theta} = -3\sin\theta + 6\cos^2\theta\sin\theta$$
$$\frac{dx}{d\theta} = 3\sin\theta(2\cos^2\theta - 1)$$

Using the standard double-angle identity $\cos 2\theta = 2\cos^2\theta - 1$:

$$\frac{dx}{d\theta} = 3\sin\theta\cos 2\theta$$

Differentiating $y$ with respect to $\theta$:

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(3\sin\theta - 2\sin^3\theta)$$
$$\frac{dy}{d\theta} = 3\cos\theta - 2\cdot 3\sin^2\theta\cdot(\cos\theta)$$
$$\frac{dy}{d\theta} = 3\cos\theta - 6\sin^2\theta\cos\theta$$
$$\frac{dy}{d\theta} = 3\cos\theta(1 - 2\sin^2\theta)$$

Using the standard double-angle identity $\cos 2\theta = 1 - 2\sin^2\theta$:

$$\frac{dy}{d\theta} = 3\cos\theta\cos 2\theta$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$
$$\frac{dy}{dx} = \frac{3\cos\theta\cos 2\theta}{3\sin\theta\cos 2\theta}$$
$$\frac{dy}{dx} = \frac{\cos\theta}{\sin\theta}$$
$$\frac{dy}{dx} = \cot\theta$$

**Final Answer**

$\dfrac{dy}{dx} = \cot\theta$

## Question 10 (i)

If $x = 2\cos t - \cos 2t$ and $y = 2\sin t - \sin 2t$, find $\dfrac{dy}{dx}$ at $t = \dfrac{\pi}{4}$.

**TYPE 52** **Parametric Differentiation with Evaluation**

**Given:** $x = 2\cos t - \cos 2t$, $y = 2\sin t - \sin 2t$, and $t = \dfrac{\pi}{4}$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \dfrac{dy}{dx} = \dfrac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = 2\cos t - \cos 2t$$
$$y = 2\sin t - \sin 2t$$

Differentiating $x$ with respect to $t$:

$$\frac{dx}{dt} = -2\sin t - (-2\sin 2t)$$
$$\frac{dx}{dt} = 2\sin 2t - 2\sin t$$

Differentiating $y$ with respect to $t$:

$$\frac{dy}{dt} = 2\cos t - 2\cos 2t$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{2\cos t - 2\cos 2t}{2\sin 2t - 2\sin t}$$
$$\frac{dy}{dx} = \frac{\cos t - \cos 2t}{\sin 2t - \sin t}$$

Evaluating the derivative at $t = \frac{\pi}{4}$:

$$\cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$
$$\cos\left(2 \cdot \frac{\pi}{4}\right) = \cos\left(\frac{\pi}{2}\right) = 0$$
$$\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$
$$\sin\left(2 \cdot \frac{\pi}{4}\right) = \sin\left(\frac{\pi}{2}\right) = 1$$

Substituting these values into the expression for $\frac{dy}{dx}$:

$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} = \frac{\frac{1}{\sqrt{2}} - 0}{1 - \frac{1}{\sqrt{2}}}$$
$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} = \frac{\frac{1}{\sqrt{2}}}{\frac{\sqrt{2}-1}{\sqrt{2}}}$$
$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} = \frac{1}{\sqrt{2} - 1}$$

Rationalizing the denominator:

$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} = \frac{1}{\sqrt{2} - 1} \cdot \frac{\sqrt{2}+1}{\sqrt{2}+1}$$
$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} = \frac{\sqrt{2}+1}{2-1}$$
$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} = \sqrt{2}+1$$

**Final Answer**

$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} = \sqrt{2}+1$$

## Question 10 (ii)

If $x = 3\sin t - \sin 3t$ and $y = 3\cos t - \cos 3t$, find $\frac{dy}{dx}$ at $t = \frac{\pi}{3}$.

**TYPE 52 Parametric Differentiation with Evaluation**

**Given:** $x = 3\sin t - \sin 3t$, $y = 3\cos t - \cos 3t$, and $t = \frac{\pi}{3}$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = 3\sin t - \sin 3t$$
$$y = 3\cos t - \cos 3t$$

Differentiating $x$ with respect to $t$:

$$\frac{dx}{dt} = 3\cos t - 3\cos 3t$$

Differentiating $y$ with respect to $t$:

$$\frac{dy}{dt} = -3\sin t - (-3\sin 3t)$$
$$\frac{dy}{dt} = 3\sin 3t - 3\sin t$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{3\sin 3t - 3\sin t}{3\cos t - 3\cos 3t}$$
$$\frac{dy}{dx} = \frac{\sin 3t - \sin t}{\cos t - \cos 3t}$$

Evaluating the derivative at $t = \dfrac{\pi}{3}$:

$$\sin\left(3 \cdot \frac{\pi}{3}\right) = \sin\pi = 0$$
$$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$
$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$
$$\cos\left(3 \cdot \frac{\pi}{3}\right) = \cos\pi = -1$$

Substituting these values into the expression for $\dfrac{dy}{dx}$:

$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{3}} = \frac{0 - \frac{\sqrt{3}}{2}}{\frac{1}{2} - (-1)}$$
$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{3}} = \frac{-\frac{\sqrt{3}}{2}}{\frac{3}{2}}$$
$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{3}} = -\frac{\sqrt{3}}{3}$$
$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{3}} = -\frac{1}{\sqrt{3}}$$

**Final Answer**

$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{3}} = -\frac{1}{\sqrt{3}}$$

## Question 10 (iii)

If $x = a\sin 2t$ and $y = a(\cos 2t + \log\tan t)$, find $\frac{dy}{dx}$.

**TYPE 51** **Parametric Differentiation**

**Given:** $x = a\sin 2t$ and $y = a(\cos 2t + \log\tan t)$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = a\sin 2t$$
$$y = a(\cos 2t + \log\tan t)$$

Differentiating $x$ with respect to $t$:

$$\frac{dx}{dt} = 2a\cos 2t$$

Differentiating $y$ with respect to $t$:

$$\frac{dy}{dt} = a\left(-2\sin 2t + \frac{1}{\tan t}\cdot\sec^2 t\right)$$

Simplifying the term $\frac{\sec^2 t}{\tan t}$:

$$\begin{aligned}\frac{\sec^2 t}{\tan t} &= \frac{\frac{1}{\cos^2 t}}{\frac{\sin t}{\cos t}}\\ &= \frac{1}{\sin t\cos t}\\ &= \frac{2}{2\sin t\cos t}\\ &= \frac{2}{\sin 2t}\end{aligned}$$

Substituting this back into $\frac{dy}{dt}$:

$$\frac{dy}{dt} = a\left(-2\sin 2t + \frac{2}{\sin 2t}\right)$$
$$\frac{dy}{dt} = 2a\left(\frac{1-\sin^2 2t}{\sin 2t}\right)$$
$$\frac{dy}{dt} = 2a\frac{\cos^2 2t}{\sin 2t}$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{2a\frac{\cos^2 2t}{\sin 2t}}{2a\cos 2t}$$
$$\frac{dy}{dx} = \frac{\cos 2t}{\sin 2t}$$
$$\frac{dy}{dx} = \cot 2t$$

**Final Answer**

$\frac{dy}{dx} = \cot 2t$

## Question 11 (i)

If $x = \frac{2bt}{1+t^2}$ and $y = \frac{a(1-t^2)}{1+t^2}$, find $\frac{dy}{dx}$ at $t = 2$.

**TYPE 52** **Parametric Differentiation with Evaluation**

**Given:** $x = \frac{2bt}{1+t^2}$, $y = \frac{a(1-t^2)}{1+t^2}$, and $t = 2$

**Concept / Formula Used:** $x = f(t) \qquad y = g(t) \qquad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

**Solution:**

We are given the parametric equations:

$$x = \frac{2bt}{1+t^2}$$
$$y = \frac{a(1-t^2)}{1+t^2}$$

Differentiating $x$ with respect to $t$ using the quotient rule:

$$\frac{dx}{dt} = 2b \cdot \frac{(1+t^2)(1) - t(2t)}{(1+t^2)^2}$$
$$\frac{dx}{dt} = 2b \cdot \frac{1+t^2-2t^2}{(1+t^2)^2}$$
$$\frac{dx}{dt} = 2b\frac{1-t^2}{(1+t^2)^2}$$

Differentiating $y$ with respect to $t$ using the quotient rule:

$$\frac{dy}{dt} = a \cdot \frac{(1+t^2)(-2t) - (1-t^2)(2t)}{(1+t^2)^2}$$
$$\frac{dy}{dt} = a \cdot \frac{-2t-2t^3-2t+2t^3}{(1+t^2)^2}$$
$$\frac{dy}{dt} = -\frac{4at}{(1+t^2)^2}$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$
$$\frac{dy}{dx} = \frac{-\frac{4at}{(1+t^2)^2}}{2b\frac{1-t^2}{(1+t^2)^2}}$$
$$\frac{dy}{dx} = -\frac{4at}{2b(1-t^2)}$$
$$\frac{dy}{dx} = -\frac{2at}{b(1-t^2)}$$

Evaluating the derivative at $t = 2$:

$$\left.\frac{dy}{dx}\right|_{t=2} = -\frac{2a(2)}{b(1-2^2)}$$
$$\left.\frac{dy}{dx}\right|_{t=2} = -\frac{4a}{b(1-4)}$$
$$\left.\frac{dy}{dx}\right|_{t=2} = -\frac{4a}{-3b}$$
$$\left.\frac{dy}{dx}\right|_{t=2} = \frac{4a}{3b}$$

**Final Answer**

$$\left.\frac{dy}{dx}\right|_{t=2} = \frac{4a}{3b}$$

## Question 11 (ii)

If $x = ae^{\theta}(\sin\theta - \cos\theta)$ and $y = ae^{\theta}(\sin\theta + \cos\theta)$, find $\dfrac{dy}{dx}$ at $\theta = \dfrac{\pi}{4}$.

**TYPE 52 Parametric Differentiation with Evaluation**

**Given:** $x = ae^{\theta}(\sin\theta - \cos\theta)$, $y = ae^{\theta}(\sin\theta + \cos\theta)$, and $\theta = \dfrac{\pi}{4}$

**Concept / Formula Used:** $x = f(\theta)$ $\quad y = g(\theta)$ $\quad \dfrac{dy}{dx} = \dfrac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

**Solution:**

We are given the parametric equations:

$$x = ae^{\theta}(\sin\theta - \cos\theta)$$
$$y = ae^{\theta}(\sin\theta + \cos\theta)$$

Differentiating $x$ with respect to $\theta$ using the product rule:

$$\frac{dx}{d\theta} = ae^{\theta}\frac{d}{d\theta}(\sin\theta - \cos\theta) + a(\sin\theta - \cos\theta)\frac{d}{d\theta}(e^{\theta})$$
$$\frac{dx}{d\theta} = ae^{\theta}(\cos\theta - (-\sin\theta)) + ae^{\theta}(\sin\theta - \cos\theta)$$
$$\frac{dx}{d\theta} = ae^{\theta}(\cos\theta + \sin\theta + \sin\theta - \cos\theta)$$
$$\frac{dx}{d\theta} = 2ae^{\theta}\sin\theta$$

Differentiating $y$ with respect to $\theta$ using the product rule:

$$\frac{dy}{d\theta} = ae^{\theta}\frac{d}{d\theta}(\sin\theta + \cos\theta) + a(\sin\theta + \cos\theta)\frac{d}{d\theta}(e^{\theta})$$
$$\frac{dy}{d\theta} = ae^{\theta}(\cos\theta - \sin\theta) + ae^{\theta}(\sin\theta + \cos\theta)$$
$$\frac{dy}{d\theta} = ae^{\theta}(\cos\theta - \sin\theta + \sin\theta + \cos\theta)$$
$$\frac{dy}{d\theta} = 2ae^{\theta}\cos\theta$$

Now, using the parametric differentiation formula:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$
$$\frac{dy}{dx} = \frac{2ae^{\theta}\cos\theta}{2ae^{\theta}\sin\theta}$$
$$\frac{dy}{dx} = \cot\theta$$

Evaluating the derivative at $\theta = \dfrac{\pi}{4}$:

$$\left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{4}} = \cot\left(\frac{\pi}{4}\right) = 1$$

**Final Answer**

$$\left.\frac{dy}{dx}\right|_{\theta=\frac{\pi}{4}} = 1$$

## Question 12 (i)

Differentiate $\dfrac{x^2}{1-x^2}$ with respect to $x^2$.

**TYPE 53** **Differentiation of One Function with Respect to Another**

**Given:** Let $u = \dfrac{x^2}{1-x^2}$ and $v = x^2$

**Concept / Formula Used:** $u$ $v$ $\dfrac{du}{dv}$

**Solution:**

Let:

$$u = \frac{x^2}{1-x^2}$$
$$v = x^2$$

We can express $u$ directly in terms of $v$:

$$u = \frac{v}{1-v}$$

Differentiating $u$ with respect to $v$ using the quotient rule:

$$\frac{du}{dv} = \frac{(1-v)\frac{d}{dv}(v) - v\frac{d}{dv}(1-v)}{(1-v)^2}$$
$$\frac{du}{dv} = \frac{(1-v)(1) - v(-1)}{(1-v)^2}$$
$$\frac{du}{dv} = \frac{1-v+v}{(1-v)^2}$$
$$\frac{du}{dv} = \frac{1}{(1-v)^2}$$

Back-substituting $v = x^2$:

$$\frac{du}{dv} = \frac{1}{(1-x^2)^2}$$

**Final Answer**

$$\frac{d}{d(x^2)}\left(\frac{x^2}{1-x^2}\right) = \frac{1}{(1-x^2)^2}$$

## Question 12 (ii)

Differentiate $\sin x^2$ with respect to $x^3$.

**TYPE 53** Differentiation of One Function with Respect to Another

**Given:** Let $u = \sin x^2$ and $v = x^3$

**Concept / Formula Used:** $u$ $v$ $\frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}}$

**Solution:**

Let:

$$u = \sin x^2$$
$$v = x^3$$

Differentiating $u$ with respect to $x$ using the chain rule:

$$\frac{du}{dx} = \cos x^2 \cdot \frac{d}{dx}(x^2)$$
$$\frac{du}{dx} = 2x\cos x^2$$

Differentiating $v$ with respect to $x$:

$$\frac{dv}{dx} = 3x^2$$

Now, using the derivative formula:

$$\frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}}$$
$$\frac{du}{dv} = \frac{2x\cos x^2}{3x^2}$$
$$\frac{du}{dv} = \frac{2\cos x^2}{3x}$$

**Final Answer**

$$\frac{d(\sin x^2)}{d(x^3)} = \frac{2\cos x^2}{3x}$$

## Question 12 (iii)

Differentiate $\cot^3(2x+1)$ with respect to $x^2+1$.

**TYPE 53** **Differentiation of One Function with Respect to Another**

**Given:** Let $u = \cot^3(2x+1)$ and $v = x^2+1$

**Concept / Formula Used:** $u$ $v$ $\dfrac{du}{dv} = \dfrac{\frac{du}{dx}}{\frac{dv}{dx}}$

**Solution:**

Let:

$$u = \cot^3(2x+1)$$
$$v = x^2+1$$

Differentiating $u$ with respect to $x$ using the chain rule:

$$\frac{du}{dx} = 3\cot^2(2x+1)\cdot\frac{d}{dx}(\cot(2x+1))$$
$$\frac{du}{dx} = 3\cot^2(2x+1)\cdot\left(-\csc^2(2x+1)\cdot\frac{d}{dx}(2x+1)\right)$$
$$\frac{du}{dx} = 3\cot^2(2x+1)\cdot\left(-\csc^2(2x+1)\cdot 2\right)$$
$$\frac{du}{dx} = -6\cot^2(2x+1)\csc^2(2x+1)$$

Differentiating $v$ with respect to $x$:

$$\frac{dv}{dx} = 2x$$

Now, using the derivative formula:

$$\frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}}$$
$$\frac{du}{dv} = \frac{-6\cot^2(2x+1)\csc^2(2x+1)}{2x}$$
$$\frac{du}{dv} = -\frac{3\cot^2(2x+1)\csc^2(2x+1)}{x}$$

**Final Answer**

$$\frac{du}{dv} = -\frac{3\cot^2(2x+1)\csc^2(2x+1)}{x}$$

## Question 12 (iv)

Differentiate $\sin^2 x$ with respect to $e^{\cos x}$.

**TYPE 53** **Differentiation of One Function with Respect to Another**

**Given:** Let $u = \sin^2 x$ and $v = e^{\cos x}$

**Concept / Formula Used:** $u$ $v$ $\dfrac{du}{dv} = \dfrac{\frac{du}{dx}}{\frac{dv}{dx}}$

**Solution:**

Let:

$$u = \sin^2 x$$
$$v = e^{\cos x}$$

Differentiating $u$ with respect to $x$:

$$\frac{du}{dx} = 2 \sin x \cos x$$

Differentiating $v$ with respect to $x$ using the chain rule:

$$\frac{dv}{dx} = e^{\cos x} \cdot \frac{d}{dx}(\cos x)$$
$$\frac{dv}{dx} = e^{\cos x} \cdot (-\sin x)$$
$$\frac{dv}{dx} = -\sin x e^{\cos x}$$

Now, using the derivative formula:

$$\frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}}$$
$$\frac{du}{dv} = \frac{2 \sin x \cos x}{-\sin x e^{\cos x}}$$
$$\frac{du}{dv} = -\frac{2 \cos x}{e^{\cos x}}$$

**Final Answer**

$$\frac{du}{dv} = -\frac{2 \cos x}{e^{\cos x}}$$

## Question 13 (i)

Differentiate $\sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$ with respect to $\tan^{-1} x$.

**TYPE 54** **Differentiation of One Function with Respect to Another with Substitution**

**Given:** Let $u = \sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$ and $v = \tan^{-1} x$

**Concept / Formula Used:** $x = \tan\theta \qquad \sin 2\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$

**Solution:**

Let:

$$u = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$
$$v = \tan^{-1} x$$

Let $x = \tan\theta$, which implies $\theta = \tan^{-1} x$. Substituting $x = \tan\theta$ into the expression for $u$:

$$u = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$$

Using the standard identity $\sin 2\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$:

$$u = \sin^{-1}(\sin 2\theta)$$
$$u = 2\theta$$

Back-substituting $\theta = \tan^{-1} x$:

$$u = 2\tan^{-1} x$$

Since $v = \tan^{-1} x$, we can write:

$$u = 2v$$

Differentiating $u$ with respect to $v$:

$$\frac{du}{dv} = \frac{d}{dv}(2v)$$
$$\frac{du}{dv} = 2$$

**Final Answer**

$\dfrac{du}{dv} = 2$

## Question 13 (ii)

Differentiate $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ with respect to $\tan^{-1}x$.

**TYPE 54** **Differentiation of One Function with Respect to Another with Substitution**

**Given:** Let $u = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ and $v = \tan^{-1}x$

**Concept / Formula Used:** $x = \tan\theta \qquad \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

**Solution:**

Let:

$$u = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$
$$v = \tan^{-1}x$$

Let $x = \tan\theta$, which implies $\theta = \tan^{-1}x$. Substituting $x = \tan\theta$ into the expression for $u$:

$$u = \tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)$$

Using the standard identity $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$:

$$u = \tan^{-1}(\tan 2\theta)$$
$$u = 2\theta$$

Back-substituting $\theta = \tan^{-1}x$:

$$u = 2\tan^{-1}x$$

Since $v = \tan^{-1}x$, we can write:

$$u = 2v$$

Differentiating $u$ with respect to $v$:

$$\frac{du}{dv} = \frac{d}{dv}(2v)$$
$$\frac{du}{dv} = 2$$

**Final Answer**

$\frac{du}{dv} = 2$

## Question 14

Differentiate $\cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$ with respect to $\sin^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$.

**TYPE 53** **Differentiation of One Function with Respect to Another**

**Given:** Let $u = \cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$ and $v = \sin^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$

**Concept / Formula Used:** $\sin^{-1} z + \cos^{-1} z = \frac{\pi}{2} \qquad z \in [-1, 1]$

**Solution:**

Let:

$$u = \cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$$
$$v = \sin^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$$

Let $z = \frac{1}{\sqrt{1+t^2}}$. Since $t^2 \geq 0$, we have $1+t^2 \geq 1$, which implies $0 < \frac{1}{\sqrt{1+t^2}} \leq 1$. Thus, $z \in (0, 1]$.

Using the standard inverse trigonometric identity:

$$\sin^{-1} z + \cos^{-1} z = \frac{\pi}{2}$$

Substituting $z = \frac{1}{\sqrt{1+t^2}}$:

$$v + u = \frac{\pi}{2}$$
$$u = \frac{\pi}{2} - v$$

Differentiating $u$ with respect to $v$:

$$\frac{du}{dv} = \frac{d}{dv}\left(\frac{\pi}{2} - v\right)$$
$$\frac{du}{dv} = -1$$

**Final Answer**

$$\frac{du}{dv} = -1$$

**Common Mistake:** The textbook answer key lists 1 for this question. This is a common sign error in the textbook key; since $u + v = \frac{\pi}{2}$, their derivatives must be equal in magnitude but opposite in sign, yielding $\frac{du}{dv} = -1$.

## Question 15 (i)

Differentiate $\sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$ with respect to $\cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$.

**TYPE 54** **Differentiation of One Function with Respect to Another with Substitution**

**Given:** Let $u = \sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$ and $v = \cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$ for $x \geq 0$

**Concept / Formula Used:** $x = \tan\theta$ $\sin 2\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$ $\cos 2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta}$

### Solution:

Let:

$$u = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$
$$v = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

Let $x = \tan\theta$, which implies $\theta = \tan^{-1} x$. For $x \geq 0$, we have $\theta \in [0, \pi/2)$. Substituting $x = \tan\theta$ into the expression for $u$:

$$u = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$$
$$u = \sin^{-1}(\sin 2\theta)$$
$$u = 2\theta$$

Substituting $x = \tan\theta$ into the expression for $v$:

$$v = \cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)$$
$$v = \cos^{-1}(\cos 2\theta)$$
$$v = 2\theta$$

Thus, we have:

$$u = v$$

Differentiating $u$ with respect to $v$:

$$\frac{du}{dv} = 1$$

**Final Answer**

$\dfrac{du}{dv} = 1$

## Question 15 (ii)

Differentiate $\tan^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right)$ with respect to $\tan^{-1}\left(\dfrac{2x}{1-x^2}\right)$.

**TYPE 54 Differentiation of One Function with Respect to Another with Substitution**

**Given:** Let $u = \tan^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right)$ and $v = \tan^{-1}\left(\dfrac{2x}{1-x^2}\right)$

**Concept / Formula Used:** $x = \tan\theta$ $\quad \tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}$ $\quad \tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$

### Solution:

Let:

$$u = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$$
$$v = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

Let $x = \tan\theta$, which implies $\theta = \tan^{-1} x$. Substituting $x = \tan\theta$ into the expression for $u$:

$$u = \tan^{-1}\left(\frac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}\right)$$
$$u = \tan^{-1}(\tan 3\theta)$$
$$u = 3\theta$$
$$u = 3\tan^{-1} x$$

Substituting $x = \tan\theta$ into the expression for $v$:

$$v = \tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)$$
$$v = \tan^{-1}(\tan 2\theta)$$
$$v = 2\theta$$
$$v = 2\tan^{-1} x$$

We can express $u$ in terms of $v$:

$$u = \frac{3}{2}v$$

Differentiating $u$ with respect to $v$:

$$\frac{du}{dv} = \frac{3}{2}$$

**Final Answer**

$$\frac{du}{dv} = \frac{3}{2}$$

## Question 16 (i)

Differentiate $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ with respect to $\tan^{-1} x$.

**TYPE 54** **Differentiation of One Function with Respect to Another with Substitution**

**Given:** Let $u = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ and $v = \tan^{-1} x$

**Concept / Formula Used:** $x = \tan\theta \qquad 1 + \tan^2\theta = \sec^2\theta$

### Solution:

Let:

$$u = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$$
$$v = \tan^{-1} x$$

Let $x = \tan\theta$, which implies $\theta = \tan^{-1} x$. Substituting $x = \tan\theta$ into the expression for $u$:

$$u = \tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right)$$
$$u = \tan^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right)$$
$$u = \tan^{-1}\left(\frac{\frac{1}{\cos\theta} - 1}{\frac{\sin\theta}{\cos\theta}}\right)$$
$$u = \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

Using the standard half-angle identities $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$:

$$u = \tan^{-1}\left(\frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right)$$
$$u = \tan^{-1}\left(\tan\frac{\theta}{2}\right)$$
$$u = \frac{\theta}{2}$$

Back-substituting $\theta = \tan^{-1} x$:

$$u = \frac{1}{2}\tan^{-1} x$$

Since $v = \tan^{-1} x$, we can write:

$$u = \frac{1}{2}v$$

Differentiating $u$ with respect to $v$:

$$\frac{du}{dv} = \frac{1}{2}$$

**Final Answer**

$\frac{du}{dv} = \frac{1}{2}$

## Question 16 (ii)

Differentiate $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ with respect to $\cos^{-1}(2x\sqrt{1-x^2})$.

**TYPE 54** **Differentiation of One Function with Respect to Another with Substitution**

**Given:** Let $u = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ and $v = \cos^{-1}(2x\sqrt{1-x^2})$

**Concept / Formula Used:** $x = \cos\theta$ $\qquad \sin 2\theta = 2\sin\theta\cos\theta$

**Solution:**

Let:

$$u = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$$
$$v = \cos^{-1}(2x\sqrt{1-x^2})$$

Let $x = \cos\theta$, which implies $\theta = \cos^{-1} x$. Substituting $x = \cos\theta$ into the expression for $u$:

$$u = \tan^{-1}\left(\frac{\sqrt{1-\cos^2\theta}}{\cos\theta}\right)$$
$$u = \tan^{-1}\left(\frac{\sin\theta}{\cos\theta}\right)$$
$$u = \tan^{-1}(\tan\theta)$$
$$u = \theta$$

Substituting $x = \cos\theta$ into the expression for $v$:

$$v = \cos^{-1}(2\cos\theta\sqrt{1-\cos^2\theta})$$
$$v = \cos^{-1}(2\cos\theta\sin\theta)$$
$$v = \cos^{-1}(\sin 2\theta)$$

Using the standard identity $\sin 2\theta = \cos\left(\frac{\pi}{2} - 2\theta\right)$:

$$v = \cos^{-1}\left(\cos\left(\frac{\pi}{2} - 2\theta\right)\right)$$
$$v = \frac{\pi}{2} - 2\theta$$

Since $u = \theta$, we can write:

$$v = \frac{\pi}{2} - 2u$$
$$2u = \frac{\pi}{2} - v$$
$$u = \frac{\pi}{4} - \frac{v}{2}$$

Differentiating $u$ with respect to $v$:

$$\frac{du}{dv} = \frac{d}{dv}\left(\frac{\pi}{4} - \frac{v}{2}\right)$$
$$\frac{du}{dv} = -\frac{1}{2}$$

**Final Answer**

$\frac{du}{dv} = -\frac{1}{2}$

## Question 16 (iii)

Differentiate $\sec^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)$ with respect to $\sin^{-1}(2x\sqrt{1-x^2})$.

**TYPE 54 Differentiation of One Function with Respect to Another with Substitution**

**Given:** Let $u = \sec^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)$ and $v = \sin^{-1}(2x\sqrt{1-x^2})$

**Concept / Formula Used:** $x = \sin\theta$ $\quad \sin 2\theta = 2\sin\theta\cos\theta$

### Solution:

Let:

$$u = \sec^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)$$
$$v = \sin^{-1}(2x\sqrt{1-x^2})$$

Let $x = \sin\theta$, which implies $\theta = \sin^{-1} x$. Substituting $x = \sin\theta$ into the expression for $u$:

$$\begin{aligned}
u &= \sec^{-1}\left(\frac{1}{\sqrt{1-\sin^2\theta}}\right)\\
u &= \sec^{-1}\left(\frac{1}{\cos\theta}\right)\\
u &= \sec^{-1}(\sec\theta)\\
u &= \theta
\end{aligned}$$

Substituting $x = \sin\theta$ into the expression for $v$:

$$\begin{aligned}
v &= \sin^{-1}(2\sin\theta\sqrt{1-\sin^2\theta})\\
v &= \sin^{-1}(2\sin\theta\cos\theta)\\
v &= \sin^{-1}(\sin 2\theta)\\
v &= 2\theta
\end{aligned}$$

Since $u = \theta$ and $v = 2\theta$, we can write:

$$u = \frac{1}{2}v$$

Differentiating $u$ with respect to $v$:

$$\frac{du}{dv} = \frac{1}{2}$$

**Final Answer**

$\dfrac{du}{dv} = \dfrac{1}{2}$

## Question 17

Prove that the derivative of $\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$ with respect to $\sin^{-1} x$ is independent of $x$.

**TYPE 54** **Differentiation of One Function with Respect to Another with Substitution**

**To Prove:** The derivative of $u = \tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$ with respect to $v = \sin^{-1} x$ is a constant.

**Concept / Formula Used:** $x = \sin\theta$ $1+\cos\theta = 2\cos^2\frac{\theta}{2}$ $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$

**Solution:**

Let:

$$u = \tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$$
$$v = \sin^{-1} x$$

Let $x = \sin\theta$, which implies $\theta = \sin^{-1} x$. Substituting $x = \sin\theta$ into the expression for $u$:

$$u = \tan^{-1}\left(\frac{\sin\theta}{1+\sqrt{1-\sin^2\theta}}\right)$$
$$u = \tan^{-1}\left(\frac{\sin\theta}{1+\cos\theta}\right)$$

Using the standard half-angle identities $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$ and $1+\cos\theta = 2\cos^2\frac{\theta}{2}$:

$$u = \tan^{-1}\left(\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}}\right)$$
$$u = \tan^{-1}\left(\tan\frac{\theta}{2}\right)$$
$$u = \frac{\theta}{2}$$

Since $v = \sin^{-1} x = \theta$, we can write:

$$u = \frac{1}{2}v$$

Differentiating $u$ with respect to $v$:

$$\frac{du}{dv} = \frac{d}{dv}\left(\frac{1}{2}v\right)$$
$$\frac{du}{dv} = \frac{1}{2}$$

Since the derivative is $\frac{1}{2}$, which is a constant, it is independent of $x$.

**Hence Proved**

The derivative of $\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$ with respect to $\sin^{-1} x$ is $\frac{1}{2}$, which is independent of $x$.

## Question 18

Differentiate $x^x$ with respect to $x\log x$.

**TYPE 53** **Differentiation of One Function with Respect to Another**

**Given:** Let $u = x^x$ and $v = x\log x$

**Concept / Formula Used:** $u$ $v$ $\dfrac{du}{dv} = \dfrac{\frac{du}{dx}}{\frac{dv}{dx}}$

**Solution:**

Let:

$$u = x^x$$
$$v = x \log x$$

To differentiate $u = x^x$ with respect to $x$, we write $u$ as:

$$u = e^{x \log x}$$

Differentiating $u$ with respect to $x$ using the chain rule and product rule:

$$\frac{du}{dx} = e^{x \log x} \cdot \frac{d}{dx}(x \log x)$$
$$\frac{du}{dx} = x^x \cdot \left(x \cdot \frac{1}{x} + \log x \cdot 1\right)$$
$$\frac{du}{dx} = x^x(1 + \log x)$$

Differentiating $v = x \log x$ with respect to $x$ using the product rule:

$$\frac{dv}{dx} = \frac{d}{dx}(x \log x)$$
$$\frac{dv}{dx} = x \cdot \frac{1}{x} + \log x \cdot 1$$
$$\frac{dv}{dx} = 1 + \log x$$

Now, using the derivative formula:

$$\frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}}$$
$$\frac{du}{dv} = \frac{x^x(1 + \log x)}{1 + \log x}$$
$$\frac{du}{dv} = x^x$$

**Final Answer**

$$\frac{d(x^x)}{d(x \log x)} = x^x$$

## Question 19

If $x = \sqrt{a^{\tan^{-1} t}}$ and $y = \sqrt{a^{\cot^{-1} t}}$, show that $x\dfrac{dy}{dx} + y = 0$.

**TYPE 43** **Simplification before Implicit Differentiation**

**To Prove:** $x\dfrac{dy}{dx} + y = 0$

**Concept / Formula Used:** $\tan^{-1} t + \cot^{-1} t = \dfrac{\pi}{2}$

### Solution:

We are given:

$$x = \sqrt{a^{\tan^{-1} t}}$$
$$y = \sqrt{a^{\cot^{-1} t}}$$

Multiplying $x$ and $y$:

$$x \cdot y = \sqrt{a^{\tan^{-1} t}} \cdot \sqrt{a^{\cot^{-1} t}}$$
$$x \cdot y = \sqrt{a^{\tan^{-1} t} \cdot a^{\cot^{-1} t}}$$
$$x \cdot y = \sqrt{a^{\tan^{-1} t + \cot^{-1} t}}$$

Using the standard inverse trigonometric identity $\tan^{-1} t + \cot^{-1} t = \dfrac{\pi}{2}$:

$$x \cdot y = \sqrt{a^{\pi/2}}$$

Since $a$ is a constant, $\sqrt{a^{\pi/2}}$ is also a constant. Let $C = \sqrt{a^{\pi/2}}$:

$$xy = C$$

Differentiating both sides with respect to $x$ using the product rule:

$$\frac{d}{dx}(xy) = \frac{d}{dx}(C)$$
$$x \cdot \frac{dy}{dx} + y \cdot \frac{d}{dx}(x) = 0$$
$$x\frac{dy}{dx} + y(1) = 0$$
$$x\frac{dy}{dx} + y = 0$$

**Hence Proved**

LHS $= x\dfrac{dy}{dx} + y = 0 =$ RHS. Hence Proved.

# 5.11 – Second-Order Derivatives (Miscellaneous)

## Question 1(i)

Find the second order derivative of $x^3 + \tan x$.

**TYPE 63** **Second Order Derivative – Direct Differentiation**

**Given:** Let $y = x^3 + \tan x$

**Concept / Formula Used:** $\frac{d}{dx}(x^n) = nx^{n-1}$ $\quad \frac{d}{dx}(\tan x) = \sec^2 x$

**Solution:**

Differentiating both sides with respect to $x$ to find the first derivative:

$$\frac{dy}{dx} = \frac{d}{dx}(x^3) + \frac{d}{dx}(\tan x)$$

$$\frac{dy}{dx} = 3x^2 + \sec^2 x$$

Differentiating again with respect to $x$ to find the second order derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(3x^2) + \frac{d}{dx}(\sec^2 x)$$

Using the chain rule for $\sec^2 x = (\sec x)^2$:

$$\frac{d}{dx}(\sec^2 x) = 2\sec x \cdot \frac{d}{dx}(\sec x) = 2\sec x \cdot (\sec x \tan x) = 2\sec^2 x \tan x$$

Thus:

$$\frac{d^2y}{dx^2} = 6x + 2\sec^2 x \tan x$$

**Final Answer**

$\frac{d^2y}{dx^2} = 6x + 2\sec^2 x \tan x$

**Common Mistake:** Forgetting to apply the chain rule when differentiating $\sec^2 x$ with respect to $x$.

## Question 1(ii)

Find the second order derivative of $\log x$.

**TYPE 63** **Second Order Derivative – Direct Differentiation**

**Given:** Let $y = \log x$

**Concept / Formula Used:** $\dfrac{d}{dx}(\log x) = \dfrac{1}{x}$

**Solution:**

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = \frac{d}{dx}(\log x) = \frac{1}{x} = x^{-1}$$

Differentiating again with respect to $x$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(x^{-1})$$

Using the power rule $\dfrac{d}{dx}(x^n) = nx^{n-1}$:

$$\frac{d^2y}{dx^2} = -1 \cdot x^{-2} = -\frac{1}{x^2}$$

**Final Answer**

$\dfrac{d^2y}{dx^2} = -\dfrac{1}{x^2}$

**Common Mistake:** Applying the power rule incorrectly for negative exponents by writing $x^0$ instead of $x^{-2}$.

## Question 1(iii)

Find the second order derivative of $\tan^{-1} x$.

**TYPE 63** **Second Order Derivative – Direct Differentiation**

**Given:** Let $y = \tan^{-1} x$

**Concept / Formula Used:** $\dfrac{d}{dx}(\tan^{-1} x) = \dfrac{1}{1+x^2}$

**Solution:**

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2} = (1+x^2)^{-1}$$

Differentiating again with respect to $x$ using the chain rule and power rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left((1+x^2)^{-1}\right)$$

$$\frac{d^2y}{dx^2} = -1(1+x^2)^{-2} \cdot \frac{d}{dx}(1+x^2)$$

$$\frac{d^2y}{dx^2} = -1(1+x^2)^{-2} \cdot (0+2x)$$
$$\frac{d^2y}{dx^2} = -\frac{2x}{(1+x^2)^2}$$

**Final Answer**

$\frac{d^2y}{dx^2} = -\frac{2x}{(1+x^2)^2}$

**Common Mistake:** Omitting the derivative of the inner function $(1+x^2)$, which is $2x$, due to missing chain rule application.

## Question 2(i)

If $y = \log(x-2)$, $x > 2$, find $\frac{d^2y}{dx^2}$.

**TYPE 63** **Second Order Derivative – Direct Differentiation**

**Given:** $y = \log(x-2)$ for $x > 2$

**Concept / Formula Used:** $\frac{d}{dx}(\log u) = \frac{1}{u} \cdot \frac{du}{dx}$

**Solution:**

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = \frac{d}{dx}[\log(x-2)]$$
$$\frac{dy}{dx} = \frac{1}{x-2} \cdot \frac{d}{dx}(x-2)$$
$$\frac{dy}{dx} = \frac{1}{x-2} \cdot (1-0) = (x-2)^{-1}$$

Differentiating again with respect to $x$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left((x-2)^{-1}\right)$$
$$\frac{d^2y}{dx^2} = -1(x-2)^{-2} \cdot \frac{d}{dx}(x-2)$$
$$\frac{d^2y}{dx^2} = -1(x-2)^{-2} \cdot (1) = -\frac{1}{(x-2)^2}$$

**Final Answer**

$\frac{d^2y}{dx^2} = -\frac{1}{(x-2)^2}$

**Common Mistake:** Forgetting to multiply by the derivative of the inner function $(x-2)$, though it evaluates to 1 here.

## Question 2(ii)

If $y = \cot x$, find $\dfrac{d^2y}{dx^2}$ at $x = \dfrac{\pi}{4}$.

**TYPE 63** **Second Order Derivative – Direct Differentiation**

**Given:** $y = \cot x$

**Concept / Formula Used:** $\dfrac{d}{dx}(\cot x) = -\csc^2 x \qquad \dfrac{d}{dx}(\csc x) = -\csc x \cot x$

**Solution:**

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = \frac{d}{dx}(\cot x) = -\csc^2 x$$

Differentiating again with respect to $x$ using the chain rule for $(\csc x)^2$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(-(\csc x)^2\right)$$

$$\frac{d^2y}{dx^2} = -2(\csc x) \cdot \frac{d}{dx}(\csc x)$$

$$\frac{d^2y}{dx^2} = -2(\csc x) \cdot (-\csc x \cot x)$$

$$\frac{d^2y}{dx^2} = 2\csc^2 x \cot x$$

Now, substituting $x = \dfrac{\pi}{4}$:

$$\csc\left(\frac{\pi}{4}\right) = \frac{1}{\sin(\pi/4)} = \sqrt{2}$$

$$\cot\left(\frac{\pi}{4}\right) = \frac{\cos(\pi/4)}{\sin(\pi/4)} = 1$$

Substituting these values into the second derivative expression:

$$\left.\frac{d^2y}{dx^2}\right|_{x=\pi/4} = 2(\sqrt{2})^2(1) = 2(2)(1) = 4$$

**Final Answer**

$$\left.\frac{d^2y}{dx^2}\right|_{x=\pi/4} = 4$$

**Common Mistake:** Incorrectly evaluating trigonometric values at $\pi/4$ or missing the negative sign when differentiating cosecant.

## Question 2(iii)

If $x = at^2$ and $y = 2at$, then find $\dfrac{d^2y}{dx^2}$.

**TYPE 64a** **Parametric Second-Order Derivatives – General Form**

**Given:** $x = at^2$, $y = 2at$

**Concept / Formula Used:** $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} \qquad \dfrac{d^2y}{dx^2} = \dfrac{d}{dt}\left(\dfrac{dy}{dx}\right) \Big/ \dfrac{dx}{dt}$

**Solution:**

First, find the derivatives with respect to $t$:

$$\frac{dx}{dt} = \frac{d}{dx}(at^2) = 2at$$

$$\frac{dy}{dt} = \frac{d}{dt}(2at) = 2a$$

Now, find the first derivative $\dfrac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2a}{2at} = \frac{1}{t} = t^{-1}$$

Now, find the second order derivative $\dfrac{d^2y}{dx^2}$ with respect to $x$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

$$\frac{d}{dt}(t^{-1}) = -1 \cdot t^{-2} = -\frac{1}{t^2}$$

Dividing by $\dfrac{dx}{dt} = 2at$:

$$\frac{d^2y}{dx^2} = \frac{-\frac{1}{t^2}}{2at} = -\frac{1}{2at^3}$$

**Final Answer**

$$\frac{d^2y}{dx^2} = -\frac{1}{2at^3}$$

**Common Mistake:** Differentiating $\dfrac{dy}{dx}$ directly with respect to $x$ without dividing by $\dfrac{dx}{dt}$ (forgetting the chain rule in parametric form).

## Question 3(i)

Find the second order derivatives of $\sin^{-1} x$.

**TYPE 63** **Second Order Derivative – Direct Differentiation**

**Given:** Let $y = \sin^{-1} x$

**Concept / Formula Used:** $\dfrac{d}{dx}(\sin^{-1} x) = \dfrac{1}{\sqrt{1-x^2}}$

**Solution:**

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-1/2}$$

Differentiating again with respect to $x$ using the chain rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left((1-x^2)^{-1/2}\right)$$

$$\frac{d^2y}{dx^2} = -\frac{1}{2}(1-x^2)^{-3/2} \cdot \frac{d}{dx}(1-x^2)$$

$$\frac{d^2y}{dx^2} = -\frac{1}{2}(1-x^2)^{-3/2} \cdot (0-2x)$$

$$\frac{d^2y}{dx^2} = -\frac{1}{2}(1-x^2)^{-3/2} \cdot (-2x)$$

$$\frac{d^2y}{dx^2} = \frac{x}{(1-x^2)^{3/2}} = \frac{x}{(1-x^2)\sqrt{1-x^2}}$$

**Final Answer**

$$\frac{d^2y}{dx^2} = \frac{x}{(1-x^2)^{3/2}}$$

**Common Mistake:** Forgetting the negative sign from the power rule when differentiating $(1-x^2)^{-1/2}$.

## Question 3(ii)

Find the second order derivatives of $x \cos x$.

**TYPE 65** **Product Rule Differentiation**

**Given:** Let $y = x \cos x$

**Concept / Formula Used:** $\dfrac{d}{dx}(u \cdot v) = u\dfrac{dv}{dx} + v\dfrac{du}{dx}$

**Solution:**

Differentiating both sides with respect to $x$ using the product rule:

$$\frac{dy}{dx} = x \cdot \frac{d}{dx}(\cos x) + \cos x \cdot \frac{d}{dx}(x)$$

$$\frac{dy}{dx} = x(-\sin x) + \cos x(1)$$

$$\frac{dy}{dx} = -x\sin x + \cos x$$

Differentiating again with respect to $x$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(-x\sin x) + \frac{d}{dx}(\cos x)$$

For the first term, apply the product rule again:

$$\frac{d}{dx}(-x\sin x) = -\left[x \cdot \frac{d}{dx}(\sin x) + \sin x \cdot \frac{d}{dx}(x)\right]$$

$$= -\left[x\cos x + \sin x(1)\right] = -x\cos x - \sin x$$

For the second term:

$$\frac{d}{dx}(\cos x) = -\sin x$$

Combining both terms:

$$\frac{d^2y}{dx^2} = -x\cos x - \sin x - \sin x = -x\cos x - 2\sin x$$

**Final Answer**

$$\frac{d^2y}{dx^2} = -x\cos x - 2\sin x$$

**Common Mistake:** Distributing the negative sign incorrectly while applying the product rule to the second derivative.

## Question 3(iii)

Find the second order derivatives of $x\sin 2x$.

**TYPE 65** **Product Rule Differentiation**

**Given:** Let $y = x\sin 2x$

**Concept / Formula Used:** $\frac{d}{dx}(\sin 2x) = 2\cos 2x$

**Solution:**

Differentiating both sides with respect to $x$ using the product rule:

$$\frac{dy}{dx} = x \cdot \frac{d}{dx}(\sin 2x) + \sin 2x \cdot \frac{d}{dx}(x)$$

$$\frac{dy}{dx} = x(2\cos 2x) + \sin 2x(1)$$

$$\frac{dy}{dx} = 2x\cos 2x + \sin 2x$$

Differentiating again with respect to $x$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(2x\cos 2x) + \frac{d}{dx}(\sin 2x)$$

Applying the product rule to the first term:

$$\frac{d}{dx}(2x\cos 2x) = 2x \cdot \frac{d}{dx}(\cos 2x) + \cos 2x \cdot \frac{d}{dx}(2x)$$

$$= 2x(-2\sin 2x) + \cos 2x(2)$$

$$= -4x\sin 2x + 2\cos 2x$$

Adding the derivative of the second term, $\frac{d}{dx}(\sin 2x) = 2\cos 2x$:

$$\frac{d^2y}{dx^2} = (-4x\sin 2x + 2\cos 2x) + 2\cos 2x$$

$$\frac{d^2y}{dx^2} = -4x\sin 2x + 4\cos 2x$$

**Final Answer**

$$\frac{d^2y}{dx^2} = -4x\sin 2x + 4\cos 2x$$

**Common Mistake:** Forgetting the factor of 2 from the chain rule when differentiating $\sin 2x$.

## Question 3(iv)

Find the second order derivatives of $e^x \sin 5x$.

**TYPE 65** **Product Rule Differentiation**

**Given:** Let $y = e^x \sin 5x$

**Concept / Formula Used:** $\frac{d}{dx}(u \cdot v) = u\frac{dv}{dx} + v\frac{du}{dx}$

**Solution:**

Differentiating both sides with respect to $x$ using the product rule:

$$\frac{dy}{dx} = e^x \cdot \frac{d}{dx}(\sin 5x) + \sin 5x \cdot \frac{d}{dx}(e^x)$$

$$\frac{dy}{dx} = e^x(5\cos 5x) + \sin 5x(e^x)$$

$$\frac{dy}{dx} = e^x(5\cos 5x + \sin 5x)$$

Differentiating again with respect to $x$ using the product rule:

$$\frac{d^2y}{dx^2} = e^x \cdot \frac{d}{dx}(5\cos 5x + \sin 5x) + (5\cos 5x + \sin 5x) \cdot \frac{d}{dx}(e^x)$$

Evaluating the derivative of the bracket:

$$\frac{d}{dx}(5\cos 5x + \sin 5x) = 5(-5\sin 5x) + 5\cos 5x = -25\sin 5x + 5\cos 5x$$

Substituting back into the product rule expression:

$$\frac{d^2y}{dx^2} = e^x(-25\sin 5x + 5\cos 5x) + (5\cos 5x + \sin 5x)e^x$$

Factorizing $e^x$:

$$\frac{d^2y}{dx^2} = e^x\left[-25\sin 5x + 5\cos 5x + 5\cos 5x + \sin 5x\right]$$

$$\frac{d^2y}{dx^2} = e^x\left[-24\sin 5x + 10\cos 5x\right]$$

**Final Answer**

$$\frac{d^2y}{dx^2} = e^x(10\cos 5x - 24\sin 5x)$$

**Common Mistake:** Arithmetic errors when collecting and combining like terms of $\sin 5x$ and $\cos 5x$.

## Question 3(v)

Find the second order derivatives of $e^{2x}\sin 3x$.

**TYPE 65 Product Rule Differentiation**

**Given:** Let $y = e^{2x}\sin 3x$

**Concept / Formula Used:** $\frac{d}{dx}(u \cdot v) = u\frac{dv}{dx} + v\frac{du}{dx}$

**Solution:**

Differentiating both sides with respect to $x$ using the product rule:

$$\frac{dy}{dx} = e^{2x} \cdot \frac{d}{dx}(\sin 3x) + \sin 3x \cdot \frac{d}{dx}(e^{2x})$$

$$\frac{dy}{dx} = e^{2x}(3\cos 3x) + \sin 3x(2e^{2x})$$
$$\frac{dy}{dx} = e^{2x}(3\cos 3x + 2\sin 3x)$$

Differentiating again with respect to $x$ using the product rule:

$$\frac{d^2y}{dx^2} = e^{2x} \cdot \frac{d}{dx}(3\cos 3x + 2\sin 3x) + (3\cos 3x + 2\sin 3x) \cdot \frac{d}{dx}(e^{2x})$$

Evaluating the derivative of the bracket:

$$\frac{d}{dx}(3\cos 3x + 2\sin 3x) = 3(-3\sin 3x) + 2(3\cos 3x) = -9\sin 3x + 6\cos 3x$$

And $\frac{d}{dx}(e^{2x}) = 2e^{2x}$. Substituting these:

$$\frac{d^2y}{dx^2} = e^{2x}(-9\sin 3x + 6\cos 3x) + (3\cos 3x + 2\sin 3x)(2e^{2x})$$

Factorizing $e^{2x}$:

$$\frac{d^2y}{dx^2} = e^{2x}\left[-9\sin 3x + 6\cos 3x + 6\cos 3x + 4\sin 3x\right]$$
$$\frac{d^2y}{dx^2} = e^{2x}\left[-5\sin 3x + 12\cos 3x\right]$$

**Final Answer**

$$\frac{d^2y}{dx^2} = e^{2x}(12\cos 3x - 5\sin 3x)$$

**Common Mistake:** Missing the coefficient multipliers from the chain rule when differentiating exponential and trigonometric functions.

## Question 3(vi)

Find the second order derivatives of $\log(\log x)$.

**TYPE 63** **Second Order Derivative – Direct Differentiation**

**Given:** Let $y = \log(\log x)$

**Concept / Formula Used:** $\frac{d}{dx}[\log(\log x)] = \frac{1}{\log x} \cdot \frac{1}{x}$

**Solution:**

Differentiating both sides with respect to $x$ using the chain rule:

$$\frac{dy}{dx} = \frac{1}{\log x} \cdot \frac{d}{dx}(\log x)$$

$$\frac{dy}{dx} = \frac{1}{x \log x} = (x \log x)^{-1}$$

Differentiating again with respect to $x$ using the product rule and chain rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x \log x}\right)$$

Using the quotient rule $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$ with $u = 1$ and $v = x \log x$:

$$\frac{d^2y}{dx^2} = \frac{(x \log x)(0) - 1 \cdot \frac{d}{dx}(x \log x)}{(x \log x)^2}$$

$$\frac{d}{dx}(x \log x) = x \cdot \frac{1}{x} + \log x \cdot 1 = 1 + \log x$$

Thus:

$$\frac{d^2y}{dx^2} = \frac{-(1 + \log x)}{(x \log x)^2}$$

**Final Answer**

$\frac{d^2y}{dx^2} = -\frac{1 + \log x}{(x \log x)^2}$

**Common Mistake:** Applying the chain rule incorrectly on nested logarithmic functions.

## Question 3(vii)

Find the second order derivatives of $\frac{\log x}{x}$.

**TYPE 66** **Quotient Rule Differentiation**

**Given:** Let $y = \frac{\log x}{x}$

**Concept / Formula Used:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

**Solution:**

Differentiating both sides with respect to $x$ using the quotient rule:

$$\frac{dy}{dx} = \frac{x \cdot \frac{d}{dx}(\log x) - \log x \cdot \frac{d}{dx}(x)}{x^2}$$

$$\frac{dy}{dx} = \frac{x\left(\frac{1}{x}\right) - \log x(1)}{x^2}$$

$$\frac{dy}{dx} = \frac{1 - \log x}{x^2}$$

Differentiating again with respect to $x$ using the quotient rule:

$$\frac{d^2y}{dx^2} = \frac{x^2 \cdot \frac{d}{dx}(1 - \log x) - (1 - \log x) \cdot \frac{d}{dx}(x^2)}{(x^2)^2}$$

$$\frac{d}{dx}(1 - \log x) = 0 - \frac{1}{x} = -\frac{1}{x}$$

$$\frac{d}{dx}(x^2) = 2x$$

Substituting these into the quotient rule formula:

$$\frac{d^2y}{dx^2} = \frac{x^2\left(-\frac{1}{x}\right) - (1 - \log x)(2x)}{x^4}$$

$$\frac{d^2y}{dx^2} = \frac{-x - 2x(1 - \log x)}{x^4}$$

Factorizing $x$ from the numerator:

$$\frac{d^2y}{dx^2} = \frac{-x[1 + 2(1 - \log x)]}{x^4} = \frac{-[1 + 2 - 2\log x]}{x^3} = \frac{2\log x - 3}{x^3}$$

**Final Answer**

$$\frac{d^2y}{dx^2} = \frac{2\log x - 3}{x^3}$$

**Common Mistake:** Algebraic simplification errors when expanding and canceling terms in the numerator.

## Question 3(viii)

Find the second order derivatives of $x^2 \log|\cos x|$.

**TYPE 65** **Product Rule Differentiation**

**Given:** Let $y = x^2 \log|\cos x|$

**Concept / Formula Used:** $\frac{d}{dx}[\log|\cos x|] = \frac{-\sin x}{\cos x} = -\tan x$

**Solution:**

Differentiating both sides with respect to $x$ using the product rule:

$$\frac{dy}{dx} = x^2 \cdot \frac{d}{dx}(\log|\cos x|) + \log|\cos x| \cdot \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = x^2\left(\frac{-\sin x}{\cos x}\right) + \log|\cos x|(2x)$$

$$\frac{dy}{dx} = -x^2 \tan x + 2x\log|\cos x|$$

Differentiating again with respect to $x$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(-x^2 \tan x) + \frac{d}{dx}(2x \log|\cos x|)$$

Applying the product rule to the first term:

$$\frac{d}{dx}(-x^2 \tan x) = -\left[x^2 \cdot \frac{d}{dx}(\tan x) + \tan x \cdot \frac{d}{dx}(x^2)\right]$$

$$= -\left[x^2 \sec^2 x + \tan x(2x)\right] = -x^2 \sec^2 x - 2x \tan x$$

Applying the product rule to the second term:

$$\frac{d}{dx}(2x \log|\cos x|) = 2x \cdot \frac{d}{dx}(\log|\cos x|) + \log|\cos x| \cdot \frac{d}{dx}(2x)$$

$$= 2x(-\tan x) + \log|\cos x|(2) = -2x \tan x + 2\log|\cos x|$$

Combining all terms:

$$\frac{d^2y}{dx^2} = -x^2 \sec^2 x - 2x \tan x - 2x \tan x + 2\log|\cos x|$$

$$\frac{d^2y}{dx^2} = -x^2 \sec^2 x - 4x \tan x + 2\log|\cos x|$$

**Final Answer**

$\frac{d^2y}{dx^2} = 2\log|\cos x| - 4x \tan x - x^2 \sec^2 x$

**Common Mistake:** Forgetting to apply the product rule separately to both terms of the first derivative.

## Question 3(ix)

Find the second order derivatives of $\frac{2x+1}{2x+3}$.

**TYPE 66** **Quotient Rule Differentiation**

**Given:** Let $y = \frac{2x+1}{2x+3}$

**Concept / Formula Used:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

**Solution:**

Differentiating both sides with respect to $x$ using the quotient rule:

$$\frac{dy}{dx} = \frac{(2x+3) \cdot \frac{d}{dx}(2x+1) - (2x+1) \cdot \frac{d}{dx}(2x+3)}{(2x+3)^2}$$

$$\frac{dy}{dx} = \frac{(2x+3)(2)-(2x+1)(2)}{(2x+3)^2}$$

$$\frac{dy}{dx} = \frac{4x+6-4x-2}{(2x+3)^2} = \frac{4}{(2x+3)^2} = 4(2x+3)^{-2}$$

Differentiating again with respect to $x$ using the chain rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(4(2x+3)^{-2}\right)$$

$$\frac{d^2y}{dx^2} = 4\cdot(-2)(2x+3)^{-3}\cdot\frac{d}{dx}(2x+3)$$

$$\frac{d^2y}{dx^2} = -8(2x+3)^{-3}\cdot(2) = -\frac{16}{(2x+3)^3}$$

**Final Answer**

$$\frac{d^2y}{dx^2} = -\frac{16}{(2x+3)^3}$$

**Common Mistake:** Failing to multiply by the inner derivative 2 when differentiating $(2x+3)^{-2}$.

## Question 4(i)

Find the second order derivative of $\sec ax$.

**TYPE 63** Second Order Derivative – Direct Differentiation

**Given:** Let $y = \sec ax$

**Concept / Formula Used:** $\frac{d}{dx}(\sec x) = \sec x \tan x$

**Solution:**

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = \frac{d}{dx}(\sec ax) = a\sec ax\tan ax$$

Differentiating again with respect to $x$ using the product rule and chain rule:

$$\frac{d^2y}{dx^2} = a\cdot\frac{d}{dx}(\sec ax\tan ax)$$

$$\begin{aligned}\frac{d}{dx}(\sec ax\tan ax) &= \sec ax\cdot\frac{d}{dx}(\tan ax) + \tan ax\cdot\frac{d}{dx}(\sec ax)\\ &= \sec ax(a\sec^2 ax) + \tan ax(a\sec ax\tan ax)\\ &= a\sec^3 ax + a\sec ax\tan^2 ax\end{aligned}$$

Multiplying by the outer constant $a$:

$$\frac{d^2y}{dx^2} = a^2 \sec^3 ax + a^2 \sec ax \tan^2 ax = a^2 \sec ax(\sec^2 ax + \tan^2 ax)$$

Using the trigonometric identity $\tan^2 ax = \sec^2 ax - 1$:

$$\frac{d^2y}{dx^2} = a^2 \sec ax(\sec^2 ax + \sec^2 ax - 1) = a^2 \sec ax(2 \sec^2 ax - 1)$$

**Final Answer**

$$\frac{d^2y}{dx^2} = a^2 \sec ax(2 \sec^2 ax - 1)$$

**Common Mistake:** Forgetting the factor of $a^2$ produced by applying the chain rule twice.

## Question 4(ii)

Find the second order derivative of $\cot(1 - 2x)$.

**TYPE 63** Second Order Derivative – Direct Differentiation

**Given:** Let $y = \cot(1 - 2x)$

**Concept / Formula Used:** $\frac{d}{dx}[\cot(1 - 2x)] = -\csc^2(1 - 2x) \cdot (-2) = 2 \csc^2(1 - 2x)$

**Solution:**

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = -\csc^2(1 - 2x) \cdot \frac{d}{dx}(1 - 2x) = -\csc^2(1 - 2x) \cdot (-2) = 2 \csc^2(1 - 2x)$$

Differentiating again with respect to $x$ using the chain rule for $(\csc(1 - 2x))^2$:

$$\frac{d^2y}{dx^2} = 2 \cdot 2 \csc(1 - 2x) \cdot \frac{d}{dx}(\csc(1 - 2x))$$

$$\frac{d}{dx}(\csc(1 - 2x)) = -\csc(1 - 2x) \cot(1 - 2x) \cdot \frac{d}{dx}(1 - 2x)$$
$$= -\csc(1 - 2x) \cot(1 - 2x) \cdot (-2) = 2 \csc(1 - 2x) \cot(1 - 2x)$$

Substituting back into the second derivative expression:

$$\frac{d^2y}{dx^2} = 4 \csc(1 - 2x) \cdot [2 \csc(1 - 2x) \cot(1 - 2x)]$$

$$\frac{d^2y}{dx^2} = 8 \csc^2(1 - 2x) \cot(1 - 2x)$$

**Final Answer**

$$\frac{d^2y}{dx^2} = 8\csc^2(1-2x)\cot(1-2x)$$

**Common Mistake:** Dropping negative signs during successive applications of the chain rule.

## Question 4(iii)

Find the second order derivative of $\sin 3x \cos 5x$.

**TYPE 65** **Product Rule Differentiation**

**Given:** Let $y = \sin 3x \cos 5x$

**Concept / Formula Used:** $2\sin A \cos B = \sin(A+B) + \sin(A-B)$

**Solution:**

Method 1: Using the product rule directly.

$$\frac{dy}{dx} = \sin 3x \cdot \frac{d}{dx}(\cos 5x) + \cos 5x \cdot \frac{d}{dx}(\sin 3x)$$

$$\frac{dy}{dx} = \sin 3x(-5\sin 5x) + \cos 5x(3\cos 3x)$$

$$\frac{dy}{dx} = 3\cos 3x \cos 5x - 5\sin 3x \sin 5x$$

Differentiating again with respect to $x$:

$$\frac{d^2y}{dx^2} = 3 \cdot \frac{d}{dx}(\cos 3x \cos 5x) - 5 \cdot \frac{d}{dx}(\sin 3x \sin 5x)$$

Applying the product rule to each term:

$$\frac{d}{dx}(\cos 3x \cos 5x) = \cos 3x(-5\sin 5x) + \cos 5x(-3\sin 3x) = -5\cos 3x \sin 5x - 3\sin 3x \cos 5x$$

$$\frac{d}{dx}(\sin 3x \sin 5x) = \sin 3x(5\cos 5x) + \sin 5x(3\cos 3x) = 5\sin 3x \cos 5x + 3\cos 3x \sin 5x$$

Combining both parts:

$$\frac{d^2y}{dx^2} = 3(-5\cos 3x \sin 5x - 3\sin 3x \cos 5x) - 5(5\sin 3x \cos 5x + 3\cos 3x \sin 5x)$$

Expanding and collecting like terms:

$$= -15\cos 3x \sin 5x - 9\sin 3x \cos 5x - 25\sin 3x \cos 5x - 15\cos 3x \sin 5x$$

$$= -30\cos 3x \sin 5x - 34\sin 3x \cos 5x$$

**Final Answer**

$$\frac{d^2y}{dx^2} = -34\sin 3x\cos 5x - 30\cos 3x\sin 5x$$

**Common Mistake:** Sign errors while differentiating products of sine and cosine functions.

## Question 4(iv)

Find the second order derivative of $\sin^3 x$.

**TYPE 63** **Second Order Derivative – Direct Differentiation**

**Given:** Let $y = \sin^3 x = (\sin x)^3$

**Concept / Formula Used:** $\frac{d}{dx}[(\sin x)^3] = 3\sin^2 x\cos x$

**Solution:**

Differentiating both sides with respect to $x$ using the chain rule:

$$\frac{dy}{dx} = 3\sin^2 x \cdot \frac{d}{dx}(\sin x) = 3\sin^2 x\cos x$$

Differentiating again with respect to $x$ using the product rule:

$$\frac{d^2y}{dx^2} = 3\left[\sin^2 x \cdot \frac{d}{dx}(\cos x) + \cos x \cdot \frac{d}{dx}(\sin^2 x)\right]$$

$$\frac{d}{dx}(\sin^2 x) = 2\sin x\cos x$$

Substituting this back:

$$\frac{d^2y}{dx^2} = 3\left[\sin^2 x(-\sin x) + \cos x(2\sin x\cos x)\right]$$

$$\frac{d^2y}{dx^2} = 3\left[-\sin^3 x + 2\sin x\cos^2 x\right]$$

Using $\cos^2 x = 1 - \sin^2 x$:

$$\frac{d^2y}{dx^2} = 3\left[-\sin^3 x + 2\sin x(1 - \sin^2 x)\right]$$

$$\frac{d^2y}{dx^2} = 3\left[-\sin^3 x + 2\sin x - 2\sin^3 x\right] = 3(2\sin x - 3\sin^3 x) = 6\sin x - 9\sin^3 x$$

**Final Answer**

$$\frac{d^2y}{dx^2} = 6\sin x - 9\sin^3 x$$

**Common Mistake:** Forgetting to apply the product rule for the second derivative of $\sin^2 x\cos x$.

## Question 4(v)

Find the second order derivative of $\cos(2x^2-1)$.

**TYPE 63** **Second Order Derivative – Direct Differentiation**

**Given:** Let $y = \cos(2x^2-1)$

**Concept / Formula Used:** $\frac{d}{dx}[\cos(f(x))] = -\sin(f(x)) \cdot f'(x)$

**Solution:**

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = -\sin(2x^2-1) \cdot \frac{d}{dx}(2x^2-1)$$

$$\frac{dy}{dx} = -\sin(2x^2-1) \cdot (4x) = -4x\sin(2x^2-1)$$

Differentiating again with respect to $x$ using the product rule:

$$\frac{d^2y}{dx^2} = -4\left[x \cdot \frac{d}{dx}(\sin(2x^2-1)) + \sin(2x^2-1) \cdot \frac{d}{dx}(x)\right]$$

Evaluating the derivative of the sine term using the chain rule:

$$\frac{d}{dx}(\sin(2x^2-1)) = \cos(2x^2-1) \cdot \frac{d}{dx}(2x^2-1) = \cos(2x^2-1) \cdot (4x) = 4x\cos(2x^2-1)$$

Substituting this back:

$$\frac{d^2y}{dx^2} = -4\left[x(4x\cos(2x^2-1)) + \sin(2x^2-1)(1)\right]$$

$$\frac{d^2y}{dx^2} = -4\left[4x^2\cos(2x^2-1) + \sin(2x^2-1)\right]$$

$$\frac{d^2y}{dx^2} = -16x^2\cos(2x^2-1) - 4\sin(2x^2-1)$$

**Final Answer**

$$\frac{d^2y}{dx^2} = -16x^2\cos(2x^2-1) - 4\sin(2x^2-1)$$

**Common Mistake:** Omitting the product rule when differentiating the product of $-4x$ and $\sin(2x^2-1)$.

## Question 4(vi)

Find the second order derivative of $\sqrt{1-x^2}$.

**TYPE 63** **Second Order Derivative – Direct Differentiation**

**Given:** Let $y = \sqrt{1-x^2} = (1-x^2)^{1/2}$

**Concept / Formula Used:** $\dfrac{d}{dx}[u^{1/2}] = \dfrac{1}{2\sqrt{u}} \cdot \dfrac{du}{dx}$

**Solution:**

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}} \cdot \frac{d}{dx}(1-x^2) = \frac{-2x}{2\sqrt{1-x^2}} = \frac{-x}{\sqrt{1-x^2}}$$

Differentiating again with respect to $x$ using the quotient rule:

$$\frac{d^2y}{dx^2} = -\frac{d}{dx}\left(\frac{x}{\sqrt{1-x^2}}\right)$$

$$\frac{d^2y}{dx^2} = -\left[\frac{\sqrt{1-x^2} \cdot \frac{d}{dx}(x) - x \cdot \frac{d}{dx}(\sqrt{1-x^2})}{(\sqrt{1-x^2})^2}\right]$$

$$\frac{d}{dx}(x) = 1, \quad \frac{d}{dx}(\sqrt{1-x^2}) = \frac{-x}{\sqrt{1-x^2}}$$

Substituting these into the quotient rule:

$$\frac{d^2y}{dx^2} = -\left[\frac{\sqrt{1-x^2}(1) - x\left(\frac{-x}{\sqrt{1-x^2}}\right)}{1-x^2}\right]$$

Simplifying the numerator by finding a common denominator $\sqrt{1-x^2}$:

$$\frac{d^2y}{dx^2} = -\left[\frac{(1-x^2)+x^2}{(1-x^2)\sqrt{1-x^2}}\right] = -\left[\frac{1}{(1-x^2)^{3/2}}\right] = -\frac{1}{(1-x^2)^{3/2}}$$

**Final Answer**

$$\frac{d^2y}{dx^2} = -\frac{1}{(1-x^2)^{3/2}}$$

**Common Mistake:** Errors in fraction simplification within the numerator during quotient rule execution.

## Question 5

If $y = \cos^{-1} x$, find $\dfrac{d^2y}{dx^2}$ in terms of $y$ alone.

**TYPE 67** **Implicit / Inverse Trig Second Derivative in terms of $y$**

**Given:** $y = \cos^{-1} x$

**Concept / Formula Used:** $\sin y = \sqrt{1 - \cos^2 y}$ $\quad \cos y = x$

**Solution:**

From $y = \cos^{-1} x$, we have $x = \cos y$. Differentiating both sides with respect to $x$:

$$1 = -\sin y \cdot \frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{1}{\sin y} = -\csc y$$

Differentiating again with respect to $x$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(-\csc y) = -(-\csc y \cot y) \cdot \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = \csc y \cot y \cdot (-\csc y) = -\csc^2 y \cot y$$

**Final Answer**

$$\frac{d^2y}{dx^2} = -\csc^2 y \cot y$$

**Common Mistake:** Failing to substitute $\frac{dy}{dx}$ in terms of $y$ when differentiating with respect to $x$.

## Question 6(i)

If $y = \cot x$, prove that $\frac{d^2y}{dx^2} + 2y\frac{dy}{dx} = 0$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $\frac{d^2y}{dx^2} + 2y\frac{dy}{dx} = 0$ given $y = \cot x$

**Concept / Formula Used:** $\frac{dy}{dx} = -\csc^2 x$ $\quad \frac{d^2y}{dx^2} = 2\csc^2 x \cot x$

**Solution:**

Given $y = \cot x$. First derivative:

$$\frac{dy}{dx} = -\csc^2 x$$

Second derivative:

$$\frac{d^2y}{dx^2} = 2\csc^2 x \cot x$$

Substitute $y = \cot x$ and $\frac{dy}{dx} = -\csc^2 x$ into the expression $\frac{d^2y}{dx^2} + 2y\frac{dy}{dx}$:

$$\text{LHS} = 2\csc^2 x \cot x + 2(\cot x)(-\csc^2 x)$$

$$= 2\csc^2 x \cot x - 2\csc^2 x \cot x = 0 = \text{RHS}$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Sign errors when substituting the first derivative back into the expression.

## Question 6(ii)

If $y = 5\cos x - 3\sin x$, prove that $\frac{d^2y}{dx^2} + y = 0$.

**TYPE 68a** **Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $\frac{d^2y}{dx^2} + y = 0$ given $y = 5\cos x - 3\sin x$

**Concept / Formula Used:** $\frac{d}{dx}(\cos x) = -\sin x \qquad \frac{d}{dx}(\sin x) = \cos x$

**Solution:**

Given $y = 5\cos x - 3\sin x$. Differentiating with respect to $x$:

$$\frac{dy}{dx} = 5(-\sin x) - 3(\cos x) = -5\sin x - 3\cos x$$

Differentiating again with respect to $x$:

$$\frac{d^2y}{dx^2} = -5(\cos x) - 3(-\sin x) = -5\cos x + 3\sin x$$

Now substitute $\frac{d^2y}{dx^2}$ and $y$ into the LHS of the expression:

$$\text{LHS} = \frac{d^2y}{dx^2} + y = (-5\cos x + 3\sin x) + (5\cos x - 3\sin x)$$

Combining terms:

$$\text{LHS} = -5\cos x + 5\cos x + 3\sin x - 3\sin x = 0 = \text{RHS}$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Incorrectly distributing negative signs during the second differentiation.

## Question 7(i)

If $y = x + \tan x$, prove that $\cos^2 x \dfrac{d^2y}{dx^2} - 2y + 2x = 0$.

**TYPE 68a** **Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $\cos^2 x \dfrac{d^2y}{dx^2} - 2y + 2x = 0$ given $y = x + \tan x$

**Concept / Formula Used:** $\dfrac{dy}{dx} = 1 + \sec^2 x \qquad \dfrac{d^2y}{dx^2} = 2\sec^2 x \tan x$

**Solution:**

Given $y = x + \tan x$. First derivative:

$$\frac{dy}{dx} = 1 + \sec^2 x$$

Second derivative:

$$\frac{d^2y}{dx^2} = 0 + 2\sec x \cdot (\sec x \tan x) = 2\sec^2 x \tan x$$

Substitute $\dfrac{d^2y}{dx^2}$ and $y$ into the LHS:

$$\text{LHS} = \cos^2 x(2\sec^2 x \tan x) - 2(x + \tan x) + 2x$$

Since $\cos^2 x \cdot \sec^2 x = 1$:

$$\text{LHS} = 2(1)\tan x - 2x - 2\tan x + 2x$$

$$\text{LHS} = 2\tan x - 2\tan x - 2x + 2x = 0 = \text{RHS}$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Forgetting to substitute $y = x + \tan x$ fully into the expression.

## Question 7(ii)

If $y = \tan x + \sec x$, prove that $\dfrac{d^2y}{dx^2} = \dfrac{\cos x}{(1 - \sin x)^2}$.

**TYPE 68a** **Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $\dfrac{d^2y}{dx^2} = \dfrac{\cos x}{(1 - \sin x)^2}$ given $y = \tan x + \sec x$

**Concept / Formula Used:** $\tan x = \dfrac{\sin x}{\cos x} \qquad \sec x = \dfrac{1}{\cos x}$

**Solution:**

Given $y = \tan x + \sec x = \dfrac{\sin x + 1}{\cos x}$. First derivative:

$$\frac{dy}{dx} = \sec^2 x + \sec x \tan x = \frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x} = \frac{1 + \sin x}{\cos^2 x}$$

Second derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1 + \sin x}{\cos^2 x}\right)$$

Using the quotient rule:

$$\frac{d^2y}{dx^2} = \frac{\cos^2 x \cdot \frac{d}{dx}(1 + \sin x) - (1 + \sin x) \cdot \frac{d}{dx}(\cos^2 x)}{(\cos^2 x)^2}$$

$$\frac{d}{dx}(1 + \sin x) = \cos x$$

$$\frac{d}{dx}(\cos^2 x) = 2\cos x(-\sin x) = -2\sin x \cos x$$

Substituting these:

$$\frac{d^2y}{dx^2} = \frac{\cos^2 x(\cos x) - (1 + \sin x)(-2\sin x \cos x)}{\cos^4 x}$$

Factorizing $\cos x$ from the numerator:

$$\frac{d^2y}{dx^2} = \frac{\cos x\left[\cos^2 x + 2\sin x(1 + \sin x)\right]}{\cos^4 x} = \frac{\cos^2 x + 2\sin x + 2\sin^2 x}{\cos^3 x}$$

Using $\cos^2 x = 1 - \sin^2 x$:

$$\frac{d^2y}{dx^2} = \frac{1 - \sin^2 x + 2\sin x + 2\sin^2 x}{\cos^3 x} = \frac{1 + 2\sin x + \sin^2 x}{\cos^3 x} = \frac{(1 + \sin x)^2}{\cos^3 x}$$

Expressing $\cos^3 x$ as $\cos x \cdot \cos^2 x = \cos x(1 - \sin^2 x) = \cos x(1 - \sin x)(1 + \sin x)$:

$$\frac{d^2y}{dx^2} = \frac{(1 + \sin x)^2}{\cos x(1 - \sin x)(1 + \sin x)} = \frac{1 + \sin x}{\cos x(1 - \sin x)}$$

Multiplying numerator and denominator by $(1 - \sin x)$:

$$\frac{d^2y}{dx^2} = \frac{(1 + \sin x)(1 - \sin x)}{\cos x(1 - \sin x)^2} = \frac{1 - \sin^2 x}{\cos x(1 - \sin x)^2} = \frac{\cos^2 x}{\cos x(1 - \sin x)^2} = \frac{\cos x}{(1 - \sin x)^2}$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Algebraic simplification errors in trigonometric identities.

## Question 8(i)

If $y = \sec x - \tan x$, prove that $\cos x \dfrac{d^2y}{dx^2} = y^2$.

**TYPE 68a** **Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $\cos x \dfrac{d^2y}{dx^2} = y^2$ given $y = \sec x - \tan x$

**Concept / Formula Used:** $\dfrac{dy}{dx} = \sec x \tan x - \sec^2 x$

**Solution:**

Given $y = \sec x - \tan x$. First derivative:

$$\frac{dy}{dx} = \sec x \tan x - \sec^2 x = \sec x(\tan x - \sec x) = -\sec x(\sec x - \tan x) = -y \sec x$$

Second derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(-y \sec x) = -\left[y \cdot \frac{d}{dx}(\sec x) + \sec x \cdot \frac{dy}{dx}\right]$$

$$= -\left[y \sec x \tan x + \sec x(-y \sec x)\right] = -y \sec x \tan x + y \sec^2 x$$

$$= y \sec x(\sec x - \tan x) = y \sec x(y) = y^2 \sec x$$

Multiplying both sides by $\cos x$:

$$\cos x \frac{d^2y}{dx^2} = \cos x(y^2 \sec x) = y^2(\cos x \sec x) = y^2(1) = y^2$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Not substituting $y$ back into the derivative expression to simplify.

## Question 8(ii)

If $y = x + \cot x$, prove that $\sin^2 x \dfrac{d^2y}{dx^2} - 2y + 2x = 0$.

**TYPE 68a** **Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $\sin^2 x \dfrac{d^2y}{dx^2} - 2y + 2x = 0$ given $y = x + \cot x$

**Concept / Formula Used:** $\frac{dy}{dx} = 1 - \csc^2 x \qquad \frac{d^2y}{dx^2} = 2\csc^2 x \cot x$

**Solution:**

Given $y = x + \cot x$. First derivative:

$$\frac{dy}{dx} = 1 - \csc^2 x$$

Second derivative:

$$\frac{d^2y}{dx^2} = 0 - 2\csc x(-\csc x \cot x) = 2\csc^2 x \cot x$$

Substitute $\frac{d^2y}{dx^2}$ and $y$ into the LHS:

$$\text{LHS} = \sin^2 x(2\csc^2 x \cot x) - 2(x + \cot x) + 2x$$

Since $\sin^2 x \cdot \csc^2 x = 1$:

$$\text{LHS} = 2(1)\cot x - 2x - 2\cot x + 2x = 0 = \text{RHS}$$

**Hence Proved**

$LHS = RHS, hence proved.$

**Common Mistake:** Sign errors while evaluating the derivative of $\cot x$.

## Question 9(i)

If $y = ae^{mx} + be^{-mx}$, prove that $y_2 - m^2y = 0$.

**TYPE 68a** **Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $y_2 - m^2y = 0$ given $y = ae^{mx} + be^{-mx}$

**Concept / Formula Used:** $\frac{d}{dx}(e^{mx}) = me^{mx} \qquad \frac{d}{dx}(e^{-mx}) = -me^{-mx}$

**Solution:**

Given $y = ae^{mx} + be^{-mx}$. First derivative ($y_1$):

$$y_1 = a(me^{mx}) + b(-me^{-mx}) = m(ae^{mx} - be^{-mx})$$

Second derivative ($y_2$):

$$y_2 = m\left(a(me^{mx}) - b(-me^{-mx})\right) = m^2(ae^{mx} + be^{-mx})$$

Since $y = ae^{mx} + be^{-mx}$, we have:

$$y_2 = m^2y \implies y_2 - m^2y = 0$$

**Hence Proved**

$LHS = RHS, hence proved.$

**Common Mistake:** Forgetting the negative sign from the exponent in $e^{-mx}$ during differentiation.

## Question 9(ii)

If $y = ae^{2x} + be^{-x}$, prove that $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0$ given $y = ae^{2x} + be^{-x}$

**Concept / Formula Used:** $\frac{d}{dx}(e^{2x}) = 2e^{2x} \qquad \frac{d}{dx}(e^{-x}) = -e^{-x}$

**Solution:**

Given $y = ae^{2x} + be^{-x}$. First derivative:

$$\frac{dy}{dx} = 2ae^{2x} - be^{-x}$$

Second derivative:

$$\frac{d^2y}{dx^2} = 4ae^{2x} + be^{-x}$$

Substitute $y$, $\frac{dy}{dx}$, and $\frac{d^2y}{dx^2}$ into the LHS:

$$\text{LHS} = (4ae^{2x} + be^{-x}) - (2ae^{2x} - be^{-x}) - 2(ae^{2x} + be^{-x})$$

Expanding and grouping terms:

$$\text{LHS} = 4ae^{2x} + be^{-x} - 2ae^{2x} + be^{-x} - 2ae^{2x} - 2be^{-x}$$

$$= (4a - 2a - 2a)e^{2x} + (1 + 1 - 2)be^{-x} = 0 \cdot ae^{2x} + 0 \cdot be^{-x} = 0 = \text{RHS}$$

**Hence Proved**

$LHS = RHS, hence proved.$

**Common Mistake:** Algebraic errors when combining coefficients of $e^{2x}$ and $e^{-x}$.

## Question 10(i)

If $y = A\cos nx + B\sin nx$, show that $\frac{d^2y}{dx^2} + n^2y = 0$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $\dfrac{d^2y}{dx^2} + n^2y = 0$ given $y = A\cos nx + B\sin nx$

**Concept / Formula Used:** $\dfrac{d}{dx}(\cos nx) = -n\sin nx \qquad \dfrac{d}{dx}(\sin nx) = n\cos nx$

**Solution:**

Given $y = A\cos nx + B\sin nx$. First derivative:

$$\frac{dy}{dx} = -nA\sin nx + nB\cos nx$$

Second derivative:

$$\frac{d^2y}{dx^2} = -n^2A\cos nx - n^2B\sin nx = -n^2(A\cos nx + B\sin nx)$$

Since $y = A\cos nx + B\sin nx$, we have:

$$\frac{d^2y}{dx^2} = -n^2y \implies \frac{d^2y}{dx^2} + n^2y = 0$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Missing the factor of $n^2$ produced by the chain rule upon second differentiation.

## Question 10(ii)

If $y = A\sin 2x + B\cos 2x$ and $\dfrac{d^2y}{dx^2} - ky = 0$, find the value of $k$.

**TYPE 69** **Parameter Finding via Second Derivatives**

**Given:** $y = A\sin 2x + B\cos 2x$ and $\dfrac{d^2y}{dx^2} - ky = 0$

**Concept / Formula Used:** $\dfrac{d}{dx}(\sin 2x) = 2\cos 2x \qquad \dfrac{d}{dx}(\cos 2x) = -2\sin 2x$

**Solution:**

Given $y = A\sin 2x + B\cos 2x$. First derivative:

$$\frac{dy}{dx} = 2A\cos 2x - 2B\sin 2x$$

Second derivative:

$$\frac{d^2y}{dx^2} = -4A\sin 2x - 4B\cos 2x = -4(A\sin 2x + B\cos 2x) = -4y$$

Substitute $\frac{d^2y}{dx^2} = -4y$ into the given equation $\frac{d^2y}{dx^2} - ky = 0$:

$$-4y - ky = 0 \implies (-4 - k)y = 0$$

Since this holds for all $x$, the coefficient must vanish:

$$-4 - k = 0 \implies k = -4$$

**Final Answer**

$k = -4$

**Common Mistake:** Incorrectly equating coefficients or missing the negative sign in the second derivative.

## Question 11(i)

If $y = \tan^{-1} x$, prove that $(1 + x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 0$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $(1 + x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 0$ given $y = \tan^{-1} x$

**Concept / Formula Used:** $\frac{dy}{dx} = \frac{1}{1 + x^2}$

**Solution:**

Given $y = \tan^{-1} x$. First derivative:

$$\frac{dy}{dx} = \frac{1}{1 + x^2} \implies (1 + x^2)\frac{dy}{dx} = 1$$

Differentiating both sides with respect to $x$ using the product rule:

$$(1 + x^2) \cdot \frac{d}{dx}\left(\frac{dy}{dx}\right) + \frac{dy}{dx} \cdot \frac{d}{dx}(1 + x^2) = \frac{d}{dx}(1)$$

$$(1 + x^2)\frac{d^2y}{dx^2} + \frac{dy}{dx}(2x) = 0$$

$$(1 + x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 0$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Attempting to differentiate the second order derivative directly without cross-multiplying $1 + x^2$.

## Question 11(ii)

If $y = \sin^{-1} x$, prove that $(1 - x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} = 0$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $(1 - x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} = 0$ given $y = \sin^{-1} x$

**Concept / Formula Used:** $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1 - x^2}}$

**Solution:**

Given $y = \sin^{-1} x$. First derivative:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}} \implies \sqrt{1 - x^2}\frac{dy}{dx} = 1$$

Squaring both sides:

$$(1 - x^2)\left(\frac{dy}{dx}\right)^2 = 1$$

Differentiating both sides with respect to $x$ using the product rule and chain rule:

$$(1 - x^2) \cdot 2\left(\frac{dy}{dx}\right)\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 \cdot (-2x) = 0$$

Dividing both sides by $2\dfrac{dy}{dx}$ (assuming $\dfrac{dy}{dx} \neq 0$):

$$(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0$$

**Hence Proved**

$LHS = RHS, hence proved.$

**Common Mistake:** Forgetting to divide by $2\dfrac{dy}{dx}$ correctly after differentiation.

## Question 12(i)

If $y = \sin(\log x)$, prove that $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$ given $y = \sin(\log x)$

**Concept / Formula Used:** $\dfrac{dy}{dx} = \dfrac{\cos(\log x)}{x}$

**Solution:**

Given $y = \sin(\log x)$. First derivative:

$$\frac{dy}{dx} = \cos(\log x) \cdot \frac{1}{x} \implies x\frac{dy}{dx} = \cos(\log x)$$

Differentiating both sides with respect to $x$ using the product rule:

$$x \cdot \frac{d}{dx}\left(\frac{dy}{dx}\right) + \frac{dy}{dx} \cdot \frac{d}{dx}(x) = \frac{d}{dx}(\cos(\log x))$$

$$x\frac{d^2y}{dx^2} + \frac{dy}{dx}(1) = -\sin(\log x) \cdot \frac{1}{x}$$

Multiply the entire equation by $x$:

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = -\sin(\log x)$$

Since $y = \sin(\log x)$, substitute $y$ on the RHS:

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = -y \implies x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Omitting the multiplication by $x$ after the second differentiation step.

## Question 12(ii)

If $y = 3\cos(\log x) + 4\sin(\log x)$, prove that $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$ given $y = 3\cos(\log x) + 4\sin(\log x)$

**Concept / Formula Used:** $\dfrac{dy}{dx} = \dfrac{-3\sin(\log x) + 4\cos(\log x)}{x}$

**Solution:**

Given $y = 3\cos(\log x) + 4\sin(\log x)$. First derivative:

$$\frac{dy}{dx} = \frac{-3\sin(\log x) + 4\cos(\log x)}{x} \implies x\frac{dy}{dx} = -3\sin(\log x) + 4\cos(\log x)$$

Differentiating both sides with respect to $x$ using the product rule:

$$x\frac{d^2y}{dx^2} + \frac{dy}{dx}(1) = -3\cos(\log x) \cdot \frac{1}{x} - 4\sin(\log x) \cdot \frac{1}{x}$$

Multiply the entire equation by $x$:

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = -3\cos(\log x) - 4\sin(\log x) = -(3\cos(\log x) + 4\sin(\log x))$$

Since $y = 3\cos(\log x) + 4\sin(\log x)$, we get:

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = -y \implies x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$$

**Hence Proved**

$LHS = RHS, hence proved.$

**Common Mistake:** Sign errors while differentiating logarithmic trigonometric terms.

## Question 13

If $y = x\cos x$, prove that $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + (2 + x^2)y = 0$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + (2 + x^2)y = 0$ given $y = x\cos x$

**Concept / Formula Used:** $\frac{dy}{dx} = \cos x - x\sin x \qquad \frac{d^2y}{dx^2} = -2\sin x - x\cos x$

**Solution:**

Given $y = x\cos x$. First derivative:

$$\frac{dy}{dx} = x(-\sin x) + \cos x(1) = \cos x - x\sin x$$

Second derivative:

$$\frac{d^2y}{dx^2} = -\sin x - (x\cos x + \sin x) = -2\sin x - x\cos x$$

Substitute $y$, $\frac{dy}{dx}$, and $\frac{d^2y}{dx^2}$ into the LHS:

$$\text{LHS} = x^2(-2\sin x - x\cos x) - 2x(\cos x - x\sin x) + (2 + x^2)(x\cos x)$$

Expanding all terms:

$$\text{LHS} = -2x^2\sin x - x^3\cos x - 2x\cos x + 2x^2\sin x + 2x\cos x + x^3\cos x$$

Grouping and canceling opposite terms:

LHS $= (-2x^2 \sin x + 2x^2 \sin x) + (-x^3 \cos x + x^3 \cos x) + (-2x \cos x + 2x \cos x) = 0 =$ RHS

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Algebraic expansion errors when multiplying out terms with $x^2$ and $x$.

## Question 14(i)

If $y = (\cos^{-1} x)^2$, prove that $(1 - x^2)y_2 - xy_1 = 2$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $(1 - x^2)y_2 - xy_1 = 2$ given $y = (\cos^{-1} x)^2$

**Concept / Formula Used:** $\dfrac{dy}{dx} = 2(\cos^{-1} x)\left(-\dfrac{1}{\sqrt{1-x^2}}\right)$

**Solution:**

Given $y = (\cos^{-1} x)^2$. First derivative ($y_1$):

$$y_1 = 2(\cos^{-1} x) \cdot \left(-\frac{1}{\sqrt{1-x^2}}\right)$$

$$\sqrt{1-x^2}\, y_1 = -2\cos^{-1} x$$

Squaring both sides:

$$(1 - x^2)y_1^2 = 4(\cos^{-1} x)^2 = 4y$$

Differentiating both sides with respect to $x$ using the product rule:

$$(1 - x^2) \cdot 2y_1y_2 + y_1^2 \cdot (-2x) = 4y_1$$

Dividing both sides by $2y_1$ (assuming $y_1 \neq 0$):

$$(1 - x^2)y_2 - xy_1 = 2$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Forgetting to square both sides to eliminate the square root before differentiating.

## Question 14(ii)

If $y = (\tan^{-1} x)^2$, prove that $(x^2+1)^2\dfrac{d^2y}{dx^2} + 2x(x^2+1)\dfrac{dy}{dx} = 2$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $(x^2+1)^2\dfrac{d^2y}{dx^2} + 2x(x^2+1)\dfrac{dy}{dx} = 2$ given $y = (\tan^{-1} x)^2$

**Concept / Formula Used:** $\dfrac{dy}{dx} = \dfrac{2\tan^{-1} x}{1+x^2}$

**Solution:**

Given $y = (\tan^{-1} x)^2$. First derivative:

$$\frac{dy}{dx} = \frac{2\tan^{-1} x}{1+x^2} \implies (1+x^2)\frac{dy}{dx} = 2\tan^{-1} x$$

Differentiating both sides with respect to $x$ using the product rule:

$$(1+x^2)\cdot\frac{d^2y}{dx^2} + \frac{dy}{dx}\cdot(2x) = 2\cdot\frac{1}{1+x^2}$$

Multiply the entire equation by $(1+x^2)$:

$$(1+x^2)^2\frac{d^2y}{dx^2} + 2x(1+x^2)\frac{dy}{dx} = 2$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Failing to multiply through by $(1+x^2)$ at the final step to match the target equation.

## Question 14(iii)

If $y = (\sec^{-1} x)^2, x > 1$, show that $x^2(x^2-1)\dfrac{d^2y}{dx^2} + (2x^3-x)\dfrac{dy}{dx} - 2 = 0$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $x^2(x^2-1)\dfrac{d^2y}{dx^2} + (2x^3-x)\dfrac{dy}{dx} - 2 = 0$ given $y = (\sec^{-1} x)^2$

**Concept / Formula Used:** $\dfrac{dy}{dx} = \dfrac{2\sec^{-1} x}{x\sqrt{x^2-1}}$

**Solution:**

Given $y = (\sec^{-1} x)^2$. First derivative:

$$\frac{dy}{dx} = \frac{2\sec^{-1} x}{x\sqrt{x^2-1}} \implies x\sqrt{x^2-1}\frac{dy}{dx} = 2\sec^{-1} x$$

Squaring both sides:

$$x^2(x^2-1)\left(\frac{dy}{dx}\right)^2 = 4(\sec^{-1} x)^2 = 4y$$

Differentiating both sides with respect to $x$ using the product rule:

$$\frac{d}{dx}\left[x^2(x^2-1)\right]\left(\frac{dy}{dx}\right)^2 + x^2(x^2-1)\cdot 2\left(\frac{dy}{dx}\right)\frac{d^2y}{dx^2} = 4\frac{dy}{dx}$$

$$\frac{d}{dx}(x^4-x^2) = 4x^3-2x = 2x(2x^2-1)$$

Substituting this back and dividing by $2\frac{dy}{dx}$:

$$2x(2x^2-1)\frac{dy}{dx} + 2x^2(x^2-1)\frac{d^2y}{dx^2} = 4$$

Dividing the entire equation by 2:

$$x^2(x^2-1)\frac{d^2y}{dx^2} + (2x^3-x)\frac{dy}{dx} - 2 = 0$$

**Hence Proved**

$LHS = RHS, henceproved.$

**Common Mistake:** Algebraic errors when expanding and differentiating $x^2(x^2-1)$.

## Question 15(i)

If $y = \log(x+\sqrt{x^2+1})$, prove that $(x^2+1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0$.

**TYPE 68a Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $(x^2+1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0$ given $y = \log(x+\sqrt{x^2+1})$

**Concept / Formula Used:** $\frac{dy}{dx} = \frac{1}{x+\sqrt{x^2+1}}\cdot\left(1+\frac{x}{\sqrt{x^2+1}}\right)$

**Solution:**

Given $y = \log(x+\sqrt{x^2+1})$. First derivative:

$$\frac{dy}{dx} = \frac{1}{x+\sqrt{x^2+1}}\cdot\left(1+\frac{2x}{2\sqrt{x^2+1}}\right)$$

$$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2+1}} \cdot \left(\frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1}}\right) = \frac{1}{\sqrt{x^2+1}}$$

Cross-multiplying:

$$\sqrt{x^2+1}\frac{dy}{dx} = 1$$

Differentiating both sides with respect to $x$ (product rule on the left):

$$\sqrt{x^2+1} \cdot \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot \left(\frac{x}{\sqrt{x^2+1}}\right) = 0$$

Multiplying through by $\sqrt{x^2+1}$:

$$(x^2+1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 0$$

**Hence Proved**

$LHS = RHS, hence proved.$

**Common Mistake:** Failing to simplify $\frac{dy}{dx}$ to $\frac{1}{\sqrt{x^2+1}}$ before differentiating the second time.

## Question 15(ii)

If $y = \log(x + \sqrt{x^2+a^2})$, prove that $(x^2+a^2)y_2 + xy_1 = 0$.

**TYPE 68a** **Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $(x^2+a^2)y_2 + xy_1 = 0$ given $y = \log(x + \sqrt{x^2+a^2})$

**Concept / Formula Used:** $y_1 = \frac{1}{\sqrt{x^2+a^2}}$

**Solution:**

Given $y = \log(x + \sqrt{x^2+a^2})$. First derivative ($y_1$):

$$y_1 = \frac{1}{x + \sqrt{x^2+a^2}} \cdot \left(1 + \frac{x}{\sqrt{x^2+a^2}}\right) = \frac{1}{\sqrt{x^2+a^2}}$$

Cross-multiplying:

$$\sqrt{x^2+a^2}\, y_1 = 1$$

Differentiating both sides with respect to $x$ using the product rule:

$$\sqrt{x^2+a^2} \cdot y_2 + y_1 \cdot \left(\frac{x}{\sqrt{x^2+a^2}}\right) = 0$$

Multiply the entire equation by $\sqrt{x^2+a^2}$:

$$(x^2+a^2)y_2 + xy_1 = 0$$

**Hence Proved**

$LHS = RHS, hence proved.$

**Common Mistake:** Omitting the multiplication by $\sqrt{x^2 + a^2}$ to clear the denominator.

## Question 15(iii)

If $y = (\log x)^2$, prove that $x^2y'' + xy' = 2$.

**TYPE 68a** **Proving Second-Order Differential Equation Identities – Trigonometric Function**

**To Prove:** $x^2y'' + xy' = 2$ given $y = (\log x)^2$

**Concept / Formula Used:** $y' = \dfrac{2\log x}{x}$

**Solution:**

Given $y = (\log x)^2$. First derivative ($y'$):

$$y' = 2(\log x) \cdot \frac{1}{x} = \frac{2\log x}{x}$$

Cross-multiplying by $x$:

$$xy' = 2\log x$$

Differentiating both sides with respect to $x$ using the product rule:

$$x \cdot y'' + y' \cdot (1) = 2 \cdot \frac{1}{x}$$

Multiply the entire equation by $x$:

$$x^2y'' + xy' = 2$$

**Hence Proved**

$LHS = RHS, hence proved.$

**Common Mistake:** Forgetting to multiply the final equation by $x$ to clear the denominator.

## Question 16

If $y = \cos(\sin x)$, prove that $\dfrac{d^2y}{dx^2} + \tan x\dfrac{dy}{dx} + y\cos^2 x = 0$.

**TYPE 68b** **Proving Second-Order Differential Equation Identities – Composite Trigonometric Function**

**★ Likely Exam Question**

**To Prove:** (1) $\frac{d^2y}{dx^2} + \tan x\frac{dy}{dx} + y\cos^2 x = 0$

**Concept / Formula Used:** $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

**Solution:**

Given equation:

$$y = \cos(\sin x)$$

Differentiating both sides with respect to $x$ using the chain rule:

$$\frac{dy}{dx} = -\sin(\sin x) \cdot \frac{d}{dx}(\sin x)$$

$$\frac{dy}{dx} = -\sin(\sin x) \cdot \cos x$$

Differentiating again with respect to $x$ using the product rule:

$$\begin{aligned}\frac{d^2y}{dx^2} &= -\left[\sin(\sin x) \cdot \frac{d}{dx}(\cos x) + \cos x \cdot \frac{d}{dx}(\sin(\sin x))\right] \\ &= -[\sin(\sin x) \cdot (-\sin x) + \cos x \cdot \cos(\sin x) \cdot \cos x] \\ &= \sin x \sin(\sin x) - \cos^2 x \cos(\sin x)\end{aligned}$$

Since $y = \cos(\sin x)$, the second term contains $y$:

$$\frac{d^2y}{dx^2} = \sin x \sin(\sin x) - y\cos^2 x$$

From the first derivative expression, we have $\sin(\sin x) = -\frac{1}{\cos x}\frac{dy}{dx}$. Substituting this:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \sin x\left(-\frac{1}{\cos x}\frac{dy}{dx}\right) - y\cos^2 x \\ &= -\frac{\sin x}{\cos x}\frac{dy}{dx} - y\cos^2 x \\ &= -\tan x\frac{dy}{dx} - y\cos^2 x\end{aligned}$$

Rearranging the terms:

$$\frac{d^2y}{dx^2} + \tan x\frac{dy}{dx} + y\cos^2 x = 0$$

**Hence Proved**

$LHS = RHS, henceproved.$

## Question 17 (i)

If $y = e^{\tan^{-1} x}$, prove that $(1 + x^2)\frac{d^2y}{dx^2} + (2x - 1)\frac{dy}{dx} = 0.$

**TYPE 68c** **Proving Second-Order Differential Equation Identities – Exponential of Inverse Trig Function**

### ★ Likely Exam Question

**To Prove:** $(1+x^2)\dfrac{d^2y}{dx^2} + (2x-1)\dfrac{dy}{dx} = 0$

**Concept / Formula Used:** $\dfrac{d}{dx}(\tan^{-1} x) = \dfrac{1}{1+x^2}$

**Solution:**

Given equation:

$$y = e^{\tan^{-1} x}$$

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = e^{\tan^{-1} x} \cdot \frac{d}{dx}(\tan^{-1} x)$$

$$\frac{dy}{dx} = e^{\tan^{-1} x} \cdot \frac{1}{1+x^2}$$

Since $y = e^{\tan^{-1} x}$, we can write:

$$\frac{dy}{dx} = \frac{y}{1+x^2}$$

Cross-multiplying:

$$(1+x^2)\frac{dy}{dx} = y$$

Differentiating both sides again with respect to $x$ using the product rule on the left side:

$$(1+x^2)\frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot \frac{d}{dx}(1+x^2) = \frac{dy}{dx}$$

$$(1+x^2)\frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot (2x) = \frac{dy}{dx}$$

$$(1+x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} - \frac{dy}{dx} = 0$$

$$(1+x^2)\frac{d^2y}{dx^2} + (2x-1)\frac{dy}{dx} = 0$$

**Hence Proved**

$LHS = RHS, hence proved.$

## Question 17 (ii)

If $x = \tan\left(\frac{1}{a}\log y\right)$, then prove that $(1+x^2)\dfrac{d^2y}{dx^2} + (2x-a)\dfrac{dy}{dx} = 0$.

**TYPE 68d** **Proving Second-Order Differential Equation Identities – Implicit Logarithmic Substitution**

### ★ Likely Exam Question

**To Prove:** $(1+x^2)\frac{d^2y}{dx^2} + (2x-a)\frac{dy}{dx} = 0$

**Solution:**

Given equation:

$$x = \tan\left(\frac{1}{a}\log y\right)$$

Taking $\tan^{-1}$ on both sides:

$$\tan^{-1} x = \frac{1}{a}\log y$$

$$\log y = a\tan^{-1} x$$

Differentiating both sides with respect to $x$:

$$\frac{1}{y}\frac{dy}{dx} = a \cdot \frac{1}{1+x^2}$$

$$(1+x^2)\frac{1}{y}\frac{dy}{dx} = a$$

$$(1+x^2)\frac{dy}{dx} = ay$$

Differentiating both sides again with respect to $x$ using the product rule:

$$(1+x^2)\frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot \frac{d}{dx}(1+x^2) = a\frac{dy}{dx}$$

$$(1+x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = a\frac{dy}{dx}$$

$$(1+x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} - a\frac{dy}{dx} = 0$$

$$(1+x^2)\frac{d^2y}{dx^2} + (2x-a)\frac{dy}{dx} = 0$$

**Hence Proved**

$LHS = RHS, henceproved.$

## Question 17 (iii)

If $y = e^{a\cos^{-1} x}$, $-1 < x < 1$, prove that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$.

**TYPE 68e Proving Second-Order Differential Equation Identities – Inverse Trig Exponential Function**

★ **Likely Exam Question**

**To Prove:** $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$

**Concept / Formula Used:** $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$

**Solution:**

Given equation:

$$y = e^{a\cos^{-1} x}$$

Differentiating both sides with respect to $x$:

$$\frac{dy}{dx} = e^{a\cos^{-1} x} \cdot \frac{d}{dx}(a\cos^{-1} x)$$

$$\frac{dy}{dx} = e^{a\cos^{-1} x} \cdot \left(-\frac{a}{\sqrt{1-x^2}}\right)$$

$$\frac{dy}{dx} = -\frac{ay}{\sqrt{1-x^2}}$$

Squaring and cross-multiplying:

$$(1-x^2)\left(\frac{dy}{dx}\right)^2 = a^2y^2$$

Differentiating both sides with respect to $x$:

$$(1-x^2) \cdot 2\left(\frac{dy}{dx}\right)\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 \cdot (-2x) = a^2 \cdot 2y\frac{dy}{dx}$$

Dividing the entire equation by $2\frac{dy}{dx}$ (assuming $\frac{dy}{dx} \neq 0$):

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = a^2y$$

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$$

**Hence Proved**

$LHS = RHS, hence proved.$

## Question 18

If $y = \log\left(\frac{x}{a+bx}\right)^x$, prove that $x^3\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^2$.

**TYPE 68f** **Proving Second-Order Differential Equation Identities – Logarithmic Power-Rule Identity**

**★ Likely Exam Question**

**To Prove:** $x^3\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^2$

**Concept / Formula Used:** $\log(A^B) = B\log A \qquad \log\left(\frac{A}{B}\right) = \log A - \log B$

**Solution:**

Given equation:

$$y = \log\left(\frac{x}{a+bx}\right)^x$$

Using logarithm properties to simplify the function:

$$y = x\left[\log x - \log(a+bx)\right]$$

Differentiating both sides with respect to $x$ using the product rule:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot \frac{d}{dx}\left[\log x - \log(a+bx)\right] + \left[\log x - \log(a+bx)\right] \cdot \frac{d}{dx}(x) \\ &= x\left(\frac{1}{x} - \frac{b}{a+bx}\right) + \log x - \log(a+bx) \\ &= 1 - \frac{bx}{a+bx} + \log\left(\frac{x}{a+bx}\right)\end{aligned}$$

Since $y = x\log\left(\frac{x}{a+bx}\right)$, the last term can be replaced by $\frac{y}{x}$:

$$\frac{dy}{dx} = \frac{a+bx-bx}{a+bx} + \frac{y}{x}$$

$$\frac{dy}{dx} = \frac{a}{a+bx} + \frac{y}{x}$$

Rearranging to isolate the term without $y$:

$$x\frac{dy}{dx} - y = \frac{ax}{a+bx}$$

Differentiating both sides with respect to $x$:

$$\begin{aligned}\frac{d}{dx}\left(x\frac{dy}{dx} - y\right) &= \frac{d}{dx}\left(\frac{ax}{a+bx}\right) \\ x\frac{d^2y}{dx^2} + \frac{dy}{dx}\cdot 1 - \frac{dy}{dx} &= \frac{(a+bx)(a) - ax(b)}{(a+bx)^2} \\ x\frac{d^2y}{dx^2} &= \frac{a^2+abx-abx}{(a+bx)^2} \\ x\frac{d^2y}{dx^2} &= \frac{a^2}{(a+bx)^2}\end{aligned}$$

From earlier, we had $\frac{ax}{a+bx} = x\frac{dy}{dx} - y$, which gives $\frac{a}{a+bx} = \frac{1}{x}\left(x\frac{dy}{dx} - y\right)$. Substituting this into our expression for $x\frac{d^2y}{dx^2}$:

$$x\frac{d^2y}{dx^2} = \left[\frac{1}{x}\left(x\frac{dy}{dx} - y\right)\right]^2$$

$$x\frac{d^2y}{dx^2} = \frac{1}{x^2}\left(x\frac{dy}{dx} - y\right)^2$$

Multiply both sides by $x^2$:

$$x^3\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^2$$

**Hence Proved**

$LHS = RHS, henceproved.$

## Question 19

If $\sqrt{x} + \sqrt{y} = \sqrt{a}$, find $\dfrac{d^2y}{dx^2}$ at $x = a$.

**TYPE 75 Implicit differentiation with radical substitution evaluation**

**To Find:** $\dfrac{d^2y}{dx^2}$ at $x = a$

**Concept / Formula Used:** $\dfrac{d}{dx}(x^n) = nx^{n-1}$

### Solution:

Given equation:

$$\sqrt{x} + \sqrt{y} = \sqrt{a}$$

Differentiating implicitly with respect to $x$:

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0$$

$$\frac{1}{2\sqrt{y}}\frac{dy}{dx} = -\frac{1}{2\sqrt{x}}$$

$$\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$$

To find the value of $y$ when $x = a$:

$$\sqrt{a} + \sqrt{y} = \sqrt{a} \implies \sqrt{y} = 0 \implies y = 0$$

Differentiating $\dfrac{dy}{dx} = -\dfrac{y^{1/2}}{x^{1/2}}$ a second time would require substituting $y = 0$ into a term containing $\dfrac{1}{\sqrt{y}}$, which is undefined. Since the implicit form breaks down at this point, we instead solve the original equation explicitly for $y$ near $x = a$.

$$\sqrt{y} = \sqrt{a} - \sqrt{x} \implies y = (\sqrt{a} - \sqrt{x})^2$$

Differentiating with respect to $x$:

$$\frac{dy}{dx} = 2(\sqrt{a} - \sqrt{x}) \cdot \left(-\frac{1}{2\sqrt{x}}\right) = -\frac{\sqrt{a} - \sqrt{x}}{\sqrt{x}} = 1 - \frac{\sqrt{a}}{\sqrt{x}}$$

Differentiating again with respect to $x$:

$$\frac{d^2y}{dx^2} = 0 - a \cdot \left(-\frac{1}{2}x^{-3/2}\right) = \frac{a}{2x^{3/2}}$$

Evaluating at $x = a$:

$$\left.\frac{d^2y}{dx^2}\right|_{x=a} = \frac{a}{2a^{3/2}} = \frac{1}{2\sqrt{a}}$$

**Final Answer**

$\frac{1}{2\sqrt{a}}$

## Question 20

If $x^m y^n = (x+y)^{m+n}$, then prove that: (i) $\frac{dy}{dx} = \frac{y}{x}$ (ii) $\frac{d^2y}{dx^2} = 0$

**TYPE 68g Proving Second-Order Differential Equation Identities – Homogeneous Logarithmic Power Functions**

**★ Likely Exam Question**

**To Prove:** (i) $\frac{dy}{dx} = \frac{y}{x}$ (ii) $\frac{d^2y}{dx^2} = 0$

**Concept / Formula Used:** $\log(AB) = \log A + \log B$

### Solution:

Given equation:

$$x^m y^n = (x+y)^{m+n}$$

Taking natural logarithm on both sides:

$$\log(x^m y^n) = \log((x+y)^{m+n})$$

$$m \log x + n \log y = (m+n)\log(x+y)$$

Differentiating both sides implicitly with respect to $x$:

$$\frac{m}{x} + \frac{n}{y}\frac{dy}{dx} = \frac{m+n}{x+y}\left(1 + \frac{dy}{dx}\right)$$

$$\frac{m}{x} + \frac{n}{y}\frac{dy}{dx} = \frac{m+n}{x+y} + \frac{m+n}{x+y}\frac{dy}{dx}$$

Grouping the terms containing $\frac{dy}{dx}$ on one side:

$$\left(\frac{n}{y} - \frac{m+n}{x+y}\right)\frac{dy}{dx} = \frac{m+n}{x+y} - \frac{m}{x}$$

Simplifying both sides by taking common denominators:

$$\left(\frac{n(x+y)-y(m+n)}{y(x+y)}\right)\frac{dy}{dx}=\frac{x(m+n)-m(x+y)}{x(x+y)}$$

$$\left(\frac{nx+ny-my-ny}{y(x+y)}\right)\frac{dy}{dx}=\frac{mx+nx-mx-my}{x(x+y)}$$

$$\left(\frac{nx-my}{y(x+y)}\right)\frac{dy}{dx}=\frac{nx-my}{x(x+y)}$$

Multiplying both sides by $\frac{y(x+y)}{nx-my}$ (assuming $nx-my\neq 0$):

$$\frac{dy}{dx}=\frac{y}{x}$$

This proves part (i).

Now, differentiating $\frac{dy}{dx}=\frac{y}{x}$ with respect to $x$ to find $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2}=\frac{x\cdot\frac{dy}{dx}-y\cdot 1}{x^2}$$

Substitute $\frac{dy}{dx}=\frac{y}{x}$:

$$\frac{d^2y}{dx^2}=\frac{x\left(\frac{y}{x}\right)-y}{x^2}$$

$$\frac{d^2y}{dx^2}=\frac{y-y}{x^2}=0$$

This proves part (ii).

**Hence Proved**

(i) $\frac{dy}{dx}=\frac{y}{x}$ and (ii) $\frac{d^2y}{dx^2}=0$, hence proved.

## Question 21

If $x=a(\theta-\sin\theta)$, $y=a(1+\cos\theta)$, find $\frac{d^2y}{dx^2}$.

**TYPE 64b** Parametric Second-Order Derivatives – Cycloid-Type Parameters

**★ Likely Exam Question**

**To Find:** $\frac{d^2y}{dx^2}$

**Concept / Formula Used:** $\frac{dy}{dx}=\frac{dy/d\theta}{dx/d\theta}\qquad \frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{\frac{d}{d\theta}\left(\frac{dy}{dx}\right)}{dx/d\theta}$

**Solution:**

Given parametric equations:

$$x = a(\theta - \sin\theta)$$
$$y = a(1 + \cos\theta)$$

Differentiating $x$ and $y$ with respect to $\theta$:

$$\frac{dx}{d\theta} = a(1 - \cos\theta)$$

$$\frac{dy}{d\theta} = a(0 - \sin\theta) = -a\sin\theta$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{-a\sin\theta}{a(1 - \cos\theta)} = -\frac{\sin\theta}{1 - \cos\theta}$$

Using half-angle trigonometric identities ($\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$ and $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$):

$$\frac{dy}{dx} = -\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}} = -\cot\frac{\theta}{2}$$

Now, finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{\frac{d}{d\theta}\left(-\cot\frac{\theta}{2}\right)}{dx/d\theta}$$

Using the derivative of $\cot u$: $\frac{d}{d\theta}\left(-\cot\frac{\theta}{2}\right) = -\left(-\csc^2\frac{\theta}{2}\cdot\frac{1}{2}\right) = \frac{1}{2}\csc^2\frac{\theta}{2}$. Thus:

$$\frac{d^2y}{dx^2} = \frac{\frac{1}{2}\csc^2\frac{\theta}{2}}{a(1 - \cos\theta)}$$

Since $1 - \cos\theta = 2\sin^2\frac{\theta}{2} = \frac{2}{\csc^2(\theta/2)}$:

$$\frac{d^2y}{dx^2} = \frac{\frac{1}{2}\csc^2\frac{\theta}{2}}{a\cdot 2\sin^2\frac{\theta}{2}} = \frac{1}{4a}\csc^2\frac{\theta}{2}\cdot\csc^2\frac{\theta}{2} = \frac{1}{4a}\csc^4\frac{\theta}{2}$$

**Final Answer**

$\frac{1}{4a}\csc^4\frac{\theta}{2}$

## Question 22

If $x = \log t$ and $y = \frac{1}{t}$, prove that $\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$.

**TYPE 64c Parametric Second-Order Derivatives – Logarithmic/Reciprocal Parameters**

★ **Likely Exam Question**

**To Prove:** $\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} = 0$

**Solution:**

Given parametric equations:

$$x = \log t \implies \frac{dx}{dt} = \frac{1}{t}$$
$$y = \frac{1}{t} \implies \frac{dy}{dt} = -\frac{1}{t^2}$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{-1/t^2}{1/t} = -\frac{1}{t}$$

Since $y = \frac{1}{t}$, we have $\frac{dy}{dx} = -y$. Finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(-\frac{1}{t}\right) = \frac{d}{dt}\left(-\frac{1}{t}\right) \cdot \frac{dt}{dx}$$
$$\frac{d^2y}{dx^2} = \left(\frac{1}{t^2}\right) \cdot t = \frac{1}{t}$$

Substituting $\frac{d^2y}{dx^2}$ and $\frac{dy}{dx}$:

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} = \frac{1}{t} + \left(-\frac{1}{t}\right) = 0$$

**Hence Proved**

$LHS = RHS, hence proved.$

## Question 23

If $x = a\left(\dfrac{1-t^2}{1+t^2}\right)$ and $y = \dfrac{2at}{1+t^2}$, prove that $\dfrac{dy}{dx} = -\dfrac{x}{y}$. Hence, prove that $\dfrac{d^2y}{dx^2} = -\left(\dfrac{x^2+y^2}{y^3}\right)$.

**TYPE 64d** **Parametric Second-Order Derivatives – Trigonometric Substitution Proof**

**★ Likely Exam Question**

**To Prove:** $\dfrac{dy}{dx} = -\dfrac{x}{y}$ and $\dfrac{d^2y}{dx^2} = -\left(\dfrac{x^2+y^2}{y^3}\right)$

**Solution:**

Given parametric equations:

$$x = a\left(\frac{1-t^2}{1+t^2}\right), \quad y = \frac{2at}{1+t^2}$$

Differentiating both $x$ and $y$ with respect to $t$:

$$\begin{aligned}\frac{dx}{dt} &= a \cdot \frac{(1+t^2)(-2t) - (1-t^2)(2t)}{(1+t^2)^2} \\ &= a \cdot \frac{-2t - 2t^3 - 2t + 2t^3}{(1+t^2)^2} \\ &= -\frac{4at}{(1+t^2)^2}\end{aligned}$$

$$\begin{aligned}\frac{dy}{dt} &= a \cdot \frac{(1+t^2)(2) - (2t)(2t)}{(1+t^2)^2} \\ &= a \cdot \frac{2 + 2t^2 - 4t^2}{(1+t^2)^2} \\ &= \frac{2a(1-t^2)}{(1+t^2)^2}\end{aligned}$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\frac{2a(1-t^2)}{(1+t^2)^2}}{-\frac{4at}{(1+t^2)^2}} = -\frac{2a(1-t^2)}{4at} = -\frac{1-t^2}{2t}$$

Now, consider $-\frac{x}{y}$:

$$-\frac{x}{y} = -\frac{a\left(\frac{1-t^2}{1+t^2}\right)}{\frac{2at}{1+t^2}} = -\frac{1-t^2}{2t}$$

Thus, $\frac{dy}{dx} = -\frac{x}{y}$. This proves the first part.

Now, finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(-\frac{x}{y}\right) = -\frac{y \cdot 1 - x \cdot \frac{dy}{dx}}{y^2}$$

Substitute $\frac{dy}{dx} = -\frac{x}{y}$:

$$\frac{d^2y}{dx^2} = -\frac{y - x\left(-\frac{x}{y}\right)}{y^2} = -\frac{y + \frac{x^2}{y}}{y^2} = -\frac{\frac{y^2+x^2}{y}}{y^2} = -\frac{x^2+y^2}{y^3}$$

**Hence Proved**

$\dfrac{dy}{dx} = -\dfrac{x}{y}$ and $\dfrac{d^2y}{dx^2} = -\left(\dfrac{x^2+y^2}{y^3}\right)$, hence proved.

## Question 24

If $x = a\sin pt$ and $y = b\cos pt$, find the value of $\dfrac{d^2y}{dx^2}$ at $t = 0$.

TYPE 64e Parametric Second-Order Derivatives – Evaluation at a Point

★ Likely Exam Question

**To Find:** Value of $\dfrac{d^2y}{dx^2}$ at $t = 0$

**Solution:**

Given parametric equations:

$$x = a\sin pt \implies \frac{dx}{dt} = ap\cos pt$$

$$y = b\cos pt \implies \frac{dy}{dt} = -bp\sin pt$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-bp\sin pt}{ap\cos pt} = -\frac{b}{a}\tan pt$$

Finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(-\frac{b}{a}\tan pt\right) = \frac{\frac{d}{dt}\left(-\frac{b}{a}\tan pt\right)}{dx/dt}$$

$$\frac{d}{dt}\left(-\frac{b}{a}\tan pt\right) = -\frac{b}{a}\cdot p\sec^2 pt = -\frac{bp}{a}\sec^2 pt$$

Thus:

$$\frac{d^2y}{dx^2} = \frac{-\frac{bp}{a}\sec^2 pt}{ap\cos pt} = -\frac{b}{a^2p}\cdot\frac{1}{\cos^3 pt} = -\frac{b}{a^2}\sec^3 pt$$

Evaluating at $t = 0$:

$$\left.\frac{d^2y}{dx^2}\right|_{t=0} = -\frac{b}{a^2}\sec^3(0) = -\frac{b}{a^2}(1)^3 = -\frac{b}{a^2}$$

**Final Answer**

$-\dfrac{b}{a^2}$

## Question 25

If $x = a\sec^3\theta$ and $y = a\tan^3\theta$, find $\dfrac{d^2y}{dx^2}$ at $\theta = \dfrac{\pi}{3}$.

TYPE 64f Parametric Second-Order Derivatives – Trigonometric Evaluation

★ Likely Exam Question

**To Find:** $\dfrac{d^2y}{dx^2}$ at $\theta = \dfrac{\pi}{3}$

**Solution:**

Given parametric equations:

$$x = a\sec^3\theta \implies \frac{dx}{d\theta} = 3a\sec^2\theta \cdot (\sec\theta\tan\theta) = 3a\sec^3\theta\tan\theta$$

$$y = a\tan^3\theta \implies \frac{dy}{d\theta} = 3a\tan^2\theta \cdot \sec^2\theta$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{3a\tan^2\theta\sec^2\theta}{3a\sec^3\theta\tan\theta} = \frac{\tan\theta}{\sec\theta} = \sin\theta$$

Finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(\sin\theta) = \frac{\frac{d}{d\theta}(\sin\theta)}{dx/d\theta} = \frac{\cos\theta}{3a\sec^3\theta\tan\theta} = \frac{\cos\theta\cdot\cos^3\theta}{3a\tan\theta} = \frac{\cos^4\theta}{3a\left(\frac{\sin\theta}{\cos\theta}\right)} = \frac{\cos^5\theta}{3a\sin\theta}$$

Evaluating at $\theta = \dfrac{\pi}{3}$:

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}, \quad \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$\left.\frac{d^2y}{dx^2}\right|_{\theta=\pi/3} = \frac{(1/2)^5}{3a\left(\frac{\sqrt{3}}{2}\right)} = \frac{1/32}{3\sqrt{3}a/2} = \frac{1}{32}\cdot\frac{2}{3\sqrt{3}a} = \frac{1}{48\sqrt{3}a}$$

**Final Answer**

$\dfrac{1}{48\sqrt{3}a}$

## Question 26

If $x = a(1+\cos t)$, $y = a(t+\sin t)$, find $\dfrac{d^2y}{dx^2}$ at $t = \dfrac{\pi}{2}$.

**TYPE 64g Parametric Second-Order Derivatives – Trigonometric Sum Evaluation**

★ **Likely Exam Question**

**To Find:** $\dfrac{d^2y}{dx^2}$ at $t = \dfrac{\pi}{2}$

**Solution:**

Given parametric equations:

$$x = a(1+\cos t) \implies \frac{dx}{dt} = -a\sin t$$

$$y = a(t + \sin t) \implies \frac{dy}{dt} = a(1 + \cos t)$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a(1 + \cos t)}{-a \sin t} = -\frac{1 + \cos t}{\sin t}$$

Finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(-\frac{1+\cos t}{\sin t}\right)}{dx/dt}$$

Using the quotient rule for $\frac{d}{dt}\left(-\frac{1+\cos t}{\sin t}\right)$:

$$\begin{aligned}
\frac{d}{dt}\left(-\frac{1 + \cos t}{\sin t}\right) &= -\frac{\sin t(-\sin t) - (1 + \cos t)(\cos t)}{\sin^2 t} \\
&= -\frac{-\sin^2 t - \cos t - \cos^2 t}{\sin^2 t} \\
&= -\frac{-(\sin^2 t + \cos^2 t) - \cos t}{\sin^2 t} \\
&= -\frac{-1 - \cos t}{\sin^2 t} = \frac{1 + \cos t}{\sin^2 t}
\end{aligned}$$

Thus:

$$\frac{d^2y}{dx^2} = \frac{\frac{1+\cos t}{\sin^2 t}}{-a \sin t} = -\frac{1 + \cos t}{a \sin^3 t}$$

Evaluating at $t = \frac{\pi}{2}$:

$$\cos\left(\frac{\pi}{2}\right) = 0, \quad \sin\left(\frac{\pi}{2}\right) = 1$$

$$\left.\frac{d^2y}{dx^2}\right|_{t=\pi/2} = -\frac{1 + 0}{a(1)^3} = -\frac{1}{a}$$

**Final Answer**

$-\frac{1}{a}$

## Question 27

If $x = \cos t + \log\left(\tan \frac{t}{2}\right)$, $y = \sin t$, then find $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$.

**TYPE 64h** **Parametric Second-Order Derivatives – Complex Logarithmic-Trigonometric Evaluation**

**★ Likely Exam Question**

**To Find:** $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$

**Solution:**

Given parametric equations:

$$x = \cos t + \log\left(\tan\frac{t}{2}\right)$$

Differentiating $x$ with respect to $t$:

$$\begin{aligned}\frac{dx}{dt} &= -\sin t + \frac{1}{\tan\frac{t}{2}} \cdot \sec^2\frac{t}{2} \cdot \frac{1}{2} \\ &= -\sin t + \frac{\cos\frac{t}{2}}{\sin\frac{t}{2}} \cdot \frac{1}{\cos^2\frac{t}{2}} \cdot \frac{1}{2} \\ &= -\sin t + \frac{1}{2\sin\frac{t}{2}\cos\frac{t}{2}} \\ &= -\sin t + \frac{1}{\sin t} = \frac{1-\sin^2 t}{\sin t} = \frac{\cos^2 t}{\sin t}\end{aligned}$$

Differentiating $y = \sin t$ with respect to $t$:

$$\frac{dy}{dt} = \cos t$$

Finding the first derivative $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\cos t}{\frac{\cos^2 t}{\sin t}} = \frac{\sin t}{\cos t} = \tan t$$

Finding the second derivative $\frac{d^2y}{dx^2}$:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(\tan t)}{dx/dt} = \frac{\sec^2 t}{\frac{\cos^2 t}{\sin t}} = \frac{\frac{1}{\cos^2 t}}{\frac{\cos^2 t}{\sin t}} = \frac{\sin t}{\cos^4 t} = \tan t \sec^3 t$$

Evaluating at $t = \frac{\pi}{4}$:

$$\tan\left(\frac{\pi}{4}\right) = 1, \quad \sec\left(\frac{\pi}{4}\right) = \sqrt{2}$$

$$\left.\frac{d^2y}{dx^2}\right|_{t=\pi/4} = (1)(\sqrt{2})^3 = 2\sqrt{2}$$

**Final Answer**

General result: $\dfrac{d^2y}{dx^2} = \tan t \sec^3 t$

At $t = \dfrac{\pi}{4}$: $\dfrac{d^2y}{dx^2} = 2\sqrt{2}$